# Continuum Structural Complexity: Target-Aligned Source Algebra, Physical Expenditure, and Singular Reachability

Guillem Duran Ballester
guillem@fragile.tech

**Abstract**

A public continuum presentation consists of a weak PDE or flow, its datum and boundary conditions, a rational approximation grammar, its continuation and native-operation syntax, a canonical weak test core, and the formal physical carrier identity. No target-relative estimate, compactness statement, rigidity theorem, or continuation conclusion is input. The presentation generates an Archimedean ordered cylinder algebra. Its state space contains ordinary solutions, every approximation limit, and the first graph value at which an equation, carrier, chart, trace, or covariance relation fails.

The continuum extension preserves the five source families of Part I. They are $\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}$, and $\mathsf{Bdry}$. Limits, concentrations, oscillations, defect measures, tails, traces, and blow-up germs retain the ancestry of the records that generate them. The residual is taken only after this sourcewise completion and is exactly the common kernel of the prospective target functionals. A target-visible residual source is therefore impossible. Finite-cylinder target shells carry an exact derived Stieltjes-current identity. Contact forces unit shell progress, hence a positive contribution from at least one inherited source on every sufficiently small pre-contact shell. A canonical disjoint cascade then converts finite total physical expenditure into either divergent source price, which excludes contact, or a zero-price target-aligned successor path.

A distinguished nonreversible subfamily applies a Wick transform to the self-adjoint $\mathsf{Abs}$ quotient field of an existing Part I compensated record and leaves its real $\mathsf{Geom}$ and authorized $\mathsf{Caus}$ fields unchanged. The transformed causal bridge produces an exact Feshbach self-energy, a Mori–Zwanzig memory kernel, and a completed-square structural action. Its positive response kernel transfers the effective-dimension, Rademacher, PAC–Bayes, sequential, control, and continuum-learning bounds of Part I. Empirical trace terms use the finite Part I evaluation carriers; the population effective dimension is the extended supremum of those finite compressions. Structure-preserving likelihood, inverse-response kernel ridge, and action-safe fitted Bellman/Gibbs losses train the dynamics, unrestricted bridge, and RL branches, respectively. The inverse-response norm is the exact minimum energy of a latent response lift. The Part I exact-cell Rademacher union and learned-operator cover, or the cell-mixture PAC–Bayes divergence and coefficient-selection charge, select the model family without an additional simplicity criterion. The learner restricts the Part I saturated usefulness profile to records at a requested risk tolerance, then returns the Pareto-minimal points of its strict inverse at the inherited visibility threshold. Minimum-capacity covering readout remains an inner coordinate within each fixed geometry; it does not replace saturated-profile selection.

The remaining path problem is evaluated by an explicit ordered calculus. Every identity is first restricted to the same-origin target/source stratum that invoked it and decomposed into signed five-channel remainders, carrier coboundary, admissible physical production, exterior input, target-null current, and total-graph defects. No equation-wide identity has an independent reachability consequence. For each source support and rational target floor, a quadratic

module contains the weak equation, signed carrier balance, total graph relations, same-origin successor relations, nonnegative physical production, and shift coboundaries. Exactly one algebraic alternative holds: either a finite rational identity represents $-1$ in the ordered cone and excludes that target-aligned cell, or an Archimedean separation theorem produces a positive shift-invariant functional satisfying all retained relations. A nested cylinder-moment hierarchy enumerates the finite identities and reconstructs the positive functional from compatible finite moments. Hamiltonian resolvents, signed carrier cohomology, entropy cocycles, commutator squares, and compactness–rigidity formulae are consequence generators inside this calculus, not additional hypotheses. A complementary positive functional is an equality successor: it re-enters the target-cyclic compiler generated by the selected shell covector. The compiler restricts every operation to the same-origin target cell. Projection-first saturation proves that quotienting by the common target kernel commutes with every constructor; a failed descent becomes a projected graph defect. The resulting target-cyclic spine assembles one feeding–storage current before a canonical zero/signed-dyadic/graph-boundary polarity split. A new coordinate increases resolved-prefix rank; an equivalent positive cell is followed by its induced first-return law. On each recurrent cell, a primal–dual circulation hierarchy computes the sharp physical price of unit target return. Positive price produces a carrier-potential identity; zero price reconstructs the invariant circulation that generates the next target-cyclic refinement. Adjoint transport, constraint descent, defect polarization, commutator production, and covariant scale valuation are internal operations of this one queue. Its only outputs are divergent physical charge, target-nullity, or a closed reduced public equation with a same-origin path that contacts the continuation target. A formal model, equality kernel, failed analytic property, or equation-wide estimate is an internal coordinate rather than a terminal.

# Contents

# Introduction

Let $u$ be an ordinary maximal branch of a fixed continuum evolution and let $\Sigma$ be a target in its state or germ space. The central question is whether the equation can reach $\Sigma$ while spending only a finite amount of the physical quantity controlled by its evolution law. Each orbit carries one nonnegative countably additive expenditure measure $\mu_u$ on a generated event space $\mathfrak{Q}_u$. The finite-budget stratum is

$$\mu_u(\mathfrak{Q}_u) \le B_{\mathrm{phys}} < \infty.$$

The public carrier identity selects, among action, dissipation, entropy production, forcing work, and boundary flux terms, the mutually compatible positive variations that constitute this one measure. Signed internal currents are glued first, and only carrier-dominated target-local production may spend it. Energy and conservation identities then bound its total mass before the ceiling $B_{\mathrm{phys}}$ is set.

The source alphabet, ancestry discipline, saturation rule, and target-relative residual extend the structural algebra of Part I [2]. The continuum construction below derives its physical budget, source-preserving limit closure, prospective contact identity, signed carrier calculus, successor recurrence, Hamiltonian formulation, and saturation equations within this paper. The remaining citations identify standard analytic results used at explicitly named steps. These references and Part I constitute the complete logical dependency set.

The structural source alphabet records how target-directed behavior is generated. The native dissipative or reversible mechanism is $\mathsf{Geom}$; the fixed directed or nonreversible transport is $\mathsf{Caus}$; quotient and symmetry reduction are $\mathsf{Abs}$; charts, renormalizations, and augmented state variables are $\mathsf{Lift}$; and boundary, terminal, killing, and exterior-flux records are $\mathsf{Bdry}$. A composite continuum object may carry several labels, but every elementary source belongs to this alphabet.

The completion is sourcewise. A limit object is placed in the closure of the cell that generated it. Thus a concentration measure retains the ancestry of the concentrating quantity, a blow-up profile retains the ancestry of the physical orbit together with its $\mathsf{Abs}/\mathsf{Lift}$ chart, and a terminal tail trace acquires $\mathsf{Bdry}$ ancestry. Every source cell has an exact structural row value; its complementary value remains in the same cell and does not create an additional continuum source.

The target-relative residual is correspondingly strict. Let $\mathscr{N}_\Sigma$ be the common kernel of the target shell derivatives, target fluxes, and other declared prospective contact functionals. The residual coordinate is contained in $\mathscr{N}_\Sigma$. Hence

$$\boxed{\text{target-visible} \Longrightarrow \text{ancestry in } \{\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}\}; \qquad J_{\mathrm{res}} \Longrightarrow \text{target-null.}}$$

Every target-visible limit therefore retains ancestry in one of the five source families.

The central target-relative question is therefore not whether the entire solution is large, but which source current contributes positively to the formation of the singularity. For a normalized shell crossing $I_r$, set

$$Q_{c,r} := \int_{I_r} \langle D\Phi_r(u_t), J_t^c \rangle \, dt.$$

The compiled current identity gives the exact pre-contact identity on the ordinary shell branch

$$1 = \sum_{c \in \mathfrak{C}} Q_{c,r}, \qquad \sum_{c \in \mathfrak{C}} (Q_{c,r})_+ \geq 1, \qquad \max_{c \in \mathfrak{C}} (Q_{c,r})_+ \geq \frac{1}{5}.$$

The residual source term vanishes. Thus contact itself supplies the unit progress and proves the existence of target-aligned structure. If all source currents vanish in the target-active quotient, their magnitude is irrelevant: every shell observable is constant along the orbit, so contact is excluded without compactness, a physical-price gap, or a global estimate.

The aligned branch is simplified before any size estimate is attempted. A target-null or gauge direction is passed to the generated quotient and never reconsidered. Projection-first saturation generates every descendant inside the target-cyclic spine and retains each failed descent as a graph coordinate. For every surviving total-graph value the algebra forms its positive target-current measure, compact activity lift, and witness-indexed successors. Every subsequent analytic identity is restricted to that cell and compiled into (102) before it can close or charge the branch. The topological sets $Z_{n+1} = \mathcal{V}_\eta(Z_n)$ describe whether an occurrence can sustain one more target-retaining zero-price step. Proof is supplied separately by the ordered cone: a checked finite identity $-1 \in \mathcal{K}_{\mathfrak{b}}$ excludes the active cell, while the complementary order-unit functional is a stationary equality successor satisfying every generated relation at that depth. Its first unresolved local coordinate is split into zero, signed dyadic, and graph-boundary cells; finite local identities prune target-null cells, while positive-moment cells re-enter source selection. New coordinates increase resolved-prefix rank; recurrent equivalent cells use their induced return maps. The target-aligned recurrent closure compiler performs all of these transitions internally and returns only divergent physical charge, a target-null path, or an explicit same-origin singular mechanism. Semantic emptiness is never used as a proof object.

The main expenditure argument then uses the same measure $\mu_u$. A singular contact compiler produces infinitely many pairwise disjoint pre-contact events and selects a persistent aligned source cell. A divergent-price value contradicts countable additivity and $\mu_u(\mathfrak{Q}_u) \leq B_{\mathrm{phys}}$. A finite-price value produces vanishing-price windows and activates the ordered successor calculus. It is therefore a reduction step, not a terminal singularity class.

$$\boxed{\begin{gathered} \text{contact} \xRightarrow{\text{normalized shell}} \text{target-aligned source}, \\ \text{target-aligned source} \xRightarrow{\text{one physical measure}} \begin{cases} \text{divergent expenditure}, \\ \text{zero-price successor path}, \end{cases} \\ \text{first branch contradicts } B_{\mathrm{phys}}; \qquad \text{second branch is pruned by } Z_{n+1} = \mathcal{V}_\eta(Z_n). \end{gathered}}$$

Hamiltonian, resolvent, commutator, constraint-tail, scale, and entropy identities are functorial refinements generated when their typed expressions occur in the public equation. The symmetric form yields a spectral Hamiltonian and resistance; the full fixed-Caus generator retains its nonnormal current; signed carrier gluing cancels internal transport and exposes scale-feed cohomology; and a covariant entropy cocycle telescopes production across changing charts. A failed identity is lifted as a successor with the same ancestry. These modules operate on the five source families; they neither enlarge the public presentation nor alter the source algebra.

The Wick-rotated compensated-quotient module treats a self-adjoint $\mathsf{Abs}$ generator in spectral time and a real $\mathsf{Geom}$ generator in dissipative time. The actual selected $\mathsf{Caus}$ bridge determines both the Feshbach response and the structural action. The resulting positive response kernel imports the learning theory of Part I without diagonalizing a nonnormal generator, while free, hybrid, and imposed authority remain distinct optimization domains. The unrestricted branch optimizes spectral kernel ridge risk on the zero-waste bridge fiber; the RL branch uses a frozen Bellman target, safe response compression, and a Gibbs actor calibrated to the same action-valued critic.

**Scope discipline.** The paper uses exhaustive quantification only to construct the closed five-source algebra: enumeration of tests, total graphs, sourcewise closures, and the common target kernel. Such a construction statement has no reachability conclusion. An analytic identity becomes a reachability fact only after contact has selected a same-origin active branch $\mathfrak{b}$, the identity has been restricted to $X_{\mathfrak{b}}$, and its terms have been placed in the target-local normal form (102). A physical consequence additionally requires the charging rule (90). Thus full-algebra quantifiers provide source exhaustion, whereas every exclusion, price, successor, and continuation inference is target-local and source-resolved. This distinction is formalized in Theorem II.2.18 and Corollary II.2.20.

## Visual architecture and proof-flow guide

The twenty-two diagrams are maps of proved implications and invariant interfaces. Blue nodes contain evolution data, violet nodes contain ancestry or coordinate transformations, teal nodes contain geometric or compactness data, amber nodes contain target uses or decisions, coral nodes contain reachability outputs, and gray nodes contain target-null classes or invariant branches. Solid arrows denote implications or constructions. Dashed arrows denote comparisons or derived evaluator routes.

The unique root of the reachability graph is the public continuum presentation. Rectangles are generated constructions, diamonds are exact structural tests, and ellipses are completed outputs. Every nonzero target-active value has an outgoing edge to the activity lift or successor relation. All locality, source labels, and physical balances established upstream remain active downstream. Node [6] forms the total shell identity, and node [17] performs finite-depth viability pruning.

### Diagram map

| No. | DAG node | Diagram | Principal internal sources |
|---|---|---|---|
| 1 | [1]–[18] | Constructive target-reachability proof flow | Proposition I.1.7, Definition II.2.39, and Theorem II.2.47 |
| 2 | [1]–[2] | Public PDE to physical orbit records | Definition I.1.1 and Proposition I.1.7 |
| 3 | [2] | Equation-generated physical expenditure | Definition I.1.12 and Proposition I.1.15 |

| No. | DAG node | Diagram | Principal internal sources |
|---|---|---|---|
| 4 | [2] | Five continuum source roles | Definition I.2.5 and Theorems I.2.6 and I.2.8 |
| 5 | [3] | Sourcewise orbit completion | Definition I.3.2 and Theorem I.3.7 |
| 6 | [4] | Limit ancestry and propagated reduction | Theorems I.3.4 and II.1.3 |
| 7 | [5] | Target-visible re-entry and null residual | Definition I.4.2 and Theorems I.4.6 and I.4.7 |
| 8 | [6]–[7] | Normalized shell-crossing accountability | Theorems I.5.2 and I.5.14 |
| 9 | [6]–[7] | Resolvent source-response factorization | Theorem I.5.5 |
| 10 | [6]–[7] | Stochastic no-spontaneous-contact factorization | Theorems I.5.7 and I.5.14 |
| 11 | [1], [9]–[10] | Carrier-to-physical-budget compiler | Theorems II.3.5, II.7.4 and II.7.6 |
| 12 | [4] | Three Hamiltonian meanings | Proposition II.4.2 |
| 13 | [4] | Same-current chart covariance | Theorem II.4.3 and Proposition II.4.4 |
| 14 | [6], [9] | Killed arrival, resolvent, and resistance | Proposition II.5.3 and Theorems II.5.4 to II.5.6 |
| 15 | [4], [6] | Full nonnormal target response | Definition II.6.1 and Proposition II.6.2 |
| 16 | [6]–[9] | Physical aiming geometry | Theorem II.6.4 and Proposition II.6.5 |
| 17 | [4], [14]–[17] | Branch-to-successor compiler | Definitions II.1.8 and II.2.39 and Theorem II.1.16 |
| 18 | [1], [9]–[10] | Covariant entropy-cocycle production | Definition II.10.1 and Theorem II.10.2 |
| 19 | [6]–[18] | Target-aligned scale-cascade and viability ledger | Proposition III.1.4, Theorems I.4.7, II.2.21, II.2.24, III.4.28, III.4.31, III.4.32 and III.4.34, Definitions III.4.23, III.4.24 and III.4.30, and Corollary III.4.35 |
| 20 | [1]–[18] | Constructive structural regularity compiler | Theorems I.4.7, II.2.2, II.2.29, II.2.41, II.2.47, III.4.5, III.4.6, III.4.11, III.4.17, III.4.18, III.4.20, III.4.22, III.4.28, III.4.31, III.4.32, III.4.34 and III.4.36, Propositions III.4.25 and II.2.26, Definitions III.4.1, III.4.3, III.4.4, III.4.23, III.4.24, III.4.27, III.4.29 and III.4.30, and Corollaries III.4.7 and III.4.35 |
| 21 | [4], [6], [9] | Wick-rotated compensated learning | Theorems II.11.7, II.11.12, II.11.22, II.11.24, II.11.31, II.11.35 and II.11.37 and Propositions II.11.15 and II.11.30 |

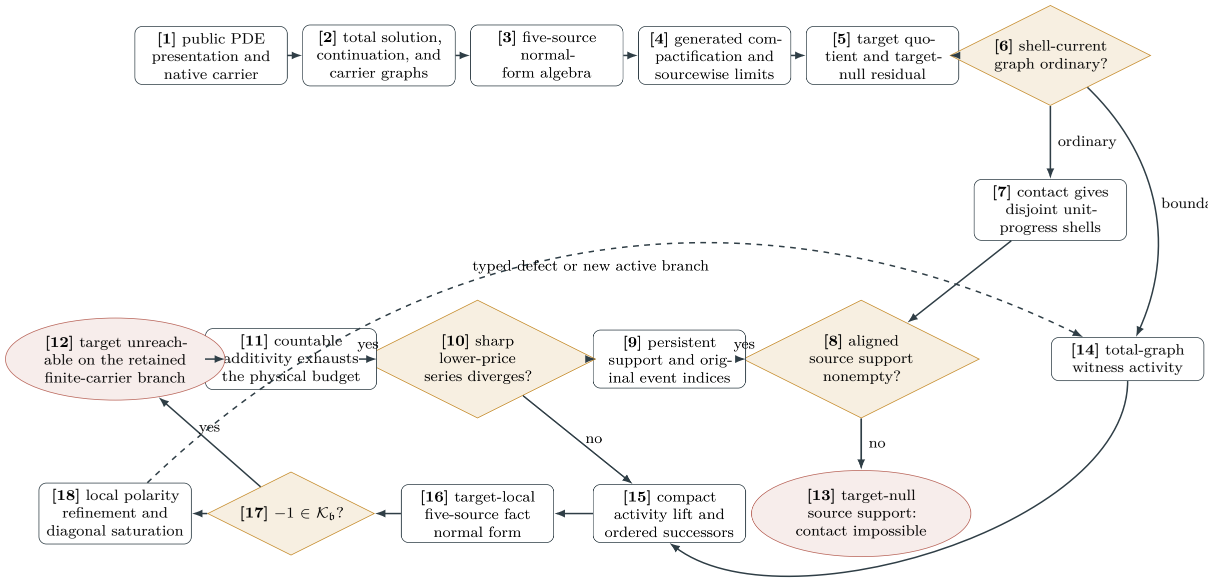


Figure 1: Target-reachability proof DAG. Public textbook data enter only at [1]. Nodes [2]–[6] generate the total graphs, source algebra, compactification, target quotient, and total shell graph. Its ordinary value gives unit progress at [7], while every boundary value enters [14]. Thus [8] forces aligned support. Nodes [9]–[11] close the divergent-price branch. Every ordinary or graph-boundary value also enters [14]; finite price joins it at [15]. Every invoked relation is restricted and source-factorized at [16] before it feeds the ordered cone at [17]. A checked finite identity gives [12]; the complementary order-unit functional gives the zero-price resaturation state at [18]. Target-local polarity splitting, target-cyclic adjoint generation, covariant valuation, defect polarization, and prospective reselection return it to [14] with strictly larger resolved-prefix rank whenever a target-active cell remains. A cofinal refinement is evaluated by diagonal target-aligned saturation. Only a closed same-origin realization of every generated relation is an actual singular mechanism. No semantic compact-set assertion is used as an exclusion proof, and no live node is an unexpanded analytic label.

| No. | DAG node | Diagram | Principal internal sources |
|---|---|---|---|
| 22 | [7]–[18] | Navier–Stokes specialization of the covariant microscope | Propositions III.5.2, III.5.3, III.5.7, III.5.9, III.5.11, III.5.13, III.5.19 and III.5.32, Theorems III.5.14, III.5.20, III.5.22, III.5.23, III.5.25, III.5.28, III.5.33 and III.5.36, Lemma III.5.16, Definitions III.5.8, III.5.10 and III.5.27, and Corollaries III.5.24, III.5.26 and III.5.29 |

## Proof-dependency diagram

## Node-by-node source audit

| Node | State or test | Exact source | Successor |
|---|---|---|---|
| [1] | Public PDE, datum class, test core, operations, and native formal carrier | Definition I.1.1 and Proposition I.1.7 | [2] |
| [2] | Total solution, continuation, and signed carrier graphs | Definitions I.1.1 and I.1.12 | [3] |
| [3] | Five-source normal-form algebra | Definition I.2.1 and Theorem I.4.5 | [4] |
| [4] | Generated evaluation compactification and sourcewise completion | Definition I.3.1 and Theorem I.3.4 | [5] |
| [5] | Target-active quotient and null residual | Definition I.4.2 and Theorem I.4.6 | [6] |
| [6] | Ordinary/boundary split of the canonical total shell-current relation | Definition I.5.1 and Theorem III.2.3 | [7] or [14] |
| [7] | Contact-generated disjoint unit-progress shell family | Theorems I.5.2 and III.1.5 | [8] |
| [8] | Singularity-formation support test | Theorem I.5.14 and Proposition III.1.4 | [9] or [13] |
| [9] | Persistent source support with original disjoint event indices | Proposition III.1.10 and Lemma II.2.33 | [10] |
| [10] | Canonical sharp lower-price series | Definition III.1.11 and Theorem III.1.7 | [11] or [15] |
| [11] | Divergent series exhausts the finite physical measure | Theorems III.1.12 and III.1.14 | [12] |
| [12] | Target unreachable on the retained ordinary finite-carrier branch | Theorems II.2.47 and III.2.11 | completed output |
| [13] | Every structural source current is target-null | Corollary I.5.3 | completed output |
| [14] | Positive target activity of every total-graph witness | Definitions III.2.2 and II.2.30 | [15] |
| [15] | Compact activity lift and witness-indexed successor relation | Definitions II.2.30 and II.2.32 and Theorem II.2.31 | [16] |
| [16] | Target-local five-source fact normal form and generated nonnegative zero-price descendants | Definitions II.2.17 and II.2.42, Theorems II.2.18 and II.2.43, and Corollary II.2.20 | [17] |
| [17] | Finite-depth viability sets and canonical rational-cylinder pruning transcript | Definition II.2.39 and Theorems II.2.41 and III.2.8 | [12] or [18] |

| Node | State or test | Exact source | Successor |
|---|---|---|---|
| [18] | Target-local polarity splitting, resolved-prefix refinement, target-cyclic covector generation, covariant production valuation, prospective reselection, diagonal saturation, and same-origin realization | Theorems II.2.15, II.2.16, II.2.29, II.2.34, II.2.47, III.4.11, III.4.18, III.4.20, III.4.22 and III.4.36 and Definitions III.4.1, II.2.12 to III.4.13 and III.4.15 | [14], [12], or an explicit singular mechanism |

### Monotone retained-fact table

| Retained fact | Persistence rule |
|---|---|
| Physical orbit | Every normalization, window, limit, and prospective functional is attached to the same ordinary maximal branch or to a witnessed boundary sequence from that branch. |
| Physical budget | Restrictions to subintervals and pairwise disjoint event families use one nonnegative expenditure measure and cannot increase its total mass. |
| Source ancestry | Algebraic operations take unions of source words; sourcewise limits remain in the same closed cell. |
| Structural reduction | A closed target-null or otherwise excluded direction is propagated through every zero descent-defect quotient and is never re-estimated downstream. |
| Caus authority | Charts transport an imposed current without substitution. Free and hybrid envelopes optimize only over their declared represented control families, retaining the selected current and its cost. |
| Target residual | The residual remains in the common target kernel under closure; a nonzero target pairing triggers re-entry into a source cell. |
| Target alignment | Every normalized contact shell has unit total progress, so at least one source pairing is at least 1/5; target-null motion may be arbitrarily large without entering later estimates. |
| Source evaluation | Each persistent source cell initializes one target-aligned current queue. Zero kernels, signed-dyadic returns, and graph boundaries are internal transitions; the queue returns divergent physical expenditure, target-nullity, or a realized same-origin singular path. |
| Locality | Compactness, chain rules, lower semicontinuity, and price relations are evaluated on the generated windows and evaluation topology; failures are typed graph defects. |
| Invariant status | Every row returns its closing or complementary value; its source label remains fixed. |

### Branch-closure audit

| Route | Branch | Constructive source | Output |
|---|---|---|---|
| [6]–[14]–[15] | an interface has a target-visible graph-boundary value | Theorem III.2.3 and Definition II.2.30 | compact activity profile with witnessed successor |
| [8]–[13] | every compiled source pairing is target-null | Corollary I.5.3 | target unreachable directly |
| [10]–[15]–[16]–[17] | the persistent aligned source has finite sharp lower-price series | Definitions III.1.11 and II.2.39 and Theorem II.2.46 | finite target-local exclusion or an equality successor |
| [10]–[11]–[12] | the persistent aligned source has divergent sharp lower-price series | Theorems III.1.14 and III.2.11 | target unreachable |
| [17]–[12] | $-1 \in \mathcal{K}_{\mathfrak{b}}$ for every reachable active branch | Theorems II.2.47 and III.2.8 | target unreachable on the retained ordinary finite-carrier branch |
| [17]–[18] | a reachable active branch has no finite exclusion at the current depth | Theorems II.2.36 and III.4.34 and Definition III.4.23 | internal queue transition or realized same-origin singular evolution |

# Part I
# Closed Continuum Structural Algebra

## I.1 Physical continuum presentations

**Definition I.1.1** (Canonical physical continuum presentation)**.** A canonical physical continuum presentation is the public tuple

$$\mathfrak{P}_{\mathrm{phys}} = (\mathsf{Eq}, u_0, \mathcal{D}, \mathsf{Appr}, \mathsf{Cont}, \mathsf{Op}, \mathcal{K})$$

with the following data, all transcribed directly from a public textbook or standard reference statement of the initial–boundary value problem. A nonliteral entry below is admissible only when it is target-blind and already part of the standard local theory of the displayed equation; establishing a new estimate or theorem is not part of the specialization datum.

(P1) $\mathsf{Eq}$ is the displayed differential equation with its domain, coefficients, boundary conditions, and declared regularity class; $u_0$ is its datum; and $\mathcal{D} = \{\varphi_n\}_{n\geq 1}$ is the canonical rational smooth test core. A standard local solution theorem may identify an ordinary component of the generated solution graph, but it is not needed to define that graph. Any local theorem used for this identification is target-blind in the sense of Definition I.1.8.

(P2) The *total local-solution graph* contains every maximal path $u : [0, T_*(u)) \to X_{\mathrm{reg}}$ satisfying $\mathsf{Eq}$ against every $\varphi_n$, with the prescribed datum and physical boundary data. If the fiber over $u_0$ is empty, its value is the explicit record $\perp_{\mathrm{loc}}$. Nonuniqueness retains all maximal paths. Consequently an empty solution fiber is a form defect and can never be read as target avoidance. The approximation graph below independently tests whether the ordinary fiber is nonempty.

(P3) $\mathsf{Appr}$ is the least countable grammar generated by rational Galerkin projections, rational mollifiers, rational implicit time steps, rational domain exhaustions, and the typed gauge or boundary operations in $\mathsf{Op}$. A grammar word is retained with its finite-dimensional or finite-step residual relation against the first finitely many tests and with the signed carrier identity obtained by applying the same word to $\mathcal{K}$. Every compatible grammar word is enumerated; no convergence property or preferred scheme belongs to the presentation.

(P4) $\mathsf{Cont}$ is the syntactic extension relation in the declared regularity class: two local paths are related when their restrictions agree. No continuation theorem is supplied. Its ordinary values are actual regular extensions and its graph-boundary values are terminal germs. The singular target is generated from this graph by Definition III.1.1; it is not an independently chosen favorable subset.

(P5) $\mathsf{Op} = \{O_n\}_{n \geq 1}$ is a countable typed list of the restriction, rational truncation, quotient, chart, augmentation, trace, and boundary operations syntactically available for $\mathsf{Eq}$. An operation acts as a symmetry or gauge only on the ordinary value of its total covariance graph, where it preserves the weak equation, the carrier, and the continuation target. Thus the generated gauge family is the equalizer of these three actions; it is not declared independently.

(P6)

$$\mathfrak{C} = \{\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}\}$$

is the inherited source alphabet of Definition I.2.1. Native terms are routed by the sector/gradient compiler and subsequent orbit-record operations acquire their source words by the constructor grammar; this is proved in Theorem I.4.5.

(P7) $\mathcal{K}$ is the native textbook energy, action, or entropy density displayed with the equation before the continuation target is generated. It is not selected after a singular scenario has been inspected. Only its formal multiplier identity—obtained by testing $\mathsf{Eq}$, integration by parts, and cancellation of equal internal faces—belongs to the public presentation. Testing $\mathsf{Eq}$, summing with its native signs, and cancelling common internal faces generates a total signed carrier relation. Positivity, a lower carrier bound, finite exterior input, and finite total expenditure are values of that relation, not presentation data.

(P8) Let $\mathcal{A}_{\mathbb{Q}}(\mathfrak{P})$ be the unital commutative rational cylinder algebra generated by bounded truncations of all weak pairings, continuation coordinates, approximation residuals, carrier coordinates, and input–output coordinates of the typed operations in (P1)–(P7). For each typed operation it also contains graph-defect, oscillation, concentration, and exterior-tail coordinates. Let $M_{\mathfrak{P}}$ be the quadratic module generated by squares, the bounds $1 - e_n^2$, nonnegative defect masses, carrier-production atoms, and the positive and negative forms of every exact rational cylinder identity. The graph-complete state space is

$$X_{\mathrm{gc}} := \{L : \mathcal{A}_{\mathbb{Q}}(\mathfrak{P}) \to \mathbb{R} : L(1) = 1, \quad L(M_{\mathfrak{P}}) \subseteq [0, \infty)\}. \tag{1}$$

It carries pointwise convergence on the countable cylinder algebra. Ordinary states and germs embed by evaluation. The notation $X_{\mathrm{ev}}$ below means $X_{\mathrm{gc}}$, and

$$d_{\mathrm{ev}}(L_1, L_2) := \sum_{n \geq 1} 2^{-n} |L_1(e_n) - L_2(e_n)| \tag{2}$$

for a fixed enumeration of normalized generators and rational products.
The presentation is therefore fixed by public equation data and a formal multiplier calculation. The specialization step consists only of recording these statements and their public references; it contains no new mathematical argument. It may not contain a compactness theorem, coercive estimate, price lower bound, Liouville statement, target-exclusion lemma, special blow-up theorem, or exact invariant value. Such an insertion changes the equation signature and is rejected by the compiler.

**Theorem I.1.2** (Graph-complete compactness and separation)**.** *The module $M_{\mathfrak{P}}$ is Archimedean. Its state space $X_{\mathrm{gc}}$ is compact and metrizable. The cylinder coordinates separate its points, and every ordinary state, germ, orbit segment, or approximation record defines an evaluation state in $X_{\mathrm{gc}}$. If $X_{\mathrm{gc}} = \varnothing$, then $-1 \in M_{\mathfrak{P}}$; this finite algebraic inconsistency is the total form-defect value.*

*Proof.* Every cylinder polynomial contains only finitely many normalized generators. The relations $1 - e_n^2 \in M_{\mathfrak{P}}$, together with closure of a quadratic module under addition and multiplication by squares, give $N \pm a \in M_{\mathfrak{P}}$ for some rational $N$ and every $a \in \mathcal{A}_{\mathbb{Q}}(\mathfrak{P})$. Thus the module is Archimedean. Positivity gives $|L(a)| \leq N$ whenever $N \pm a \in M_{\mathfrak{P}}$. The state space is therefore a closed subset of a countable product of compact intervals; it is compact and metrizable. Pointwise equality on the cylinder algebra is equality of states. Ordinary records satisfy every defining graph and positivity relation, so evaluation is positive and normalized.

For the final assertion, the Archimedean separation theorem for quadratic modules says that a proper Archimedean module admits a normalized positive linear functional [15]. Hence an empty state space forces $M_{\mathfrak{P}}$ to be improper, equivalently $-1 \in M_{\mathfrak{P}}$. Membership uses one finite algebraic expression and hence only finitely many presentation relations. ☐

**Definition I.1.3** (Exhaustive approximation tower)**.** An approximation record is a tuple

$$a = (g, N, h, \mathsf{r}_a, \mathsf{k}_a),$$

where $g$ is a word of $\mathsf{Appr}$, $N$ is the number of tested weak coordinates, $h \in \mathbb{Q}_+$ is its resolution, $\mathsf{r}_a$ is the vector of equation and operation-graph residuals, and $\mathsf{k}_a$ is its exact signed carrier record. Write $a' \preceq a$ when $a'$ is obtained from $a$ by forgetting tests, coarsening resolution, or restricting its time or spatial domain. The exhaustive approximation tower is the countable inverse graph

$$\mathfrak{A}(\mathfrak{P}) := \{a : a \text{ is generated by } \mathsf{Appr}\}$$

with all maps $a \mapsto a'$ for $a' \preceq a$. A compatible tower path is *zero defect* when every fixed residual coordinate tends to zero and the signed carrier coordinates converge in the graph-complete topology. For each enumerated normalized residual $r_k \in [-1, 1]$, the grammar also contains the monotone tail coordinate

$$d_{k,N} := \sup\{|r_k(a)| : a \text{ occurs after refinement level } N\}, \qquad d_k := \inf_N d_{k,N}.$$

Suprema are totalized through bounded rational upper-envelope coordinates. Thus $d_k = \limsup |r_k|$, and failure of convergence to zero is a nonzero closed graph coordinate rather than an unrecorded oscillation.

**Theorem I.1.4** (Approximation realization and first failure)**.** *Every zero-defect compatible tower path has a graph-complete limit satisfying the public weak equation and native carrier relation in every rational test coordinate. Conversely, every ordinary weak solution embeds in $X_{\mathrm{gc}}$ by evaluation and is a zero-defect value of the total weak solution graph. For any compatible tower path which is not zero defect, there is a least operation index whose residual, oscillation, concentration, tail, or carrier-defect coordinate is nonzero. That coordinate is the first-failure record of the path.*

*Proof.* Compactness gives a convergent subnet, and metrizability gives a convergent subsequence. Fix a rational test coordinate. Compatibility makes that coordinate present on every sufficiently refined record, while zero defect makes its residual tend to zero. Pointwise convergence of states therefore passes the tested weak equation and the signed carrier relation to the limit. Here satisfaction of the public weak equation means satisfaction on the declared separating rational test core. When the conventional formulation uses all smooth tests, the approximation grammar also records the standard distribution-order seminorm bound; on its ordinary graph that bound extends the identity by density, while failure of the bound is its typed form coordinate. An ordinary weak solution satisfies these relations directly, so its evaluation state lies on the zero-defect total graph.

The tail-limsup failure coordinates are countable and nonnegative. If the path is not zero defect, at least one $d_k$ is nonzero. The fixed enumeration therefore has a least nonzero coordinate. Its typed graph already contains its input records and constructor ancestry, which makes it a first-failure record rather than an untyped remainder. □

**Definition I.1.5** (Continuum extension functor)**.** Let $\mathfrak{C}$ send a finite cylinder record to its evaluation state, a finite current to its distributional cylinder pairings, a finite target probe to its pullback on $X_{\mathrm{gc}}$, and a constructor word to the closure of its typed graph. It sends a convergent family to its sourcewise graph-complete closure and sends stopping, restriction, signed addition, composition, and physical-ledger addition to the corresponding closed graph operations.

**Theorem I.1.6** (Part-I preservation under continuum completion)**.** *The functor $\mathfrak{C}$ preserves the five source tags, constructor composition, signed sums, restriction, first-contact stopping, and additive physical ledgers. Quotient, chart, and terminal operations append only* Abs*,* Lift*, and* Bdry*, respectively. For every cylinder record $R$,*

$$\operatorname{Anc}(\mathfrak{C}R) = \operatorname{cl}_{\mathrm{src}} \operatorname{Anc}(R), \qquad \Pi_\Sigma \mathfrak{C}R = \mathfrak{C}(\Pi_\Sigma R). \tag{3}$$

*Consequently sourcewise limit closure creates no new primitive source and the target-null kernel commutes with continuum completion. On a finite state space, $\mathfrak{C}$ is the identity and prospective contact accountability reduces to the Part-I first-hit identity.*

*Proof.* Each assertion is first checked on an atomic constructor. Native currents retain their assigned source. A quotient changes only provenance and appends Abs; a chart or augmentation appends Lift; a terminal restriction or exterior trace appends Bdry. The graph of a composite is the composition of the typed graphs, so induction on constructor length proves the assertions for finite records. Sourcewise closure is taken separately inside each tagged graph, proving the first identity in (3). Target probes are cylinder functionals and therefore commute with pointwise graph closure, proving the second identity. Their common kernel is consequently closed under $\mathfrak{C}$. Finite state spaces have no additional analytic boundary: their evaluation image is already closed, so the construction returns the original finite algebra and its first-hit decomposition. □

**Proposition I.1.7** (Public-presentation input contract)**.** *After* Eq*, its public datum class, and its native formal carrier identity are fixed, every object used by the target-reachability track* *is generated functorially: the test core by rational enumeration, the solution and continuation graphs by satisfaction of the weak relation, the operation list by the typed syntax, the topology by bounded evaluation coordinates, the target by failed continuation, the source words by normal-form compilation, and the physical measure by the positive Jordan part of the signed carrier identity. A PDE specialization therefore supplies no additional problem-specific regularity input, proof object, favorable constant, or target-relative lemma.*

*Proof.* Each construction is the one stated in items (P1)–(P8). All subsequent objects are defined by graph closure, target quotient, Jordan decomposition, normalized shell formation, compact activity lifting, or the successor operator. These operations consume only the preceding generated objects. Inspection of the dependency chain shows that none has an input slot for any of the forbidden conclusions listed above. □

**Definition I.1.8** (Target-blind public input)**.** Let $\mathsf{Pre}(\mathsf{Eq})$ be the signature formed before the continuation target, target shells, normalized bad windows, source quotients, or physical-price strata are constructed. It contains the displayed initial–boundary value problem, its declared local regularity class, weak test core, optional standard local solution theorem, native algebraic symmetries and scalings, and formal multiplier identities. It may use a fixed universal library of target-independent results such as distribution theory, integration by parts, Sobolev-space definitions, Jordan decomposition, compactness of countable products of compact metric spaces, weak compactness of probability measures on such spaces, Archimedean ordered-algebra separation, and the spectral theorem.

A fact is an admissible public input precisely when it is expressible in $\mathsf{Pre}(\mathsf{Eq})$ and remains unchanged if the continuation target and all target observables are deleted. In particular, the following are not public inputs:

$$\begin{gathered}\text{compactness of a target-approaching sequence, an epsilon-regularity or continuation conclusion,}\\ \text{a target price gap, a blow-up Liouville theorem, absence of a scale-soliton,}\\ \text{vanishing of a target quotient, or any decomposition selected after singular shells are formed.}\end{gathered} \tag{4}$$

All objects in (4) must be generated and processed by the successor algebra.

**Theorem I.1.9** (Syntactic no-smuggling criterion)**.** *The dependency graph of the target-reachability compiler has exactly one edge from user data: the inclusion of* $\mathsf{Pre}(\mathsf{Eq})$ *into the canonical presentation. Every target, source, price, compactness, equality, and recurrence object lies strictly downstream of that edge. Hence no target-relative conclusion can enter as a public input. Adding any fact in* (4) *changes the signature and does not count as an evaluation of the original presentation.*

*Proof.* The target is generated from the continuation graph, the shells from the target distance in $X_{\mathrm{ev}}$, the source cells from normal-form syntax, the price from the native carrier, and the recurrent equations from the successor fixed point. These constructions form a directed acyclic dependency chain beginning at $\mathsf{Pre}(\mathsf{Eq})$. None has a backward input slot. Every forbidden fact mentions a downstream object and therefore cannot be a term of the pre-target signature. □

**Corollary I.1.10** (Textbook-only specialization interface)**.** *The external specialization datum ends with* $\mathsf{Pre}(\mathsf{Eq})$*. Its mathematical content consists of the displayed equation, its native formal carrier identity, and any cited target-blind local theorem already attached to its standard presentation. In particular, no later row has an argument slot for a PDE-specific compactness proof, shell estimate,*

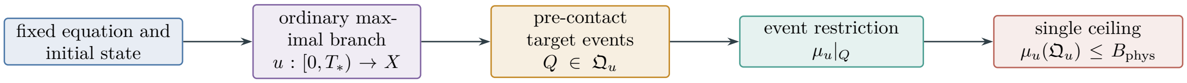


Figure 2: One physical-budget architecture. The equation generates one countably additive orbit measure, while target approach selects pre-contact events in its measurable event algebra. A selected event pays for target progress only after the target-local carrier factorization proves the charging rule; the measure's total mass is bounded by $B_{\mathrm{phys}}$.

*price bound, equality analysis, continuation criterion, or prescribed value of $Z_n$. The compiler itself generates the total graphs, the ordered cones, their finite proof identities, and their complementary positive functionals.*

*Proof.* The first sentence is Definition I.1.8. The dependency statement follows from Theorem I.1.9. Totalization and the successor relation generate both values of every downstream relation; the proof/model alternative is Theorem II.2.36. None of these constructions adds an input edge to the dependency graph. □

**Proposition I.1.11** (Non-vacuity of the total solution graph)**.** *For every datum, exactly one of the following occurs: the total local-solution fiber contains at least one maximal orbit, or it contains the nonzero form defect $\perp_{\mathrm{loc}}$. In particular, no reachability or regularity functional defined from the presentation assigns the value zero because the orbit fiber is empty.*

*Proof.* The total graph is the disjoint union of its ordinary solution values and its explicit failure value. The empty ordinary fiber therefore selects $\perp_{\mathrm{loc}}$. Subsequent quotients retain that record unless its target projection is proved to vanish. □

**Definition I.1.12** (Generated physical expenditure and budget)**.** Let $u$ be an ordinary value of the total solution graph. Evaluate the total signed carrier relation generated by $\mathcal{K}$. On its ordinary branch, signed gluing gives

$$K_u(t) + \mu_u((s,t]) + W_u^-((s,t]) = K_u(s) + W_u^+((s,t]), \qquad 0 \leq s < t < T_*(u), \tag{5}$$

where $W_u^\pm$ are the Jordan parts of true exterior work and $\mu_u$ is the sum, with the native coefficients of the identity, of all generated nonnegative production and balance-defect measures. Internal transport, chart, gauge, and interface currents have already cancelled and do not enter $\mu_u$. Every surviving summand retains its five-source ancestry. This orbitwise construction bounds the available physical measure; it does not yet assign any portion of that measure to target progress. Such an assignment is made only by Definition II.2.3 on an active branch.

If $K_u \geq K_- > -\infty$ and $W_u^+([0, T_*(u))) < \infty$, define the single orbit budget

$$B_{\mathrm{phys}}(u_0, W^+) := K_u(0) - K_- + W_u^+([0, T_*(u))) < \infty. \tag{6}$$

Then (5) derives

$$\mu_u(\mathfrak{Q}_u) \leq B_{\mathrm{phys}}(u_0, W^+). \tag{7}$$

Failure of the carrier identity, nonnegativity, the lower bound, or finite exterior input is the corresponding total-graph value. Its target-visible part enters the activity lift and ordered successor algebra. A finite ordered identity may exclude that value; otherwise it remains in the positive

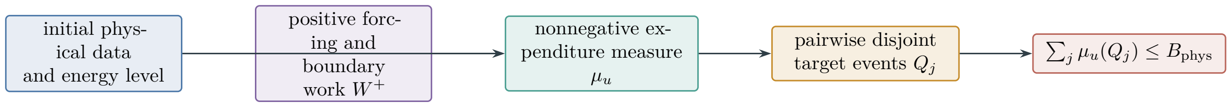


Figure 3: Physical expenditure accounting. The initial state and legitimate positive input bound one countably additive measure. Pairwise disjoint target events have disjoint restrictions of that measure. Only restrictions admitted by the local target-aligned charging rule pay for shell progress.

structural model. It is neither discarded nor misread as finite-budget regularity. The finite-budget reachability theorems apply only to the ordinary branch on which (7) has been derived.

The measure is localizable. For pairwise disjoint $Q_j \in \mathcal{A}_u$,

$$\sum_j \mu_u(Q_j) = \mu_u\left(\bigsqcup_j Q_j\right) \le B_{\text{phys}}. \tag{8}$$

No target-dependent expenditure may be appended. A Hamiltonian, entropy, commutator, or localized identity refines $\mu_u$ only after it has been proved to be an algebraic decomposition of (5). If the basic carrier assigns finite price to a singular cascade, that cascade proceeds to the zero-cost successor calculus; the budget is not strengthened after the fact.

**Example I.1.13** (Energy–work budget)**.** On the zero-defect branch of the lower-energy, dissipation-balance, and finite-work rows, an energy $\mathcal{E}$, lower value $E_-$, and nonnegative dissipation measure $\mu_{\text{diss}}$ satisfy

$$\mathcal{E}(u(t)) + \mu_{\text{diss}}([0,t]) \le \mathcal{E}(u_0) + W^+([0,t]),$$

and $W^+([0,\infty)) < \infty$. Then

$$\mu_{\text{diss}}([0,\infty)) \le \mathcal{E}(u_0) - E_- + W^+([0,\infty)).$$

Taking $\mu_u = \mu_{\text{diss}}$ gives the single budget

$$B_{\text{phys}} = \mathcal{E}(u_0) - E_- + W^+([0,\infty)).$$

If this physical measure assigns finite total price to a singular cascade, the cascade enters the zero-price successor calculus. No stronger measure or second budget is introduced after the target has been inspected.

**Definition I.1.14** (Fixed-Caus gradient-flow presentation)**.** Let $\mathcal{V} \hookrightarrow \mathcal{H} \hookrightarrow \mathcal{V}^*$ be a Gelfand triple. A fixed-Caus gradient-flow presentation consists of a differentiable energy $\mathcal{E}$, a symmetric nonnegative mobility $\mathbb{K}(u) : \mathcal{V}^* \to \mathcal{V}$, a fixed constitutive transport $A_{\mathsf{Caus}}(u) \in \mathcal{V}^*$, and an exterior input $f_{\mathsf{Bdry}} \in \mathcal{V}^*$, with equation

$$\dot{u} + A_{\mathsf{Caus}}(u) = -\mathbb{K}(u) D\mathcal{E}(u) + f_{\mathsf{Bdry}}. \tag{9}$$

The Caus term is energy-skew when

$$\langle D\mathcal{E}(u), A_{\mathsf{Caus}}(u) \rangle = 0. \tag{10}$$

Write

$$\left\|\mathbb{K}(u)^{1/2}\xi\right\|^2 := \langle \xi, \mathbb{K}(u)\xi \rangle$$

for $\xi \in \mathcal{V}^*$; this notation does not assume a bounded inverse or a globally defined square root on $\mathcal{V}^*$. Here $\mathsf{Geom}$ is the mobility current $-\mathbb{K}D\mathcal{E}$, $\mathsf{Caus}$ is the declared transport, and $\mathsf{Bdry}$ contains the exterior input and boundary flux. Quotients and symmetry reductions applied to this orbit have $\mathsf{Abs}$ ancestry; charts, blow-up rescalings, and augmented variables have $\mathsf{Lift}$ ancestry.

**Proposition I.1.15** (Fixed-Caus expenditure bound)**.** *Let $u$ satisfy Eqs.* (9) *and* (10) *on* $[0,T]$, *with the chain rule for $\mathcal{E}(u)$. Then*

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{E}(u(t)) = -\left\|\mathbb{K}(u)^{1/2}D\mathcal{E}(u)\right\|^2 + \langle D\mathcal{E}(u), f_{\mathsf{Bdry}}\rangle. \tag{11}$$

*If $\mathcal{E} \geq E_-$, then*

$$\int_0^T \left\|\mathbb{K}(u)^{1/2}D\mathcal{E}(u)\right\|^2 \mathrm{d}t \leq \mathcal{E}(u_0) - E_- + \int_0^T \langle D\mathcal{E}(u), f_{\mathsf{Bdry}}\rangle_+ \,\mathrm{d}t. \tag{12}$$

*For a weak solution with an energy inequality, the same statement holds with an additional nonnegative energy-defect measure on the left.*

*Proof.* Pair (9) with $D\mathcal{E}(u)$. The chain rule gives the left side of (11), the Caus pairing vanishes by (10), and positivity of $\mathbb{K}$ gives the squared norm. Integrate, discard the terminal energy above $E_-$, and replace exterior work by its positive part. The weak form is the increment formula for the energy inequality; its defect is nonnegative. ☐

## I.2 The five-source orbit algebra

**Definition I.2.1** (Continuum source alphabet)**.** The source alphabet is

$$\mathfrak{C} = \{\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}\}.$$

A raw continuum record is an orbit occurrence together with a finite ancestry word $\mathrm{Anc}(r) \in \mathfrak{C}^*$. Its source support $\mathrm{Src}(r) \subseteq \mathfrak{C}$ is the set of letters appearing in that word. The primitive and closure rules are:

(S1) symmetric mobility, Dirichlet-form, reversible, and dissipative current records start in $\mathsf{Geom}$;
(S2) the fixed directed transport and its current start in $\mathsf{Caus}$;
(S3) quotient, symmetry reduction, gauge removal, and coarse variables append $\mathsf{Abs}$;
(S4) charts, renormalizations, blow-up variables, auxiliary-state extensions, and Young- or defect-variable lifts append $\mathsf{Lift}$;
(S5) boundary traces, exterior fluxes, killing, terminal values, and tails read at the boundary of a compactification append $\mathsf{Bdry}$;
(S6) finite algebraic combination, restriction, and concatenation take the union of the input source supports.

The symbol $J_{\mathrm{res}}$ is absent from this source alphabet. It is introduced only as the target-null coordinate of Definition I.4.2.

*Remark* I.2.2 (Channel and source terminology)*.* The channels $\mathsf{Geom}$ and $\mathsf{Abs}$ carry intrinsic reusable structure, while $\mathsf{Caus}$, $\mathsf{Lift}$, and $\mathsf{Bdry}$ orient, reorganize, and read out that structure. A *source cell* is a provenance cell in the complete five-channel normal form; this terminology does not promote the three derived channels to independent creators of intrinsic structure.

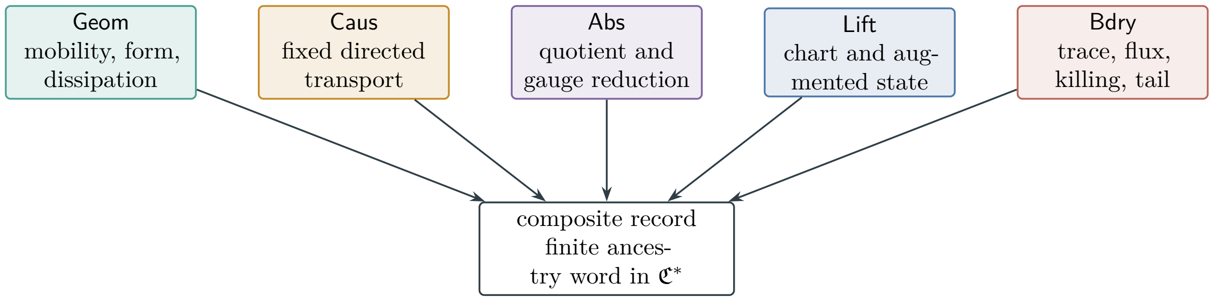


Figure 4: The five continuum source roles. Geom and Caus are native current mechanisms; Abs and Lift record transformations of those mechanisms; Bdry records exterior and terminal effects. Composite records retain the union of their ancestries.

*Remark* I.2.3 (Forcing and constraints). The source compiler routes a volume force according to its constitutive role. A fixed directed body force may be $\mathsf{Caus}$, while exterior work or a boundary supply is $\mathsf{Bdry}$. A Lagrange multiplier implementing a quotient constraint has $\mathsf{Abs}$ ancestry. The five labels are assigned from the declared weak equation, not from the sign of a later estimate.

**Definition I.2.4** (Operational weak-PDE signature)**.** An operational weak-PDE signature is formed downstream from the public presentation. It consists of a Gelfand triple $V \hookrightarrow H \hookrightarrow V^*$, a separating test core $\mathcal{D} \subset V$, an initial datum, the continuation target $\Sigma$ generated by Definition III.1.1, and a weak evolution relation

$$\langle \dot{u}, \varphi \rangle_{V^*,V} = \mathcal{A}(u;\varphi) + \mathcal{F}_{\mathrm{ext}}(u;\varphi), \qquad \varphi \in \mathcal{D}. \tag{13}$$

The native spatial relation is entered before any realization property is asserted. Its totalized realization relation has the following ordinary values.

(W1) A fixed-Caus gradient realization:

$$\mathcal{A}(u;\varphi) = -\langle \mathbb{K}(u) D\mathcal{E}(u), \varphi \rangle - \langle A_{\mathsf{Caus}}(u), \varphi \rangle,$$

with $\mathbb{K}(u)$ symmetric and nonnegative.

(W2) A sector-form realization on a real $L^2$-space:

$$\mathcal{E} = \mathcal{E}^{\mathrm{s}} + \mathcal{E}^{\mathrm{a}}, \qquad \mathcal{E}^{\mathrm{s}}(v,w) = \tfrac{1}{2}\{\mathcal{E}(v,w) + \mathcal{E}(w,v)\}, \quad \mathcal{E}^{\mathrm{a}}(v,w) = \tfrac{1}{2}\{\mathcal{E}(v,w) - \mathcal{E}(w,v)\},$$

where $\mathcal{E}^{\mathrm{s}}$ is a Dirichlet form and $\mathcal{E}^{\mathrm{a}}$ satisfies the sector bound.

If neither ordinary value exists, the realization relation returns $\perp_{\mathrm{nat}}$. The raw fixed bulk current remains a $\mathsf{Caus}$ occurrence and the failed attempted Geom–Caus split is a form-defect record with support $\{\mathsf{Geom}, \mathsf{Caus}\}$. Its target-active quotient is evaluated by the form row. Thus absence of a gradient, Dirichlet, positivity, or sector realization is admitted as structure rather than excluded from the signature. The signature also declares the symmetry, quotient, chart, auxiliary-state, and trace operations that may be applied to (13). It contains no assertion of compactness, regularity, target avoidance, positive physical price, or global continuation.

**Definition I.2.5** (Continuum source compiler)**.** Let $\mathsf{Syn}(\mathfrak{P})$ be the smallest typed syntax containing the native terms of (13) and closed under finite sum, restriction, concatenation, quotient, chart, state augmentation, weak limit, and trace. The source compiler

$$\mathsf{C}_{\mathfrak{P}} : \mathsf{Syn}(\mathfrak{P}) \longrightarrow 2^{\mathfrak{C}} \setminus \{\varnothing\}$$

is defined recursively by

$$\begin{aligned} \mathsf{C}_{\mathfrak{P}}(-\mathbb{K}D\mathcal{E}) &= \{\mathsf{Geom}\}, & \mathsf{C}_{\mathfrak{P}}(\mathcal{E}^{\mathrm{s}}_{\mathrm{local/jump}}) &= \{\mathsf{Geom}\}, \\ \mathsf{C}_{\mathfrak{P}}(-A_{\mathsf{Caus}}) &= \{\mathsf{Caus}\}, & \mathsf{C}_{\mathfrak{P}}(\mathcal{E}^{\mathrm{a}}) &= \{\mathsf{Caus}\}, \\ \mathsf{C}_{\mathfrak{P}}(\mathcal{F}_{\mathrm{ext}}) &= \{\mathsf{Bdry}\}, & \mathsf{C}_{\mathfrak{P}}(\perp_{\mathrm{nat}}) &= \{\mathsf{Geom}, \mathsf{Caus}\}. \end{aligned} \tag{14}$$

Finite sums take unions. Quotients append Abs; charts and auxiliary variables append Lift; traces, killing, terminal values, and compactification boundaries append Bdry. Restriction, concatenation, and sourcewise weak closure retain support. A constraint multiplier is compiled through the operation that generates it: quotient constraints append Abs, while an exterior constraint supply appends Bdry.

**Theorem I.2.6** (Operational native-source exhaustion)**.** *For every operational weak-PDE signature, $\mathsf{C}_{\mathfrak{P}}$ is defined on every finite dynamically generated record and has image in the five inherited source families. Its evaluation commutes with the weak equation: for every $\varphi \in \mathcal{D}$,*

$$\langle \dot{u}, \varphi \rangle = \sum_{c \in \mathfrak{C}} \langle J^c(u), \varphi \rangle. \tag{15}$$

*The source word of a transformed record is the union of the native word and the letters appended by the transformation. There is no untyped native or finite-constructor remainder.*

*Proof.* On ordinary branch (W1), the native relation has exactly the mobility, directed-transport, and exterior terms displayed in the definition. Nonnegativity of $\mathbb{K}$ identifies the first as Geom; the fixed transport and exterior relation give Caus and Bdry. On ordinary branch (W2), polarization uniquely separates the symmetric and antisymmetric forms. The Beurling–Deny representation uniquely separates the symmetric form into strongly local, jump, and killing terms; the first two are Geom and killing is Bdry. The antisymmetric form is Caus.

On the failure branch, retain the weak bulk current itself as a fixed Caus occurrence, so (15) still holds, and record the failed attempted split with composite $\{\mathsf{Geom}, \mathsf{Caus}\}$-ancestry. The form row evaluates its target-active quotient without asserting the missing realization.

Structural induction over $\mathsf{Syn}(\mathfrak{P})$ proves the remaining claim. Each constructor is covered by exactly one recursive clause in Definition I.2.5; finite combination takes unions and cannot erase an ancestor. Applying the evaluation map at each induction step gives (15). Weak closure is treated sourcewise, so it changes the analytic representative but not the cell label. □

**Corollary I.2.7** (Failure is compiled structure)**.** *If an operation in $\mathsf{Syn}(\mathfrak{P})$ lacks a classical value on a limit record, its totalized graph defect has the source support assigned to the approximating graph, together with* Lift *or* Bdry *when the failure occurs at a chart or terminal interface. Consequently lack of closed range, compactness, a classical trace, or a strong nonlinear limit cannot create a sixth source.*

*Proof.* Apply the compiler before graph closure, then use the sourcewise closure rule. The graph defect is a limit of records in that tagged cell. The chart and terminal cases use the corresponding recursive clauses. □

**Theorem I.2.8** (Sector-form source compiler)**.** *Let $X$ be locally compact and separable, let $m$ be a Radon measure of full support, and let $(\mathcal{E}^{\mathrm{s}}, D(\mathcal{E}))$ be a regular symmetric Dirichlet form on the real space $L^2(X, m)$. Write its Beurling–Deny decomposition, with normalization absorbed into $J$, as*

$$\mathcal{E}^{\mathrm{s}}(u,v) = \mathcal{E}^{\mathrm{c}}(u,v) + \int_{X \times X \setminus \mathrm{diag}} (u(x) - u(y))(v(x) - v(y))\, J(\,\mathrm{d}x,\, \mathrm{d}y) + \int_X uv\, \kappa(\,\mathrm{d}x). \tag{16}$$

*Let $\mathcal{E}^{\mathrm{a}}$ be antisymmetric on the same domain and satisfy the sector bound*

$$|\mathcal{E}^{\mathrm{a}}(u,v)| \le C_{\mathrm{sec}}\, \mathcal{E}^{\mathrm{s}}_1(u,u)^{1/2}\mathcal{E}^{\mathrm{s}}_1(v,v)^{1/2}. \tag{17}$$

*Then the form $\mathcal{E} = \mathcal{E}^{\mathrm{s}} + \mathcal{E}^{\mathrm{a}}$ is closed and sectorial on $D(\mathcal{E})$, and its source assignment is exhaustive: $\mathcal{E}^{\mathrm{c}}$ and the symmetric jump form are* Geom*, $\mathcal{E}^{\mathrm{a}}$ is the fixed* Caus *form, and the killing measure $\kappa$ is* Bdry*. The uniqueness of* (16) *leaves no additional symmetric Markovian primitive. Quotients and state-space enlargements append* Abs *and* Lift *as usual.*

*Proof.* The Beurling–Deny theorem gives (16) and uniqueness of its strongly local, jump, and killing measures [5]. Antisymmetry gives $\mathcal{E}^{\mathrm{a}}(u,u) = 0$, while (17) makes it continuous in the $\mathcal{E}^{\mathrm{s}}_1$-norm. Hence $\mathcal{E}$ has the same form norm completeness as $\mathcal{E}^{\mathrm{s}}$ and satisfies the sector condition. The stated routing is exactly the source assignment of Definition I.2.1. A quasi-regular form has the same conclusion after a declared quasi-homeomorphic regular realization. □

**Proposition I.2.9** (Exact-drift symmetrization). *Let $M$ be a Riemannian domain, let $\Psi \in C^2(M)$. On the locally-finite-weight branch let $\pi(\,\mathrm{d}x) = e^{-\Psi(x)}\,\mathrm{d}x$ is a locally finite measure. For*

$$L_\Psi g = \Delta g - \nabla\Psi\cdot\nabla g,$$

*totalize the integration-by-parts trace. On its zero boundary-defect branch, the resulting form core satisfies*

$$-\langle f, L_\Psi g\rangle_{L^2(\pi)} = \int_M \nabla f\cdot\nabla g\ \mathrm{d}\pi. \tag{18}$$

*Thus an exact drift is absorbed into a symmetric* Geom *form by a weighted* Abs*-gauge. If an exact/nonexact decomposition of a general drift is proved on the declared domain, only its nonexact remainder retains independent* Caus *ancestry. No global boundedness of $\mathrm{e}^{\pm\Psi}$ is inferred; comparison with another physical measure is evaluated by its own totalized physical-transfer row.*

*Proof.* The weighted divergence identity

$$L_\Psi g = e^{\Psi}\,\mathrm{div}(e^{-\Psi}\nabla g)$$

and the stated trace condition give (18). Symmetry and nonnegativity of the right side prove the source statement; changing the reference measure is the declared gauge operation. □

**Proposition I.2.10** (Wiener lift creates no additional source). *Let $H$ and $U$ be separable Hilbert spaces and let a legal Itô evolution have drift $b : H \to H$ and Hilbert–Schmidt noise coefficient $\sigma : H \to \mathcal{L}_2(U,H)$. Totalize the generator relation on bounded $C^2$ cylinder functions. Its ordinary branch is*

$$LF(u) = \langle DF(u), b(u)\rangle + \frac{1}{2}\,\mathrm{Tr}\big[\sigma(u)\sigma(u)^* D^2F(u)\big]. \tag{19}$$

*Then the carré du champ is*

$$\Gamma(F) := L(F^2) - 2FLF = \|\sigma^* DF\|_U^2 \ge 0. \tag{20}$$

*Thus the diffusion covariance and predictable quadratic variation are* Geom*; the drift retains its declared* Geom/Caus/Bdry *decomposition; and adjoining the Wiener path is* Lift*. Noise therefore creates no sixth continuum source.*

*Proof.* Apply the second-order chain rule to $F^2$ in (19); the drift terms cancel and the remaining trace is $\|\sigma^* DF\|_U^2$. Itô's formula gives the same integrand for the predictable quadratic variation of the martingale part. The source labels follow from positivity of $\Gamma$, the fixed drift decomposition, and the state-space augmentation rule in Definition I.2.1. □

## I.3 Physical-budget-saturated sourcewise completion

The closed continuum algebra contains every mathematical limit generated by an ordinary finite-budget orbit. Its topology is generated canonically by the equation, target, physical balance, and constructor grammar. Strong convergence of a particular nonlinear term is subsequently read as a graph invariant in this completion.

**Definition I.3.1** (Structural test system)**.** For each nonempty $A \subseteq \mathfrak{C}$ and derived physical ceiling $B_{\mathrm{phys}}$, let $\mathscr{R}^0_A(B_{\mathrm{phys}})$ be the raw orbit records with source support equal to $A$. The structural test system $\mathfrak{F}_A = (\phi^A_n)_{n\geq 1}$ is generated recursively from the countable coordinates (2). At every finite grammar depth it contains:

(T1) the weak PDE pairings and source-current pairings;

(T2) the target shells and target fluxes;

(T3) every truncation of the physical-expenditure and entropy-production coordinates generated by the fixed balance;

(T4) every chart, quotient, boundary, and tail coordinate generated by the source grammar;

(T5) the total graphs of every finite algebraic operation generated by that grammar, encoded by rational distance-to-graph truncations.

An unbounded real-valued test is represented by its rational bounded truncations. Enumerating grammar depth and rational parameters gives one countable family with values in $[-1, 1]$. No map or topology may be appended after a row value is inspected. Failure of an operation to extend as an ordinary continuous map is recorded by the boundary of its total graph.

**Definition I.3.2** (Physical-budget-saturated closed orbit algebra)**.** Let

$$\iota_A(r) = (\phi^A_n(r))_{n\geq 1} \in K_A := [-1, 1]^{\mathbb{N}}.$$

The space $K_A$ is compact metrizable in the product metric. The sourcewise saturated cell and the full physical orbit algebra are

$$\mathscr{R}^{\mathrm{phys,sat}}_A(B_{\mathrm{phys}}) := \overline{\iota_A(\mathscr{R}^0_A(B_{\mathrm{phys}}))}^{K_A}, \qquad \mathscr{R}^{\mathrm{phys,sat}}(B_{\mathrm{phys}}) := \bigsqcup_{\varnothing\neq A\subseteq\mathfrak{C}} \mathscr{R}^{\mathrm{phys,sat}}_A(B_{\mathrm{phys}}). \tag{21}$$

A point of the closure is a mathematical limit record. Its ancestry support is the cell label $A$. A finer ancestry word may also be retained when it is retained by the generated test system. The closure contains all sequential limits of ordinary orbit records seen by the generated tests. Here Src denotes provenance support, not the smallest support of an analytically simplified formula.

*Remark* I.3.3 (Why the disjoint union is retained)*.* The same analytic denotation can arise through different source histories. The closed algebra retains those histories as distinct records. Forgetting ancestry is an evaluation map from the disjoint union to the ambient analytic space; it is not part of the source decomposition.

**Theorem I.3.4** (Source-preserving continuum completion)**.** *Every sequence $(r_i)$ of ordinary orbit records with $\mathrm{Src}(r_i) = A$ that converges in the structural test topology has its limit record in $\mathscr{R}^{\mathrm{phys,sat}}_A(B_{\mathrm{phys}})$. Consequently*

$$\mathrm{Src}(\lim_i r_i) = A = \bigcup_i \mathrm{Src}(r_i). \tag{22}$$

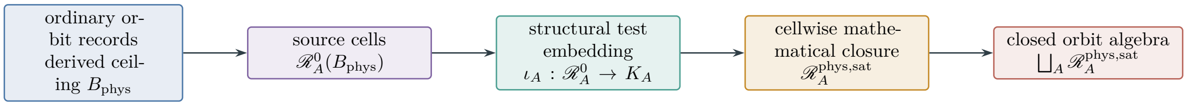


Figure 5: Sourcewise physical saturation. Closure is taken separately in each ancestry cell before the cells are assembled, and every cell retains the same physical ceiling $B_{\mathrm{phys}}$.

*If the sequence first passes through quotient or chart maps,* Abs *or* Lift*, respectively, is appended before taking the limit. If the limit is read as a boundary or tail trace,* Bdry *is appended. Thus defect measures, oscillation variables, concentration profiles, closure directions, blow-up germs, and tails create no source outside $\mathfrak{C}$.*

*Proof.* By definition, $\mathscr{R}_A^{\mathrm{phys,sat}}(B_{\mathrm{phys}})$ is the sequential closure of the test image of the raw cell with provenance support $A$; compact metrizability makes this equal to its topological closure. Hence every limit of such a sequence belongs to that cell. The algebraic rules of Definition I.2.1 take unions of source supports and append the labels of quotient, chart, and boundary operations before the closure is formed. The new coordinates of a limit record are limits of tests on the same tagged sequence; they do not supply a new primitive letter. This proves (22) and each listed case. □

**Corollary I.3.5** (Mixed sequences)**.** *Every sequence of raw records has source support in the finite set $2^{\mathfrak{C}} \setminus \{\varnothing\}$. It has a subsequence with constant source support. Each convergent such subsequence has its limit therefore contained in an existing saturated cell.*

*Proof.* There are only $2^5 - 1$ nonempty source supports. One support occurs infinitely often. Apply Theorem I.3.4 to that subsequence. □

**Definition I.3.6** (Source realization map)**.** Let $\mathcal{P}_A$ and $\mathcal{J}_A$ be Hilbert spaces of potentials and currents for a fixed source support $A \subseteq \mathfrak{C}$. A *source realization map* is a densely defined closed operator

$$G_A : D(G_A) \subseteq \mathcal{P}_A \longrightarrow \mathcal{J}_A$$

whose range consists of the currents carrying an actual potential witness in that source cell. Its saturated current space is $\overline{\operatorname{Ran} G_A}^{\mathcal{J}_A}$. The closure has the same source support $A$, independently of whether every closure direction has a potential witness.

**Theorem I.3.7** (Closed-range realization criterion)**.** *Let $G : D(G) \subseteq \mathcal{P} \to \mathcal{J}$ be a source realization map. The following conditions are equivalent.*

(R1) $\operatorname{Ran} G$ *is closed in* $\mathcal{J}$*.*

(R2) *There is $\gamma > 0$ such that*

$$\|G\phi\|_{\mathcal{J}}^2 \geq \gamma \operatorname{dist}_{\mathcal{P}}(\phi, \ker G)^2, \qquad \phi \in D(G). \tag{23}$$

(R3) *The restriction of $G^*G$ to $(\ker G)^{\perp}$ has spectrum in $[\gamma, \infty)$ for some $\gamma > 0$.*

(R4) *The Moore–Penrose inverse $G^+$ is bounded on* $\operatorname{Ran} G$*.*

*When these conditions hold, every saturated current in $\overline{\operatorname{Ran} G}$ has a potential witness, unique modulo* $\ker G$*. When they fail, elements of $\overline{\operatorname{Ran} G} \setminus \operatorname{Ran} G$ remain limit records in the same source cell by Theorem I.3.4; failure of potential realization does not define another source family. In the reachability calculus this operator theorem is activated only after restriction to the selected target-local source cell; its complementary range value is then a polarity successor of that cell.*

*Proof.* The closed-range theorem for densely defined closed operators gives the equivalence of (R1), (R2), and boundedness of $G^+$. On $(\ker G)^\perp$, $\|G\phi\|^2 = \langle G^*G\phi, \phi\rangle$, so the spectral theorem makes (R2) equivalent to (R3). Closed range gives $\overline{\operatorname{Ran} G} = \operatorname{Ran} G$, and the kernel is exactly the ambiguity in a potential witness. The final assertion is Theorem I.3.4 applied to the source cell generated by $G$. $\square$

**Theorem I.3.8** (Target-relative gradient exhaustiveness)**.** *Let $G : D(G) \subseteq \mathcal{P} \to \mathcal{J}$ be a densely defined closed realization of one inherited source cell, and let $P_{\overline{G}}$ be the orthogonal projection onto $\overline{\operatorname{Ran} G}$. Then every $j \in \mathcal{J}$ has the unique split*

$$j = P_{\overline{G}} j + (I - P_{\overline{G}})j, \qquad P_{\overline{G}} j \in \overline{\operatorname{Ran} G}, \quad (I - P_{\overline{G}})j \in \ker G^*. \tag{24}$$

*The first term belongs to the saturated cell generated by $G$, including when it lies in $\overline{\operatorname{Ran} G} \setminus \operatorname{Ran} G$. Relative to the target family $\Gamma_\Sigma$, the second term has the further canonical split into its class in $\ker G^*/(\ker G^* \cap \mathcal{N}_\Sigma)$ and its target-null part. Hence failure of closed range creates neither a new source nor a target-visible residual: it creates a closure direction in the inherited cell, while every orthogonal target-visible class remains a current of the native compiled equation and retains its native ancestry.*

*Proof.* The range closure is a closed Hilbert subspace, so the projection theorem gives (24); the identity $(\operatorname{Ran} G)^\perp = \ker G^*$ identifies its complement. The definition of the sourcewise completion and Theorem I.3.4 assign $P_{\overline{G}} j$ to the same tagged cell as the approximating gradients. Quotienting the complementary space by its intersection with $\mathcal{N}_\Sigma$ is exact. Its kernel is target-null by definition, and a nonzero quotient class has a nonzero target pairing. Such a class cannot be renamed residual; its ancestry is the ancestry of the native current from which $j$ was compiled by Theorem I.4.5. These alternatives exhaust $\mathcal{J}$. $\square$

**Corollary I.3.9** (Regularized realization)**.** *For $j \in \mathcal{J}$ and $\lambda > 0$, the functional*

$$\phi \longmapsto \|G\phi - j\|^2_{\mathcal{J}} + \lambda \|\phi\|^2_{\mathcal{P}}$$

*has a unique minimizer $\phi_\lambda \in D(G)$. The resulting current $G\phi_\lambda$ is a stable regularized realization, with the parameter $\lambda$ recording its bias. Existence of this minimizer for every $\lambda > 0$ does not imply any of the equivalent conditions in Theorem I.3.7.*

*Proof.* On $D(G)$, the form $\langle G\phi, G\psi\rangle + \lambda\langle \phi, \psi\rangle$ is closed and coercive in its form norm. The functional $\psi \mapsto \langle j, G\psi\rangle$ is continuous in that norm, so the Riesz representation theorem gives a unique minimizer. The final statement follows because adding $\lambda I$ supplies coercivity even when the unregularized range is not closed. $\square$

*Remark* I.3.10 (Structural and evaluator topologies)*.* The graph-complete state space closes every typed operation together with its defect coordinates. A stronger PDE topology is represented by adjoining its bounded seminorm and graph coordinates to the cylinder algebra. Compactness in that topology is then an ordered state-space assertion. Failure of the strong coordinate to be finite or closed is its typed graph value in the same source cell.

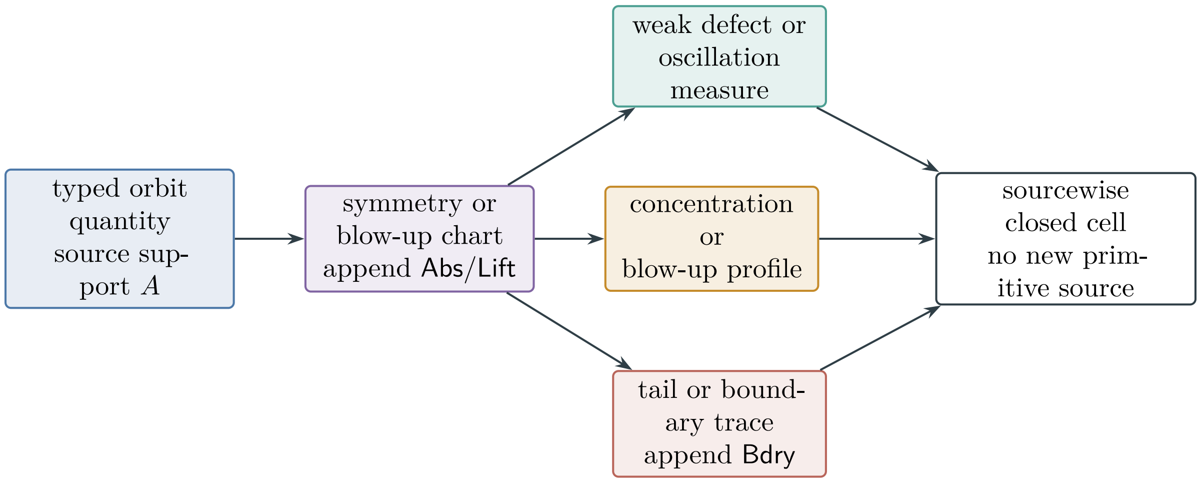


Figure 6: Continuum limit phenomena inherit ancestry. Normalization changes the source word in the declared Abs/Lift manner; a terminal trace appends Bdry. Weakness or concentration changes the analytic realization, not the source alphabet.

## I.4 Target quotient and the null residual

**Definition I.4.1** (Generated prospective target family)**.** Let $\mathscr{J}^{\mathrm{phys,sat}}(B_{\mathrm{phys}})$ be the current-occurrence part of the closed orbit algebra. The finite-current locus is replaced, when necessary, by its Hausdorff locally convex envelope under the continuously extended addition and scalar multiplication; the kernel of the envelope map is retained as a tagged closure defect. Records at an infinite compactification endpoint are treated as boundary or tail records and are tested by their bounded readouts, not inserted into this vector space. Let $\rho_\Sigma(x) = d_{\mathrm{ev}}(x, \Sigma)$, where $\Sigma$ is the complete continuation boundary of Definition III.1.1. The *target-shell algebra* is the least countable algebra containing every rational ramp of $\rho_\Sigma$, its pullback by every ordinary generated covariance map, and the rational cylindrical approximants supplied by the coordinates (2).

The prospective target family $\Gamma_\Sigma$ is the closed normalized cone generated by every derived finite-cylinder shell current, together with the killed-flux and predictable-compensator differentials when their types occur in the fixed equation. In particular it contains

$$J \longmapsto \langle D\widehat{\phi}_{r,m,I}(z), J\rangle$$

on the ordinary branch and the corresponding Stieltjes, oscillation, concentration, exterior, or variation-boundary pairing on every other branch. Thus neither the probes nor their normalization are application data: they are generated by the continuation target, approximation tower, and verified covariance grammar.

**Definition I.4.2** (Target-null residual)**.** The target-null space and target-active quotient are

$$\mathscr{N}_\Sigma := \bigcap_{\Lambda \in \Gamma_\Sigma} \ker \Lambda, \qquad \mathscr{J}_\Sigma := \mathscr{J}^{\mathrm{phys,sat}}(B_{\mathrm{phys}})/\mathscr{N}_\Sigma. \tag{25}$$

The post-saturation residual is any closed current coordinate containing zero and satisfying

$$\mathscr{J}^{\mathrm{phys,sat}}_{J_{\mathrm{res}}} \subseteq \mathscr{N}_\Sigma. \tag{26}$$

It is an audit coordinate for target-invisible transport, not a sixth source family. A current is target-visible precisely when its quotient class in $\mathscr{J}_\Sigma$ is nonzero.

**Theorem I.4.3** (Target quotient commutes with sourcewise completion)**.** *For every source support $A \subseteq \mathfrak{C}$,*

$$\overline{\mathscr{R}_A + \mathscr{N}_\Sigma}^{\,\mathrm{gc}} / \mathscr{N}_\Sigma = \overline{q_\Sigma(\mathscr{R}_A)}^{\,\mathscr{J}_\Sigma}.$$

*The kernel $\mathscr{N}_\Sigma$ is closed under sourcewise limits, signed gluing, restriction, gauge descent, and the witness-successor maps. Consequently all target-conditioned carrier, cohomology, moment, and pruning constructions may be performed on $\mathscr{J}_\Sigma$ after ancestry has been fixed.*

*Proof.* Every element of $\Gamma_\Sigma$ is a continuous cylinder-current coordinate on the graph-complete space. Its kernel is closed, and so is their intersection. The displayed identity is the standard quotient–closure identity for a continuous linear quotient by a closed kernel. Signed gluing, restriction, gauge descent, and successor formation preserve every shell pairing by their defining graph identities. They therefore preserve the common kernel. The displayed identity follows from the definition of the quotient topology after saturating the source cell by the kernel. Sourcewise completion precedes this saturation, so no ancestry tag is lost. □

**Definition I.4.4** (Compiled continuum current)**.** A continuum current is *compiled* when it is generated from the native terms of the fixed equation by the finite constructor rules (S1)–(S6) of Definition I.2.1. Its evaluation has the normal form

$$J = J^{\mathsf{Geom}} + J^{\mathsf{Caus}} + J^{\mathsf{Abs}} + J^{\mathsf{Lift}} + J^{\mathsf{Bdry}} + J^{J_{\mathrm{res}}}, \qquad J^{J_{\mathrm{res}}} \in \mathscr{N}_\Sigma, \tag{27}$$

as an equality of weak current pairings. Empty components are allowed. The residual is an optional regrouping in the common target kernel after the five source components have been evaluated; the choice $J^{J_{\mathrm{res}}} = 0$ is always admissible. It is not a constructor and carries no source letter.

**Theorem I.4.5** (Continuum normal-form compilation)**.** *For every operational weak-PDE signature, totalize the native realization relation and adjoin the quotient, chart, auxiliary-state, trace, forcing, and boundary operations generated by Definition I.2.1. Every finite orbit-current expression is compiled and has* (27)*. A failed native realization additionally generates its target-active form-defect class with* $\{\mathsf{Geom}, \mathsf{Caus}\}$*-ancestry. After sourcewise completion, every finite target pairing has the same normal form. Thus five-channel exhaustion does not presume a gradient or sector realization.*

*Proof.* On the fixed-Caus ordinary branch, positivity of $\mathbb{K}$, the directed transport, and the exterior term give $\mathsf{Geom}$, $\mathsf{Caus}$, and $\mathsf{Bdry}$, respectively. On the form ordinary branch, the sector-form compiler makes the same assignment to the symmetric mobility, antisymmetric directed, and killing terms. Structural induction on the syntax of an orbit-current expression now proves the claim. A quotient appends $\mathsf{Abs}$, a chart or auxiliary variable appends $\mathsf{Lift}$, a trace appends $\mathsf{Bdry}$, and finite combination takes the union of the already assigned supports. These exhaust the transformed ordinary branches.

On the native failure branch, retain the raw fixed bulk current in $\mathsf{Caus}$ and its totalized attempted split in the composite Geom–Caus form-defect cell, as in Theorem I.2.6. This additional case remains inside the five-source alphabet.

For a sourcewise convergent net, each prospective target functional is a coordinate of the structural test system. Its value therefore passes to the limit. By Theorem I.3.4, the limit remains in the same tagged source cell. The image in the quotient by $\mathscr{N}_\Sigma$ is therefore generated by the five cell images. Taking $J^{J_{\mathrm{res}}} = 0$, or regrouping any already identified target-null summand as $J^{J_{\mathrm{res}}}$, gives (27) without requiring a complemented kernel and without adding a primitive source. □

**Theorem I.4.6** (No post-saturation target-visible residual)**.** *Let $\mathfrak{P}_{\mathrm{phys}}$ be generated by the compiled continuum algebra of Theorem I.4.5. Every target-visible current record in $\mathscr{J}^{\mathrm{phys,sat}}(B_{\mathrm{phys}})$ has nonempty ancestry in $\mathfrak{C}$, and its target-active quotient is generated by the images of the five saturated source cells. The residual source profile vanishes:*

$$G^{J_{\mathrm{res}},\mathrm{src,sat}}_{B_{\mathrm{phys}},\Sigma} := \sup_{J\in\mathscr{J}^{\mathrm{phys,sat}}_{J_{\mathrm{res}}}} \sup_{\Lambda\in\Gamma_\Sigma} (\Lambda J)_+ = 0. \tag{28}$$

*If a provisional limit object pairs nontrivially with a target functional, then Theorem I.3.4 places it in the existing source cell inherited from its generating records. Thus every dynamically generated target-visible limit belongs to one of the five source families.*

*Proof.* For raw occurrences, apply the quotient map $q_\Sigma : \mathscr{J}^{\mathrm{phys,sat}} \to \mathscr{J}_\Sigma$ to (27). The residual term vanishes because it lies in $\mathscr{N}_\Sigma$. Continuity of every $\Lambda \in \Gamma_\Sigma$ passes the same identity to each sourcewise limit. The source-preserving theorem assigns every nonresidual limit term to the closure of its existing cell. Thus a nonzero quotient class is generated by the five cell images. Finally, every functional in $\Gamma_\Sigma$ vanishes on $\mathscr{J}^{\mathrm{phys,sat}}_{J_{\mathrm{res}}}$, so both nested suprema in (28) are zero. □

**Theorem I.4.7** (Projection-first saturation)**.** *Let $\mathsf{G}$ be the continuum constructor grammar generated by the public equation, and let $q_\Sigma : \mathscr{J}^{\mathrm{phys,sat}} \to \mathscr{J}_\Sigma$ be the quotient by $\mathscr{N}_\Sigma$. Totalize the descent graph of every constructor $T \in \mathsf{G}$. On its ordinary descent cell set $T_\Sigma(q_\Sigma J) := q_\Sigma(TJ)$; on every other cell retain the first descent defect*

$$\mathcal{D}_{T,\Sigma}(J) := q_\Sigma(TJ) - T_\Sigma(q_\Sigma J) \tag{29}$$

*with the ancestry of $T$ and the target covector that detects it. If $\mathsf{G}_\Sigma$ denotes the resulting projected grammar, then*

$$q_\Sigma(\mathrm{Sat}_{\mathsf{G}}\,\mathscr{J}) = \mathrm{Sat}_{\mathsf{G}_\Sigma}(q_\Sigma \mathscr{J})\,. \tag{30}$$

*The identity holds sourcewise and with the physical carrier and same-origin coordinates retained. Thus the reachability calculus may quotient by $\mathscr{N}_\Sigma$ before generating descendants. A failed commutation is a projected graph coordinate, so projection first loses no target-visible current.*

*Proof.* For a native current, (30) is the quotient of (27). Suppose it holds for the inputs of a constructor. On the ordinary descent graph, the definition of $T_\Sigma$ gives the identity for its output. On the complementary graph, add $\mathcal{D}_{T,\Sigma}$; then $q_\Sigma TJ = T_\Sigma q_\Sigma J + \mathcal{D}_{T,\Sigma}(J)$ by (29). Source-preserving totalization retains the ancestry of the failed operation. This proves both inclusions for finite words by structural induction.

Every prospective target functional is a coordinate of the evaluation compactification. Hence $q_\Sigma$ is continuous on each sourcewise total graph, and the two finite-word inclusions pass to sourcewise completion. Signed carrier gluing commutes with the quotient because it is performed before Jordan decomposition. Physical domination and same-origin relations are retained coordinates rather than quotient relations. This proves the completed identity. □

**Corollary I.4.8** (Source type and invariant value are independent)**.** *Each saturated source cell has a value in every applicable fast-track invariant: null or nonnull quotient, finite or infinite radius, positive or zero price, zero or nonzero equality remainder, zero or positive capacity, finite or infinite resistance, and divergent or finite cascade price. These values change neither the source word nor the target quotient. Failure of a compactness or resolvent invariant therefore returns the complementary invariant branch; it never creates an untyped or residual source.*

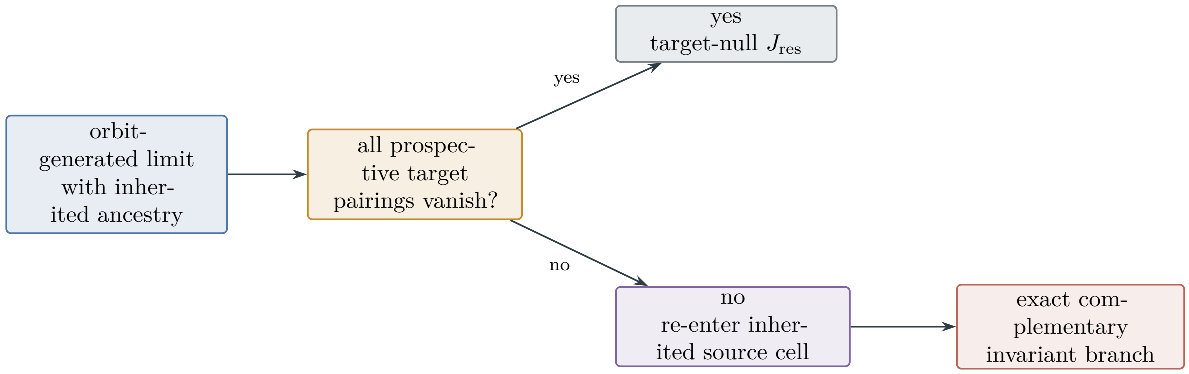


Figure 7: Target-visible limits retain their inherited source cell. The only residual output is target-null. A nonzero pairing fixes structural ancestry before any attempt is made to estimate the corresponding source profile.

## I.5 Prospective contact accountability

Target contact may produce a retrospective object, such as a singular germ or a completed hitting distribution. Source accountability is instead read from the evolution strictly before contact. The next three theorems give deterministic, semigroup, and stochastic versions of the same factorization.

**Definition I.5.1** (Canonical target shells and total shell current)**.** In the compact metric space $X_{\mathrm{ev}}$, put $\rho_\Sigma(x) = d_{\mathrm{ev}}(x, \Sigma)$, $U_r(\Sigma) = \{\rho_\Sigma < r\}$, and, for rational $r > 0$,

$$\Phi_r(x) = \begin{cases} 1, & \rho_\Sigma(x) \le r, \\ 2 - \rho_\Sigma(x)/r, & r < \rho_\Sigma(x) < 2r, \\ 0, & \rho_\Sigma(x) \ge 2r. \end{cases} \tag{31}$$

Then $\Phi_r$ is continuous and $r^{-1}$-Lipschitz, $\bigcap_{r\in\mathbb{Q}_+} U_{2r}(\Sigma) = \Sigma$, and

$$\Phi_r = 0 \quad \text{on } X_{\mathrm{ev}} \setminus U_{2r}(\Sigma), \qquad \Phi_r = 1 \quad \text{on } U_r(\Sigma). \tag{32}$$

Adjoin the bounded coordinate $\Phi_r$ to the graph-complete algebra and its defining distance-ramp graph to the ordered relations. The rational cylinder algebra separates the resulting compact state space. By Stone–Weierstrass, for every $m \ge 1$ there is a rational cylinder polynomial $\phi_{r,m}$ such that

$$\|\phi_{r,m} - \Phi_r\|_\infty \le 2^{-m}. \tag{33}$$

Choose the polynomial first with error $2^{-(m+2)}$. The two strictly positive cylinder polynomials $2^{-m} \pm (\phi_{r,m} - \Phi_r)$ then belong to the Archimedean quadratic module by the Archimedean Positivstellensatz [15]. The canonical choice is the first polynomial in the fixed enumeration accompanied by these two finite ordered identities. For a crossing interval $I = [a, b]$ and $m$ with $\delta_{r,m}(I) := \phi_{r,m}(u_b) - \phi_{r,m}(u_a) > 0$, put

$$\widehat{\phi}_{r,m,I} := \frac{\phi_{r,m} - \phi_{r,m}(u_a)}{\delta_{r,m}(I)}.$$

The reciprocal is not inserted analytically: its total graph is generated by the bounded coordinate $h_{r,m,I}$ and the polynomial relations

$$\delta_{r,m}(I) h_{r,m,I} = 1, \qquad \widehat{\phi}_{r,m,I} = h_{r,m,I}(\phi_{r,m} - \phi_{r,m}(u_a)).$$

Since a crossing gives $\delta_{r,m}(I) \geq 1 - 2^{1-m} > 0$ for $m \geq 2$, this graph has one ordinary value and a uniform bound. The event-normalized cylinder shell therefore has endpoint increment one on every crossing record.

On every finite approximation record, the weak evolution and the typed graph chain rules derive a signed Stieltjes current

$$\mathrm{d}\widehat{\phi}_{r,m,I}(u_t) = \sum_{c\in\mathfrak{C}} \mathrm{d}Q^c_{r,m,I}(t). \tag{34}$$

Discrete time steps contribute their exact jump measures. A failed product, trace, inverse, chart, or covariance rule contributes the graph current of its least failed typed operation to the corresponding inherited source cell. There is no undifferentiated shell remainder.

The current coordinates are

$$Q^c_{r,m,I}(\psi) := \int_I \psi \,\mathrm{d}Q^c_{r,m,I}, \qquad \psi \in C^1_{\mathbb{Q}}(I).$$

Their sourcewise graph closure defines distributional shell currents on $X_{\mathrm{gc}}$. On the bounded-variation branch they are finite signed Radon measures with their Lebesgue, jump, and Cantor parts retained. The provenance disintegration additionally retains oscillation, concentration, and exterior coordinates inherited from the approximating state records. If their total variation is unbounded, the normalized current and the compactified variation coordinate

$$\left(\frac{Q^c}{1+\|Q^c\|_{\mathrm{TV}}}, \frac{\|Q^c\|_{\mathrm{TV}}}{1+\|Q^c\|_{\mathrm{TV}}}\right)$$

give the typed variation-boundary value. Because (34) is an exact cylinder identity, its closure gives

$$1 = \sum_{c\in\mathfrak{C}} Q^c_{r,m,I}(1) \tag{35}$$

on every ordinary, Radon, and graph-boundary branch.

The contact time is $\tau_\Sigma(u) = \inf\{t : u(t) \in \Sigma\}$, with the germ interpretation when $T_*(u)$ is a noncontinuation time. If $u(0) \notin \Sigma$ and $u$ contacts $\Sigma$, then for every sufficiently small $r$ there are strictly pre-contact times

$$0 \leq t_r^- < t_r^+ < \tau_\Sigma(u), \qquad u(t_r^-) \notin U_{2r}(\Sigma), \qquad u(t_r^+) \in U_r(\Sigma). \tag{36}$$

The metric shells separate every point outside the closed target, while contact supplies the second time. A target at infinity is already a boundary point of $X_{\mathrm{ev}}$; an additional ordinary normalization merely appends its $\mathsf{Abs}/\mathsf{Lift}$ ancestry.

**Theorem I.5.2** (Total normalized shell-crossing accountability)**.** *Let $u$ be an ordinary maximal orbit generated by the compiled continuum algebra, with $u(0) \notin \Sigma$, on the contact branch. For every sufficiently small $r$, choose a crossing interval $I_r = [t_r^-, t_r^+]$ as in* (36) *and then $m \geq 2$. Equation* (33) *gives $\delta_{r,m}(I_r) \geq 1 - 2^{1-m} > 0$. Write the physical current as*

$$J = \sum_{c\in\mathfrak{C}} J^c + J^{J_{\mathrm{res}}},$$

*where $J^{J_{\mathrm{res}}} \in \mathscr{N}_\Sigma$. Evaluate the derived cylinder-shell current on $I_r$. Its ordinary signed source progress is*

$$Q_{c,r,m}(u) := \int_{I_r} \langle D\widehat{\phi}_{r,m,I_r}(u_t), J^c_t\rangle \,\mathrm{d}t. \tag{37}$$

*On every retained ordinary or graph-boundary realization of this same-origin crossing branch, the shell crossing has the exact accountability identity*

$$\boxed{1 = \widehat{\phi}_{r,m,I_r}(u(t_r^+)) - \widehat{\phi}_{r,m,I_r}(u(t_r^-)) = \sum_{c\in\mathfrak{C}} Q^c_{r,m,I_r}(1).} \tag{38}$$

*On the ordinary branch,*

$$\int_{I_r} \langle D\widehat{\phi}_{r,m,I_r}(u_t), J_t^{J_{\mathrm{res}}}\rangle \,\mathrm{d}t = 0. \tag{39}$$

*On every graph-complete realization of this same-origin crossing branch,*

$$\sum_{c\in\mathfrak{C}} (Q^c_{r,m,I_r}(1))_+ \geq 1, \qquad \max_{c\in\mathfrak{C}} (Q^c_{r,m,I_r}(1))_+ \geq \frac{1}{5}. \tag{40}$$

*The positive contribution may be ordinary, jump, Cantor, oscillation, concentration, exterior, or variation-boundary current, but it always has the ancestry of one of the five source cells. Thus, without requiring a classical chain rule,*

$$\boxed{\begin{gathered} u \textit{ contacts } \Sigma \quad \Longrightarrow \\ \forall r \textit{ sufficiently small } \exists \textit{ a five-channel source occurrence } R_r : \mathrm{Prog}_\Sigma(R_r) > 0. \end{gathered}} \tag{41}$$

*Thus contact itself exposes a target-aligned pre-contact occurrence in an existing source cell at every sufficiently small shell.*

*Proof.* The cylinder approximation gives the stated positivity of $\delta_{r,m}(I_r)$, and event normalization gives the unit increment. Equation (35) gives the source sum on every graph-complete branch. On the ordinary branch, insert the current normal form. The generated shell differential belongs to $\Gamma_\Sigma$, so residual target-nullity proves (39). Taking positive parts of a five-term signed sum equal to one proves (40). The first-failure theorem and sourcewise closure assign every nonordinary positive summand its inherited nonempty support. □

**Corollary I.5.3** (No spontaneous deterministic contact)**.** *Let $u$ be an ordinary maximal orbit generated by the compiled continuum algebra, with $u(0) \notin \Sigma$. Suppose that every ordinary and graph-boundary derived shell current is target-null on every pre-contact shell. On the ordinary branch this reads*

$$\langle D\widehat{\phi}_{r,m,I}(u_t), J_t^c\rangle = 0 \quad \textit{for a.e. } t < \tau_\Sigma(u), \quad c \in \mathfrak{C}. \tag{42}$$

*Then $u$ cannot contact $\Sigma$. Target avoidance also follows when every derived source-current mass is nonpositive. For a singularity at infinity, these statements apply to the canonical boundary shells in the generated evaluation compactification.*

*Proof.* If contact occurred, Theorem I.5.2 would produce a positive source-resolved derived current on every sufficiently small shell. This contradicts the stated target-nullity. Under the stronger ordinary zero-pairing condition, the same conclusion follows directly from (35). □

**Corollary I.5.4** (Normalized contact at infinity)**.** *On the branch where singular contact is defined after generated normalizations $v_r = \mathcal{S}_r u$ into a state or germ compactification carrying normalized shells $\widetilde{\Phi}_r$. If the normalized current $\widetilde{J}_r$ has the compiled normal form, including every chart derivative with its* Abs/Lift *ancestry, then every normalized crossing interval satisfies*

$$1 = \sum_{c\in\mathfrak{C}} \int_{I_r} \langle D\widetilde{\Phi}_r(v_r(t)), \widetilde{J}^c_{r,t}\rangle \,\mathrm{d}t, \qquad \max_{c\in\mathfrak{C}} \left(\int_{I_r} \langle D\widetilde{\Phi}_r(v_r(t)), \widetilde{J}^c_{r,t}\rangle \,\mathrm{d}t\right)_+ \geq \frac{1}{5}.$$

*Thus blow-up at infinity cannot occur without target-aligned structure in an existing source cell.*

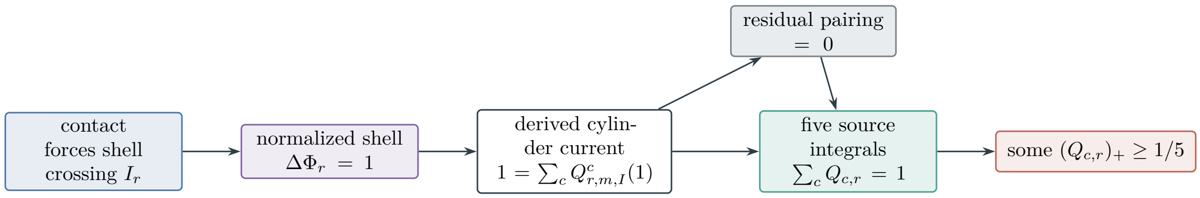


Figure 8: Normalized shell-crossing accountability. Contact forces a unit increase of the shell observable along the actual pre-contact current. The target-null residual contributes zero, so one of the five source integrals has positive part at least 1/5.

*Proof.* Apply Theorem I.5.2 in the normalized state space. Source preservation under the legal chart is Theorem I.3.4. □

**Theorem I.5.5** (Resolvent source-response comparison)**.** *Let $L$ and $L_0$ be closed killed generators on the same Banach space, and let $\lambda \in \rho(L) \cap \rho(L_0)$. Write*

$$R_L(\lambda) = (\lambda - L)^{-1}, \qquad R_0(\lambda) = (\lambda - L_0)^{-1}.$$

*Totalize domain membership of $R_0(\lambda)k_\Sigma$ and the generator difference. On their ordinary branch,*

$$L - L_0 = \sum_{c \in \mathfrak{C}} K^c + K^{J_{\text{res}}} \tag{43}$$

*on that vector. Then, for every continuous initial functional $\ell_0$,*

$$\begin{aligned}\langle \ell_0, (R_L(\lambda) - R_0(\lambda))k_\Sigma \rangle = &\sum_{c \in \mathfrak{C}} \langle R_L(\lambda)^* \ell_0, K^c R_0(\lambda) k_\Sigma \rangle \\ &+ \langle R_L(\lambda)^* \ell_0, K^{J_{\text{res}}} R_0(\lambda) k_\Sigma \rangle.\end{aligned} \tag{44}$$

*If the last pairing is target-null, it vanishes. If the real part $p$ of the left side is positive, some existing source cell has real pairing at least $p/5$. Any domain or generator-split failure is returned by the form/realization row and appears only as the ordinary value of that specialized row. The factorization is used for reachability only on the retained shell/source cell; the five displayed summands inherit the source words of* (43)*.*

*Proof.* The second resolvent identity gives

$$R_L(\lambda) - R_0(\lambda) = R_L(\lambda)(L - L_0)R_0(\lambda).$$

Pair with $\ell_0$ and $k_\Sigma$, then insert (43). Target-nullity removes the residual term. A positive sum of five real numbers has a summand at least one fifth of that sum. □

*Remark* I.5.6 (Arrival semantics). Equation (44) is an operator identity. It becomes a difference of discounted contact responses when the killed semigroups admit the target-flux realization of Theorem II.5.4. No probabilistic meaning is inferred from the algebraic resolvent alone.

**Theorem I.5.7** (Stochastic no-spontaneous-contact accountability)**.** *Let $\tau_\Sigma$ be a stopping time and $N_t = \mathbf{1}_{\{\tau_\Sigma \leq t\}}$. Totalize the Doob–Meyer and source-split relations. On the branch where $N - A$ is a martingale, the predictable compensator on $[0, T]$ has the finite-variation decomposition*

$$A = \sum_{c \in \mathfrak{C}} A^c + A^{J_{\text{res}}}, \tag{45}$$

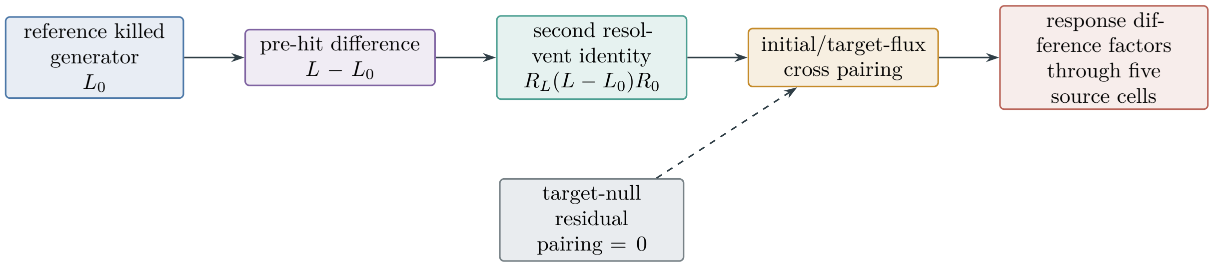


Figure 9: Resolvent source-response comparison. A discounted target-response difference is sandwiched around the pre-hit generator difference and therefore inherits its five-source decomposition. The no-spontaneous-contact theorem itself comes from normalized shell crossing.

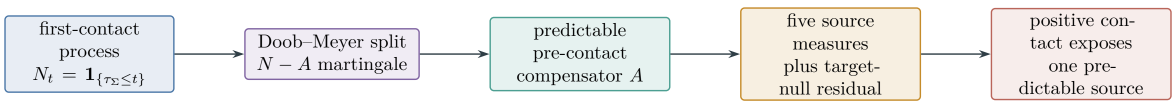


Figure 10: Stochastic no-spontaneous-contact accountability. First contact is charged by its predictable pre-contact compensator. A positive contact probability forces a positive contribution from one of the five source measures before the stopping time.

*where all measures are supported on $\{t < \tau_\Sigma\}$, the signed source terms are integrable, every term starts from zero, $\mathbb{P}\{\tau_\Sigma = 0\} = 0$, and $\mathbb{E}A_T^{J_{\mathrm{res}}} = 0$. Then*

$$\mathbb{P}\{\tau_\Sigma \le T\} = \sum_{c \in \mathfrak{C}} \mathbb{E}A_T^c. \tag{46}$$

*If $p_T := \mathbb{P}\{\tau_\Sigma \le T\} > 0$, one pre-contact source compensator satisfies $\mathbb{E}A_T^c \ge p_T/5$. In particular, all five source expectations cannot vanish when contact has positive probability. Failure of the martingale, finite-variation, support, or integrability relation is its stochastic total-graph defect and remains in the fluctuation row.*

*Proof.* Taking expectations of the integrable martingale $N - A$, using the stated zero initial values, gives $\mathbb{E}N_T = \mathbb{E}A_T$. Insert (45) and use the zero expectation of the residual term. The final assertion follows from the five-term pigeonhole argument. □

**Corollary I.5.8** (Continuum prospective source exhaustion)**.** *Normalized deterministic contact and positive-probability stochastic contact both force a positive pre-contact occurrence with ancestry in $\mathfrak{C}$. A positive killed-resolvent response difference has the same source-extraction property. The post-saturation residual contribution is zero in all three interfaces.*

**Definition I.5.9** (Saturated prospective source profile)**.** Fix one of the three prospective interfaces, restricted to records satisfying its stated target-null residual condition, and fix its normalization. Write $\mathrm{Prog}_\Sigma^c(R)$ for the signed $c$-source contribution: $Q_{c,r}$ for a deterministic shell, $\mathbb{E}A_T^c$ for a stochastic contact record, and the real $c$-pairing in (44) for a resolvent comparison. Set

$$\mathrm{Prog}_\Sigma(R) := \sum_{c \in \mathfrak{C}} \mathrm{Prog}_\Sigma^c(R). \tag{47}$$

Define

$$\mathsf{T}_{\Sigma,c}^{\mathrm{phys,sat}}(B_{\mathrm{phys}}) := \sup_{R \in \mathscr{R}^{\mathrm{phys,sat}}(B_{\mathrm{phys}})} \left(\mathrm{Prog}_\Sigma^c(R)\right)_+, \tag{48}$$

$$\mathsf{T}^{\mathrm{phys,sat}}_{\Sigma}(B_{\mathrm{phys}}) := \sum_{c\in\mathfrak{C}} \mathsf{T}^{\mathrm{phys,sat}}_{\Sigma,c}(B_{\mathrm{phys}}). \tag{49}$$

The supremum is taken over the ordinary graph values of the selected interface and may equal $+\infty$. Inputs outside that graph enter the corresponding totalized defect cell, so every input has a structural value.

**Definition I.5.10** (Source-support and realization-mode profiles)**.** For each nonempty $A \subseteq \mathfrak{C}$, set

$$\mathrm{Prog}^{A}_{\Sigma}(R) := \sum_{c\in A} \mathrm{Prog}^{c}_{\Sigma}(R)\mathbf{1}_{\{\mathrm{Src}(R)=A\}}.$$

Every compiled record also carries the subset of realization-mode flags

$$\mathfrak{M} = \{\mathsf{exact}, \mathsf{closure}, \mathsf{harmonic}, \mathsf{concentration}, \mathsf{boundary}\}$$

determined by its construction:

(M1) $\mathsf{exact}$: it has a witness in the range of its native realization map;

(M2) $\mathsf{closure}$: it lies in the sourcewise range closure, possibly outside the range;

(M3) $\mathsf{harmonic}$: its class in $\ker G_A^*/(\ker G_A^* \cap \mathscr{N}_\Sigma)$ is nonzero;

(M4) $\mathsf{concentration}$: it is read through a Young measure, defect measure, profile, or blow-up-state lift;

(M5) $\mathsf{boundary}$: it is a trace, killing, tail, terminal germ, or graph-boundary record.

Flags may coexist; they refine ancestry and do not form new sources. For bookkeeping fix the order

$$\mathsf{exact} < \mathsf{closure} < \mathsf{harmonic} < \mathsf{concentration} < \mathsf{boundary}$$

and let $m_*(R)$ be the maximal flag carried by $R$. Thus the complete flag set is retained, while every record belongs to exactly one *primary-mode* cell. Generated operations only append flags, so $m_*$ is monotone along a successor path and hence eventually constant. For $m \in \mathfrak{M}$, define

$$\mathsf{T}^{\mathrm{phys,sat}}_{\Sigma,A,m}(B_{\mathrm{phys}}) := \sup_{\substack{R\in\mathscr{R}^{\mathrm{phys,sat}}_{A}(B_{\mathrm{phys}})\\ m_*(R)=m}} \left(\mathrm{Prog}^{A}_{\Sigma}(R)\right)_+, \tag{50}$$

with supremum 0 for an empty class.

**Theorem I.5.11** (Realization-mode exhaustion)**.** *Every target-active continuum record belongs to a nonempty source cell $A \subseteq \mathfrak{C}$, has at least one flag in $\mathfrak{M}$, and has exactly one primary mode. Moreover,*

$$\mathsf{T}^{\mathrm{phys,sat}}_{\Sigma}(B_{\mathrm{phys}}) \le \sum_{\varnothing\neq A\subseteq\mathfrak{C}} \sum_{m\in\mathfrak{M}} \mathsf{T}^{\mathrm{phys,sat}}_{\Sigma,A,m}(B_{\mathrm{phys}}). \tag{51}$$

*Thus exact currents, nonclosed-range limits, harmonic classes, concentrations, and boundary defects exhaust the analytic realization modes of target progress without enlarging the five-source alphabet.*

*Proof.* Native records are exact. A sourcewise limit is in the closure mode. The orthogonal decomposition $\mathcal{J}_A = \overline{\mathrm{Ran}\, G_A} \oplus \ker G_A^*$, followed by the target quotient, assigns every target-active orthogonal record to the harmonic mode. The compiler marks every profile lift and every boundary operation with the last two flags. These cases cover the finite syntax and its sourcewise completion. Every positive source contribution is therefore bounded by at least one term on the right of (51); summing gives the result. Multiple flags only enlarge the right side. $\square$

**Theorem I.5.12** (Source-only prospective envelope)**.** *Every record in the domain of a prospective interface satisfies*

$$\left(\mathrm{Prog}_\Sigma(R)\right)_+ \le \mathsf{T}_\Sigma^{\mathrm{phys,sat}}(B_{\mathrm{phys}}). \tag{52}$$

*The envelope consists solely of the five source profiles. Any finite upper bounds on those profiles give a complete prospective progress bound. A normalized deterministic contact record satisfies* $\mathrm{Prog}_\Sigma(R) = 1$*.*

*Proof.* Each of Theorems I.5.2, I.5.5 and I.5.7 has, after removal of its target-null residual pairing, the form

$$\mathrm{Prog}_\Sigma(R) = \sum_{c\in\mathfrak{C}} \mathrm{Prog}_\Sigma^c(R).$$

Take the positive part, bound every signed summand by its positive part, and then apply (48). □

**Definition I.5.13** (Singularity-aligned structural support)**.** For a pre-contact record $R$ in any prospective interface, define

$$\mathrm{Align}_\Sigma(R) := \{c \in \mathfrak{C} : \mathrm{Prog}_\Sigma^c(R) > 0\} \cup \begin{cases} \mathrm{Src}(R), & R \text{ is a total-graph defect with } \mathrm{Prog}_\Sigma(R) > 0, \\ \varnothing, & \text{otherwise.} \end{cases} \tag{53}$$

Equivalently, in the deterministic interface an occurrence is aligned when the singularity-descending covector $D\Phi_r(u_t)$ has positive integrated pairing with its source current on the crossing interval. The word “aligned” is target-relative: it does not rename a source as $\mathsf{Caus}$, and it is evaluated only along the fixed equation.

**Theorem I.5.14** (No-spontaneous-progress principle)**.** *For every prospective record, including total-graph defect records,*

$$\mathrm{Prog}_\Sigma(R) > 0 \quad \Longrightarrow \quad \mathrm{Align}_\Sigma(R) \neq \varnothing. \tag{54}$$

*For an ordinary five-term additive record one also has*

$$\max_{c\in\mathfrak{C}} \left(\mathrm{Prog}_\Sigma^c(R)\right)_+ \ge \frac{\mathrm{Prog}_\Sigma(R)}{5}. \tag{55}$$

*If every ordinary source occurrence and every total-graph defect represented by $R$ lies in $\mathscr{N}_\Sigma$, then*

$$\mathrm{Prog}_\Sigma(R) = 0, \tag{56}$$

*regardless of the size, oscillation, or compactness of those target-null currents. More generally, if every source contribution is nonpositive, then positive target progress is impossible. For a normalized deterministic shell crossing, the derived Stieltjes-current identity yields a positive contribution of at least* $1/5$ *in one inherited source cell on every ordinary or graph-boundary branch.*

*Proof.* Equation (47) is the compiled five-term identity on ordinary records and its exact sourcewise graph closure on boundary records. A positive sum has a positive summand, and its largest positive part is at least one fifth of the sum. Source-preserving graph closure assigns each boundary summand its nonempty inherited support. If every occurrence lies in the common target kernel, every term vanishes. The shell assertion is Theorem I.5.2. □

**Corollary I.5.15** (No-spontaneous-contact regularity)**.** *Let the target have a normalized shell presentation, and let every legal orbit be generated by the compiled algebra on its regular pre-contact interval. If* $\mathrm{Align}_\Sigma(R) = \varnothing$ *for every legal shell record, then the target is unreachable. In particular, if all five source currents have zero class in* $\mathscr{J}_\Sigma$ *along every pre-contact shell, their ancestry is retained while their target contribution lies in* $\mathscr{N}_\Sigma$*, and contact is impossible:*

$$\boxed{q_\Sigma(J^c) = 0 \text{ on every pre-contact target shell for all } c \in \mathfrak{C} \quad \Longrightarrow \quad \tau_\Sigma = \infty.} \tag{57}$$

*This conclusion requires neither compactness of the full orbit nor a positive physical-price estimate.*

*Proof.* If an orbit contacted $\Sigma$, Theorem I.5.2 would provide a normalized shell record with unit progress. By Theorem I.5.14, its aligned support would be nonempty, contrary to the displayed null-source condition. Under that quotient value, every shell source integral vanishes, so the same contradiction follows directly from (38). ☐

*Remark* I.5.16 (Why the target quotient is stronger than a full norm bound). A classical energy estimate commonly controls every component of a current, including directions irrelevant to the singular target. The quotient $\mathscr{J}_\Sigma$ discards a current only after all declared prospective target pairings vanish. The discarded current may be large or noncompact; no estimate of it is needed for (56). Only the target-aligned quotient directions proceed to resistance, compactness, rigidity, or physical-price invariants. The quotient strictly reduces the regularity problem because target-null directions are absent from all downstream estimates.

# Part II
# Physical Structural-Invariant Calculus

## II.1 Structural invariant reduction of analytic arguments

*Remark* II.1.1 (Ordinary-branch convention for analytic modules). Every conditional identity in this part describes the ordinary value of a total graph generated from the public signature. Its operator, form, chart, or multiplier must already occur in the typed syntax or in the fixed target-independent textbook library. The condition is never appended as specialization data: when it does not hold, its complementary graph value is sent to the activity lift and successor relation. Hamiltonian, commutator, compactness, tail, rigidity, and entropy modules can shorten a pruning transcript, but the core contact-to-successor theorem does not require their ordinary values.

The closed source algebra separates the origin of a continuum quantity from the estimate used to control it. Once a source family has been placed in a closed target-null quotient or removed from the admissible orbit family by a proved physical exclusion, later invariants are evaluated on the remaining reduced space. The analytic calculation then has only three possible jobs: close a remaining class, compare two nonnegative forms, or resolve the equality set of a sharp comparison.

**Definition II.1.2** (Quotient-compatible structural reduction)**.** Let $Y$ be a Hausdorff locally convex space of currents, profiles, or limit records, and let $K \subseteq Y$ be a closed linear subspace already annihilated by the downstream target and invariant maps. Write

$$\pi_K : Y \longrightarrow Y/K.$$

A downstream map, form, evaluation closure, or source transport is *compatible with the reduction* when it is constant on cosets of $K$ and therefore factors continuously through $\pi_K$. A previously excluded closed family is removed from the admissible domain before the quotient is formed. Thus a reduction never discards a target-visible family without the proved invariant identity that excludes that family.

**Theorem II.1.3** (Propagation of structural reductions)**.** *Let $K_0 \subseteq K_1 \subseteq Y$ be closed subspaces. Totalize every later map relative to both reductions. On the zero descent-defect branch, the canonical map is a topological linear isomorphism*

$$(Y/K_0)/(K_1/K_0) \cong Y/K_1. \tag{58}$$

*On a nonzero descent-defect branch that defect is retained in its inherited source cell. Consequently a source direction closed at one stage may be propagated into the next quotient and omitted from every later compatible calculation. A later invariant computed after reduction has the same value as the corresponding invariant on $Y$ after quotienting by all previously closed directions.*

*Proof.* The quotient map $Y/K_0 \to Y/K_1$ is continuous, open, and has kernel $K_1/K_0$. The topological first isomorphism theorem gives (58). Every zero-defect downstream object factors through these quotient maps, so its reduced and unreduced evaluations agree under the displayed isomorphism. A failure to factor is the nonzero descent-defect branch stipulated in the theorem. □

**Theorem II.1.4** (Target-quotient coercivity)**.** *Let $Y$ be a Hilbert source space, let $K \subseteq Y$ be a closed target-null subspace, and let $T : Y \to Z_\Sigma$ be bounded with $K \subseteq \ker T$. Let $q : Y \to [0, \infty]$ be an extended nonnegative quadratic form with finite domain $D(q)$ satisfying $D(q) + K \subseteq D(q)$, and define its relaxed quotient form by*

$$q_{\mathrm{red}}([y]) := \inf_{k \in K} q(y + k), \qquad [y] \in Y/K. \tag{59}$$

*If, on the required reduced domain,*

$$\|[y]\|^2_{Y/K} \le C q_{\mathrm{red}}([y]), \tag{60}$$

*then*

$$\|Ty\|^2_{Z_\Sigma} \le \|\widetilde{T}\|^2 C q_{\mathrm{red}}([y]) \le \|\widetilde{T}\|^2 C q(y), \tag{61}$$

*where $\widetilde{T} : Y/K \to Z_\Sigma$ is the descended target map. No bound for the full norm $\|y\|_Y$ follows or is needed. When invoked by the regularity calculus, $Y/K$, $q_{\mathrm{red}}$, and $T$ are first restricted to the contact-generated active source cell; failure of* (60) *is its complementary local structural value.*

*Proof.* Because $T$ annihilates $K$, the quotient universal property gives the bounded map $\widetilde{T}$ with $T = \widetilde{T}\pi_K$. Hence

$$\|Ty\| \le \|\widetilde{T}\| \, \|[y]\|_{Y/K}.$$

Apply (60); the final inequality follows by using $k = 0$ in (59). □

**Example II.1.5** (Strict gain over full-state coercivity)**.** Let $Y = K \oplus Z_\Sigma$, let $T(k, z) = z$, and let $q(k, z) = \|z\|^2$. Then (60) holds with $C = 1$, while no inequality $\|(k, z)\|^2_Y \le C' q(k, z)$ can hold when $K \ne \{0\}$. The sequence $(nk_0, 0)$ may diverge in the full state norm but is exactly target-null. Thus the quotient theorem proves every target estimate available from $q$ without controlling that divergent motion.

**Definition II.1.6** (Structural evaluation invariants)**.** Work on a zero-descent-defect reduced space $Y_{\mathrm{red}}$, with a continuous target projection $\Pi_\Sigma : Y_{\mathrm{red}} \to Z_\Sigma$. The space and every family below are restricted to one propagated same-origin target/source cell: a fixed nonempty source support, realization mode, target-shell floor, and all earlier zero transcript coordinates. No extremum in this definition ranges over a source-transverse state. The structural invariants used below have the following forms.

(I1) For a retained target/source family $\mathcal{F} \subseteq Y_{\mathrm{red}}$, its *target closure invariant* is

$$\mathfrak{H}^{\Sigma}_{\mathcal{F}} := \overline{\operatorname{span}\Pi_\Sigma(\mathcal{F})}^{Z_\Sigma}. \tag{62}$$

(I2) For nonnegative functionals $D, N_+$ on a reduced target/source family $\mathcal{P}$, the *sharp domination radius* is

$$\alpha_*(N_+ \mid D) := \sup_{\substack{V\in\mathcal{P}\\ D(V)>0}} \frac{N_+(V)}{D(V)}. \tag{63}$$

The value is $+\infty$ when some $V$ has $D(V) = 0 < N_+(V)$. For a normalized retained family $\mathcal{N} \subseteq \mathcal{P}$, the *physical-price gap* is

$$\delta_*(D \mid \mathcal{N}) := \inf_{V\in\mathcal{N}} D(V). \tag{64}$$

(I3) When $\alpha_* < \infty$, let $\widehat{\mathcal{E}}_{\alpha_*}$ be the closure in the canonical evaluation compactification of the normalized equality family

$$\mathcal{E}_{\alpha_*} := \{V \in \mathcal{P} : D(V) > 0,\ N_+(V) = \alpha_* D(V)\}.$$

Totalize the remainder relation generated by the equality identities. Its ordinary graph image has *equality–rigidity invariant*

$$\mathfrak{H}^{\Sigma}_{\mathrm{eq}} := \overline{\operatorname{span}\Pi_\Sigma\mathscr{R}_{\mathrm{eq}}(\widehat{\mathcal{E}}_{\alpha_*})}^{Z_\Sigma}. \tag{65}$$

and its failure and graph-boundary values enter $\mathfrak{H}^{\Sigma}_{\mathrm{sat}}$. For $\alpha_* = +\infty$, the equality row is canonically inapplicable and the positive null-cost or unbounded-feeding class is already returned by the radius row. The equality identities may be saturation equations, conservation laws, modulation equations, Pohozaev identities, or Liouville constraints.

Each invariant is computed after all zero-descent-defect source, gauge, trace, and target-null reductions already proved have been propagated.

**Theorem II.1.7** (Structural normal form for invariant evaluation)**.** *Within the interfaces of this paper, every analytic input to target exclusion has one of the following exact structural meanings.*

(N1) *A retained target/source family is target-null on the reduced interface exactly when $\mathfrak{H}^{\Sigma}_{\mathcal{F}} = 0$.*

(N2) *The domination inequality $N_+ \le aD$ holds on the retained target/source family $\mathcal{P}$ exactly when $\alpha_*(N_+ \mid D) \le a$.*

(N3) *The normalized target/source family $\mathcal{N}$ has a uniform positive physical price exactly when $\delta_*(D \mid \mathcal{N}) > 0$.*

(N4) *A sharp equality branch has no target-visible remainder exactly when $\mathfrak{H}^{\Sigma}_{\mathrm{eq}} = 0$.*

*Source closure and target-null residual calculations use* (N1)*; resolvent, flux, and carrier identities generate* (N2)*; physical prices generate* (N3)*; and zero-cost equations generate* (N4)*. None is an endpoint. Its nonclosing value supplies a normalized witness to the activity lift, after which the successor and viability operators continue the calculation.*

*Proof.* Item (N1) is the definition of the closed target projection in (62). For (N2), divide by $D(V)$ when it is positive; the convention for a positive $D$-null mode gives the remaining case. Item (N3) is the definition of the infimum in (64). Item (N4) follows from (65): its vanishing is precisely the vanishing of every target projection of the equality remainder. These are the four structural coordinates occurring in Theorems I.4.6 and I.5.12, Definition II.8.2, and Lemma II.8.3. □

**Definition II.1.8** (Operational fast-track evaluator)**.** For a source/mode cell $(A, m)$, propagate all previously vanished closed subspaces and let $Y^{\Sigma}_{A,m}$ be the resulting target-active quotient. Introduce the ordered row set

$$\begin{aligned}\mathscr{R}_{\mathrm{FT}} := \{&\mathrm{form}, \mathrm{grad}, \mathrm{chain}, \mathrm{bdry}, \mathrm{tail},\\ &\mathrm{target}, \mathrm{feed}, \mathrm{price}, \mathrm{route}, \mathrm{capacity},\\ &\mathrm{scale}, \mathrm{sat}, \mathrm{fluct}, \mathrm{cov}, \mathrm{succ}\}.\end{aligned} \tag{66}$$

Its fast-track value is the row-indexed tuple

$$\mathfrak{I}^{\Sigma}_{A,m} := \left(\mathfrak{I}^{\Sigma,r}_{A,m}\right)_{r\in\mathscr{R}_{\mathrm{FT}}}, \tag{67}$$

where the first five and covariance entries are target-active total-graph defect quotients; $\mathfrak{H}^{\Sigma}_{A,m}$ is the target closure; $\alpha^*$ is the feeding radius; $\delta^*$ is the physical-price infimum; $\mathcal{R}$ is effective resistance; Cap is target capacity; $\mathscr{S}$ is the pulled-back nonnegative scale-price series; $\mathfrak{H}^{\Sigma}_{\mathrm{sat}}$ is the projected saturation or equality remainder; and $\chi^{\Sigma}$ is the target-active covariance, bracket, or fluctuation radius when the evolution is stochastic. An inapplicable row has the singleton value $\{\perp_{\mathrm{n/a}}\}$ precisely when its defining type is absent from the fixed PDE signature, as for the fluctuation row of a deterministic equation. Applicability is not a choice. Every other entry is an exact quotient, graph defect, infimum, supremum, kernel, capacity, resistance, or series value. No favorable analytic property occurs in the definition.

**Definition II.1.9** (Structural property compiler)**.** For a row $r \in \mathscr{R}_{\mathrm{FT}}$, let $F^{\mathrm{tot}}_r$ be its totalized interface on the current reduced cell and let $\mathcal{K}^{\Sigma}_r$ be the closed ledger already propagated into that cell. The structural property compiler $\mathsf{P}_{\Sigma}$ assigns the following exact invariant statements:

$$\mathsf{P}_{\Sigma}[\text{native realization or local existence}] := \mathfrak{H}^{\Sigma}_{\mathrm{form},A,m} = 0, \tag{68}$$

$$\mathsf{P}_{\Sigma}[\text{target-relative closed gradient range}] := \mathfrak{H}^{\Sigma}_{\mathrm{grad},A,m} = 0, \tag{69}$$

$$\mathsf{P}_{\Sigma}[\text{target-relevant compactness of generated limits}] := \mathfrak{H}^{\Sigma}_{\mathrm{bdry},A,m} = 0, \tag{70}$$

$$\mathsf{P}_{\Sigma}[\text{pre-contact chain rule}] := \mathfrak{H}^{\Sigma}_{\mathrm{chain},A,m} = 0, \tag{71}$$

$$\mathsf{P}_{\Sigma}[\text{constraint-tail closure}] := \mathfrak{H}^{\Sigma}_{\mathrm{tail},A,m} = 0, \tag{72}$$

$$\mathsf{P}_{\Sigma}[\text{generated operation descends}] := \mathfrak{H}^{\Sigma}_{\mathrm{cov},A,m} = 0, \tag{73}$$

$$\mathsf{P}_{\Sigma}[N_+ \le aD] := \alpha^*_{A,m}(N_+ \mid D) \le a, \tag{74}$$

$$\mathsf{P}_{\Sigma}[\inf_{\mathcal{N}} D > 0] := \delta^*_{A,m}(D \mid \mathcal{N}) > 0, \tag{75}$$

$$\mathsf{P}_{\Sigma}[\text{routable target current}] := \mathcal{R}_{A,m}(\Sigma) < \infty, \tag{76}$$

$$\mathsf{P}_{\Sigma}[\text{target polar}] := \mathrm{Cap}_{A,m}(\Sigma) = 0, \tag{77}$$

$$\mathsf{P}_{\Sigma}[\text{finite-budget scale exclusion}] := \mathscr{S}^{\Sigma}_{A,m} = +\infty, \tag{78}$$

$$\mathsf{P}_{\Sigma}[\text{sharp equality or saturation closes}] := \mathfrak{H}^{\Sigma}_{\mathrm{sat},A,m} = 0, \tag{79}$$

$$\mathsf{P}_{\Sigma}[\text{fluctuation is target-null or physically routed}] := \chi^{\Sigma}_{A,m} = 0 \text{ or its resistance is finite.} \tag{80}$$

The successor row is the finite-depth sequence $Z_{n+1,A,m} = \mathcal{V}_\eta(Z_{n,A,m})$; its two values are finite pruning with its rational-cylinder transcript and the compact inverse-limit path of Theorem III.2.8. Here every object is formed after quotienting by $\mathcal{K}_r^\Sigma$. Any algebraic identity derivable from the public weak equation may refine a row into nonnegative descendants. The compiler has no input slot for an external compactness, coercivity, Liouville, or realization statement.

**Definition II.1.10** (Propagated physical fast-track ledger)**.** Order the rows as in (66). Begin with the closed target-null ledger $\mathcal{K}_0^\Sigma = \mathscr{N}_\Sigma$. If row $i$ has closed branch $C_i$ and transport $T_i$ into the next row, set

$$\mathcal{K}_{i+1}^\Sigma := \overline{\mathcal{K}_i^\Sigma + T_i(C_i)}. \tag{81}$$

All spaces, totalized interfaces, target projections, physical forms, and subsequent invariants are then replaced by their descended objects on the quotient by $\mathcal{K}_{i+1}^\Sigma$. If a map fails to descend, its nonzero commutator with the quotient map is the next row's target-active graph defect. Thus quotient compatibility is tested by the construction rather than imposed on it.

**Theorem II.1.11** (Analytic-property compilation)**.** *Every analytic property used by the continuum argument to decide whether a source/mode cell can contribute to singular contact is equivalent to the corresponding structural invariant statement in Definition II.1.9. The opposite analytic property is equivalent to the complementary invariant value and produces a tagged activity profile and successor. In particular, the fast track never presupposes closed range, compactness, a chain rule, a coercive estimate, a positive price, rigidity, polarity, finite resistance, summable tails, fluctuation control, orbit realization, or continuation.*

*Proof.* Existence and regularity properties are membership statements in the graph of a partial map. Totalization makes their negations explicit graph values, and quotienting their closed graph boundary by $\mathscr{N}_\Sigma$ proves (68)–(72) and (73). In the compactness row only target-visible loss of compactness survives; target-null escape is already in $J_{\text{res}}$. Thus the invariant tests exactly the compactness needed downstream rather than ambient compactness of the orbit. The radius and gap equivalences follow directly from the supremum and infimum definitions, including positive $D$-null directions and zero-price sequences. The spectral theorem gives the resistance/range/harmonic trichotomy, with the harmonic part retained in the target quotient. Capacity zero is the definition of the polar branch; positive capacity is its complementary target-compatible class. A nonnegative scale-price series either diverges or is finite. Saturation equations define a remainder map, so closure of the equality branch is exactly vanishing of its target quotient. Finally, stochastic drift, bracket, covariance, and conditional-expectation contributions are currents in the lifted state space; quotienting their target-null part and applying the same radius/resistance construction gives (80). Every complementary value enters the successor row, whose finite-depth/inverse-limit alternatives are exhaustive by compactness. These cases exhaust (66). □

**Definition II.1.12** (Target-relevant analytic statement grammar)**.** Let $\mathsf{An}_\Sigma(\mathfrak{P})$ be the least grammar generated from the fixed PDE signature by:

(A1) membership in, or failure of, a native, limit, chain, trace, tail, normalization, or continuation graph;
(A2) vanishing or nonvanishing after the target quotient;
(A3) comparison of two nonnegative physical forms, including an infimum or sharp radius;
(A4) range, kernel, resistance, capacity, and harmonic statements for the generated operator or form;
(A5) convergence or divergence of a nonnegative physical-price series;

(A6) a saturation/equality remainder after sharp comparison;

(A7) covariance, bracket, compensator, or conditional-expectation statements in a generated stochastic lift;

(A8) membership in a finite-depth successor set or its compact inverse-limit path space.

The grammar is closed under finite conjunction, sourcewise limit, pullback by a generated chart, and descent through a generated quotient. These closure operations are themselves totalized.

**Theorem II.1.13** (Exhaustive fast-track compilation)**.** *The property compiler extends uniquely to a total map*

$$\mathsf{P}_\Sigma : \mathsf{An}_\Sigma(\mathfrak{P}) \longrightarrow \prod_{\substack{\varnothing\neq A\subseteq\mathfrak{C}\\ m\in\mathfrak{M}}} \prod_{r\in\mathscr{R}_{\mathrm{FT}}} \mathrm{Val}\Big(\mathfrak{I}^{\Sigma,r}_{A,m}\Big)\,. \tag{82}$$

*Every statement that can affect the contact identity or the physical expenditure contradiction occurs in exactly one primitive row, possibly with several source/mode tags. Its negation is the complementary value of that same row. Hence no analytic property, compactness assertion, or equation-specific estimate is an unregistered premise of the fast track.*

*Proof.* Clauses (A1)–(A8) map respectively to the total-graph rows; target row; feeding/price rows; gradient/routing/capacity rows; scale row; saturation row; fluctuation row; and finite-depth successor row. Source and realization tags are assigned before evaluation by Theorems I.2.6 and I.5.11. Finite conjunction gives a product value. Sourcewise limits remain in their tagged closures. Chart pullback appends $\mathsf{Abs}/\mathsf{Lift}$, and quotient descent either factors or produces its totalized descent defect. Structural induction therefore defines $\mathsf{P}_\Sigma$ on the whole grammar and leaves no unmatched constructor.

By normalized shell accountability, contact affects the argument only through a target-active current or a totalized shell/interface defect. By the physical balance, exclusion affects it only through a nonnegative expenditure comparison or series. These are clauses (A1)–(A6); stochastic and successor-path semantics are exactly (A7)–(A8). Thus the grammar covers every input to the contact-versus-expenditure proof. □

**Theorem II.1.14** (Closed-input rule)**.** *The value of $\mathfrak{I}^{\Sigma,r}_{A,m}$ is determined by the fixed PDE signature, target, physical balance, and the closures generated from them. An auxiliary lemma may prove that value but cannot be added as data defining it. Any proposed object not generated by this algebra either:*

(E1) *is derived from the fixed signature, in which case the source compiler and property compiler assign it to existing rows; or*

(E2) *changes the signature, in which case the conclusion concerns the enlarged equation/presentation and is not a conclusion about the original PDE.*

*These alternatives exhaust the generated continuum algebra.*

*Proof.* The evaluation compactification is generated by Definition I.3.1; the source syntax by Definition I.2.5; and the analytic statement syntax by Definition II.1.12. Membership in either syntax is therefore determined before an invariant is evaluated. Structural induction gives (E1). Failure of membership means that a new primitive map, topology, law, or balance has been appended, which is precisely a change of signature and gives (E2). □

**Theorem II.1.15** (Ledger propagation and no re-estimation)**.** *At each fast-track row, exactly one of the closed and complementary branches is selected by its invariant value. A closed branch propagated by* (81) *remains zero in every later descended target quotient. If a downstream object* *does not descend, its descent defect is retained in the appropriate existing source/mode cell. Therefore later rows neither assume nor re-estimate any property closed by an earlier row.*

*Proof.* For closed subspaces the third isomorphism theorem gives $(Y/\mathcal{K}_i)/(\mathcal{K}_{i+1}/\mathcal{K}_i) \cong Y/\mathcal{K}_{i+1}$. Hence every propagated class is zero in all later quotients. The same statement for closed convex ledgers follows by passing to the associated quotient seminorm. Failure of descent is precisely a nonzero commutator between the downstream map and the quotient map, which is a totalized graph defect. Sourcewise closure retains its ancestry. The exhaustive two-branch statement follows from Theorem II.1.11. $\square$

**Theorem II.1.16** (Fast-track branch-to-successor completeness)**.** *For every target-active source/mode cell, the tuple $\mathfrak{I}^{\Sigma}_{A,m}$ returns one of the following exhaustive structural branches.*

(F1) *Each form, gradient, chain, boundary, tail, and covariance defect quotient vanishes or survives as a target-visible total-graph defect with the same source ancestry.*

(F2) $\mathfrak{H}^{\Sigma}_{A,m} = 0$*: the cell is target-null and closes. Otherwise its nonzero quotient is retained.*

(F3) $\alpha^*_{A,m} \le 1$*: feeding is dominated by the generated carrier. If $\alpha^*_{A,m} > 1$, or $D = 0 < N_+$ for a member of the cell, the maximizing sequence becomes a target-feeding activity profile.*

(F4) $\delta^*_{A,m} > 0$*: the cell has positive normalized price. If $\delta^*_{A,m} = 0$, a zero-price sequence exists in the closure and is retained with the same ancestry.*

(F5) $\mathfrak{H}^{\Sigma}_{\mathrm{eq},A,m} = 0$*: the sharp equality branch closes. Otherwise its target-visible equality remainder survives.*

(F6) $\mathrm{Cap}_{A,m}(\Sigma) = 0$*: the target is polar for that cell. Positive capacity supplies a normalized target-active activity profile.*

(F7) $\mathcal{R}_{A,m}(\Sigma) < \infty$*: the source can be routed through the generated geometry. Infinite resistance supplies its harmonic component to the activity lift.*

(F8) *For a scale cascade, the pulled-back series $\sum_j r_j^{\kappa_{A,m}} \delta_{A,m,j}$ either diverges, excluding that finite-budget cascade, or is finite, yielding vanishing-price windows for the ordered successor and recurrent-saturation compiler.*

(F9) *The saturation quotient $\mathfrak{H}^{\Sigma}_{\mathrm{sat},A,m}$ vanishes or retains the target-visible equality remainder. Every Pohozaev, Noether, virial, frequency, or entropy identity already generated by the equation refines its nonnegative descendants; the remaining equality record becomes a successor.*

(F10) *The fluctuation invariant $\chi^{\Sigma}_{A,m}$ is target-null, physically routed at finite resistance, or survives as a target-visible covariance/bracket class in the existing lifted source cell.*

(F11) *The total weak-solution graph retains ordinary values and every graph-boundary witness. Both enter the successor graph; persistent boundary values survive only through an infinite zero-price witness path.*

(F12) *An inapplicable row returns $\{\perp_{\mathrm{n/a}}\}$; failure of a type-present map to descend returns its graph defect and activates the same successor operator.*

*Every complementary branch is a value of the same invariant used by the closing branch. The tuple is not a verdict: after each row, the canonical activity map sends every nonzero target quotient to $\mathcal{R}_\eta$, and the viability sets $Z_n$ either remove it at finite depth or construct its recurrent zero-price mechanism. Before either action, the row is converted to the target-local normal form* (102)*; hence no row acts on an equation-wide class or charges a source-transverse positive quantity.*

*Proof.* The graph-defect row follows from totalization. The target, radius, gap, and equality rows follow from Theorem II.1.7 and the definitions of the closed quotient, supremum, infimum, and equality remainder. In the radius row, the convention $\alpha^* = +\infty$ records a positive $D$-null feeding direction. If $\delta^* = 0$, the definition of the infimum gives a sequence with prices tending to zero; sourcewise closure retains its limit records.

Zero capacity is the polar branch of the associated form. Its complement is positive capacity and supplies an activity profile. The spectral theorem gives the finite resistance/range branch and the orthogonal harmonic remainder. Finally a nonnegative series either diverges or has finite sum. The physical pullback identity makes these precisely the excluded branch and the successor-activation branch. The latter is closed by Theorem II.2.46; it is not a terminal formation class. The saturation and fluctuation alternatives are Theorem II.1.11. The singleton inapplicable value and totalized descent defect exhaust the remaining interface cases. Applying the activity lift and Theorem III.2.8 to each nonzero quotient proves the final constructive statement. □

**Corollary II.1.17** (Estimate compression after structural derivation)**.** *Once a graph relation, positive-form identity, target-null quotient, or zero-set relation has been derived from the symbolic presentation, subsequent rows use only that structural output. Formal identities and universal functional calculus refine the generated positive cone directly. When they do not remove a row, its graph-boundary, positive-null, or zero-price witnesses are generated and passed to the successor calculus. Closed outputs are propagated by Theorem II.1.15.*

**Definition II.1.18** (Canonical symbolic structural transcript)**.** For the fixed PDE, complete continuation target, carrier, and derived ceiling, define

$$\Theta_\Sigma^{\mathrm{syn}}(\mathfrak{P}, B_{\mathrm{phys}}) := \Big(\mathsf{Syn}(\mathfrak{P}), \mathsf{Graph}^{\mathrm{tot}}(\mathfrak{P}), \mathsf{Pos}(\mathcal{K}), \Pi_\Sigma, \mathsf{Cov}(\mathsf{Op}), \mathsf{Anc}\Big)\,. \tag{83}$$

The entries are respectively the typed equation syntax, total graphs of its generated operations, the cone of nonnegative terms occurring in the proved carrier identity, the generated target morphism, the total covariance graphs, and the event/source ancestry maps. Exact extrema, capacities, resistances, realization images, contact orbits, continuation verdicts, and Liouville conclusions are not entries. They are semantic outputs of rows constructed from $\Theta_\Sigma^{\mathrm{syn}}$.

**Theorem II.1.19** (Public-signature structural sufficiency)**.** *The symbolic transcript functorially determines the source cells, totalized interfaces, target-null quotient, carrier complex, activity measures, and successor relation used by the regularity argument:*

$$\mathsf{Compiler}_\Sigma = \overline{\mathsf{Compiler}}_\Sigma \circ \Theta_\Sigma^{\mathrm{syn}}. \tag{84}$$

*Every row is constructed together with its ordinary and complementary witness graphs. No exact row value or actual-contact image is consulted in this construction. Two presentations with isomorphic symbolic transcripts therefore have isomorphic prospective successor and zero-cost fixed-point algebras. An analytic derivation can only prove a relation already present in the transcript or refine its generated positive cone; it cannot insert a favorable value.*

*Proof.* Proceed by structural induction over the typed syntax and then by the fixed row order (66). Each constructor supplies its ordinary graph, explicit failure value, source transport, and event ancestry. Graph closure supplies the corresponding boundary witnesses. The target morphism forms the null quotient, and the positive carrier cone supplies the expenditure coordinate. Quotient descent either factors or produces its commutator graph. These operations construct both branches of every row without deciding which branch contains a given future orbit record.

The activity and successor constructions use only the generated metric, target shell, graph witnesses, and event ancestry. The prospective fixed-point construction below uses their closed images, not the image of actual contact orbits. This proves (84) and the isomorphism statement. □

**Corollary II.1.20** (Closed structural input). *A PDE specialization supplies only the target-blind public signature of Definition I.1.8. The continuum algebra generates* (83)*. Every identity obtainable by formal manipulation of that signature or by the fixed universal library is inserted automatically as a graph relation or positive-descendant refinement. The ordinary graph and every first-failure witness are inputs to the canonical activity lift and successor operator of Definitions II.2.30 and II.2.32. Their descending zero-cost closure is computed by Definition II.2.39. Thus every generated branch is reduced to a target-null class or to an explicit recurrent zero-cost mechanism; there is no unevaluated row and no additional invariant value to be supplied.*

**Proposition II.1.21** (Positive-separator evaluation of an equality branch). *Totalize the separator relation on $\widehat{\mathcal{E}}_{\alpha_*}$. On its zero graph-defect branch the identity is*

$$0 = Q_\Sigma(\Pi_\Sigma \mathscr{R}_{\mathrm{eq}}(Z)) + \sum_{j=1}^{m} q_j(Z), \tag{85}$$

*where $Q_\Sigma \geq 0$ is positive definite on $Z_\Sigma$ and every $q_j \geq 0$. Then $\mathfrak{H}^{\Sigma}_{\mathrm{eq}} = 0$.*

*Proof.* Every term on the right of (85) is nonnegative, so each term vanishes. Positive definiteness gives $\Pi_\Sigma \mathscr{R}_{\mathrm{eq}}(Z) = 0$ for every $Z$. Taking the closed linear span proves the claim. □

## II.2 Coherent carrier passivity and structural successor closure

The scalar price infimum of Definition II.1.6 is not the terminal evaluator for a finite-price cascade. A finite measure on infinitely many disjoint target windows produces a vanishing-price subsequence. The continuum analogue of the residual closure in the discrete algebra is to follow that subsequence through all compact, concentrating, escaping, diffuse, gauge, and equality outputs. This section constructs that closure directly.

**Definition II.2.1** (Coherent carrier complex). Let $u$ be one physical orbit. A *coherent carrier complex* consists of a locally finite rooted tree $\mathcal{T}$, cells $Q_\alpha$, and nonnegative localization tests $\psi_\alpha$ with the following properties.

(C1) Every child is a subevent of its parent and, on the region where the local balance is evaluated,

$$\psi_\alpha = \sum_{\beta \in \mathrm{ch}(\alpha)} \psi_\beta. \tag{86}$$

(C2) All cells are restrictions, chart images, or normalized descendants of the same orbit. Their transition maps satisfy the cocycle composition law. Galilean, gauge, and quotient representatives are fixed compatibly on overlaps.

(C3) The weak equation gives a local carrier relation, linear in the test,

$$K_\psi(t) + D_I(\psi) + M_I(\psi) = K_\psi(s) + J_I(\psi) + E_I(\psi). \tag{87}$$

Here $D_I, M_I \geq 0$ are respectively the intrinsic production and the weak energy-defect measure, $J_I$ is the signed transport current, and $E_I$ is true exterior input. An inequality is written as equality by adjoining its nonnegative defect to $M_I$.

(C4) The normalized target observable is constant under the allowed transition maps. Consequently target retention is inherited by a child selected from a target-retaining parent.

The complex is generated from the shell cascade by taking the maximal locally finite family of compatible target-retaining descendants. A failure of local finiteness, the partition relation, transition coherence, or gauge compatibility is a totalized $\mathsf{Lift}/\mathsf{Bdry}$ successor and is not an additional admissibility condition.

**Theorem II.2.2** (Signed carrier gluing)**.** *Let $\mathcal{T}_0 \subset \mathcal{T}$ be a finite rooted subtree. Sum* (87) *over $\mathcal{T}_0$ before taking positive parts. Every current through an internal face cancels with the same current with opposite orientation in the adjacent cell. In particular, conservative transport, pressure or constraint transport, chart translation, gauge motion, cutoff motion, and exact scale motion contribute only through the root, the leaves, or the true exterior boundary. The result has the form*

$$K_{\mathrm{leaf}}(\mathcal{T}_0) + \mu_{\mathsf{Geom}}(\mathcal{T}_0) + \mu_{\mathrm{def}}(\mathcal{T}_0) = K_{\mathrm{root}} + I_{\mathrm{ext}}(\mathcal{T}_0) + H_{\mathrm{sc}}(\mathcal{T}_0), \tag{88}$$

*where the first two measures on the left are nonnegative. The term $H_{\mathrm{sc}}$ is the cohomology class of scale motion that is not an edge coboundary in the chosen scale-critical carrier. It has inherited* $\mathsf{Caus}/\mathsf{Lift}$ *ancestry. It vanishes when the scale current admits a coherent carrier potential. No sum of the absolute values of the internal currents occurs in* (88)*.*

*Proof.* Linearity in $\psi$ and (86) imply that the carrier, production, defect, transport, and exterior terms of a parent are the sums of the corresponding child terms. Give every common face the orientation induced by its parent. The child has the opposite induced orientation, so its signed flux cancels the parent flux. Pressure-gauge constants cancel because their weak pairing with the divergence-free or constraint-compatible test is zero. Transition coherence makes translation, gauge, and exact scale terms edge coboundaries; the sum of a coboundary over a finite tree is its root value minus its leaf values. The only possible scale term left after this cancellation is its class modulo the range of the edge coboundary, which is $H_{\mathrm{sc}}$. The intrinsic production and weak balance defect retain their signs and add over the cells. This proves (88). $\square$

**Definition II.2.3** (Target-aligned physical charge)**.** Let $Q$ be a pre-contact event selected by a shell covector $D\Phi$. First glue every compatible internal face and form the signed target current

$$q_{\Phi,Q} := \sum_{c\in\{\mathsf{Geom},\mathsf{Caus},\mathsf{Abs},\mathsf{Lift},\mathsf{Bdry}\}} \int_Q \langle D\Phi, J^c\rangle. \tag{89}$$

A nonnegative measure $P$ is an *admissible charge* for this event only if a generated local carrier identity proves

$$(q_{\Phi,Q})^+ \le \Delta K_Q + P(Q) + I^+_{\mathrm{ext}}(Q) + n_Q, \qquad n_Q = \langle D\Phi, N_Q\rangle = 0, \quad N_Q \in \mathscr{N}_\Sigma, \tag{90}$$

with the carrier increments telescoping on same-origin concatenation and $P$ dominated by the public physical ledger. Positive quantities not appearing in such an identity are structural coordinates, not expenditure. Positive parts of individual internal currents are never charged before signed gluing.

**Theorem II.2.4** (Local target-aligned charging rule)**.** *Every physical charge used by the successor algebra is obtained by Definition II.2.3. On a disjoint same-origin shell cascade, summing* (90) *cancels internal carrier increments and target-null terms. Therefore repeated target progress is paid only by the restriction of the public physical measure and true positive exterior input to those target-active events. If no generated identity dominates a nonzero target current, that current is passed to the polarity stratifier as a target-active successor; it is not assigned zero price and is not charged to an unrelated positive invariant.*

*Proof.* The shell identity exposes $q_{\Phi,Q}$ before contact. Signed carrier gluing gives the aggregate current and the root–leaf form (88). Restriction to the target quotient annihilates $N_Q$, hence $n_Q = 0$. Add the identities over pairwise disjoint same-origin events: interior carrier faces telescope, while countable additivity bounds the $P$-terms by the single physical ledger and the exterior terms by its declared input component. The grammar admits no other rule for inserting a quantity into that ledger. Failure to derive the displayed domination has a nonzero total-graph coordinate, whose polarity split gives the stated successor. □

**Definition II.2.5** (Carrier cohomology and aggregate feeding)**.** Let $\mathscr{W}_\eta$ be the saturated closure of normalized windows whose target observable is at least $\eta > 0$, after all generated symmetry and gauge quotients. Include the values of the carrier, production, exterior input, and every totalized graph coordinate in the evaluation topology. Let $S$ be the successor or unit-window shift on its ordinary graph.

On a coherent ordinary branch, write $F_{\mathrm{sc}}$ for the aggregate signed inward scale-space feed left after Theorem II.2.2. Its target-active carrier cohomology is

$$\mathfrak{H}^{\Sigma}_{\mathrm{car}} := \overline{\Pi_\Sigma F_{\mathrm{sc}}} / \overline{\Pi_\Sigma\{\Psi - \Psi \circ S : \Psi \in C(\mathscr{W}_\eta)\}}. \tag{91}$$

The quotient is totalized when $S$, the carrier, or a transition is partial. Its graph-boundary values retain their source ancestry. The ordinary carrier-deficit cocycle is

$$g_{\mathrm{car}} := F_{\mathrm{sc}} - D - M. \tag{92}$$

For a finite block, its sum is the aggregate inward feeding minus production, up to root–leaf storage and true exterior input. Coboundaries disappear in block averages and under invariant measures. If $[F_{\mathrm{sc}}] = 0$ in (91), then for every $\varepsilon > 0$ there is a continuous $\Psi_\varepsilon$ whose coboundary approximates the feed class within $\varepsilon$; no exact correction is asserted.

**Theorem II.2.6** (Target-relative passivity)**.** *Let $\mathcal{K}_\eta \subseteq \mathscr{W}_\eta$ be a compact $S$-invariant ordinary carrier class generated by the activity lift. On the ordinary signed-carrier branch, finite exterior input is already part of the derived physical budget. Then every $S$-invariant Borel probability measure $\nu$ on $\mathcal{K}_\eta$ satisfies*

$$\int_{\mathcal{K}_\eta} g_{\mathrm{car}} \, \mathrm{d}\nu \le 0. \tag{93}$$

*Since every invariant measure annihilates continuous coboundaries, this is equivalent to*

$$\int F_{\mathrm{sc}} \, \mathrm{d}\nu \le \int (D + M) \, \mathrm{d}\nu. \tag{94}$$

*Equality in* (93) *is the zero carrier-deficit branch. It enters the zero-cost successor fixed point. It implies vanishing of a particular production term only when the generated consequence algebra contains a domination identity $p = q + r$ with that production as $q$. Every measure and current in this statement is restricted to $\mathcal{K}_\eta$, generated by one retained target/source cell; the theorem has no consequence for invariant measures outside that cell.*

*Proof.* Apply (88) to the first $N$ consecutive cells of a coherent path. Internal currents cancel and the carrier terms telescope. The carrier is bounded on the compact evaluation class. The exterior input has finite total mass, so its contribution divided by $N$ tends to zero. Integrating the block identity against an invariant measure cancels the endpoint terms exactly and gives (93), which is (94). Continuous coboundaries and their uniform closure have zero integral against every invariant measure, justifying the cohomological quotient without producing an exact potential. □

**Corollary II.2.7** (Totalized passivity branch)**.** *For every target-retaining carrier route, exactly one of the following is generated:*

(T1) *the coherent ordinary branch of Theorem II.2.6;*

(T2) *a first failure of local finiteness, carrier partition, transition coherence, gauge descent, compact retention, or carrier correction, together with its closed-graph witness descendants;*

(T3) *a target-null output removed by the generated quotient.*

*The second output enters $\mathcal{R}_\eta$; the third closes. Positive exterior variation has* Bdry *ancestry and is eligible for charge only through the local target-aligned carrier identity. Consequently compactness, coherence and carrier cohomology are processed by the successor operator.*

*Proof.* Totalize each partial relation in the order displayed in Definitions II.2.1 and II.2.5. If all ordinary values occur, Theorem II.2.6 applies. The first nonordinary value has a witness sequence in its closed total graph and therefore defines the successor in Definition II.2.39. Positive exterior variation enters the physical measure through Theorem II.2.4. The first-failure alternatives are exhaustive because all graph coordinates belong to the compact activity lift. □

**Definition II.2.8** (Carrier saturation algebra)**.** The *carrier saturation algebra* is the closed target projection

$$\mathfrak{H}^{\Sigma}_{\mathrm{eq,car}} := \overline{\operatorname{span}}\left\{ \Pi_\Sigma \operatorname{supp}\nu : \begin{array}{l} \nu \text{ is invariant on a generated compact successor core,} \\ \int g_{\mathrm{car}}\, \mathrm{d}\nu = 0 \end{array} \right\}. \tag{95}$$

The support is evaluated through all five source tags and all totalized zero-deficit equations. It may contain equilibria, relative equilibria, or nonstationary recurrent conservative dynamics. The subsequent zero-cost fixed point, rather than an assumed rigidity statement, decides its target projection.

**Theorem II.2.9** (Ergodic strict passivity or saturation)**.** *Let $\mathcal{K}_\eta$ be as in Theorem II.2.6, and set*

$$\beta_\eta := \sup_{\nu \in \mathcal{P}_S(\mathcal{K}_\eta)} \int g_{\mathrm{car}}\, \mathrm{d}\nu, \tag{96}$$

*where $\mathcal{P}_S(\mathcal{K}_\eta)$ is the compact set of $S$-invariant Borel probability measures. Then $\beta_\eta \le 0$, and exactly one of the following holds.*

(P1) *$\beta_\eta < 0$. There exist an integer $N_\eta \ge 1$ and $\delta_\eta > 0$ such that every $N_\eta$-block retained in $\mathcal{K}_\eta$ satisfies*

$$-\sum_{k=0}^{N_\eta - 1} g_{\mathrm{car}}(S^k w) \ge \delta_\eta. \tag{97}$$

(P2) *$\beta_\eta = 0$. The class $\mathcal{K}_\eta$ carries an invariant maximizing measure and its target projection belongs to $\mathfrak{H}^{\Sigma}_{\mathrm{eq,car}}$.*

*If $\mathfrak{H}^{\Sigma}_{\mathrm{eq,car}} = 0$, only* (P1) *is possible. The positive block price is therefore derived from the absence of a target-visible equality state; no contact margin or compactness gap is inserted.*

*Proof.* Let $h = g_{\mathrm{car}}$. Compactness of $\mathcal{P}_S(\mathcal{K}_\eta)$ and continuity of the integral give a maximizing measure. By Theorem II.2.6, $\beta_\eta \le 0$.

Suppose $\beta_\eta < 0$ but (P1) fails. Then there are $w_n \in \mathcal{K}_\eta$ with block length $N_n = n$ and total carrier deficit smaller than $n^{-1}$. In particular the mean of $h$ has lower limit at least zero. Form the empirical measures

$$\nu_n := \frac{1}{N_n} \sum_{k=0}^{N_n - 1} \delta_{S^k w_n}.$$

Compactness of probability measures on $\mathcal{K}_\eta$ gives a weakly convergent subsequence. The endpoint identity

$$\int f \circ S \, \mathrm{d}\nu_n - \int f \, \mathrm{d}\nu_n = \frac{f(S^{N_n} w_n) - f(w_n)}{N_n}$$

shows that its limit $\nu$ is invariant. Continuity of the evaluation coordinates and the lower bound give $\int h \, \mathrm{d}\nu \geq 0$, while Theorem II.2.6 gives the reverse inequality. Hence $\int h \, \mathrm{d}\nu = 0$, contradicting $\beta_\eta < 0$. Thus (P1) holds. If $\beta_\eta = 0$, a maximizing invariant measure is a saturation measure and gives (P2). Conversely, an invariant saturation measure contradicts any uniform positive block deficit after integration. Vanishing of $\mathfrak{H}^{\Sigma}_{\mathrm{eq,car}}$ excludes a target-retaining instance of (P2). □

### II.2.1 Target-aligned dual-carrier compilation

The preceding passivity theorem is used only after contact has selected a source and a target covector. This is the continuum selector cut. It avoids an equation-wide classification of equality states.

**Definition II.2.10** (Active branch signature)**.** An *active branch signature* is a tuple

$$\mathfrak{b} = (A, m, \Gamma, \eta, \mathbf{R}), \tag{98}$$

where $A \neq \varnothing$ is a persistent source support, $m$ is its eventual realization mode, $\Gamma$ is a generated target-shell family, $\eta > 0$ is its retained source floor, and $\mathbf{R}$ is the same-origin cascade tag. The branch ideal $I_{\mathfrak{b}}$ is generated by the public equation and by the relations already established on this branch: source support $A$, mode $m$, shell pairing at least $\eta$, chart and gauge choices, event order, and every earlier zero graph coordinate. No relation outside this list is inserted into $I_{\mathfrak{b}}$.

**Proposition II.2.11** (Branch-local consequence rule)**.** *Let $q = 0$, $q \geq 0$, or $q_1 \leq q_2$ be derivable from the public weak equation modulo $I_{\mathfrak{b}}$. Then that relation holds on every ordinary or zero-defect realization of $\mathfrak{b}$. Its failure on a total-graph boundary is the next typed successor. Consequently a relation need only be proved on the active branch on which its antecedent has already been generated; no assertion about states outside that branch is required.*

*Proof.* Every generator of $I_{\mathfrak{b}}$ vanishes on an ordinary realization of the branch, so substitution in the finite weak-equation derivation proves the relation there. The closed total graph records the first generator that fails under passage to a limit. Source-preserving completion assigns that failure its inherited source and realization mode. These two cases exhaust the totalized branch. □

**Definition II.2.12** (Target-local structural stratum)**.** For an active branch $\mathfrak{b} = (A, m, \Gamma, \eta, \mathbf{R})$, let $\mathbb{G}_{\mathfrak{b}}$ be the total evaluation graph generated by the public weak equation, its five source currents, the shell family, and all operations already used on the branch. Its *target-local structural stratum* is

$$X_{\mathfrak{b}} := \{z \in \mathbb{G}_{\mathfrak{b}} : I_{\mathfrak{b}}(z) = 0,\ Q_{A,\Phi}(z) \geq \eta,\ \mathrm{Origin}(z) = \mathbf{R}\}. \tag{99}$$

All consequence, compactness, production, and equality operations below are performed after restriction to $X_{\mathfrak{b}}$. Values outside this stratum are irrelevant to the contact branch and are never quantified over.

**Definition II.2.13** (Generated polarity stratification)**.** Fix once and for all a countable enumeration $(F_j)_{j \geq 0}$ of the scalar rational-cylinder coordinates generated by the public equation and the target-cyclic grammar. Vector, tensor, measure, and operator coordinates enter through their rational scalar probes. Replace an unbounded real coordinate by $\widehat{F}_j = F_j/(1+|F_j|)$, and retain nonordinary values in its total graph. On a target-local stratum, the $j$-th *polarity split* is the disjoint ordered partition

$$\{\widehat{F}_j = 0\}, \qquad \{s\widehat{F}_j \in [2^{-k}, 2^{-k+1})\}\ (s \in \{-1,1\},\ k \geq 1), \qquad \partial_{\mathrm{gr}} F_j. \tag{100}$$

Endpoints are assigned to the leftmost displayed cell, and $\partial_{\mathrm{gr}} F_j$ is further split by the first failed public graph operation and its inherited source word. Intersect every cell with $X_{\mathfrak{b}}$. First dovetail the finite identities and cylinder moments of the parent local Archimedean cone. A finite derivation of $-1$ prunes the parent cell. Otherwise compatible moments reconstruct a positive local functional; disintegrate it over the countable partition (100) and select the least cell of positive target mass. That cell is a target-active successor with the same origin and with its shell current renormalized to the universal source floor. Thus no semantic emptiness or exact-zero oracle is used.

Thus both possession and failure of an analytic property are structural values. For example, closed descent is the zero cell of its cokernel coordinate; nonclosed descent is a signed dyadic or graph-boundary cell, not an omitted hypothesis.

**Definition II.2.14** (Resolved-prefix rank)**.** The enumeration in Definition II.2.13 is dovetailed with the enumerations of weak words, source descendants, quotient operations, carrier rows, and total-graph defects. A branch has *resolved-prefix rank*

$$\rho(\mathfrak{b}) := sup\{N : \text{the polarity cell of every } F_j,\ j < N,\ \text{belongs to } I_{\mathfrak{b}}\}. \tag{101}$$

When $\rho(\mathfrak{b}) < \infty$ and $F_{\rho(\mathfrak{b})}$ is split, every retained proper successor adjoins its selected cell to the branch ideal and therefore has larger resolved prefix. If that split generates new operations, they enter later in the same universal dovetailing. The rank is consequently constructed by the algebra; it is not presentation data and no well-foundedness premise is required.

**Theorem II.2.15** (Target-local stratification exhaustiveness)**.** *Let $\mathfrak{b}$ be a same-origin active contact branch. At its next unresolved coordinate exactly one of the following occurs:*

(L1) *a finite local ordered identity proves that the current target cell is empty or target-null;*

(L2) *a generated carrier identity charges the retained cell to the finite physical ledger;*

(L3) *a zero, signed dyadic, or graph-boundary cell has nonzero target projection and becomes a proper successor $\mathfrak{b}'$ with $\rho(\mathfrak{b}') > \rho(\mathfrak{b})$.*

*The alternatives exhaust ordinary values, loss of every analytic property used by the derivation, and all limit values of the coordinate. No global estimate and no target-nullity statement is an input.*

*Proof.* The total graph makes (100) exhaustive, and the endpoint convention makes it disjoint. Apply the Archimedean proof/model alternative to the parent target cell. Its finite $-1$-identity gives (L1). Otherwise its compatible moments give a positive Borel law on the compact cylinder state space. Countable disintegration selects a polarity cell of positive target mass. It retains a pre-contact shell current; the five-source identity selects a source occurrence and normalization restores the fixed floor. If its mass is dominated by a public carrier row it is (L2). Otherwise the selected polarity relation is appended to the ideal, producing (L3) and increasing the resolved prefix. Total graph ancestry proves the assertion for failed analytic properties. □

**Theorem II.2.16** (Diagonal target-aligned saturation)**.** *An infinite chain of nonempty proper successors selected from one contact branch has a compatible same-origin inverse limit on which every generated scalar coordinate has a resolved ordinary or graph-boundary value. Prospective accountability on that limit gives exactly one of two conclusions:*

(D1) *all induced source pairings with the retained shell covectors vanish along the retained path, and shell contact is impossible;*

(D2) *a pairing is nonzero; its polarity and ancestry already occur in the limit ideal, and the compatible coordinates reconstruct a closed five-channel subsystem with an actual same-origin path crossing the cofinal shell family.*

*The second conclusion is an explicit singular mechanism. It is not an unresolved model, an equality-kernel label, or a request for an additional analytic theorem.*

*Proof.* Proper succession strictly increases the resolved prefix. Compactness of each bounded scalar total graph and compatibility of the restrictions give an inverse limit. Every fixed coordinate is resolved after finitely many successors, including every coordinate generated by an earlier failure. The origin and normalized target coordinates are preserved by construction. Apply the shell evolution identity to the induced path. If all five pairings vanish, its shell observable is constant, proving (D1). Otherwise choose the least nonzero pairing. Its dyadic cell, detecting covector, source ancestry, and all descendant graph operations are already resolved. Hence the induced weak equation is closed; a missing operation would be a later unresolved coordinate. The retained contact coordinates then give the path required in (D2). □

**Definition II.2.17** (Target-local structural fact)**.** Fix a same-origin active branch $\mathfrak{b}$, its shell covector $\lambda = D\Phi$, and its target-local stratum $X_{\mathfrak{b}}$. A relation is an *admissible target-local structural fact* when its restriction to $X_{\mathfrak{b}}$ has the signed form

$$Q_{\mathfrak{b},\lambda} = \delta K_{\mathfrak{b}} + \sum_{c\in\mathfrak{C}} R^{c}_{\mathfrak{b},\lambda} + P_{\mathfrak{b}} + B_{\mathfrak{b}} + N_{\mathfrak{b}} + E_{\mathfrak{b}}. \tag{102}$$

Here $Q_{\mathfrak{b},\lambda}$ is the selected target progress; $R^{c}_{\mathfrak{b},\lambda}$ is the unabsorbed signed remainder with inherited $c$-ancestry; $\delta K_{\mathfrak{b}}$ is a same-origin carrier coboundary; $P_{\mathfrak{b}} \geq 0$ is admitted only through a carrier identity dominated by the public physical measure; $B_{\mathfrak{b}}$ is true exterior input; $N_{\mathfrak{b}} \in \mathscr{N}_{\Sigma}$; and $E_{\mathfrak{b}}$ is the ordered list of nonordinary or failed graph coordinates. Compatible internal currents are added with their signs before any positive part is formed, and each expanded term occurs exactly once: either in a signed remainder, in the generated production, in exterior input, in the null term, or in the defect list.

The fact may close or charge the branch only on $X_{\mathfrak{b}}$. A nonzero component of $E_{\mathfrak{b}}$, or a nonzero uncharged component of the signed source sum, is passed to the local polarity successor with the same origin and ancestry.

**Theorem II.2.18** (Target-local factorization of continuum facts)**.** *Every relation used by the continuum reachability or regularity calculus is an admissible target-local structural fact after restriction to the active branch that invoked it. This applies to weak-equation, gradient, resolvent, Hamiltonian, commutator, compactness, constraint, entropy, covariance, carrier, and continuation relations. Consequently:*

(F1) *a relation on states outside $X_{\mathfrak{b}}$ has no effect on that branch;*

(F2) *a target-null term closes no nonnull source cell and contributes no physical charge;*

(F3) *a nonnegative quantity pays for target progress only through Definition II.2.3; and*

(F4) *failure of localization, descent, ancestry, signed gluing, carrier domination, or realization is a target-local polarity successor rather than an omitted antecedent.*

*Proof.* Pair the invoked relation with the selected shell covector and restrict every input and graph coordinate to $X_{\mathfrak{b}}$. The current normal form (27) gives the five source terms and the target-null term. Structural induction over restriction, quotient, chart, trace, adjoint, polarization, and limit constructors preserves their source words by Theorem I.3.4. The carrier constructor separates a coboundary, generated nonnegative production, and true exterior input; Theorem II.2.2 cancels compatible internal faces before taking positive parts. Totalization places the first failed operation in $E_{\mathfrak{b}}$. These are exactly the terms of (102). The local charging rule gives (F2)–(F3), while target-local polarity stratification gives (F1) and (F4). $\square$

**Corollary II.2.19** (Activation rule for universal identities)**.** *An equation-wide functional-analytic identity is a library relation. It enters the regularity calculus only through its factorization* (102) *on a contact-generated active cell. Thus a global operator bound, compactness statement, spectral gap, entropy law, or coercive inequality has no independent regularity consequence; only its target-local source components and admissible physical charge do.*

**Corollary II.2.20** (No unscoped reachability inference)**.** *Every derivation that changes a physical reachability profile factors through an active branch signature* $\mathfrak{b}$ *and a structural fact of the form* (102)*. More precisely, its final nonidentity step is exactly one of:*

(R1) *the target-null quotient annihilates every source remainder on* $X_{\mathfrak{b}}$*;*

(R2) *the signed local carrier identity charges the selected shell progress to the restriction of the public physical measure;*

(R3) *a nonordinary coordinate creates a same-origin polarity successor; or*

(R4) *diagonal saturation reconstructs a same-origin path with actual target contact.*

*Consequently no theorem about an unselected state family, no transverse positive form, and no equation-wide analytic estimate can remove, charge, or continue a contact branch.*

*Proof.* The ordered reachability calculus consumes a relation only through the local cone generated by $I_{\mathfrak{b}}$, the retained shell floor, and the factorization (102). Its zero target projection is (R1). Its admitted production and exterior terms are processed by Theorem II.2.4, giving (R2). Every nonzero defect or unabsorbed signed remainder is sent by Theorem II.2.15 to (R3). Compatible cofinal successors are processed by Theorem II.2.16, giving (R4). These are the complete outgoing edges of the target-local fact compiler, so there is no fifth inference rule with equation-wide scope. $\square$

**Theorem II.2.21** (Canonical record-scale selector)**.** *Suppose a same-origin target-retaining cascade carries scale coordinates* $r_n > 0$*. Exactly one of the following generated alternatives holds:*

(S1) $\inf_n r_n > 0$*, and the cascade remains in a fixed-scale stratum;*

(S2) $\inf_n r_n = 0$*, and it has a canonical subsequence* $n_0 < n_1 < \cdots$ *defined by*

$$n_{j+1} := \min\{n > n_j : r_n \leq \tfrac{1}{2} r_{n_j}\}. \tag{103}$$

*For the second alternative, interpolate* $\log r_{n_j}$ *affinely along the ordered normalized windows. The resulting absolutely continuous chart scale satisfies*

$$a := \frac{\dot{r}}{r} \leq 0 \quad \textit{a.e.}, \qquad r(\tau) \downarrow 0. \tag{104}$$

*The selected centers, charts, and transitions retain the origin tag* $\mathbf{R}$*. Thus* (104) *is a consequence of scale collapse, not an analytic hypothesis on the original orbit.*

*Proof.* If the first alternative fails, choose $n_0$. Since the tail infimum is zero, the set in (103) is nonempty; its least element defines $n_{j+1}$. The selected scales decrease by at least a factor two and therefore tend to zero. Affine interpolation of their logarithms is nonincreasing, which gives (104). All coordinates are restrictions and chart images of the original cascade, so same-origin ancestry is preserved. $\square$

**Definition II.2.22** (Source-adapted dual-carrier grammar)**.** Fix $\mathfrak{b} = (A, m, \Gamma, \eta, \mathbf{R})$. For $\Phi \in \Gamma$, begin with its descending covector $D\Phi$ and generate the smallest countable rational test module $\mathscr{W}_{\mathfrak{b}}$ closed under:

(D1) adjoints of the source operators occurring in $A$, including killed generator and predictable-compensator pullbacks when present;

(D2) constraint projection, gauge quotient, chart pullback, record-scale restriction, and multiplication by the generated rational cutoff algebra;

(D3) signed parent–child gluing, temporal summation by parts, carrier coboundaries, and cancellation of target-null directions;

(D4) polarization and completion of squares using the native positive $\mathsf{Geom}$ form and every nonnegative defect generated by the weak equation.

For a word $w \in \mathscr{W}_{\mathfrak{b}}$, expansion against the weak equation produces a *dual-carrier row*

$$Q_{A,\Phi} = \delta K_w + P_w + B_w + N_w + E_w. \tag{105}$$

Here $Q_{A,\Phi}$ is the selected source pairing, $\delta K_w$ is a bounded carrier coboundary, $P_w \geq 0$ is generated physical production, $B_w$ is true exterior input, $N_w \in \mathscr{N}_\Sigma$, and $E_w$ is the ordered list of total-graph defects. Paired internal currents are added with their signs before $P_w$ is formed. A production row is admitted only when the public carrier identity proves that its total mass belongs to the single physical ledger.

*Remark* II.2.23 (Constructive enumeration)*.* The grammar is generated from the public equation and the selected shell covector. At depth $k$, enumerate weak multipliers, adjoint pulls, cutoffs, partitions, chart words, and square completions of syntactic length at most $k$, then reduce them modulo $I_{\mathfrak{b}}$. This is the continuum counterpart of saturating all prospective selectors in Part I. A specialization supplies the equation and its textbook weak identities; it does not supply a dual carrier or an equality theorem.

**Theorem II.2.24** (Target-aligned carrier duality)**.** *For every active zero-price successor branch $\mathfrak{b}$, the generated dual-carrier hierarchy has exactly one of the following outcomes.*

(C1) *A finite ordered identity shows that the active branch is target-null.*

(C2) *At a finite depth it derives a row* (105) *whose graph defects are zero on the branch and whose exterior positive variation is in the physical ledger. Repeated target progress is then charged only to $P_w$, up to a bounded telescoping carrier and finite exterior input.*

(C3) *The ordered dual cone has a positive normalized functional $L$ with*

$$L(I_{\mathfrak{b}}) = 0, \qquad L(P_w) = 0 \quad (w \in \mathscr{W}_{\mathfrak{b}}), \qquad L(Q_{A,\Phi}) \geq \eta. \tag{106}$$

*It is inserted into the equality-successor graph with its source, mode, target, chart, and origin coordinates.*

*No conclusion requires control of a source or equality state outside the target-active branch $\mathfrak{b}$.*

*Proof.* Form the Archimedean ordered cylinder algebra generated by $I_{\mathfrak{b}}$, the nonnegative productions $P_w$, the exterior ledger, and the inequality $Q_{A,\Phi} \geq \eta$. If its quadratic module contains $-1$ at a finite level, it gives (C1). Otherwise dovetail the dual-carrier words. A word whose exact telescoping identity charges the retained crossings gives (C2). If neither finite event occurs, the ordered Hahn–Banach theorem applied to the cofinal cylinder cones gives a positive normalized functional annihilating the branch ideal. Since this is a zero-price successor branch and every admitted $P_w$ is dominated by the physical ledger, it also annihilates every $P_w$, while the defining shell inequality retains the target floor. This is (C3). Weak duality makes the alternatives disjoint after selecting the first finite closing row; the ordered proof/model theorem makes them exhaustive. □

**Definition II.2.25** (Target-local equality stratum)**.** The equality kernel of $\mathfrak{b}$ is

$$\mathcal{E}_{\mathfrak{b}} := \{z : I_{\mathfrak{b}}(z) = 0,\ P_w(z) = 0 \text{ for all } w \in \mathscr{W}_{\mathfrak{b}}\}\,. \tag{107}$$

It is restricted immediately to the retained shell cell $Q_{A,\Phi} \geq \eta$. Its target-null cell is precisely

$$\mathcal{E}_{\mathfrak{b}} \cap \{Q_{A,\Phi} \geq \eta\} = \varnothing. \tag{108}$$

The finite-level quadratic modules are dovetailed with the generated polarity stratification. A derivation of $-1$ after adjoining $Q_{A,\Phi} - \eta$ closes the local cell; compatible positive functionals are not accepted as a verdict but reconstruct the next target-retaining equality successor. Thus (108) is an output of local saturation, not a property to be established separately and not a classification of all solutions of the equation.

**Proposition II.2.26** (Covariant modulation conjugation)**.** *Suppose a dual-carrier word on an active moving-chart branch gives*

$$K' = -D + 2aK + R, \qquad K, D \geq 0, \tag{109}$$

*where $R$ is the remaining signed source current. Let $\theta > 0$ solve the scalar terminal or initial-value problem*

$$\theta' + 2a\theta = -1. \tag{110}$$

*Then the generated carrier $\mathcal{G} = \theta K$ satisfies the exact identity*

$$\mathcal{G}' = -K - \theta D + \theta R. \tag{111}$$

*No sign or integrability condition on $a$ is used. If $R$ is canceled by an adjoint word, a signed interface gluing, or a target-null quotient, then $K + \theta D$ is a nonnegative production generator in $\mathscr{W}_{\mathfrak{b}}$.*

*Proof.* Differentiate $\theta K$ and substitute Eqs. (109) and (110):

$$(\theta K)' = (-1 - 2a\theta)K + \theta(-D + 2aK + R) = -K - \theta D + \theta R.$$

All operations belong to the scalar adjoint and carrier-coboundary clauses of Definition II.2.22. □

**Theorem II.2.27** (Branch-local modulation stratification)**.** *Let $\mathfrak{b}$ be a record-scale active branch, so that $a \leq 0$ by Theorem II.2.21. Suppose its generated stationary equations contain*

$$\int (P - aO)\, \mathrm{d}\varpi = 0, \qquad P, O \geq 0, \tag{112}$$

*Then every retained stationary law is supported on*

$$\mathcal{E}_{\mathfrak{b}}^{P=0} := \mathcal{E}_{\mathfrak{b}} \cap \{P = 0\}. \tag{113}$$

*Restrict the local ordered cone to $\mathcal{E}_{\mathfrak{b}}^{P=0}$. A finite $-1$-identity closes its retained target cell; otherwise its compatible positive moments give the next zero-production successor. In the second case the shell covector, five source currents, and every generated operation are restricted to* (113)*, its next polarity coordinate is recorded, and prospective accountability is rerun there. No property or semantic emptiness decision for the zero set is supplied as a premise.*

*Proof.* Both terms in $P - aO = P + (-a)O$ are nonnegative. Equation (112) therefore gives $P = 0$ $\varpi$-almost surely, proving (113). Apply the local Archimedean proof/model alternative after adjoining $P = 0$ and the retained shell floor. A finite contradiction closes the cell; otherwise the reconstructed positive law retains the branch origin and target floor and therefore satisfies the definition of a successor. Restriction, polarity refinement, and prospective reselection are generated operations in Definitions II.2.13 and II.2.22. □

**Theorem II.2.28** (Exhaustive loss-of-compactness routing)**.** *Let a sequence in an active branch fail to converge in an ordinary topology used by a dual-carrier row. Its image in the generated evaluation compactification has a subsequence with exactly one primary successor mode:*

$$\mathsf{atom},\ \mathsf{separated},\ \mathsf{nested},\ \mathsf{diffuse},\ \mathsf{tail},\ \mathsf{rough},\ \mathsf{gauge},\ \mathsf{constraint},\ \mathsf{equality}. \tag{114}$$

*The successor retains the activity probability measure and therefore the target floor. Atom, separated, nested, and diffuse values are disintegrations of oscillation or concentration and append* $\mathsf{Lift}$*; tail values append* $\mathsf{Bdry}$*; gauge values append* $\mathsf{Abs}$*; rough equation and constraint values retain the ancestry of the failed operation. If a scale coordinate tends to zero, Theorem II.2.21 supplies the nonincreasing record chart before the successor is evaluated. A mode whose target pairing vanishes closes in* $\mathscr{N}_\Sigma$*; every other mode is processed by Theorem II.2.24 with the same origin tag.*

*Proof.* The normalized activity measures are probabilities on the compact observer space. Sequential compactness and disintegration give the mutually ordered readouts in (114); the first nonordinary total-graph coordinate fixes the primary mode. Source-preserving completion gives the stated ancestry. Weak convergence preserves the integral of the stored target coordinate, so loss of an individual atom cannot erase the target floor: its mass remains in the separated, diffuse, tail, or boundary coordinate. The target quotient closes exactly the zero-pairing modes. All other modes satisfy the definition of a new active branch and re-enter the dual-carrier compiler. □

**Theorem II.2.29** (Continuous selector-accountability closure)**.** *Let an ordinary finite-physical-budget orbit contact* $\Sigma$*. Then there is an active branch signature* $\mathfrak{b}$ *with* $\eta \geq 1/5$*. Along that branch, repeated contact has exactly one of the following consequences:*

(A1) *a generated dual carrier charges infinitely many disjoint crossings to the single finite physical ledger, a contradiction;*

(A2) *a generated target-null, regular, continuable, or class-incompatible successor contradicts retained contact;*

(A3) *target-local polarity stratification produces a proper successor with strictly larger resolved-prefix rank;*

(A4) *diagonal saturation reconstructs a closed same-origin five-channel subsystem whose path crosses the cofinal target-shell family.*

*The fourth alternative is an explicit singular mechanism. A stationary law, equality kernel, or failed analytic property is processed by the third alternative and is never a terminal conclusion. Hence contact is possible only through a target-aligned path of the fully saturated five-source algebra; if no such path is reconstructed, then* $\mathrm{Reach}_{\Sigma}^{\mathrm{phys}}(B_{\mathrm{phys}}) = 0$*.*

*Proof.* Shell accountability and finite source exhaustion select a persistent support and give the floor $1/5$. Eventual constancy of the realization mode gives $m$, while the cascade supplies $\Gamma$ and the origin tag. This constructs $\mathfrak{b}$. Apply Theorem II.2.24. A finite carrier row telescopes over disjoint same-origin events; bounded storage and finite exterior input leave its nonnegative physical production. Infinite retained progress then contradicts the physical ledger. Every zero target projection or terminal regularity relation gives (A2). Otherwise Theorem II.2.28 produces the next active branch. Any stationary equality law is restricted to its target-local structural stratum and passed through Theorem II.2.15. A finite nonnull cell gives the proper successor in (A3). If proper successors continue cofinally, Theorem II.2.16 gives either the target-null contradiction (A2) or the explicit subsystem in (A4). Thus no equality test or analytic classification remains as a premise. □

**Definition II.2.30** (Canonical activity lift and ordered successor rows)**.** Let $\mathfrak{C}_\eta$ be the closure, in the countable product of the generated evaluation compactification, of all indexed normalized shell cascades

$$\mathbf{R} = ((R_n, Q_n, p_n))_{n\geq 1}, \qquad \mathrm{Prog}_\Sigma(R_n) \geq \eta,$$

that satisfy the total weak-equation, shell, and event-order graphs. On an ordinary contact value, the $Q_n$ are the canonical pairwise disjoint physical events, $p_n = \mu_u(Q_n)$, and the progress is one. Here $p_n$ is the raw event-expenditure coordinate. It becomes an admissible price for shell progress only when the target-local factorization satisfies (90); otherwise the carrier-domination graph value is retained in $R_n$ as a successor coordinate. At a graph boundary, $p_n$ is the lower-semicontinuous event-expenditure coordinate. Boundary points of $\mathfrak{C}_\eta$ retain the sequences that approximate them. The compact cascade coordinate $\mathbf{R}$ is an *origin tag*; every descendant below retains it, so successor chains cannot splice different orbits or cascades.

Within the successor algebra, "zero price" abbreviates zero restriction of the public physical measure on a retained target-aligned event. It does not assert a target-progress domination inequality. When that inequality is ordinary and admitted, zero price is also zero admissible target charge; when it fails, its total-graph coordinate remains part of the same zero-expenditure successor and is processed before any exclusion inference.

The generated compact observer space $\Omega_{\mathrm{ev}}$ is the product of the state, spacetime, scale, source, chart, constraint, origin, and total-graph coordinates occurring in the symbolic transcript.

On the bounded-variation branch, let $q_n^c$ be the finite signed Radon measure containing all ordinary, jump, Cantor, oscillation, concentration, and exterior parts of the derived $c$-source shell current. After pushing them to $\Omega_{\mathrm{ev}}$, set

$$a_n := \sum_{c\in\mathfrak{C}} (q_n^c)^+, \qquad \nu_n := \frac{a_n}{a_n(\Omega_{\mathrm{ev}})}. \tag{115}$$

The target floor and (35) imply $a_n(\Omega_{\mathrm{ev}}) \geq \eta$, and an ordinary normalized contact cascade has the stronger lower bound one. If a current has no finite Radon value, its normalized variation-boundary value is placed at the corresponding typed boundary coordinate; rational truncations of its variation give the same probability-valued lift. The lifted record consists of

$$\widehat{R}_n = \left(\mathbf{R}, R_n, \nu_n, \frac{a_n(\Omega_{\mathrm{ev}})}{1 + a_n(\Omega_{\mathrm{ev}})}, \frac{p_n}{1+p_n}, n\right). \tag{116}$$

Run the following total graphs in order on this lift:

$$\begin{gathered} \text{form/constraint realization} \longrightarrow \text{chart/gauge/scale coherence}, \\ \text{activity-measure lift} \longrightarrow \text{atomic/non-atomic/boundary disintegration}, \\ \text{state/scale disintegration} \longrightarrow \text{Young, profile-tree, and tail coordinates}, \\ \text{carrier cohomology} \longrightarrow \text{strict passivity/saturation} \longrightarrow \text{zero-cost recurrent closure}. \end{gathered} \tag{117}$$

The grammar includes the cofinal tail shift $(R_n, Q_n, p_n) \mapsto (R_{n+1}, Q_{n+1}, p_{n+1})$. Hence a target-retaining zero-price witness sequence always has a successor; an analytic failure is carried by its total-graph coordinate rather than by an artificial failure self-loop. The familiar realization modes

$$\mathsf{atom},\quad \mathsf{separated},\quad \mathsf{nested},\quad \mathsf{diffuse},\quad \mathsf{tail},\quad \mathsf{rough},\quad \mathsf{gauge},\quad \mathsf{constraint},\quad \mathsf{equality}. \tag{118}$$

are canonical readouts of $\nu_n$ and its coordinate disintegrations; they are not asserted to be mutually exclusive alternatives. Every ordinary or graph-boundary output retains its five-channel ancestry, original event index, and target floor. Hence a failed analytic relation is immediately another lifted successor input, not an unevaluated row.

**Theorem II.2.31** (Compact activity-profile exhaustion)**.** *Every sequence of lifted records $\widehat{R}_n$ has a convergent subsequence in*

$$\widehat{X}_\Sigma := \mathfrak{C}_\eta \times X_{\mathrm{ev}} \times \mathcal{P}(\Omega_{\mathrm{ev}}) \times [0,1]^2 \times \overline{\mathbb{N}}, \tag{119}$$

*where $\mathcal{P}(\Omega_{\mathrm{ev}})$ carries the narrow topology and $\overline{\mathbb{N}} = \mathbb{N} \cup \{\infty\}$. Its limit retains the target floor (unit progress for a contact origin) and source ancestry. The limiting activity law has the canonical decomposition*

$$\nu = \sum_{j \ge 1} m_j \delta_{z_j} + \nu_{\mathrm{na}}, \qquad m_j > 0, \tag{120}$$

*and its projections to the compactified scale, spatial-boundary, state-law, and total-graph coordinates record, respectively, nested or separated profiles, tails, Young oscillation, and rough/gauge/constraint failures. These coordinates exhaust the limit because they are coordinates of the single probability law $\nu$; no separate compactness input or remainder class is required.*

*Proof.* The spaces $\mathfrak{C}_\eta$, $X_{\mathrm{ev}}$, and $\Omega_{\mathrm{ev}}$ are compact metric. Hence $\mathcal{P}(\Omega_{\mathrm{ev}})$ is compact metric, and so is the product in (119). Sequential compactness gives a convergent subsequence. Unit target progress and the finitely many source supports are closed coordinates; pass first to a constant-support subsequence.

Every probability measure has a unique at most countable atomic part and a nonatomic remainder, which proves (120). Coordinate projection and disintegration record all spatial, scale, state-law, and graph-boundary behavior. Boundary escape is mass on the compactification boundary, rather than failure of tightness. Since every generated observation is one of these countable coordinates, equality of all coordinates is equality of lifted records. Thus no unrecorded limit output remains. □

**Definition II.2.32** (Witness-indexed retained successor relation)**.** Let $\mathfrak{A}_\eta \subset \widehat{X}_\Sigma$ be the closed set of lifted records with target progress at least $\eta > 0$. An elementary successor witness is a tuple

$$(w, w', \mathbf{R}, \sigma, O), \tag{121}$$

where $\mathbf{R} = ((R_n, Q_n, p_n))_n \in \mathfrak{C}_\eta$, $\sigma : \mathbb{N} \to \mathbb{N}$ is strictly increasing, and $O$ is one of the ordered total-graph operations in (117). Enumerate these operations and compactify their index by $\overline{\mathbb{N}}$; an operation-index boundary is represented by the corresponding total-graph limit. The two endpoints must be limits along the *same* selector:

$$w = \lim_{k\to\infty} \widehat{R}_{\sigma(k)}, \qquad w' = \lim_{k\to\infty} O^{\mathrm{tot}}(\widehat{R}_{\sigma(k)}), \tag{122}$$

and both retain the origin coordinate $\mathbf{R}$. Let $\mathfrak{W}_\eta^0$ be the set of these tuples. Compactify the selector by its graph in $\overline{\mathbb{N}}^{\mathbb{N}}$, retain the compactified operation index, take the closure $\mathfrak{W}_\eta = \overline{\mathfrak{W}_\eta^0}$, and define

$$\mathcal{R}_\eta := \pi_{w,w'}(\mathfrak{W}_\eta) \subseteq \mathfrak{A}_\eta \times \mathfrak{A}_\eta. \tag{123}$$

The witness space is compact, so $\mathcal{R}_\eta$ is closed and every edge has a nonempty compact witness fiber. Every witness carries the common origin, approximating sequence, source tags, unit target coordinate, and original event indices. A graph-boundary value is therefore another edge limit. It is never promoted to an actual orbit state without its witness sequence, and edges from different origin cascades cannot be concatenated.

**Lemma II.2.33** (Diagonal event ancestry and no double counting)**.** *Let a finite or infinite chain of successors have one ordinary contact origin* $\mathbf{R} = ((R_n, Q_n, p_n))_n$. *For every* $k$ *one can choose an original index* $N_k$ *such that* $N_{k+1} > N_k$ *and the* $N_k$*-th original record, after the first* $k$ *ordered operations in the witness chain, approximates the* $k$*-th successor coordinate within* $2^{-k}$*. Consequently the physical events* $Q_{N_k}$ *are pairwise disjoint and*

$$\sum_k \mu_u(Q_{N_k}) \le \mu_u\left(\bigsqcup_n Q_n\right) \le B_{\text{phys}}. \tag{124}$$

*Proof.* Choose a witness from the compact fiber of every edge. At depth $k$, use the same-origin limits (122) and a diagonal approximation of the first $k$ witness tuples by elementary witnesses. Their selectors have cofinal image in the one origin sequence. Choose $N_k > N_{k-1}$ sufficiently far out that the composed depth- $k$ graph value is within $2^{-k}$ of $w_k$. No single event is used to approximate two different depths. The events therefore belong to distinct members of the original disjoint shell family, and countable additivity gives (124). □

**Theorem II.2.34** (Successor recurrence)**.** *The relation* $\mathcal{R}_\eta$ *is closed in the compact metric lifted topology. If there is an infinite chain*

$$w_0 \mathcal{R}_\eta w_1 \mathcal{R}_\eta w_2 \mathcal{R}_\eta \cdots, \tag{125}$$

*then the path space of all such chains is compact and invariant under the left shift. The orbit closure of the given path carries a shift-invariant Borel probability measure. Every point in the support of its coordinate projection is a limit generated from one retained cascade origin with a diagonal witness sequence, retains the target floor, and retains the union of source ancestries. No claim of pointwise realization by a new PDE orbit is used.*

*Proof.* The witness bundle is compact and coordinate projection is continuous, so $\mathcal{R}_\eta$ is compact and hence closed. The set $\mathfrak{A}_\eta$ is compact in the lifted topology, so $\mathfrak{A}_\eta^{\mathbb{N}}$ is compact. The path space is the intersection, over all $j$, of the closed conditions $(w_j, w_{j+1}) \in \mathcal{R}_\eta$; it is therefore compact and is nonempty by (125). Empirical measures of the shifted given path have a weakly convergent subsequence, and the endpoint calculation proves shift invariance. Apply Lemma II.2.33 to obtain the diagonal original-event witnesses. Sourcewise closure preserves ancestry and the target floor. □

**Definition II.2.35** (Target-aligned ordered path calculus)**.** Fix an active branch $\mathfrak{b} = (A, m, \Gamma, \eta, \mathbf{R})$. Let $\mathfrak{A}_\mathfrak{b}$ be the closed lifted stratum with that source support, realization mode, shell family and floor, origin tag, and all prior zero transcript coordinates. Let $\mathcal{B}_\mathfrak{b}^{\mathbb{Q}}$ be the unital rational cylinder algebra generated by finitely supported path coordinates on $\mathfrak{A}_\mathfrak{b}^{\mathbb{N}}$, including:

(a) the graph-complete equation, carrier, operation-defect, and source coordinates at every rational path time;
(b) the closed successor-witness coordinates for every adjacent pair;
(c) the bounded price coordinate $\bar{p} = p/(1+p)$;
(d) the normalized target pairing and the positive sourcewise shell masses;
(e) the shift pullback $S^*$.

Generators annihilated by every element of $\Gamma_\Sigma$ are removed before $\mathcal{B}_{\mathfrak{b}}^{\mathbb{Q}}$ is formed.

Let $I_{\mathfrak{b}}$ be the ideal generated by the weak equation, signed carrier, typed graph, same-origin, successor, zero-price, target-floor, and source-support equalities together with the complete active-branch signature. Let $M_{\mathfrak{b}}$ be the quadratic module generated by squares, normalized coordinate bounds, nonnegative production and defect masses, and the positive target-alignment coordinates. Finally, let

$$\mathcal{D}_S := \operatorname{span}_{\mathbb{Q}}\{f \circ S - f : f \in \mathcal{B}_{\mathfrak{b}}^{\mathbb{Q}}\}$$

and define the ordered proof cone

$$\mathcal{K}_{\mathfrak{b}} := I_{\mathfrak{b}} + M_{\mathfrak{b}} + \mathcal{D}_S. \tag{126}$$

A finite ordered derivation is an explicit equality

$$-1 = i + m + d, \qquad i \in I_{\mathfrak{b}}, \quad m \in M_{\mathfrak{b}}, \quad d \in \mathcal{D}_S,$$

written with rational cylinder polynomials.

**Theorem II.2.36** (Ordered proof or stationary structural model)**.** *For every reachable active branch $\mathfrak{b}$, exactly one of the following holds.*

(O1) $-1 \in \mathcal{K}_{\mathfrak{b}}$*. Then a finite ordered derivation proves that no stationary target-aligned zero-price successor law in $X_{\mathfrak{b}}$ exists.*

(O2) *There is a linear functional $L : \mathcal{B}_{\mathfrak{b}}^{\mathbb{Q}} \to \mathbb{R}$ such that*

$$L(1) = 1, \qquad L(M_{\mathfrak{b}}) \geq 0, \qquad L(I_{\mathfrak{b}}) = L(\mathcal{D}_S) = 0. \tag{127}$$

*Its continuous extension is integration against a shift-invariant Borel probability law on the graph-complete path space. That law has zero physical price and satisfies the full signature $\mathfrak{b}$, including target pairing at least $\eta$, source support $A$, mode $m$, origin $\mathbf{R}$, and all same-origin successor coordinates.*

*The alternatives are exclusive. Every assertion in* (O1) *is checked by finite rational algebra; every object in* (O2) *is reconstructed from the public graph relations.*

*Proof.* Pass first to the rational vector-space quotient by $I_{\mathfrak{b}} + \mathcal{D}_S$. The normalized coordinate bounds make the image of $M_{\mathfrak{b}}$ an Archimedean order cone with order unit [1]. If $-[1]$ is not in this cone, the order-unit separation theorem gives a positive linear functional with value one on [1]. Pulling it back gives (127). Positivity makes the functional bounded in the order-unit norm. The Archimedean representation theorem for quadratic modules, followed by the Riesz representation theorem, therefore represents its continuous extension by a Borel probability measure supported on the joint positivity set of the graph-complete path relations [15]. Annihilation of $\mathcal{D}_S$ is precisely shift invariance. Annihilation of $I_{\mathfrak{b}}$ imposes the equation, successor, same-origin, zero-price, target-floor, source, mode, origin, and prior-zero relations. This proves (O2).

Conversely, a functional satisfying (127) sends every $i + m + d \in \mathcal{K}_{\mathfrak{b}}$ to a nonnegative number. It therefore cannot coexist with $-1 \in \mathcal{K}_{\mathfrak{b}}$. If no such functional exists, order-unit separation gives $-1 \in \mathcal{K}_{\mathfrak{b}}$, proving exhaustiveness. An element of the algebraic sum is, by definition, a finite expression, so the derivation uses finitely many rational relations. $\square$

**Definition II.2.37** (Finite cylinder-moment hierarchy)**.** Fix an active branch $\mathfrak{b}$ and enumerations of its generators and proof rules. At level $\ell = (n, d, N)$, retain the first $n$ generators, monomials of degree at most $2d$, and successor coordinates through time $N$. The primal cone $\mathsf{P}_{\mathfrak{b}}^{\ell}$ consists of truncated functionals whose moment and localizing matrices for $M_{\mathfrak{b}}$ are positive semidefinite and which annihilate the retained ideal and shift-difference rows. The dual cone $\mathsf{D}_{\mathfrak{b}}^{\ell}$ consists of rational identities

$$-1 = i_\ell + \sum_j s_j^2 m_j + d_\ell \tag{128}$$

using only the retained rows. Exact rational matrix arithmetic and rational interval enclosures check every dual identity.

**Theorem II.2.38** (Proof-producing hierarchy completeness)**.** *For a fixed reachable active branch* $\mathfrak{b}$*:*

(i) *every feasible primal truncation gives valid lower moment bounds and every dual derivation is a sound exclusion of the source cell;*
(ii) *if* $-1 \in \mathcal{K}_{\mathfrak{b}}$*, then* $\mathsf{D}_{\mathfrak{b}}^{\ell}$ *contains a derivation at some finite level;*
(iii) *if every finite level is primal feasible, a diagonal subsequence of normalized moment tables converges to a functional satisfying* (127)*.*

*Thus semantic membership and compact-set inclusion are absent from the pruning interface.*

*Proof.* Weak duality follows by applying a feasible positive truncated functional to (128). A finite expression for $-1 \in \mathcal{K}_{\mathfrak{b}}$ contains finitely many generators, a finite degree, and a finite shift depth, so it occurs at some level, proving (ii). For (iii), the Archimedean bounds make each scalar moment sequence bounded. Repeated diagonal extraction gives a value on every cylinder monomial. Positivity of every finite moment and localizing matrix, and annihilation of every retained relation at all sufficiently large levels, pass to the limit. The resulting normalized positive functional satisfies (127); apply Theorem II.2.36. □

**Definition II.2.39** (Prospective zero-cost successor fixed point)**.** Fix a reachable active branch $\mathfrak{b}$. In this definition and the following pruning results, $\mathfrak{A}_\eta$, $\mathcal{R}_\eta$, $Z_n$, and $Z_\infty^\Sigma$ denote their restrictions to the closed same-origin stratum $\mathfrak{A}_{\mathfrak{b}}$; the branch index is suppressed only to keep the formulas readable. Let $p : \mathfrak{A}_{\mathfrak{b}} \to [0, \infty]$ be the lower-semicontinuous relaxation of the original-event price coordinate. For a closed $K \subseteq \mathfrak{A}_{\mathfrak{b}}$, define the viability operator

$$\mathcal{V}_\eta(K) := \{w \in K : p(w) = 0,\ \mathcal{R}_\eta(w) \cap K \neq \varnothing\}. \tag{129}$$

It maps closed sets to closed sets, because it is the projection of the compact set $\mathcal{R}_\eta \cap (K \times K) \cap (\{p = 0\} \times K)$. Starting with the target-active zero-price set $Z_0 = \mathfrak{A}_\eta \cap \{p = 0\}$, set

$$Z_{n+1} = \mathcal{V}_\eta(Z_n), \qquad Z_\infty^\Sigma = \bigcap_{n=0}^{\infty} Z_n. \tag{130}$$

Thus $Z_n$ consists exactly of the records from which a target-retaining zero-price successor path of length $n$ can be generated. Compactness and closedness give

$$\mathcal{V}_\eta(Z_\infty^\Sigma) = Z_\infty^\Sigma. \tag{131}$$

Indeed, for $w \in Z_\infty^\Sigma$, choose $w_n \in Z_n$ with $w \mathcal{R}_\eta w_n$; a convergent subsequence has a limit in every $Z_n$, and the closed successor graph retains the edge. Consequently the countable finite-depth iteration is the complete topological description of viability. It is not, by itself, a proof procedure. If $Z_\infty^\Sigma = \varnothing$, nested compactness implies $Z_N = \varnothing$ for some finite $N$. This semantic fact becomes a checkable exclusion only when the ordered hierarchy below produces an explicit identity $-1 \in \mathcal{K}_{\mathfrak{b}}$ on each active component. If every $Z_n$ is nonempty, their intersection contains an infinite zero-price path.

Let $\mathsf{Path}(Z_\infty^\Sigma)$ be the compact path space of $\mathcal{R}_\eta$ restricted to this fixed point. Let $\mathcal{P}^{0,\Sigma}_{\mathcal{R}_\eta}$ be all shift-invariant Borel probability laws on this path space. The prospective successor-saturation algebra is

$$\mathfrak{H}^\Sigma_{\mathrm{succ}} := \overline{\operatorname{span}}\left\{\Pi_\Sigma(\pi_0)_\# \boldsymbol{\nu} : \boldsymbol{\nu} \in \mathcal{P}^{0,\Sigma}_{\mathcal{R}_\eta}\right\}. \tag{132}$$

Every object in this definition is constructed prospectively from $\Theta_\Sigma^{\mathrm{syn}}$. Neither actual contact orbits nor exact invariant values occur. A total-graph failure contributes only when it belongs to an infinite target-retaining zero-cost path; naming the failure supplies no self-loop.

**Definition II.2.40** (Finite ordered pruning transcript)**.** Put

$$E_n := \mathfrak{A}_\eta \setminus Z_n. \tag{133}$$

A *finite ordered pruning transcript* for active branch $\mathfrak{b}$ is the complete finite expression

$$-1 = i + \sum_{j=1}^{M} s_j^2 m_j + d, \qquad i \in I_{\mathfrak{b}}, \quad m_j \in M_{\mathfrak{b}}^{\mathrm{gen}}, \quad d \in \mathcal{D}_S, \tag{134}$$

including the rational coefficients and the indices of every equation, carrier, graph, source, target-floor, same-origin, successor, positivity, and shift relation used. Here $M_{\mathfrak{b}}^{\mathrm{gen}}$ is the enumerated set of quadratic-module generators. Verification consists only of expanding (134), checking the cited ideal and shift rows, and checking that each multiplier is a square. The transcript is accepted only after this finite calculation succeeds. Its existence is never inferred from semantic emptiness of $Z_n$. Among valid transcripts use the first in the fixed enumeration.

**Theorem II.2.41** (Ordered pruning or same-origin structural model)**.** *The elimination sets are open and satisfy*

$$E_{n+1} = E_n \cup \{w \in Z_n : \mathcal{R}_\eta(w) \cap Z_n = \varnothing\}. \tag{135}$$

*The recursion is the topological shadow of the ordered hierarchy. Exactly one of the following proof-producing alternatives holds for every reachable active branch $\mathfrak{b}$.*

(P1) *At some finite moment level, the dual hierarchy produces $-1 \in \mathcal{K}_{\mathfrak{b}}$. The identity itself is the finite proof-valid pruning transcript and excludes the cell.*

(P2) $-1 \notin \mathcal{K}_{\mathfrak{b}}$. *Order-unit separation gives the positive functional of Theorem II.2.36, hence an infinite same-origin zero-price path law with target floor $\eta$. Equivalently, a compatible family of all finite moment restrictions reconstructs this law.*

*Thus every pruning assertion is verified by the ordered proof kernel. A presently unsuccessful finite search is neither an exclusion nor a model; the model branch is the mathematical alternative $-1 \notin \mathcal{K}_{\mathfrak{b}}$. It is successor data: zero-price resaturation must still restrict its kernel, reselect its prospective source, and totalize its realization graph.*

*Proof.* The set $E_0$ is open because $Z_0$ is closed. If $E_n$ is open, then

$$\{w : \mathcal{R}_\eta(w) \cap Z_n \neq \varnothing\} = \pi_1(\mathcal{R}_\eta \cap (\mathfrak{A}_\eta \times Z_n))$$

is compact and hence closed. This proves (135) and openness of $E_{n+1}$.

Apply Theorem II.2.38. In its dual branch, the finite identity is exactly (134); applying any putative positive stationary functional gives $-1 \geq 0$, so the cell is excluded. In the other algebraic branch, Theorem II.2.36 supplies a normalized positive functional annihilating the successor and shift-difference relations and reconstructs its stationary path law. Same-origin coordinates belong to the ideal and hence hold almost surely. The alternatives are exclusive by weak duality. $\square$

**Definition II.2.42** (Generated zero-cost dynamics)**.** Let $\boldsymbol{\nu}$ be a stationary target-retaining successor law with zero mean physical expenditure. Let $\mathsf{Conseq}(\mathfrak{P})$ be the least countable consequence algebra containing the public weak equation and native carrier identity and closed under rational linear combination, localization by the generated test core, integration by parts already encoded in the weak formulation, polarization, adjoint formation, quotient descent, chart pullback, Jordan decomposition, completion of squares, composition of total graphs, and sourcewise graph closure. If a product, chain, trace, or covariance operation lacks an ordinary value, the algebra inserts its total-graph defect rather than the desired identity. This grammar is enumerated from $\mathsf{Pre}(\mathsf{Eq})$ and has no additional input slot.

Let $\mathscr{P}_{\mathrm{gen}}$ be the countable family of all generated nonnegative functions $q$ for which $\mathsf{Conseq}(\mathfrak{P})$ contains a domination identity

$$p = q + r_q, \qquad q \geq 0, \quad r_q \geq 0. \tag{136}$$

Include $q = p$ and $r_p = 0$. This definition takes the union of all valid positive descendants and does not require a choice of a maximal or unique decomposition. The family is generated by the fixed equation and carrier: mobility squares, weak energy defects, positive commutator squares, Fisher-information or Hessian squares, constraint defects, and target-ray defects occur only when their defining identity is already a relation in $\mathsf{Conseq}(\mathfrak{P})$. No extra positive term is inserted.

The *saturation equations* are

$$q = 0 \quad (q \in \mathscr{P}_{\mathrm{gen}}), \qquad \dot{u} = J_{\mathsf{Geom}} + J_{\mathsf{Caus}} + J_{\mathsf{Abs}} + J_{\mathsf{Lift}} + J_{\mathsf{Bdry}}, \tag{137}$$

together with the totalized constitutive, constraint, chart, and covariance relations. No tangency to a symmetry orbit is asserted. Equilibria, relative equilibria, periodic or recurrent conservative orbits, scale breathers, and persistent graph-boundary dynamics are retained whenever they solve these relations. If $\mathcal{F}_{\mathrm{gen}}$ is the generated finite or countable family of partial relations used on the current branch, let $\partial_\infty F$ denote the recurrent graph-boundary values obtained from infinite witness-successor paths for $F$, rather than the entire graph boundary. Define the totalized saturation set

$$\mathscr{Z}_{\mathrm{sat}}^{\mathrm{tot}} := \left( \bigcap_{q \in \mathscr{P}_{\mathrm{gen}}} \ker q \cap \{\dot{u} = J_{\mathrm{tot}}\} \right) \sqcup \bigsqcup_{F \in \mathcal{F}_{\mathrm{gen}}} \partial_\infty F. \tag{138}$$

Thus a failed relation enters the saturation set only through a recurrent zero-cost graph-boundary path. The definition imposes only identities already derived from the equation and does not convert stationarity of a law into stationarity of a state.

**Theorem II.2.43** (Saturation-equation compiler)**.** *Every generator of $\mathfrak{H}_{\mathrm{succ}}^{\Sigma}$ is supported on the totalized solution set of* (137) *and on the greatest viable set $Z_\infty^\Sigma$. Conversely, every stationary law on the prospective path space supported on this solution set generates a class in $\mathfrak{H}_{\mathrm{succ}}^{\Sigma}$. Hence*

$$\mathfrak{H}_{\mathrm{succ}}^{\Sigma} = \overline{\mathrm{span}}\, \Pi_\Sigma \left( Z_\infty^\Sigma \cap \mathscr{Z}_{\mathrm{sat}}^{\mathrm{tot}} \right), \tag{139}$$

*where the right side denotes the coordinate supports of stationary path laws on the displayed set. It is generated entirely by algebraic kernels, the closed successor graph, source quotients, and the descending viability operator.*

*Proof.* For a stationary zero-expenditure law, $\int p \, \mathrm{d}\boldsymbol{\nu} = 0$, so $p = 0$ almost surely. For every $q \in \mathscr{P}_{\mathrm{gen}}$, (136) gives $0 \le q \le p$, hence $q = 0$ almost surely on the ordinary branch. The full weak current identity remains one of the total graph coordinates, so its ordinary support solves $\dot{u} = J_{\mathrm{tot}}$ without any symmetry-tangency inference. On a graph-boundary branch, membership in the path fixed point supplies the recurrent value in $\partial_\infty F$. This proves the first inclusion in (139). Conversely, a stationary path law in the displayed ordinary joint kernel has $p = \mathcal{P}_0 = 0$; a recurrent graph-boundary law has the same zero price by membership in $Z^\Sigma_\infty$. Both are generators of the successor-saturation algebra. Taking target projections and closed spans proves equality. $\square$

**Corollary II.2.44** (Complete zero-cost structural reduction)**.** *The finite-depth recursion* (130) *has as its intersection the largest target-retaining zero-cost successor set. Its recurrent projection satisfies the exact alternative*

$$\mathfrak{H}^\Sigma_{\mathrm{succ}} = 0 \quad \Longleftrightarrow \quad \Pi_\Sigma \operatorname{supp} \boldsymbol{\nu} = 0 \textit{ for every stationary law on } Z^\Sigma_\infty. \tag{140}$$

*At each successor stage the compiler applies the five-source quotient to the full zero-cost weak equation and to every recurrent graph-boundary coordinate. A target-null output is removed at the next stage. Any nonzero fixed-point output is a positive structural model of recurrent conservative, relative-equilibrium, profile, or graph-boundary dynamics satisfying the relations retained so far. It is not asserted to be an actual singular PDE orbit and is not terminal. It is passed to the zero-price resaturation operator of Definition III.4.13; only the exhaustive alternatives of Theorem III.4.36 may terminate that route. No compactness gap or separate rigidity datum appears in this reduction. A checked identity $-1 \in \mathcal{K}_{\mathfrak{b}}$ closes a finite stage; the complementary positive functional supplies the next kernel and prospective-source calculation.*

**Theorem II.2.45** (Zero-mean successor reduction)**.** *Let an infinite witness-indexed successor chain originate from the disjoint shell sequence of a finite-expenditure contact orbit. Then every coordinate of its diagonal path has relaxed original-event expenditure zero, and its path closure carries a stationary successor law with zero mean physical expenditure. Consequently its target projection lies in $\mathfrak{H}^\Sigma_{\mathrm{succ}}$.*

*Proof.* By countable additivity, the original event expenditures $p_n = \mu_u(Q_n)$ are summable and hence tend to zero. Every successor uses a subsequence of these same indexed events. The diagonal selection in Lemma II.2.33 therefore approximates each path coordinate by records whose expenditures tend to zero. Lower semicontinuity and nonnegativity give zero expenditure at every path coordinate. Whenever that coordinate is used as target price, admissibility follows separately from Theorem II.2.4; otherwise its failed domination relation remains in the successor graph.

Theorem II.2.34 gives a compact shift-invariant path closure. A weak limit of its empirical path measures is shift invariant and has zero mean price. Target retention and source ancestry are closed coordinates, so the law is one of the generators in (132). $\square$

**Theorem II.2.46** (Estimate-free finite-budget contact routing)**.** *Let a contact orbit with finite physical expenditure generate infinitely many pairwise disjoint target-retaining windows $I_j$. The canonical activity lift and cofinal tail shift generate an infinite path in $Z^\Sigma_\infty$, and*

$$\Pi_\Sigma Z^\Sigma_\infty \neq 0, \qquad \mathfrak{H}^\Sigma_{\mathrm{succ}} \neq 0. \tag{141}$$

*Thus finite price is completely reduced to an explicit recurrent zero-cost mechanism; it is not a terminal scalar branch or an outstanding analytic row. No lower price, compactness estimate in the original state topology, coercivity constant, or Liouville input is used: the only numerical fact is countable additivity of the equation-generated physical measure.*

*Proof.* Countable additivity gives $\sum_j \mu_u(I_j) < \infty$, hence after a subsequence $\mu_u(I_j) \to 0$. Apply the compact activity lift. Its limit has relaxed price zero, unit target progress, and inherited source support. Apply each ordered total graph to cofinal tails. Ordinary and boundary values are both coordinates of the compact lift, and the tail shift supplies a successor at every stage. This gives an infinite target-retaining zero-cost path.

The descending viability recursion retains exactly the points admitting all finite continuations of such a path; compactness supplies an infinite path in their intersection. Hence the path lies in $Z_\infty^\Sigma$. Theorem II.2.45 supplies a stationary zero-cost law. Its unit target coordinate makes its target projection nonzero, proving (141). □

**Theorem II.2.47** (Ordered prospective regularity alternative)**.** *For a fixed canonical presentation, run the total local-solution and carrier graphs, source compiler, canonical shells, signed carrier compiler, source-adapted dual-carrier grammar, activity lift, and zero-cost successor fixed point. Every target-visible local-solution or carrier defect is inserted into the same successor graph. Define the generated regularity mechanism algebra*

$$\mathfrak{H}_{\mathrm{out}}^{\Sigma} := \overline{\mathrm{span}}\{\Pi_\Sigma u_* : u_* \text{ is an explicit singular mechanism}\}, \tag{142}$$

*where explicit realization is understood in the sense of Definition III.4.33. Put $\eta_0 = 1/5$. Exactly one of the following algebraic alternatives holds:*

(R1) *every algebraically reachable active branch is eliminated by a local target-null, continuation, class-incompatibility, or divergent-charge relation produced by polarity refinement and re-saturation; every ordinary maximal finite-expenditure branch is then global and avoids the complete continuation target;*

(R2) *zero-price resaturation reconstructs an explicit singular mechanism $u_*$. It supplies the closed reduced weak equation, source ancestry, vanishing productions, resolved total-graph values with ordinary induced operations, physical and origin ledgers, and actual contact with a cofinal target-shell family.*

*Independently, if the public local-solution or carrier graph has no ordinary value, its explicit first-failure graph coordinate is retained. This is an existence or carrier-realization output, not a regularity conclusion, and the regularity alternative is applied only to ordinary fibers. Every actual finite-expenditure contact orbit constructs an output of type* (R2) *with its own diagonal event witnesses; stationary positive laws and equality kernels encountered on the way are nonterminal resaturation states. Consequently*

$$\boxed{\mathfrak{H}_{\mathrm{out}}^{\Sigma} = 0 \quad \Longrightarrow \quad \text{every ordinary maximal branch with derived finite carrier is globally regular.}} \tag{143}$$

*This is an output identity of the ordered algebra. A closing branch carries the finite transcript* (134)*; the complementary branch is iterated by Theorem III.4.36 until it closes or produces the explicit data listed in* (R2).

*Proof.* Insert ordinary values of the total local-solution and carrier graphs into the activity lift. The form value $\perp_{\mathrm{loc}}$ is retained separately. On an ordinary finite-expenditure branch, suppose a maximal orbit contacts the complete target. Canonical shells give the disjoint unit progress cascade. A divergent generated lower-price series contradicts countable additivity; the finite-price tail instead enters the successor graph. Theorem II.2.46 constructs a stationary law in $Z_\infty^\Sigma$ with nonzero target coordinate and original shell-event witnesses. Theorem II.2.29 either closes its active branch or reprocesses its typed equality successor. Theorem III.4.36 turns every nonclosing compatible route into (R2). Otherwise each finite local elimination in (R1) contradicts the retained target coordinate; diagonal saturation supplies the same conclusion for a cofinal target-null route. Terminal coverage then makes every ordinary maximal branch global. The contrapositive proves (143). □

**Proposition II.2.48** (Covariant entropy as a carrier correction)**.** *Let a covariant entropy cocycle $\mathcal{M}$ satisfy the gluing law of Theorem II.10.2. If its local balance has the form*

$$\mathcal{M}(Sw) - \mathcal{M}(w) = F_{\mathrm{sc}}(w) - D(w) - M(w) - R(w), \qquad R \geq 0, \tag{144}$$

*then $\Psi = -\mathcal{M}$ is the canonical carrier correction of Definition II.2.5, and*

$$D + M - F_{\mathrm{sc}} + \Psi - \Psi \circ S = R \geq 0. \tag{145}$$

*Every invariant saturation measure is supported on $R = 0$. It is supported on $D = M = 0$ only when the generated carrier transcript supplies a nonnegative domination identities expressing the zero-price deficit with $D$ and $M$ as positive descendants. Thus a Perelman-type formula refines the carrier cohomology and zero-price equations. Its equality models are exactly the ordinary or recurrent graph-boundary solutions of those generated equations; relative equilibria are one possible subclass, not an automatic conclusion.*

*Proof.* Rearrange (144) to obtain (145). The entropy cocycle glues on one coherent orbit, so its increments telescope and define a carrier coboundary. For an invariant saturation measure the nonnegative right side has zero integral and therefore $R = 0$ almost surely. Any further vanishing follows only by applying the generated positive-descendant rule of Definition II.2.42; the full weak equation remains in the support coordinates. □

**Corollary II.2.49** (Gradient-kernel equality reduction)**.** *Let $\Omega$ be connected and let the ordinary saturation equations on a normalized successor core contain*

$$D_m(V) := \int_{K_m} |\nabla V|^2 \,\mathrm{d}y = 0, \qquad m \geq 1, \tag{146}$$

*for an increasing compact exhaustion $K_m \Subset \Omega$. Then every ordinary zero-expenditure stationary law is supported on spatially constant states. If the generated gauge quotient sends spatial constants to $\mathscr{N}_\Sigma$, its ordinary equality contribution to $\mathfrak{H}^{\Sigma}_{\mathrm{succ}}$ vanishes. Every nonordinary value of the gradient, exhaustion, connected-core, or gauge relation remains a witness-successor and is reduced by $\widehat{\mathcal{R}}_\eta$.*

*Proof.* For a zero-expenditure stationary law $\boldsymbol{\nu}$, nonnegativity and the saturation-equation compiler give $\int D_m(V)\,\mathrm{d}\boldsymbol{\nu} = 0$ for every $m$. Hence $D_m(V) = 0$ for $\boldsymbol{\nu}$-almost every state, simultaneously for all $m$. The exhaustion gives $\nabla V = 0$ almost everywhere on $\Omega$. The weak fundamental theorem on connected domains makes $V$ spatially constant. The target projection then vanishes by the stated gauge quotient. Totalization of each relation gives the last assertion. □

## II.3 Covariant action and the physical budget

The fixed-Caus formulation separates the current selected by the PDE from the metric used to measure irreversible expenditure. This separation is physical: the action below is the quantity appearing in the energy identity. The explicit inverse notation in this section is used only when the mobility has the Hilbert realization stated next. In a general Gelfand triple, the same identities are read at the level of the closed mobility form. These action identities remain library relations until a shell covector selects an active source cell. Their reachability content is exactly the target-local five-source factorization of Theorem II.2.18; only the resulting carrier-dominated production may enter the physical ledger.

**Definition II.3.1** (Mobility current space)**.** Fix a state $u$, a real Hilbert tangent space $\mathcal{H}_u$, and a bounded self-adjoint nonnegative mobility $\mathbb{K}(u) \in \mathcal{L}(\mathcal{H}_u)$. Let $\mathbb{K}(u)^{-1/2}$ denote the Moore–Penrose inverse on $(\ker \mathbb{K}(u))^{\perp}$. The mobility current space is the Cameron–Martin space

$$\mathfrak{J}_u := \operatorname{Ran} \mathbb{K}(u)^{1/2}$$

with inner product and norm

$$(J_1, J_2)_{u,*} := \langle \mathbb{K}(u)^{-1/2} J_1, \mathbb{K}(u)^{-1/2} J_2 \rangle_{\mathcal{H}_u}, \qquad \|J\|_{u,*}^2 := \|\mathbb{K}(u)^{-1/2} J\|_{\mathcal{H}_u}^2.$$

Equivalently, this is the completion of $\operatorname{Ran} \mathbb{K}(u)$ in the displayed norm. Covectors are represented in $\mathcal{H}_u$, with components in $\ker \mathbb{K}(u)$ identified when they act on currents. For a velocity $v$, put

$$W(u, v) := v + A_{\mathsf{Caus}}(u) - f_{\mathsf{Bdry}}(u).$$

The dissipation Lagrangian is

$$\mathscr{L}(u, v) := \begin{cases} \frac{1}{2}\|W(u, v)\|_{u,*}^2, & W(u, v) \in \mathfrak{J}_u, \\ +\infty, & \text{otherwise.} \end{cases} \tag{147}$$

Its physical action on an interval $I$ is $\mathcal{A}_I(u) = \int_I 2\mathscr{L}(u, \dot{u})\, \mathrm{d}t$.

**Definition II.3.2** (Covariant dissipation Hamiltonian)**.** For $p \in \mathcal{H}_u$, define

$$\mathscr{H}_{\mathrm{cov}}(u, p) := -\langle p, A_{\mathsf{Caus}}(u) - f_{\mathsf{Bdry}}(u) \rangle + \frac{1}{2}\|\mathbb{K}(u)^{1/2} p\|^2. \tag{148}$$

This is the Legendre–Fenchel dual of $v \mapsto \mathscr{L}(u, v)$.

**Proposition II.3.3** (Fenchel equality and the selected current)**.** *Whenever the displayed quantities are finite,*

$$\mathscr{L}(u, v) + \mathscr{H}_{\mathrm{cov}}(u, p) - \langle p, v \rangle = \frac{1}{2}\|W(u, v) - \mathbb{K}(u)p\|_{u,*}^2 \geq 0. \tag{149}$$

*Equality holds precisely when $W(u, v) = \mathbb{K}(u)p$ in the current quotient. For the fixed-Caus gradient flow* (9)*, the selected velocity has*

$$W(u, \dot{u}) = -\mathbb{K}(u) D\mathcal{E}(u),$$

*so equality holds at $p = -D\mathcal{E}(u)$, modulo the mobility kernel.*

*Proof.* Substitute (147) and (148); since $v = W - A_{\mathsf{Caus}} + f_{\mathsf{Bdry}}$, the linear terms combine to $-\langle p, W \rangle$. Completing the square in the $\mathbb{K}^{-1}$-metric gives (149). The equality and selected-current statements follow directly. □

**Definition II.3.4** (Physical carrier balance)**.** Let $u : [0, T_*) \to X$ be an ordinary maximal branch. Expand the native signed carrier graph of Definition I.1.12 and take the Jordan decomposition of its signed currents. Denote the resulting scalar by $\mathcal{K}_u$, the gross positive production by $\mathfrak{D}_u$, the returned internal feeding by $\Phi_u^+$, and true positive exterior input by $R_u^+$. Its ordinary graph value is, for $0 \le s < t < T_*$,

$$\mathcal{K}_u(t) + \mathfrak{D}_u((s,t]) \le \mathcal{K}_u(s) + \Phi_u^+((s,t]) + R_u^+((s,t]). \tag{150}$$

Here all four quantities are generated from the one signed identity. The generated passivity radius is the least extended-real $\beta$ for which the measure domination

$$\Phi_u^+ \le \beta\, \mathfrak{D}_u. \tag{151}$$

holds. Whether $\beta \le 1$, whether $\mathcal{K}_u$ has a lower bound, and whether the exterior input is finite are values of generated relations, not additional data. All terms are expressed in the physical clock. A normalized-clock identity is routed through the totalized clock-comparison relation in Proposition II.10.5.

**Theorem II.3.5** (Carrier compiler for the single physical budget)**.** *If the generated values satisfy $\beta \le 1$, $\inf_t \mathcal{K}_u(t) = K_- > -\infty$, and $R_u^+([0, T_*)) < \infty$, then*

$$\mu_u := \mathfrak{D}_u - \Phi_u^+ \tag{152}$$

*is a nonnegative countably additive physical expenditure measure and*

$$\mu_u([0, T_*)) \le \mathcal{K}_u(0) - K_- + R_u^+([0, T_*)) =: B_{\mathrm{phys}} < \infty. \tag{153}$$

*If $\beta < 1$, then*

$$\mu_u \ge (1 - \beta)\mathfrak{D}_u. \tag{154}$$

*For $\beta = 1$, $\mu_u$ is the exact flux-deficit measure. If any displayed relation has the opposite value, its total-graph witness is sent directly to the activity lift and successor recursion. No alternative expenditure or independent budget is appended later. This theorem constructs the available orbitwise physical measure. A target event spends that measure only after its source-resolved shell current satisfies the local charging rule* (90)*.*

*Proof.* The domination $\Phi_u^+ \le \mathfrak{D}_u$ makes the difference in (152) a nonnegative Radon measure. Rearrange (150) with $s = 0$, use $\mathcal{K}_u(t) \ge K_-$, and let $t \uparrow T_*$. Monotone convergence gives (153). Finally, $\Phi_u^+ \le \beta\mathfrak{D}_u$ gives (154). □

**Definition II.3.6** (Reduced passivity radius)**.** Let $\mathcal{P}_{\mathrm{car}}^{\mathrm{red}}$ be a normalized carrier profile space obtained after zero-descent-defect source reductions. Let $D(V) \ge 0$ and $\Phi_+(V) \ge 0$ be the descended gross-expenditure and feeding-flux densities in the same physical units. The reduced passivity radius is

$$\beta_*^{\mathrm{red}} := \alpha_*(\Phi_+ \mid D) = \sup_{\substack{V \in \mathcal{P}_{\mathrm{car}}^{\mathrm{red}} \\ D(V) > 0}} \frac{\Phi_+(V)}{D(V)}, \tag{155}$$

with the null-mode convention of Definition II.1.6. A bound $\beta_*^{\mathrm{red}} \le 1$ supplies (151) whenever the carrier identity is local in the normalized profiles and its physical-clock pullback is proved.

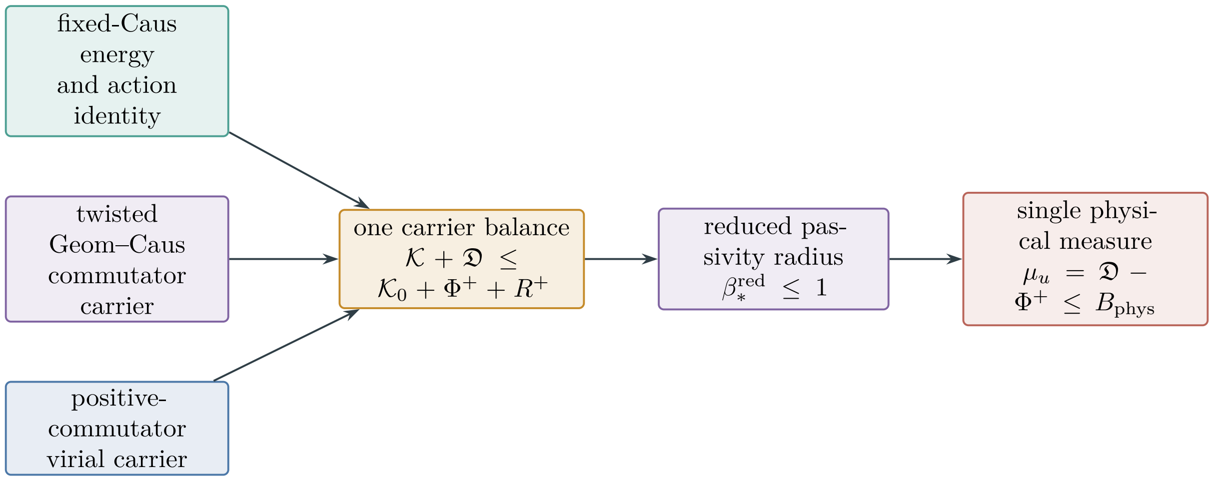


Figure 11: Carrier-to-budget compiler. Gradient-flow, hypocoercive, and positive-commutator identities are different ways to expose the same structural object: a lower-bounded carrier whose feeding flux is dominated by gross expenditure. The output is the sole physical budget measure.

**Proposition II.3.7** (Quadratic passivity is a spectral invariant)**.** *On the quadratic-form realization branch of the reduced Hilbert profile space,*

$$D(V) = \langle AV, V\rangle, \qquad \Phi_+(V) = \max\{\langle BV, V\rangle, 0\},$$

*where $A \ge 0$, $B = B^*$, all $D$-null target-visible feeding modes have been separated, and $B$ is bounded in the $A$-form norm. Let $S_{\mathrm{flux}} = A^{-1/2}BA^{-1/2}$ be the descended self-adjoint form operator on the completion modulo* $\ker A$*. Then*

$$\beta_*^{\mathrm{red}} = \max\{\sup \sigma(S_{\mathrm{flux}}), 0\}. \tag{156}$$

*Thus every multiplier or casewise proof of $\Phi_+ \le D$ is, on this reduced quadratic interface, an upper bound for one spectral radius.*

*Proof.* The $A$-form completion is a Hilbert space after quotienting by $\ker A$. The form induced by $B$ is represented there by $S_{\mathrm{flux}}$. Rayleigh–Ritz identifies the supremum of $\langle BV, V\rangle/D(V)$ with $\sup \sigma(S_{\mathrm{flux}})$; taking the positive part gives (156). □

## II.4 Hamiltonian meanings and chart covariance

The constructions in this section are universal operator identities, not equation-wide regularity assertions. In a reachability argument they are activated only after restriction to a contact-generated stratum $X_{\mathfrak{b}}$, where Theorem II.2.18 separates their $\mathsf{Geom}/\mathsf{Caus}/\mathsf{Abs}/\mathsf{Lift}/\mathsf{Bdry}$ components, target-null part, carrier coboundary, and total-graph defects. Only that factorized local value may enter a physical charge or close a branch.

**Definition II.4.1** (Geometric spectral Hamiltonian)**.** Let $\mathfrak{q}$ be a densely defined closed nonnegative symmetric form on a real Hilbert space $\mathcal{H}$. Its geometric Hamiltonian is the unique self-adjoint operator $H \ge 0$ satisfying

$$\mathfrak{q}(u, v) = \langle H^{1/2}u, H^{1/2}v\rangle, \qquad u, v \in D(\mathfrak{q}).$$

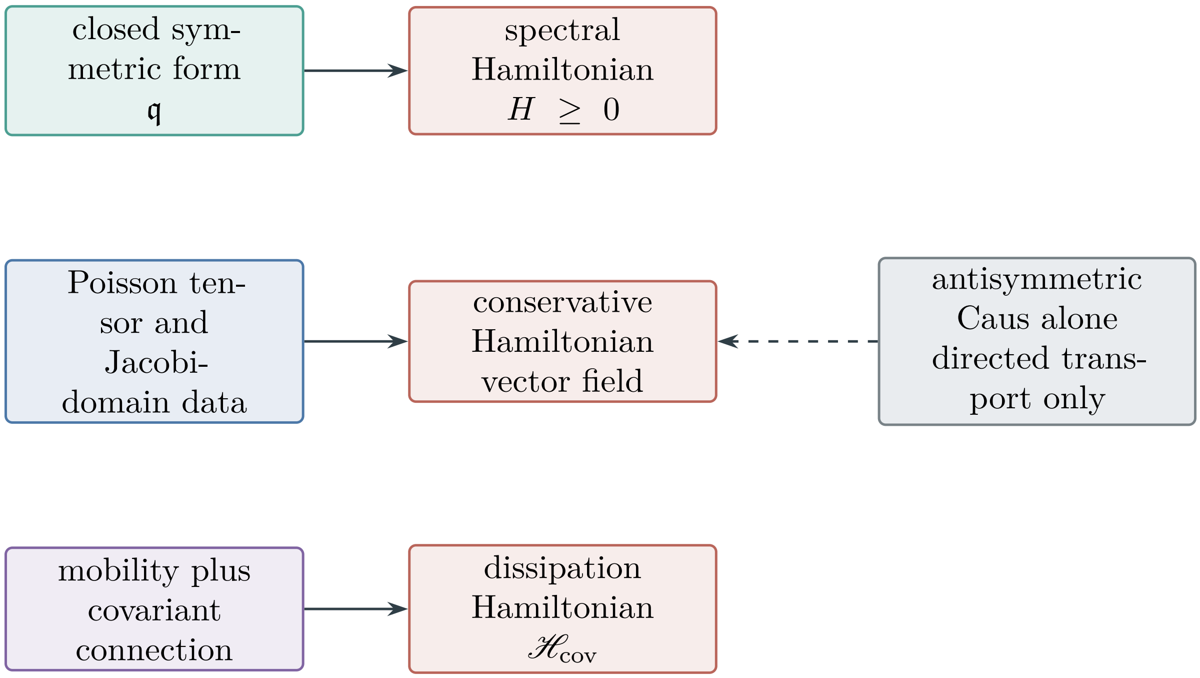


Figure 12: Three noninterchangeable Hamiltonian constructions. The dashed line marks a missing implication: skew transport alone does not establish a Poisson structure.

This Hamiltonian records only the $\mathsf{Geom}$ form; the fixed antisymmetric current remains in $\mathsf{Caus}$.

**Proposition II.4.2** (Separation of Hamiltonian meanings)**.** *The following constructions require distinct data.*

(i) *A closed symmetric form determines the spectral Hamiltonian $H$ of Definition II.4.1.*

(ii) *A canonical conservative Hamiltonian vector field requires a Poisson tensor $\mathbb{J}$, a domain on which the Jacobi identity holds, and a functional $\mathcal{H}$; the field is $\mathbb{J}D\mathcal{H}$.*

(iii) *A positive mobility and a covariant connection determine the Fenchel Hamiltonian $\mathscr{H}_{\mathrm{cov}}$ of Definition II.3.2.*

*An antisymmetric Caus form alone supplies directed transport but does not provide the Poisson or Jacobi data in item* (ii)*.*

*Proof.* The representation theorem for closed nonnegative forms gives item (i) [7, Chapter VI]. Item (ii) is the definition of a Poisson Hamiltonian system; antisymmetry is only one of the Poisson requirements. The square completion in Proposition II.3.3 proves item (iii). □

**Theorem II.4.3** (Same-current covariance)**.** *Let $v = \Phi_t(u)$ be a generated $C^1$ chart relation. Totalize its diffeomorphism and tangent-map relations. On the branch where $T_t = D\Phi_t(u) : \mathcal{H}_u \to \mathcal{H}_v$ is a bounded linear isomorphism with bounded inverse, with every expression on the right evaluated at $u = \Phi_t^{-1}(v)$, define*

$$\widetilde{\mathbb{K}} = T_t \mathbb{K} T_t^*, \qquad \widetilde{A}_{\mathsf{Caus}} = T_t A_{\mathsf{Caus}}, \qquad A^{\Phi}_{\mathsf{Lift}} = -\partial_t \Phi_t, \qquad \widetilde{f}_{\mathsf{Bdry}} = T_t f_{\mathsf{Bdry}}. \tag{157}$$

*Then*

$$\dot{v} + \widetilde{A}_{\mathsf{Caus}}(v) + A^{\Phi}_{\mathsf{Lift}}(v) - \widetilde{f}_{\mathsf{Bdry}}(v) = T_t(\dot{u} + A_{\mathsf{Caus}}(u) - f_{\mathsf{Bdry}}(u)). \tag{158}$$

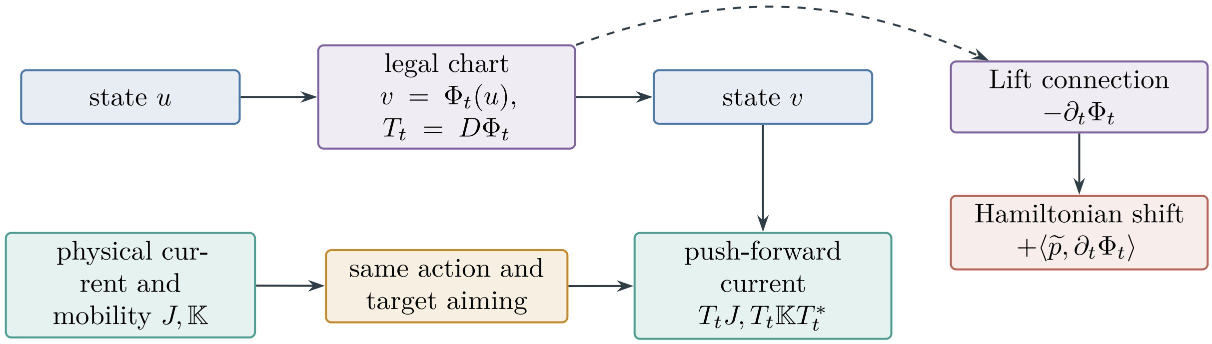


Figure 13: Same-current chart covariance. A chart transports the physical current and metric; its time dependence is a Lift connection with the exact Hamiltonian shift shown.

*The map $T_t$ is an isometry between the corresponding mobility-current spaces equipped with their transported metrics, so the Lagrangian and physical action are invariant. The chart has* Lift *ancestry, and quotienting any declared symmetry additionally has* Abs *ancestry.*

*Proof.* Differentiate $v = \Phi_t(u)$ and substitute (157); the $\partial_t\Phi_t$ terms cancel, giving (158). For $\widetilde{J} = T_tJ$, the variational identity

$$\|J\|^2_{\mathbb{K}^{-1}} = \sup_{p\in\mathcal{H}_u} \{2\langle p, J\rangle - \langle p, \mathbb{K}p\rangle\}$$

and the bijective change of covector $p = T_t^*\widetilde{p}$ give $\|\widetilde{J}\|_{\widetilde{\mathbb{K}}^{-1}} = \|J\|_{\mathbb{K}^{-1}}$. Integrating proves action invariance. The source labels follow from Definition I.2.1. □

**Proposition II.4.4** (Chart covariance of the dissipation Hamiltonian)**.** *On the ordinary covariance-graph branch of Theorem II.4.3, let $p = T_t^*\widetilde{p}$, and define the transformed Hamiltonian using the total connection $\widetilde{A}_{\mathsf{Caus}} + A^{\Phi}_{\mathsf{Lift}}$. Then*

$$\widetilde{\mathscr{H}}_{\mathrm{cov}}(t, v, \widetilde{p}) = \mathscr{H}_{\mathrm{cov}}(u, p) + \langle\widetilde{p}, \partial_t\Phi_t(u)\rangle. \tag{159}$$

*The additional term is the time-dependent Lift connection. It is not a new physical Caus term.*

*Proof.* Insert (157) into (148). The quadratic term is invariant because $\widetilde{\mathbb{K}} = T_t\mathbb{K}T_t^*$, while

$$-\langle\widetilde{p}, T_t(A_{\mathsf{Caus}} - f_{\mathsf{Bdry}}) - \partial_t\Phi_t\rangle = -\langle p, A_{\mathsf{Caus}} - f_{\mathsf{Bdry}}\rangle + \langle\widetilde{p}, \partial_t\Phi_t\rangle.$$

□

*Remark* II.4.5 (A change of physical clock is additional data)*.* Theorem II.4.3 keeps the physical time variable fixed. If a blow-up chart also introduces $\tau = \theta(t)$, the current, mobility, and action density must be transformed with the exact Jacobian $\mathrm{d}t/\mathrm{d}\tau$. Action invariance, or comparison with the original physical measure, is not automatic; the required one-sided comparison is the content of Proposition II.10.5.

## II.5 Killed resolvents, resistance, and target realization

Every resolvent, capacity, and resistance relation below is a library relation in the sense of Corollary II.2.19. Its regularity use is confined to the active target/source cell selected by contact. A failed domain, closedness, killing, flux, or range relation is therefore a typed local successor; it is not an equation-wide obstruction and supplies no physical charge by itself.

**Definition II.5.1** (Killed geometric Hamiltonian)**.** Let $\mathfrak{q}$ be a densely defined closed nonnegative symmetric Markovian form and let $\Sigma$ be quasi-closed for a regular or quasi-regular realization. The killed form is the restriction

$$D(\mathfrak{q}_\Sigma) = \{u \in D(\mathfrak{q}) : \widetilde{u} = 0 \text{ q.e. on } \Sigma\}, \qquad \mathfrak{q}_\Sigma = \mathfrak{q}|_{D(\mathfrak{q}_\Sigma)},$$

and its self-adjoint operator, when the restricted form is closed, is $H_\Sigma \geq 0$. Adding an admitted nonnegative killing potential instead defines a *soft*-killed form. It is a separate construction unless convergence to the hard restriction is proved.

**Proposition II.5.2** (Existence of the killed Hamiltonian)**.** *If $D(\mathfrak{q}_\Sigma)$ is closed in the form norm, then $\mathfrak{q}_\Sigma$ is closed and determines a unique self-adjoint $H_\Sigma \geq 0$.*

*Proof.* A form-norm closed subspace of the domain of a closed form is complete in that norm. The representation theorem for closed nonnegative forms gives $H_\Sigma$. □

**Proposition II.5.3** (Committor is the canonical binary-contact quotient)**.** *Let $A$ and $B$ be disjoint quasi-closed sets for an associated right process, and let*

$$h(x) := \mathbb{P}_x\{\tau_A < \tau_B\}$$

*be its quasi-continuous committor. Define $q_h(x) = h(x)$ outside the exceptional set. If a quotient $q : X \to Q$ preserves this binary contact law in the precise sense that $h = \bar{h} \circ q$ quasi-everywhere for some $\bar{h} : Q \to [0, 1]$, then*

$$q_h = \bar{h} \circ q \qquad \textit{quasi-everywhere.} \tag{160}$$

*Hence $q_h$ is the coarsest quotient preserving the scalar event $\{\tau_A < \tau_B\}$: every other such quotient is finer than its level-set partition. This assertion concerns the binary hitting probability, not the full hitting-time distribution.*

*Proof.* The preservation branch is exactly (160). Therefore $q(x) = q(y)$ implies $h(x) = h(y)$, so every fiber of $q$ lies in a fiber of $q_h$. Conversely, $q_h$ itself preserves the event by the defining identity for $h$. □

**Theorem II.5.4** (Killed-resolvent stopwatch)**.** *Let $L_\Sigma$ generate a killed semigroup $P_t^\Sigma$ with growth bound $\omega$. Let $\ell_0$ be an initial functional and let $k_\Sigma$ be a terminal-flux vector such that*

$$\mathbb{P}_{\ell_0}\{\tau_\Sigma \in \mathrm{d}t, \tau_\Sigma < \infty\} = \langle \ell_0, P_t^\Sigma k_\Sigma \rangle \,\mathrm{d}t \tag{161}$$

*as finite measures. Then, for $\lambda > \omega$,*

$$\mathbb{E}_{\ell_0}[e^{-\lambda \tau_\Sigma}; \tau_\Sigma < \infty] = \langle \ell_0, (\lambda - L_\Sigma)^{-1} k_\Sigma \rangle. \tag{162}$$

*Without (161), the right side remains a cross-resolvent pairing and has no automatic first-contact interpretation.*

*Proof.* Integrate (161) against $e^{-\lambda t}$. The Bochner Laplace transform of the semigroup is $\int_0^\infty e^{-\lambda t} P_t^\Sigma \,\mathrm{d}t = (\lambda - L_\Sigma)^{-1}$, which proves the identity. $\square$

**Theorem II.5.5** (Resolvent Fenchel dual and resistance)**.** *Let $K \geq 0$ be self-adjoint with closed form $\mathfrak{q}$, and let $e \in \mathcal{H}$. For $\lambda > 0$,*

$$\langle e, (K+\lambda)^{-1} e\rangle = \sup_{v \in D(\mathfrak{q})} \left\{ 2\operatorname{Re}\langle e, v\rangle - \mathfrak{q}(v,v) - \lambda \|v\|^2 \right\}. \tag{163}$$

*If $P_{\ker K} e = 0$, then monotone convergence gives*

$$\lim_{\lambda \downarrow 0} \langle e, (K+\lambda)^{-1} e\rangle = \int_{(0,\infty)} s^{-1} \,\mathrm{d}\nu_e(s) =: \mathcal{R}_K(e) \in [0, \infty]. \tag{164}$$

*Finiteness of $\mathcal{R}_K(e)$ is equivalent to boundedness of $v \mapsto \langle e, v\rangle$ on the energy space with norm $\mathfrak{q}(v,v)^{1/2}$; in that case it has a unique weak energy representer modulo* $\ker K$*.*

*Proof.* For $R_\lambda = (K+\lambda)^{-1}$, completing the square in the form norm of $K + \lambda$ shows that the supremum is attained at $v = R_\lambda e$ and equals $\langle e, R_\lambda e\rangle$. The spectral theorem gives $\langle e, R_\lambda e\rangle = \int (\lambda + s)^{-1} \,\mathrm{d}\nu_e(s)$. Monotone convergence proves (164). The last equivalence is the Riesz theorem on the completion of $D(\mathfrak{q})/\ker K$. $\square$

**Theorem II.5.6** (Harmonic defect equals resistance error)**.** *Let $K \geq 0$ be self-adjoint with form $\mathfrak{q}$, and let*

$$\mathcal{E}_K := \overline{D(\mathfrak{q})/\ker K}^{\mathfrak{q}}$$

*be its energy space. For $\ell \in \mathcal{E}_K^*$, totalize the weak solution relation $\mathfrak{q}(x, \phi) = \ell(\phi)$ modulo* $\ker K$*. On its ordinary branch let $x$ denote the solution and let $v \in D(\mathfrak{q})$ be a trial state with residual*

$$e_v(\phi) := \mathfrak{q}(v, \phi) - \ell(\phi). \tag{165}$$

*Then*

$$\|e_v\|^2_{\mathcal{E}_K^*} = \|[v - x]\|^2_{\mathcal{E}_K} = \mathfrak{q}(v - x, v - x). \tag{166}$$

*If $e_v$ is represented by a vector $e \perp \ker K$ in $\mathcal{H}$, then*

$$\|e_v\|^2_{\mathcal{E}_K^*} = \langle e, K^+ e\rangle = \mathcal{R}_K(e). \tag{167}$$

*Thus an approximate committor, elliptic corrector, or resolvent state is controlled by one resistance invariant of its harmonic defect; no* $\ker K$ *component of the trial state enters the error. When that kernel is a closed gauge or target-null space, the corresponding discarded motion requires no estimate.*

*Proof.* The weak equation and (165) give

$$e_v(\phi) = \mathfrak{q}(v - x, \phi).$$

The Riesz map of $\mathcal{E}_K$ is an isometry onto its dual, proving (166). When the residual is represented by $e$, the spectral theorem identifies its squared energy-dual norm with $\int_{(0,\infty)} s^{-1} \,\mathrm{d}\nu_e(s)$, which is (167); the middle pairing is understood as this spectral quadratic-form value. $\square$

**Corollary II.5.7** (Defect-aligned drift bound)**.** *If $e \perp \ker K$ and $\mathcal{R}_K(e) < \infty$, then for every $w \in D(\mathfrak{q})$,*

$$|\langle e, w\rangle| \le \mathcal{R}_K(e)^{1/2}\mathfrak{q}(w,w)^{1/2}. \tag{168}$$

*Only the component of $w$ paired with the target-selected routable defect enters a reachability charge, and it does so through the local carrier rule.*

*Proof.* Use the energy representer from Theorem II.5.5 and apply Cauchy–Schwarz in $\mathcal{E}_K$. □

**Proposition II.5.8** (Routable and zero-mode resistance split)**.** *For arbitrary $e \in \mathcal{H}$, write*

$$e = e_0 + e_\perp, \qquad e_0 = P_{\ker K}e, \qquad e_\perp = P_{(\ker K)^\perp}e.$$

*Then*

$$\langle e, (K+\lambda)^{-1}e\rangle = \frac{\|e_0\|^2}{\lambda} + \int_{(0,\infty)} \frac{1}{s+\lambda}\,\mathrm{d}\nu_{e_\perp}(s). \tag{169}$$

*Consequently the zero-discount resistance is finite precisely when $e_0 = 0$ and $\int_{(0,\infty)} s^{-1}\,\mathrm{d}\nu_{e_\perp}(s) < \infty$. A nonzero target-visible zero mode is an inherited Geom/Caus zero-frequency occurrence, not a residual coordinate. It is removed only when it belongs to a declared gauge or to $\mathscr{N}_\Sigma$; otherwise a separate carrier or price row evaluates it.*

*Proof.* The spectral projection of $K$ at 0 contributes $\lambda^{-1}\|e_0\|^2$; the remaining spectral measure is supported on $(0,\infty)$, which gives (169). Monotone convergence yields the finiteness criterion. The source statement follows from Theorems I.3.4 and I.4.6. □

**Proposition II.5.9** (Capacity and committor realization)**.** *Let $A$ and $B$ be disjoint quasi-closed sets for a regular or quasi-regular Dirichlet form. The condenser capacity*

$$\mathrm{Cap}(A,B) = \inf\{\mathfrak{q}(v,v) : v \in D(\mathfrak{q}),\ \widetilde{v} = 1 \text{ on } A,\ \widetilde{v} = 0 \text{ on } B\}$$

*is an analytic Geom invariant. On the zero realization-defect branch, if the minimizer is unique and the associated right process realizes the weak Dirichlet problem, then its quasi-continuous version is the committor $x \mapsto \mathbb{P}_x\{\tau_A < \tau_B\}$ quasi-everywhere. The probabilistic conclusion is the ordinary value of the right-process realization row; positive capacity alone is only its prospective value [5, 9].*

**Theorem II.5.10** (Polar-target capacity criterion)**.** *Let $\mathfrak{q}$ be a symmetric quasi-regular Dirichlet form with associated right process, and let $\Sigma$ be quasi-closed. Write*

$$\mathrm{Cap}_{\mathfrak{q}}(\Sigma) := \inf\{\mathfrak{q}(v,v) + \|v\|_2^2 : v \in D(\mathfrak{q}),\ \widetilde{v} \ge 1 \text{ q.e. on a neighborhood of } \Sigma\}.$$

*If $\mathrm{Cap}_{\mathfrak{q}}(\Sigma) = 0$, then $\Sigma$ is polar:*

$$\mathbb{P}_x\{\tau_\Sigma < \infty\} = 0 \qquad \text{for quasi-every } x. \tag{170}$$

*This is a direct Geom nonreachability criterion for the Markov realization and requires no cascade argument. Positive capacity selects the complementary row but supplies no hitting lower bound without a starting-law or nonpolarity statement.*

*Proof.* For a quasi-regular Dirichlet form, sets of zero 1-capacity are properly exceptional up to a capacity-zero enlargement. The associated right process avoids every properly exceptional set for quasi-every starting point, which gives (170) [5, 9]. □

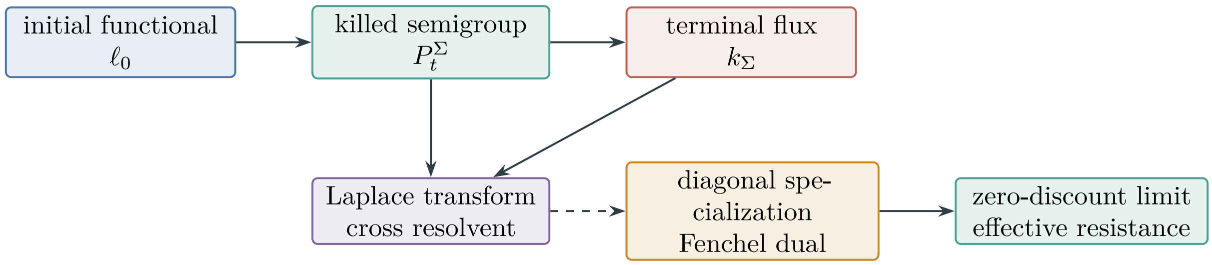


Figure 14: Killed arrival and resistance are related but distinct. Arrival is a cross pairing that requires a flux realization. The Fenchel and resistance identities concern a diagonal self-adjoint specialization.

**Proposition II.5.11** (Quotient pullback of zero capacity)**.** *Let $q : X \to Q$ identify a target $\Sigma = q^{-1}(\Sigma_Q)$ quasi-everywhere, and let $\mathfrak{q}_X, \mathfrak{q}_Q$ be symmetric quasi-regular Dirichlet forms. Totalize the lift relation which, for every admissible quotient potential $f$ with $\widetilde{f} \geq 1$ near $\Sigma_Q$, there is a lifted potential $\psi_f$ with $\widetilde{\psi}_f \geq 1$ near $\Sigma$ and*

$$(\mathfrak{q}_X)_1(\psi_f, \psi_f) \leq C(\mathfrak{q}_Q)_1(f, f) + \varepsilon(f). \tag{171}$$

*If a minimizing family for $\operatorname{Cap}_{\mathfrak{q}_Q}(\Sigma_Q)$ can be chosen with $\varepsilon(f) \to 0$, then*

$$\operatorname{Cap}_{\mathfrak{q}_X}(\Sigma) \leq C \operatorname{Cap}_{\mathfrak{q}_Q}(\Sigma_Q). \tag{172}$$

*In particular, zero quotient capacity pulls back to zero capacity on $X$.*

*Proof.* Every $\psi_f$ is an admissible competitor for the capacity of $\Sigma$. Apply (171) along the stated minimizing family and take the liminf. □

## II.6 Full-generator response and physical aiming

Full-generator formulas enter the reachability calculus only through the same target-local activation rule. The input and output projections below are those generated by the active branch signature, and every differentiated or factorized term retains its five-source ancestry. Amplification outside that target-cyclic cell is transverse to the branch under evaluation.

**Definition II.6.1** (Target-active full resolvent)**.** Let $L = -H + A_{\mathsf{Caus}} + B_{\mathsf{Bdry}}$ be a closed generator with a declared resolvent point $\lambda$. For closed mathematical input and target-output subspaces $\mathscr{I}$ and $\mathscr{O}_\Sigma$, with bounded projections $P_{\mathscr{I}}$ and $P_{\mathscr{O}}$, define

$$\Psi_\Sigma(\lambda; L) := \|P_{\mathscr{O}}(\lambda - L)^{-1} P_{\mathscr{I}}\|. \tag{173}$$

This quantity measures nonnormal amplification visible to the fixed target interface. It evaluates the existing Geom–Caus–Bdry generator, not an additional source.

**Proposition II.6.2** (Nonnormal source invariance)**.** *Let $L_0$ be closed on a Banach space and let every $K^c$ be bounded. Put $L_\theta = L_0 + \sum_c \theta_c K^c$ on $D(L_0)$. On the branch $\lambda \in \rho(L_\theta)$ locally in $\theta$, this is the open ordinary value of the totalized resolvent relation; on it the resolvent is norm differentiable and*

$$\partial_{\theta_c}(\lambda - L_\theta)^{-1} = (\lambda - L_\theta)^{-1} K^c (\lambda - L_\theta)^{-1}. \tag{174}$$

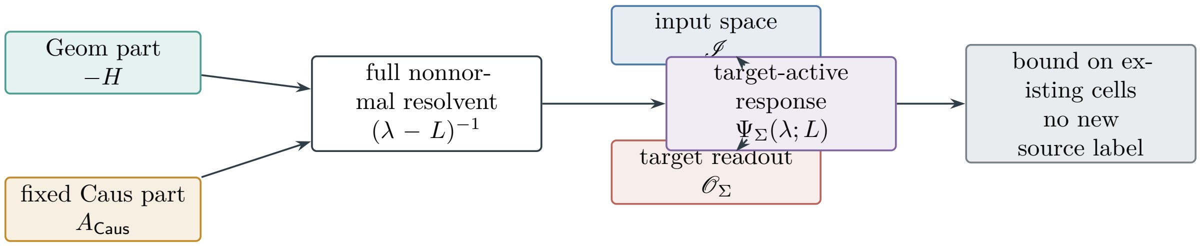


Figure 15: Full nonnormal response retains both Geom and fixed Caus. Input and target compression select a response coefficient; they do not change the operator's source ancestry.

*Thus every first variation of target-active response is factored through the same source term $K^c$. Large pseudospectral response can expose a large Geom–Caus interaction, but it creates no new nonnormal source.*

*Proof.* Differentiate $(\lambda - L_\theta)(\lambda - L_\theta)^{-1} = I$ on the common domain and solve for the derivative. Bounded input/output compression preserves the factorization. □

**Definition II.6.3** (Target-gradient ray)**.** Let $F$ be a differentiable target shell. Totalize membership of its mobility gradient

$$G_F := \mathbb{K}(u)DF(u)$$

in $\mathfrak{J}_u$. On the membership branch, for a physical current $J \in \mathfrak{J}_u$, define its positive target power and aiming contribution by

$$P_F(J) := (J, G_F)_{u,*}, \qquad \mathrm{Align}_F(J) := \frac{(P_F(J)_+)^2}{\|G_F\|_{u,*}^2}, \tag{175}$$

with value zero when $G_F = 0$. By the Cameron–Martin pairing,

$$P_F(J) = \langle DF(u), J\rangle$$

for every $J \in \mathfrak{J}_u$; components of $DF$ in $\ker \mathbb{K}(u)$ annihilate that current space.

**Theorem II.6.4** (Projection onto the target ray)**.** *For every $J, G \in \mathfrak{J}_u$,*

$$\|J\|_{u,*}^2 = \frac{\left((J,G)_{u,*}\right)_+^2}{\|G\|_{u,*}^2} + \mathrm{dist}_{u,*}^2(J, \mathbb{R}_+ G), \tag{176}$$

*with the first term zero when $G = 0$. Hence*

$$(P_F(J)_+)^2 \le \|J\|_{u,*}^2 \|G_F\|_{u,*}^2. \tag{177}$$

*Proof.* The orthogonal projection of $J$ onto the closed ray $\mathbb{R}_+G$ has coefficient $\left((J,G)_{u,*}\right)_+/\|G\|_{u,*}^2$. Pythagoras gives (176); dropping the nonnegative distance gives (177). □

**Proposition II.6.5** (Integrated action–aiming bound)**.** *Let $I$ be a pre-contact interval. On the finite ordinary branch of the two totalized squared-norm integrals below,*

$$\int_I P_F(J(t))_+ \,\mathrm{d}t \le \left(\int_I \|J(t)\|_{u(t),*}^2 \,\mathrm{d}t\right)^{1/2} \left(\int_I \|G_F(u(t))\|_{u(t),*}^2 \,\mathrm{d}t\right)^{1/2}. \tag{178}$$

*For the selected Geom current of Definition I.1.14, the square of the first factor is the physical action and is bounded by the right-hand side of* (12)*.*

*Proof.* Apply (177) and then Cauchy–Schwarz in time. The final assertion is Proposition I.1.15. □

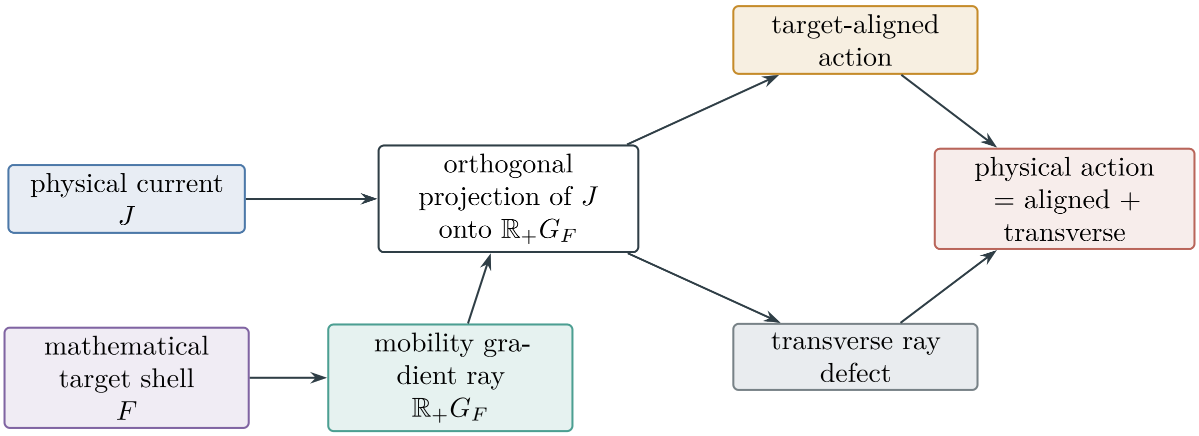


Figure 16: Physical aiming geometry. The target shell is a mathematical test of the already selected current. Projection splits the physical action into target-aligned and transverse expenditure.

## II.7 Degenerate geometry and commutator-generated carriers

Direct resistance controls only the positive spectral subspace of the Geom operator. Fixed Caus transport may nevertheless move a Geom-null direction into a dissipative direction. The correct reduced invariant is then a positive commutator Gramian together with a carrier identity realizing that Gramian along the actual evolution. These relations are evaluated only on the contact-generated $\mathsf{Geom}$–$\mathsf{Caus}$ cell. A commutator square becomes physical expenditure only after its target-local factorization proves domination by the public carrier; every complementary graph value remains a same-origin successor.

**Definition II.7.1** (Degenerate Geom–Caus core)**.** Let $\mathcal{H}$ be a complex Hilbert space and

$$L = -S + A, \qquad S = S^* \geq 0, \qquad A^* = -A,$$

on a common invariant core $\mathcal{D}$. Let $\mathcal{G} \subseteq \ker S$ be a closed $A$-invariant gauge subspace, and write $P_{\mathcal{G}}^{\perp}$ for its orthogonal complement. Put $C_0 = S^{1/2}$. Totalize the iterated commutator relations on $\mathcal{D}$; their ordinary graph branch sets

$$C_{k+1} = [A, C_k] \quad \text{on } \mathcal{D}.$$

For fixed $K$ and $c_k > 0$, totalize the quadratic form

$$q_K(v) := \sum_{k=0}^{K} c_k \|C_k v\|^2. \tag{179}$$

On its closable graph branch its positive self-adjoint operator is denoted by $Q_K$, and its reduced bracket gap is

$$\gamma_K := \inf_{\substack{v \in D(q_K) \cap \mathcal{G}^{\perp} \\ \|v\|=1}} q_K(v). \tag{180}$$

Failure of a domain, closure, or gauge relation is its totalized graph defect; algebraic commutator span alone does not select the closing branch.

**Proposition II.7.2** (Bracket resistance invariant). *If $\gamma_K > 0$, then $Q_K \geq \gamma_K I$ on $\mathcal{G}^\perp$, and for every $e \in \mathcal{G}^\perp$,*

$$\mathcal{R}_K^{\mathrm{br}}(e) := \langle e, Q_K^{-1} e\rangle \leq \gamma_K^{-1}\|e\|^2. \tag{181}$$

*The quotient kernel*

$$\mathfrak{h}_K := \mathcal{G}^\perp \cap \ker Q_K$$

*is the structural rank deficit when $\gamma_K = 0$. Every target-visible element of $\mathfrak{h}_K$ retains its Geom–Caus ancestry and remains an existing source occurrence for the routing, carrier, and price rows.*

*Proof.* The first assertion is the definition of the form lower bound in (180); functional calculus gives (181). The kernel is the intersection of the kernels of the closed commutator forms in (179). Source preservation and the target-null residual theorem give the final assertion. □

**Definition II.7.3** (Twisted-carrier realization evaluator). A bracket Gramian $Q_K$ has a twisted-carrier realization when there is a bounded self-adjoint operator $M$ on $\mathcal{G}^\perp$, constants $0 < m_- \leq m_+ < \infty$ and $c_M > 0$, and a common evolution core on which

$$m_-\|v\|^2 \leq \langle Mv, v\rangle \leq m_+\|v\|^2, \tag{182}$$

$$2\,\mathrm{Re}\langle Mv, Lv\rangle \leq -c_M q_K(v). \tag{183}$$

For a forced equation, the positive part of $2\,\mathrm{Re}\langle Mu, f_{\mathsf{Bdry}}\rangle\,\mathrm{d}t$ is the candidate $\mathsf{Bdry}$-input current. It enters $R_u^+$ only when the public carrier identity identifies it as true exterior input; otherwise its target-local projection is a boundary successor.

**Theorem II.7.4** (Hypocoercive carrier produces physical expenditure). *On the zero graph-defect branch of Definition II.7.3, the evolution and carrier chain-rule relations for an ordinary maximal branch have ordinary values $\dot{u} = Lu + f_{\mathsf{Bdry}}$ and $\mathrm{d}\langle Mu, u\rangle/\,\mathrm{d}t$. On this branch,*

$$\mathcal{K}_u(t) := \langle Mu(t), u(t)\rangle, \qquad \mathfrak{D}_u(\,\mathrm{d}t) := c_M q_K(u(t))\,\mathrm{d}t, \qquad \Phi_u^+ = 0$$

*satisfy the carrier balance of Definition II.3.4. Hence*

$$\mu_u([0,T_*)) \leq \langle Mu_0, u_0\rangle + R_u^+([0,T_*)) \leq m_+\|u_0\|^2 + R_u^+([0,T_*)) \tag{184}$$

*after subtracting any known lower carrier level. If $\gamma_K > 0$, this expenditure controls every nongauge direction in $\mathcal{G}^\perp$. The commutator calculation is therefore a structural compiler for a physical budget in a partially dissipative system [17]. Its regularity use is restricted to the active target/source cell selected by the shell current; nongauge directions outside that cell are not charged to its target progress.*

*Proof.* Differentiate $\langle Mu, u\rangle$ and use (183); the positive forcing pairing gives the declared input measure. Apply Theorem II.3.5. When $\gamma_K > 0$, $q_K(u) \geq \gamma_K\|P_{\mathcal{G}}^\perp u\|^2$. □

**Definition II.7.5** (Positive-commutator carrier). Let the evolution be $\dot{u} = -iHu$, with $H = H^*$, and let $P = \mathbf{1}_\Delta(H)$ be a reducing spectral projection. A self-adjoint conjugate operator $C$ supplies a positive-commutator carrier when the form

$$B := i[H, C]$$

is well defined and nonnegative on a common invariant core in $P\mathcal{H}$, and the expectation $\langle u(t), Cu(t)\rangle$ has a finite upper bound $C_+$ along the orbit. The strict Mourre branch has $PBP \geq \theta P$ modulo the declared gauge for some $\theta > 0$.

**Theorem II.7.6** (Conservative carrier produces physical expenditure)**.** *Under Definition II.7.5, let $u_0 \in P\mathcal{H}$. On the zero chain-defect branch, the totalized commutator relation gives*

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle u(t), Cu(t)\rangle = \langle u(t), Bu(t)\rangle. \tag{185}$$

*Consequently*

$$\mu_u(Q) := \int_Q \langle u(t), Bu(t)\rangle \,\mathrm{d}t, \qquad \mu_u([0,T_*)) \le C_+ - \langle u_0, Cu_0\rangle. \tag{186}$$

*When $PBP \ge \theta P$, this budget controls $\theta \int_0^{T_*} \|Pu(t)\|^2 \,\mathrm{d}t$. Thus a virial, Morawetz, or Mourre identity contributes to the same physical-budget calculus whenever its carrier variation is finite [6]. In the reachability calculus, the contribution is restricted to the retained target-local spectral/source cell and is charged only through Theorem II.2.4.*

*Proof.* Differentiate the expectation on the common core to obtain (185). Nonnegativity of $B$ makes the right side a measure density. Integrate and use the upper carrier bound. The strict estimate follows from $PBP \ge \theta P$. On the complementary branch the distributional difference between the two sides of (185) is the retained $\mathfrak{H}^{\Sigma}_{\mathrm{chain}}$-class, so the theorem does not presume the chain rule in the fast track. □

**Corollary II.7.7** (Complete routing alternatives for a zero mode)**.** *Let $e_0$ be the zero mode in Proposition II.5.8. Direct resistance controls $e_\perp$. A target-visible $e_0$ may be controlled by a twisted-carrier evaluator or a positive-commutator carrier, with its price eligible for the same $\mu_u$ only through its target-local carrier factorization. The complementary values of both evaluators select the harmonic nonroutable branch of Theorem II.1.16; it remains a Geom–Caus source class and does not move to $J_{\mathrm{res}}$.*

## II.8 Compactness and rigidity as a price compiler

Compactness is not an additional continuum source and is not part of the definition of the residual. Its role is quantitative: it converts a qualitative zero-price rigidity statement into a uniform positive price on a closed class of target-approaching windows. The topology used in this conversion must be generated from seminorms or graph coordinates derived from the fixed PDE. The following lemma is one method for evaluating $\delta_*$; its tuple is not input to the fast track. It is invoked only on the target/source stratum that produced the window family. Each failed clause is totalized into the boundary, chain, or price row of that same stratum.

**Definition II.8.1** (Canonical compactness–rigidity row)**.** Fix a reachable active branch $\mathfrak{b}$ at physical ceiling $B_{\mathrm{phys}}$, and let $\mathcal{W}_{\mathfrak{b}}$ be the normalized pre-contact windows generated by the equation's $\mathsf{Abs}/\mathsf{Lift}$ grammar that have the origin, source support, realization mode, shell floor, and earlier zero coordinates recorded in $\mathfrak{b}$. Inside the closure of its target-local stratum $X_{\mathfrak{b}}$, define

$$K_{\mathfrak{b}} := \overline{\mathcal{W}_{\mathfrak{b}}}, \qquad B_{\Sigma,\mathfrak{b}} := \overline{\mathcal{W}_{\mathfrak{b},\Sigma}}, \tag{187}$$

where $\mathcal{W}_{\mathfrak{b},\Sigma}$ consists of those same-origin windows carrying the selected normalized target-shell crossing. Thus $K_{\mathfrak{b}}$ is compact and $B_{\Sigma,\mathfrak{b}}$ is closed by construction; neither is an analytic premise and neither contains a source-transverse window.

Let $d_0$ be the nonnegative window expenditure read from the fixed physical balance. Its canonical lower-semicontinuous relaxation is

$$\overline{d}(w) := \sup_{U \ni w} \inf\{d_0(v) : v \in U \cap \mathcal{W}_{\mathfrak{b}}\} \in [0, \infty], \tag{188}$$

where $U$ ranges over neighborhoods of $w$, and the infimum of an empty set is $+\infty$. Equivalently, this is the largest lower-semicontinuous functional whose restriction to $\mathcal{W}_{\mathfrak{b}}$ is bounded above by $d_0$. The compactness–rigidity value is the exact extended number

$$\delta_{\mathrm{rig}}(\mathfrak{b}) := \inf_{w \in B_{\Sigma,\mathfrak{b}}} \overline{d}(w), \tag{189}$$

with value $+\infty$ when $B_{\Sigma,\mathfrak{b}} = \varnothing$. The branches $\delta_{\mathrm{rig}} > 0$ and $\delta_{\mathrm{rig}} = 0$ are exhaustive; the latter is the closed zero-price target-approach class. Every topology, normalization, target class, and price in this row is therefore generated from the fixed PDE, target, and physical balance.

**Definition II.8.2** (Canonical physical-transfer radius)**.** For each generated event $Q$, write $w_Q = \mathcal{N}_Q u$ and let $\mu_u$ be the orbit's physical expenditure measure. Define

$$c^*_{\mathrm{pay}}(\mathfrak{b}) := \inf_{\substack{Q:\ w_Q \in B_{\Sigma,\mathfrak{b}} \\ 0 < \overline{d}(w_Q) < \infty}} \frac{\mu_u(Q)}{\overline{d}(w_Q)} \in [0, \infty], \tag{190}$$

using $+\infty$ for an empty indexing family. A window with $\overline{d}(w_Q) = \infty$ and finite $\mu_u(Q)$, or a failure of the normalization/pullback relation, is retained by the price graph-defect row. For every ordinary finite-price window,

$$\mu_u(Q) \geq c^*_{\mathrm{pay}}(\mathfrak{b})\, \overline{d}(w_Q). \tag{191}$$

Consequently the physical-transfer question has the exact complementary values $c^*_{\mathrm{pay}} > 0$ and $c^*_{\mathrm{pay}} = 0$, together with its typed graph-defect value. No comparison constant is supplied as data.

**Lemma II.8.3** (Compactness–rigidity row evaluation)**.** *The value in* (189) *is positive exactly when the canonical closed target-approach class contains no sequence with relaxed physical price tending to zero. If $B_{\Sigma,\mathfrak{b}} \neq \varnothing$, the infimum is attained.*

*Proof.* The set $B_{\Sigma,\mathfrak{b}}$ is compact and $\overline{d}$ is lower semicontinuous by relaxation. Hence the infimum is attained. Its value is zero exactly when a minimizing sequence has price tending to zero; compactness then supplies a convergent subsequence in $B_{\Sigma,\mathfrak{b}}$, and lower semicontinuity identifies the zero-price branch. The converse follows from the constant sequence at a zero-price minimizer. □

**Theorem II.8.4** (Uniform physical price branch)**.** *On the invariant branch $c^*_{\mathrm{pay}}(\mathfrak{b}) > 0$ and $\delta_{\mathrm{rig}}(\mathfrak{b}) > 0$, every ordinary generated event $Q$ whose normalized window belongs to $B_{\Sigma,\mathfrak{b}}$ satisfies*

$$\mu_u(Q) \geq \delta_{\mathrm{phys}}(\mathfrak{b}), \qquad \delta_{\mathrm{phys}}(\mathfrak{b}) := c^*_{\mathrm{pay}}(\mathfrak{b})\delta_{\mathrm{rig}}(\mathfrak{b}) > 0. \tag{192}$$

*On either zero branch, the corresponding zero-transfer or zero-price sequence is retained in the same source/mode cell.*

*Proof.* Apply (191) and then (189). The complementary statements follow from the two infimum definitions and sourcewise closure. □

**Theorem II.8.5** (Borderline passivity price compiler). *On the invariant branch $\beta_*^{\mathrm{red}} = 1$ in Definition II.3.6, define the normalized carrier deficit*

$$\Delta_{\mathrm{car}}(V) := D(V) - \Phi_+(V) \geq 0. \tag{193}$$

*Let $\overline{\Delta}_{\mathrm{car}}$ be its canonical relaxation* (188) *on the generated compact sets* (187)*, and define*

$$\delta_{\mathrm{car}}^*(\mathfrak{b}) := \inf_{V \in B_{\Sigma,\mathfrak{b}}} \overline{\Delta}_{\mathrm{car}}(V). \tag{194}$$

*Define the clock-transfer radius by the same infimum construction as* (190)*, with $\overline{\mathrm{d}} = \overline{\Delta}_{\mathrm{car}}$; call it $c_{\mathrm{clock}}^*(\mathfrak{b})$. On the positive values of both rows, physical pullback gives*

$$\mu_u(Q) \geq c_{\mathrm{clock}}^* \overline{\Delta}_{\mathrm{car}}(\mathcal{N}_Q u), \tag{195}$$

*and every selected target-bad event has price*

$$\mu_u(Q) \geq c_{\mathrm{clock}}^* \delta_{\mathrm{car}}^*(\mathfrak{b}) > 0. \tag{196}$$

*Thus the critical value $\beta_*^{\mathrm{red}} = 1$ is closed by an equality–rigidity invariant on the reduced saturation class. If either infimum is zero, its minimizing sequence is the complementary price class.*

*Proof.* The canonical relaxation is lower semicontinuous on the compact generated class, so Lemma II.8.3 evaluates $\delta_{\mathrm{car}}^*$. The transfer-radius definition gives (195); multiplication yields (196). □

**Corollary II.8.6** (Separator closure of the borderline branch). *On the zero separator-defect branch, the target projection of every zero-deficit profile vanishes by Proposition II.1.21. Hence the canonical target-bad closure contains no zero of $\overline{\Delta}_{\mathrm{car}}$, so compactness gives $\delta_{\mathrm{car}}^* > 0$. The positive clock-transfer branch of Theorem II.8.5 then gives the uniform event price; its zero branch retains the exact zero-transfer sequence.*

**Proposition II.8.7** (A standard local compactness compiler). *Let $X_0, X, X_1$ be Banach spaces with $X_0 \Subset X \hookrightarrow X_1$, the first embedding compact and the second continuous. If $1 \leq p < \infty$, $1 < q \leq \infty$, and a family of windows is bounded in*

$$L^p(0,1;X_0) \cap W^{1,q}(0,1;X_1),$$

*then it is relatively compact in $L^p(0,1;X)$. Consequently, its closure agrees with the corresponding compact branch of $K_{\mathfrak{b}}$. The target and price rows are still evaluated by the canonical target closure and lower-semicontinuous relaxation; the lemma supplies no separate closedness or price premise.*

*Proof.* The relative compactness is the Aubin–Lions lemma in the stated range [1, 8, 16]. The final assertion follows by mapping that relatively compact family into the generated evaluation closure. □

*Remark* II.8.8 (Quotient compactness row). If a compact group acts continuously on a compact normalized set, its quotient is compact. For a noncompact symmetry group, fixing a gauge only defines a slice. The generated closure of that slice and the totalized gauge graph select the compact and graph-defect branches, respectively. In either case the quotient operation has Abs-ancestry and does not alter the source of the limiting physical record.

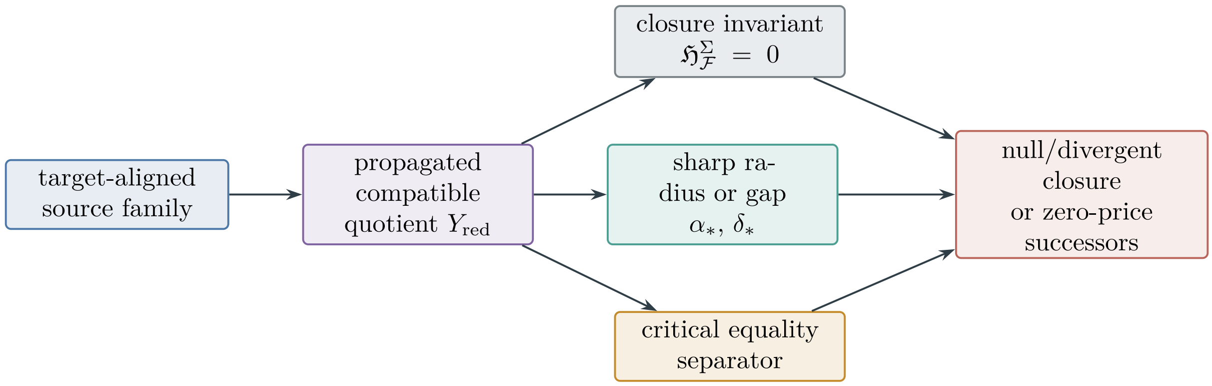


Figure 17: Identity-to-successor compiler. Once closed null directions are propagated out, generated identities expose a target closure, domination radius, physical-price gap, or critical equality relation. Every nonclosing value is normalized and passed to the successor operator. Each evaluator returns both its closing and complementary structural relation to the ordered algebra.

*Remark* II.8.9 (What failure of compactness means)*.* A nonordinary value of a normalization, target-persistence, lower-semicontinuity, or physical-pullback graph is pushed into the compact activity lift with the relevant $\mathsf{Geom}/\mathsf{Caus}/\mathsf{Abs}/\mathsf{Lift}/\mathsf{Bdry}$ ancestry. A zero value of either sharp infimum supplies its minimizing sequence to the same lift. Finite-depth viability then removes the descendants or constructs their recurrent zero-price path. No target-visible element is transferred to $J_{\mathrm{res}}$, and this optional price accelerator creates no new obligation in the core proof.

## II.9 Constraint tails as finite structural obstructions

Nonlocal constraints often enter a local evolution through an elliptic multiplier, potential, or pressure. A direct estimate of the whole nonlocal field obscures the smaller structural question: which part is fixed by the local source, which part is a gauge, which part decays from the exterior, and which finite-dimensional harmonic mode remains visible to the target? The following interface makes that separation explicit. It is formed only after restriction to the active branch. Its local, harmonic, gauge, and exterior coordinates are the $\mathsf{Geom}/\mathsf{Caus}/\mathsf{Abs}/\mathsf{Lift}/\mathsf{Bdry}$ factorization of that branch's constraint current; no norm of the full constraint field has an independent reachability consequence.

**Definition II.9.1** (Elliptic constraint–tail interface)**.** Let $U_0 \Subset U$ be open sets, let

$$\mathcal{C}p = S(V) \quad \text{in } U$$

be a linear elliptic constraint in a declared weak sense, and let $\mathcal{G}_{U_0}$ be its gauge space on $U_0$. The compiler forms the product of the following five total graphs. Its ordinary value consists of:
(T1) a local solution operator $\mathcal{L}_U$ satisfying $\mathcal{C}\mathcal{L}_U S(V) = S(V)$ on $U$;
(T2) a decomposition on $U_0$,

$$p|_{U_0} = p_{\mathrm{loc}} + h, \qquad p_{\mathrm{loc}} := \mathcal{L}_U S(V)|_{U_0}, \qquad \mathcal{C}h = 0 \text{ in } U_0; \tag{197}$$

(T3) a Banach space $\mathcal{H}(U_0)$ of homogeneous solutions with a continuous splitting

$$\mathcal{H}(U_0) = \mathcal{H}_{\mathrm{low}} \oplus \mathcal{H}_{\mathrm{dec}}, \tag{198}$$

where $\mathcal{H}_{\mathrm{low}}$ is finite dimensional;

(T4) a dyadic exterior expansion $P_{\mathrm{dec}}h = \sum_{k\geq 0} h_k$ convergent in $\mathcal{H}_{\mathrm{dec}}$, with

$$\mathfrak{s}_{\mathrm{tail}}(h) := \sum_{k\geq 0} \|h_k\|_{\mathcal{H}_{\mathrm{dec}}} < \infty; \tag{199}$$

(T5) a linear gauge realization $\Gamma : \mathcal{G}_{U_0} \to \mathcal{H}_{\mathrm{low}}$ with closed range.

The residual low-mode obstruction is the cokernel class

$$\mathfrak{o}_{\mathrm{tail}}(p) := [P_{\mathrm{low}}h] \in \mathcal{H}_{\mathrm{low}}/\operatorname{Ran}\Gamma. \tag{200}$$

The local term retains the ancestry of $S(V)$; gauge removal appends $\mathsf{Abs}$, and the exterior expansion and its limiting trace append $\mathsf{Bdry}$. None of (T1)–(T5) is additional specialization data. Candidate operators, splittings, and dyadic pieces are enumerated from the typed weak constraint and the fixed universal elliptic library. The first relation without an ordinary value is retained as its total-graph successor with the displayed ancestry.

**Theorem II.9.2** (Constraint–tail structural normal form)**.** *On the ordinary value of the elliptic constraint–tail product graph, every field has, on $U_0$, the decomposition*

$$p = p_{\mathrm{loc}} + \Gamma g + h_{\mathrm{dec}} + h_{\mathrm{obs}}, \tag{201}$$

*where $g \in \mathcal{G}_{U_0}$, $h_{\mathrm{dec}} \in \mathcal{H}_{\mathrm{dec}}$, and $h_{\mathrm{obs}}$ is any representative of the class $\mathfrak{o}_{\mathrm{tail}}(p)$. The first three terms are closed by, respectively, the local source law, a propagated quotient, and the summability invariant* (199)*. Consequently:*

(C1) *if $\mathfrak{o}_{\mathrm{tail}}(p) = 0$, the entire constraint field closes using only its inherited source,* $\mathsf{Abs}$*, and* $\mathsf{Bdry}$*;*

(C2) *if $\mathfrak{o}_{\mathrm{tail}}(p) \neq 0$ and its target projection vanishes, its remaining class lies in $\mathscr{N}_\Sigma$;*

(C3) *if $\mathfrak{o}_{\mathrm{tail}}(p) \neq 0$ and is target-visible, it is a finite-dimensional* $\mathsf{Bdry}/\mathsf{Abs}$*-ancestry occurrence evaluated by the carrier and price rows. It is not a new residual source.*

*Thus the nonlocal constraint is reduced internally to the local source graph, the summability graph, and a finite-dimensional target-projected cokernel. Every nonordinary value is already a witness-indexed successor rather than an additional analytic task.*

*Proof.* Apply the continuous projections in (198) to $h$. Closed range and finite dimensionality give $\mathcal{H}_{\mathrm{low}} = \operatorname{Ran}\Gamma \oplus H_{\mathrm{obs}}$ for a finite-dimensional complement $H_{\mathrm{obs}}$. This gives (201); changing the complement changes $h_{\mathrm{obs}}$ only by a gauge term and hence preserves the cokernel class. Absolute convergence in (199) closes the decaying series. The three routing conclusions follow from quotient propagation, Theorem I.3.4, and the definition of $\mathscr{N}_\Sigma$. □

**Proposition II.9.3** (Constraint tails enter the physical budget only through a carrier law)**.** *On the ordinary constraint-tail branch, let the signed tail contribution be $\Psi_{\mathrm{tail}}$ to a localized carrier balance. It is charged to the physical expenditure measure only if the equation supplies a measure inequality of one of the forms*

$$(\Psi_{\mathrm{tail}})_+ \leq \Phi_u^+, \qquad \textit{or} \qquad (\Psi_{\mathrm{tail}})_+ \leq R_u^+, \tag{202}$$

*with the right side appearing in* (150)*. Tail summability alone proves analytic closure but supplies no physical price and no budget bound. In a target argument these inequalities are first restricted to the active shell/source cell and then passed through* (90)*.*

*Proof.* The carrier compiler uses only the measure terms displayed in (150). Either inequality in (202) places the positive tail contribution in that balance. Without such a comparison, (199) is a norm statement on a homogeneous field and has no implication for $\mu_u$. □

## II.10 Covariant entropy cocycles and physical production

An entropy enters the reachability argument through the physical production that it controls along one orbit. Local entropy formulas in changing blow-up charts must therefore be joined by an explicit cocycle law. Independent windowwise monotonicity does not telescope. Perelman's entropy formula is a classical model of covariant monotonicity with a rigid equality case [13]. The formulation below derives the required equation-independent cocycle contracts directly from a local balance and its reset relation. For reachability, each local balance and reset is first compiled by Theorem II.2.18. Entropy production is therefore not a second budget and is not charged across unrelated windows: it contributes to the sole physical measure only on the same-origin target/source cell on which the public carrier proves domination.

**Definition II.10.1** (Physical entropy cocycle)**.** Let $I_j = [a_j, b_j]$, $j \geq 1$, be ordered normalized windows obtained from pairwise disjoint physical events $Q_j$. A physical entropy cocycle on this family consists of
(E1) scalar primitives $\mathcal{M}_j : I_j \to \mathbb{R}$;
(E2) nonnegative Radon production measures $\mathfrak{D}_j$ and nonnegative error measures $r_j$ on $I_j$;
(E3) reset errors $e_j \geq 0$ and a constant $c_{\mathcal{M}} > 0$;
(E4) constants $M_+ < \infty$ and $M_- > -\infty$;
such that, for $a_j \leq s < t \leq b_j$,

$$\mathcal{M}_j(t) + c_{\mathcal{M}}\mathfrak{D}_j((s,t]) \leq \mathcal{M}_j(s) + r_j((s,t]), \tag{203}$$

$$\mathcal{M}_{j+1}(a_{j+1}) \leq \mathcal{M}_j(b_j) + e_j, \tag{204}$$

$$\mathcal{M}_1(a_1) \leq M_+, \qquad \mathcal{M}_j(t) \geq M_-. \tag{205}$$

The measures $\mathfrak{D}_j$ are coordinate expressions of one physical production law on the events $Q_j$; chart Jacobians and time changes are included in that law. The half-open convention in (203) prevents endpoint atoms from being counted twice.

**Theorem II.10.2** (Entropy-cocycle gluing)**.** *For every $N \geq 1$, a physical entropy cocycle satisfies*

$$c_{\mathcal{M}} \sum_{j=1}^{N} \mathfrak{D}_j(I_j) \leq M_+ - M_- + \sum_{j=1}^{N} r_j(I_j) + \sum_{j=1}^{N-1} e_j. \tag{206}$$

*If the two error series are finite, then $\sum_j \mathfrak{D}_j(I_j) < \infty$. If in addition every selected bad window satisfies $\mathfrak{D}_j(I_j) \geq \delta > 0$, their number is at most*

$$\frac{M_+ - M_- + \sum_j r_j(I_j) + \sum_j e_j}{c_{\mathcal{M}}\delta}. \tag{207}$$

*Proof.* Apply (203) on every whole window. Summing gives

$$c_{\mathcal{M}} \sum_{j=1}^{N} \mathfrak{D}_j(I_j) \leq \sum_{j=1}^{N} (\mathcal{M}_j(a_j) - \mathcal{M}_j(b_j)) + \sum_{j=1}^{N} r_j(I_j).$$

The reset inequalities telescope the first sum from above:

$$\sum_{j=1}^{N} (\mathcal{M}_j(a_j) - \mathcal{M}_j(b_j)) \leq \mathcal{M}_1(a_1) - \mathcal{M}_N(b_N) + \sum_{j=1}^{N-1} e_j.$$

Use (205). This proves (206). Monotone convergence gives the countable statement, and $N\delta \leq \sum_{j=1}^{N} \mathfrak{D}_j(I_j)$ gives (207). ∎

**Definition II.10.3** (Perelman-type covariant invariant row)**.** A Perelman-type construction is evaluated by the totalized tuple $\mathfrak{I}_{\mathrm{Per}}$ whose six coordinates are the graph defects of the following relations for the fixed PDE and its generated window atlas:

(P1) Each local primitive is a quotient-invariant scalar, typically a localized energy or variance plus an adjoint entropy and explicit pressure, drift, modulation, and boundary corrections.

(P2) Differentiation along the *fixed* PDE gives (203); all unsigned remainders are placed in the displayed error measure, not suppressed formally.

(P3) The target-visible quotient of production by its nonnegative physical floor $\mathfrak{d}$ is recorded by the price-gap row.

(P4) The primitives have the lower bound in (205). Its sign must match the orientation of (203).

(P5) Transport along the same physical orbit, chart pullback, and any change of a finite-horizon adjoint datum satisfy (204), with summable reset errors.

(P6) Production either defines the expenditure measure $\mu_u$, as in Corollary II.10.6, or is dominated by the declared $\mu_u$ through an explicit change-of-variables inequality of the form (191).

The entropy-cocycle branch is the common zero set of these six target-active graph defects. Any nonzero coordinate is retained with its inherited source ancestry and sent to the form, chain, price, tail, or boundary row indicated by the failed relation. Thus the construction neither presumes the entropy scalar nor accepts it as additional favorable data. These coordinates isolate the production-and-cocycle mechanism of Perelman's entropy formula [13].

**Proposition II.10.4** (Weighted zero-production rigidity)**.** *Let $D \subset \mathbb{R}^d$ be connected and open, let $v \in L^2(I; W^{1,2}_{\mathrm{loc}}(D))$, and let $m \ge m_* > 0$ almost everywhere on $D \times I$. If a nonnegative production functional obeys*

$$\mathfrak{d}(w) \ge c_0 \int_I \int_D m(x,t) |\nabla v(x,t)|^2 \, \mathrm{d}x \, \mathrm{d}t \tag{208}$$

*with $c_0 > 0$, then $\mathfrak{d}(w) = 0$ implies that $v(\cdot, t)$ is spatially constant on $D$ for almost every $t$. Consequently, any closed bad class carrying a strictly positive mean-subtracted oscillation is disjoint from $\{\mathfrak{d} = 0\}$. The regularity compiler invokes this conclusion only on the target-local stratum whose selected production is $\mathfrak{d}$; it gives no exclusion for an unrelated bad class or source cell.*

*Proof.* Nonnegativity and (208) imply $\nabla v = 0$ almost everywhere. A Sobolev function with zero weak gradient on a connected domain is almost everywhere constant. Its mean-subtracted oscillation is therefore zero, which excludes membership in the stated bad class. □

**Proposition II.10.5** (Physical-clock comparison)**.** *Let $\mu$ be a physical measure on an event. On the absolutely-continuous clock branch, let a normalized production measure have density $w \ge 0$: $\mathrm{d}\nu = w \, \mathrm{d}\mu$. On a measurable set $Q$,*

$$w \le C, \quad 0 < C < \infty, \quad \text{a.e. on } Q \Longrightarrow \mu(Q) \ge C^{-1} \nu(Q), \tag{209}$$

$$w \ge c > 0 \quad \text{a.e. on } Q \Longrightarrow \mu(Q) \le c^{-1} \nu(Q). \tag{210}$$

*Thus a lower bound for normalized production yields a lower physical price only from an upper bound such as* (209)*; the opposite density bound has the opposite implication.*

*Proof.* Integrate the pointwise bounds in $\nu(Q) = \int_Q w \, \mathrm{d}\mu$. □

**Corollary II.10.6** (Entropy as the physical expenditure measure)**.** *On the finite-error branch of Definition II.10.1 and* (206)*, on the atomic $\sigma$-algebra generated by the disjoint events $Q_j$, define*

$$\nu_u \left( \bigsqcup_{j \in S} Q_j \right) := \sum_{j \in S} \mathfrak{D}_j(I_j), \qquad S \subseteq \mathbb{N}.$$

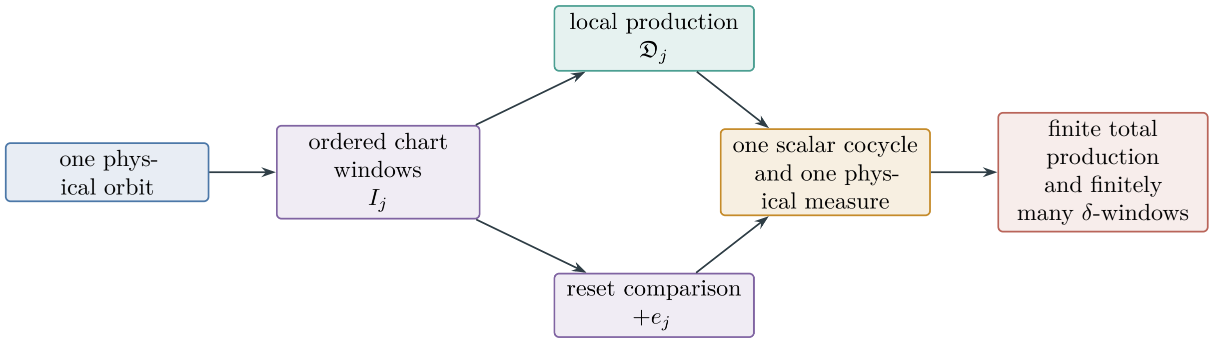


Figure 18: Covariant entropy-cocycle accounting. Local production estimates become orbitwise additive only after chart and adjoint-reset comparisons identify them along the same-origin physical orbit. The output is a finite physical production measure that can serve as the orbit expenditure $\mu_u$.

*Then $\nu_u$ is a finite, countably additive physical measure on that event family. It coincides with the restriction of $\mu_u$ in Definition I.1.12 precisely when the public carrier identity identifies this entropy production as physical expenditure. Otherwise it is a generated nonnegative structural coordinate and cannot replace or enlarge $\mu_u$. Even in the coincident case, a shell event is charged only through its target-local factorization* (102).

*Proof.* Countable additivity follows directly from the nonnegative series defining $\nu_u$; finiteness is Theorem II.10.2. □

*Remark* II.10.7 (Equality models are row-local). A Gaussian may be the equality state of an adjoint Fisher-information term, while a constant field is the equality state of a weighted Dirichlet floor. These statements concern different nonnegative terms. A valid rigidity argument uses the equality set of the particular floor that is both compactly stable and linked to target badness; it does not merge the two equality models.

## II.11 Wick-rotated compensated quotients and structural-action learning

This section acts on the general nonreversible compensated-quotient record already defined in Part I. On its spectral-Abs branch, the Wick map replaces the real quotient evolution by unitary spectral evolution and fixes the represented real Geom operator, the selected Caus operator, their authority record, and every saturated charge. The resulting generator is nonnormal in general, while its quotient response has a positive self-adjoint learning kernel.

Thus the only new datum is the choice to apply the Wick map. All source typing, quotient and fiber maps, geometric forms, causal provenance, authority restrictions, compensation inputs, and resource coordinates are those of the input Part I record. The transformed response stability, response kernel, memory kernel, action, and learning rules below are consequences evaluated on that same record. Its representation, confidence, authority, and model-selection coordinates belong to the inherited Part I computational ledger; they never augment the continuum physical budget. Conversely, any balance in this section used for PDE reachability must first pass through Theorems II.2.4 and II.2.18 on the contact-generated source cell. The learning bounds themselves make no continuum regularity claim.

### II.11.1 Spectral quotient and real geometric complement

**Definition II.11.1** (Wick transform of an inherited compensated record)**.** Let $\mathscr{C}_I$ be a dynamically qualified Part I master compensated quotient–geometry record in its represented spectral-$\mathsf{Abs}$ realization. Write $\mathcal{Q}_{\mathbb{R}}$ and $\mathcal{Y}_{\mathbb{R}}$ for its quotient and complementary real Hilbert spaces. In the notation of that record, $H_{\mathsf{Abs}}$ is the nonnegative self-adjoint quotient operator, $G$ is the real nonnegative geometric operator, and

$$K_C = \begin{pmatrix} 0 & C^* \\ -C & A_{\mathcal{Y}} \end{pmatrix}$$

is the actual selected skew $\mathsf{Caus}$ block with its existing authority and provenance certificate. These are fields of $\mathscr{C}_I$, not new structural data.

Let $\mathcal{Q} = \mathcal{Q}_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ and $\mathcal{Y} = \mathcal{Y}_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$, and extend every field of $\mathscr{C}_I$ complex-linearly. The Wick transform acts only on the quotient evolution. The complex extension of $G$ commutes with conjugation and restricts to the original operator on $\mathcal{Y}_{\mathbb{R}}$, so the $\mathsf{Geom}$ generator remains the complexification of the same real geometric operator (although a coupled trajectory need not remain in $\mathcal{Y}_{\mathbb{R}}$):

$$\mathfrak{W}_{\mathsf{Abs}} : -H_{\mathsf{Abs}} \oplus (-G) + K_C \longmapsto -iH_{\mathsf{Abs}} \oplus (-G) + K_C. \tag{211}$$

Equivalently, on $\mathcal{H}_W := \mathcal{Q} \oplus \mathcal{Y}$, its transformed generator is

$$\mathcal{L}_W(C) := \begin{pmatrix} -iH_{\mathsf{Abs}} & C^* \\ -C & -G + A_{\mathcal{Y}} \end{pmatrix} = -\begin{pmatrix} 0 & 0 \\ 0 & G \end{pmatrix} + \begin{pmatrix} -iH_{\mathsf{Abs}} & C^* \\ -C & A_{\mathcal{Y}} \end{pmatrix}. \tag{212}$$

The *positive self-energy normal form* is the inherited subbranch $A_{\mathcal{Y}} = 0$; the general inherited branch retains the full nonnormal resolvent. The spectral calculus gives

$$e^{-\tau H_{\mathsf{Abs}}} = \int_{[0,\infty)} e^{-\tau\omega}\, \mathrm{d}E_{H_{\mathsf{Abs}}}(\omega), \qquad e^{-itH_{\mathsf{Abs}}} = \int_{[0,\infty)} e^{-it\omega}\, \mathrm{d}E_{H_{\mathsf{Abs}}}(\omega). \tag{213}$$

The two spectral integrals are ordinary for every vector in their displayed time domains. Their identification by the notation $\tau = it$ is evaluated by the Part I analytic-continuation relation: it is an ordinary equality on the record's continuation domain and otherwise carries that relation's nonordinary boundary value. No continuation domain is added to structural admissibility.

The map $\mathfrak{W}_{\mathsf{Abs}}$ preserves the source cell and the complete record. In particular, $H_{\mathsf{Abs}}$, $-iH_{\mathsf{Abs}}$, and their spectral projectors have the same $\mathsf{Abs}$ ancestry; $G, C, A_{\mathcal{Y}}$, the authority envelope, and their costs are unchanged. Equation (213) is spectral functional calculus, rather than a formal substitution in a learned trajectory.

**Proposition II.11.2** (Exact inheritance under the Wick map)**.** *Let $\mathscr{C}_I^W = \mathfrak{W}_{\mathsf{Abs}}(\mathscr{C}_I)$. Under the canonical scalar-extension embeddings,*

$$\mathrm{Src}(\mathscr{C}_I^W) = \mathrm{Src}(\mathscr{C}_I), \qquad \mathrm{Anc}(\mathscr{C}_I^W) = \mathrm{Anc}(\mathscr{C}_I), \qquad \mathfrak{A}(\mathscr{C}_I^W) = \mathfrak{A}(\mathscr{C}_I), \qquad \mathbf{c}_{\mathrm{str}}(\mathscr{C}_I^W) = \mathbf{c}_{\mathrm{str}}(\mathscr{C}_I). \tag{214}$$

*The represented $\mathsf{Geom}$ form, selected $\mathsf{Caus}$ current, action metric, and quotient/fiber transports agree with their real restrictions in $\mathscr{C}_I$. Consequently the source, structural-admissibility, and authority gates have the same truth value before and after the map. The inherited compensation data remain fixed, while every response or dynamic- stability quantity containing the quotient time generator is reevaluated with $-H_{\mathsf{Abs}}$ replaced by $-iH_{\mathsf{Abs}}$.*

| Wick quantity | Part I source | Exhaustive evaluation |
|---|---|---|
| Generator realization | dynamic and sectorial realization field | bounded or ordinary closed-form value; otherwise typed realization defect |
| Resolvent and self-energy | full-generator dynamic record | Hille–Yosida value on $\Re z > 0$; totalized graph value elsewhere |
| Response dimension | finite evaluation carrier and effective-dimension machinery | finite carrier trace; population extended trace in $[0, \infty]$ |
| Spectral gap and commutation | represented operators and spectral calculus | computed gap/status; modal formula or exact singular-value formula |
| Legal bridge and action | Caus authority envelope and current metric | distance to the legal-current closure; empty family gives $+\infty$ |
| Identification | represented dynamics chart and score candidates | derived design range/kernel; full or partial identification bound |
| Statistical law | i.i.d., mixing, sequential, martingale, and channel cells | recordwise minimum ordinary radius; every unavailable candidate equals $+\infty$ |
| RL conversion | safe-action, behavior/deployment, and actor records | extended critic, converter, support, and policy coordinates |
| Model choice | exact source cells, charged nets, Rademacher union, and PAC–Bayes mixture | finite-stage saturated profile and Pareto strict inverse; empty/nonattained limit is retained and typed |
| Continuum energy | operator-range and approximation records | Moore–Penrose quadratic form and extended suprema; no closed-range or compactness gate |

Table 5: Inheritance and totalization ledger for the Wick specialization. Every row is an evaluation of Part I data or a consequence of it.

*Proof.* Complexification is an isometric scalar extension and commutes with sums, adjoints, compositions, projections, and restrictions to the original real spaces. Equation (211) fixes every block except the quotient diagonal. The source, ancestry, authority, structural cost, current, metric, and transport fields therefore remain identical. Their algebraic and authority gates are preserved. A resolvent, self-energy, memory, stability factor, or learning kernel contains the transformed diagonal and is therefore recomputed as a derived Wick response rather than copied from the input record. □

**Proposition II.11.3** (No-new-premise closure of the Wick specialization)**.** *The input domain of $\mathfrak{W}_{\mathsf{Abs}}$ is exactly the represented, dynamically qualified spectral-*$\mathsf{Abs}$ *cell of the Part I master record. Within that inherited domain, the Wick chapter adds no structural, analytic, statistical, or model-selection premise. More precisely:*

(i) *bounded, sectorial, unbounded-bridge, authority, source, channel, dependence, control, and model-family alternatives are values of existing Part I record fields;*

(ii) *spectral gaps, commutators, design excitation, operator errors, source residuals, converter constants, policy divergences, and optimization errors are computed coordinates in* $[0, \infty]$*, not admission conditions;*

(iii) *a missing analytic or statistical certificate has its Part I nonordinary value. Every dependent upper-error or risk candidate has value* $+\infty$*, every dependent lower-usefulness candidate has value zero, and a derived coordinate without an ordered extension retains its typed nonordinary value;*

(iv) *an empty legal, zero-action, or risk-feasible family produces its typed authority, representation, confidence, or budget status; and*

(v) *every finite-rate formula is the restriction of a general extended bound to the named Part I rate cell, while all complementary cells retain the general bound.*

*Consequently no favorable branch is used to define admissibility for this specialization. The only new operation is $-H_{\mathsf{Abs}} \mapsto -iH_{\mathsf{Abs}}$, after which all quantities are derived or totalized.*

*Proof.* Exact inheritance is Proposition II.11.2. Part I already partitions represented records by realization, authority, source, channel, dependence, and control status and supplies charged finite stages for family selection. Evaluation of a field gives either its ordinary value or its existing nonordinary value. Extended-real arithmetic, the conventions $\inf \varnothing = +\infty$ and $\sup \varnothing = 0$, and typed empty-family statuses make every construction total. The propositions below derive the new response, action, loss, and transfer coordinates from those fields and make each finite specialization an explicit record cell. □

**Proposition II.11.4** (Generation and physical budget)**.** *On the bounded realization branch of the inherited Part I record, $\mathcal{L}_W(C)$ generates a contraction semigroup and a solution $u = (x, y)$ of $\dot{u} = \mathcal{L}_W(C)u$ satisfies*

$$\frac{\mathrm{d}}{\mathrm{d}t}\|x\|^2 = 2\,\mathrm{Re}\langle Cx, y\rangle, \tag{215}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\|y\|^2 = -2\,\mathrm{Re}\langle Cx, y\rangle - 2\|G^{1/2}y\|^2, \tag{216}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(\|x\|^2 + \|y\|^2) = -2\|G^{1/2}y\|^2. \tag{217}$$

*Consequently the Wick-rotated* $\mathsf{Abs}$ *block and the authorized* $\mathsf{Caus}$ *block transfer carrier between the two sectors without expenditure. The real* $\mathsf{Geom}$ *block supplies the unique nonnegative expenditure in this model.*

*Proof.* The second matrix in (212) is skew-adjoint. Taking real parts in the two component equations gives (215) and (216); their transfer terms cancel. Thus $\mathcal{L}_W(C)$ is dissipative. The bounded perturbation theorem, or direct finite-dimensional exponentiation, gives the contraction semigroup. □

**Proposition II.11.5** (Closed-operator and sectorial-complement realization)**.** *On the closed sectorial branch of the inherited Part I record, the complexified complement form defines its maximal dissipative operator $-G + A_{\mathcal{Y}}$. On its inherited bounded-bridge cell, the block operator $\mathcal{L}_W(C)$ on $D(H_{\mathsf{Abs}}) \oplus D(-G + A_{\mathcal{Y}})$ is maximal dissipative. The same conclusion holds on the ordinary unbounded-bridge realization cell produced by the Part I sectorial Caus-authority interface. Its complementary nonordinary cell retains the typed realization defect. Identity* (217) *holds in integrated form,*

$$\|u(t)\|^2 + 2\int_0^t \|G^{1/2}y(s)\|^2\,\mathrm{d}s = \|u(0)\|^2, \tag{218}$$

*on the equality branch, and with $\leq$ for weak limits carrying a nonnegative energy defect.*

*Proof.* The complement sectorial form gives $-G + A_{\mathcal{Y}}$, while $-iH_{\mathsf{Abs}}$ generates a unitary group. Their direct sum is maximal dissipative. The off-diagonal bridge is bounded and skew-adjoint, so the bounded dissipative perturbation theorem preserves maximal dissipativity. The ordinary unbounded-bridge conclusion is exactly the closed-operator perturbation conclusion already certified by the Part I sectorial interface [7, 12]. Approximation by vectors in a common core and lower semicontinuity give the integrated identity or its defect version. □

### II.11.2 Lumpability defect, self-energy, and memory

Let $U : \mathcal{Q} \to \mathcal{H}_W$ be $Ux = (x, 0)$, let $P = U^*$, and let $P_\Delta = \mathbf{1}_\Delta(H_{\mathsf{Abs}})$ be a declared spectral band.

**Proposition II.11.6** (Band-lumpability defect)**.** *The compressed quotient generator is $P\mathcal{L}_W U = -iH_{\mathsf{Abs}}$, and its full-operator defect on $\Delta$ is*

$$E_{W,\Delta} := (I - UP)\mathcal{L}_W U P_\Delta = \begin{pmatrix} 0 \\ -CP_\Delta \end{pmatrix}. \tag{219}$$

*Hence the spectral band is exactly lumpable if and only if $CP_\Delta = 0$. Its intrinsic quasi-lumpability size is $\varepsilon_{W,\Delta} = \|CP_\Delta\|$, supplemented in an empirical record by the separately measured model and compression residuals. Under the canonical complexification isometry this is exactly the norm of the corresponding cross-sector defect in $\mathscr{C}_I$; Wick rotation changes the quotient diagonal and introduces no new lumpability hypothesis.*

*Proof.* Apply the two block rows of $\mathcal{L}_W$ to $(x, 0)$, compress the first component, and subtract its lift. The off-diagonal block is fixed by (211), so the resulting vector is the complexification of the inherited Part I defect. Complexification is isometric, which proves the norm statement. $\square$

**Theorem II.11.7** (Feshbach response and geometric self-energy)**.** *The Part I realization field is exhaustive. On either ordinary realization cell of Propositions II.11.4 and II.11.5, every $z \in \mathbb{C}$ with $\operatorname{Re} z > 0$ satisfies, by maximal dissipativity,*

$$\|(z + G - A_{\mathcal{Y}})^{-1}\| \le \frac{1}{\operatorname{Re} z}, \qquad \|(z - \mathcal{L}_W(C))^{-1}\| \le \frac{1}{\operatorname{Re} z}. \tag{220}$$

*Define*

$$\Sigma_C(z) := C^*(z + G - A_{\mathcal{Y}})^{-1}C, \qquad F_W(z) := z + iH_{\mathsf{Abs}} + \Sigma_C(z). \tag{221}$$

*On the ordinary inherited unbounded-bridge cell, these expressions denote the closed forms and associated operators supplied by that Part I realization certificate. Then $F_W(z)$ is invertible and*

$$P(z - \mathcal{L}_W(C))^{-1}U = F_W(z)^{-1}. \tag{222}$$

*In the positive self-energy normal form, $z = \lambda > 0$ gives*

$$\Sigma_C(\lambda) = C^*(\lambda + G)^{-1}C \ge 0. \tag{223}$$

*Thus real geometric relaxation compensates the imaginary quotient defect by a positive frequency-dependent self-energy. For $\operatorname{Re} z \le 0$, the same block formula is the ordinary value of the existing totalized resolvent relation whenever that value exists. A pole, domain failure, or noninvertible Schur complement is its nonordinary graph value. Hence every $z$ is accounted for, while the learning kernel uses the automatically ordinary half-line $\lambda > 0$ on the ordinary realization cells. On the complementary nonordinary realization cell, the self-energy, Schur complement, and learning kernel inherit the nonordinary graph value and every response-dependent upper certificate equals $+\infty$; the record is retained with its realization status.*

*Proof.* Both the complement generator $-G + A_{\mathcal{Y}}$ and the full Wick generator are maximal dissipative by Propositions II.11.4 and II.11.5. The right half-plane therefore belongs to both resolvent sets and the Hille–Yosida estimate gives (220). The block operator $z - \mathcal{L}_W$ is

$$\begin{pmatrix} z + iH_{\mathsf{Abs}} & -C^* \\ C & z + G - A_{\mathcal{Y}} \end{pmatrix}.$$

The lower-right block and the full block are invertible. Block Gaussian elimination therefore shows that their Schur complement $F_W(z)$ is invertible and gives (222). On the ordinary inherited unbounded-bridge cell the same factorization is first taken on the common form core and then closed by its Part I realization theorem. When $A_{\mathcal{Y}} = 0$ and $z = \lambda > 0$, the middle resolvent in (223) is positive self-adjoint. Outside the right half-plane, totalization records exactly the failure of one of these ordinary elimination steps. ☐

**Corollary II.11.8** (Mori–Zwanzig memory equation)**.** *On the bounded-bridge cell, for classical initial data $(x_0, y_0)$, the quotient coordinate satisfies*

$$\dot{x}(t) = -iH_{\mathsf{Abs}}x(t) - \int_0^t K_C(t-s)x(s)\,\mathrm{d}s + C^* e^{t(-G+A_{\mathcal{Y}})}y_0, \qquad K_C(t) := C^* e^{t(-G+A_{\mathcal{Y}})}C. \tag{224}$$

*For every $\operatorname{Re} z > 0$, the Laplace transform of $K_C$ is $\Sigma_C(z)$. Equations (222) and (224) are respectively the Feshbach and Mori–Zwanzig forms of the same quotient elimination [4, 10, 18]. On the ordinary unbounded-bridge cell, the same display is interpreted in the mild input–output or closed-form sense supplied by the Part I realization record, first on its common core and then by closure. A datum outside that recorded admissibility domain has the inherited nonordinary evolution value; no additional observation-domain premise is imposed here.*

*Proof.* Variation of constants in the $y$-equation gives

$$y(t) = e^{t(-G+A_{\mathcal{Y}})}y_0 - \int_0^t e^{(t-s)(-G+A_{\mathcal{Y}})}Cx(s)\,\mathrm{d}s.$$

Substitute this identity in $\dot{x} = -iH_{\mathsf{Abs}}x + C^*y$. Laplace transformation gives (221). On the ordinary unbounded-bridge cell, the Part I common-core factorization in Theorem II.11.7 gives the same identity, and closure gives the recorded mild input–output relation. ☐

**Definition II.11.9** (Dark quotient space)**.** On the bounded-bridge cell, the dark quotient space is

$$\mathcal{D}_{\mathrm{dark}} := \{x \in \mathcal{Q} : Ce^{-itH_{\mathsf{Abs}}}x = 0 \text{ for every } t \in \mathbb{R}\}. \tag{225}$$

It is the largest $H_{\mathsf{Abs}}$-reducing subspace contained in $\ker C$. On the ordinary unbounded-bridge cell, $Ce^{-itH_{\mathsf{Abs}}}$ denotes the closed observation orbit supplied by the Part I realization record and the same intersection is taken on its recorded common domain. A domain failure retains the typed realization value instead of defining a smaller dark space. Thus the dark-space coordinate is totalized by the same realization partition as the Feshbach response. In dimension $d_Q < \infty$,

$$\mathcal{D}_{\mathrm{dark}} = \bigcap_{k=0}^{d_Q-1} \ker(CH_{\mathsf{Abs}}^k). \tag{226}$$

The target-visible component of $\mathcal{D}_{\mathrm{dark}}$ is a $\mathsf{Caus}/\mathsf{Geom}$ reach obstruction. It remains available to a direct spectral readout unless that readout is separately excluded.

**Proposition II.11.10** (Totalized imaginary-axis and bracket alternatives)**.** *Define the derived geometric gap*

$$g_* := \inf \sigma(G) \in [0, \infty).$$

*Every imaginary-axis eigenpair $\mathcal{L}_W u = i\beta u$, $u = (x, y)$, satisfies*

$$y \in \ker G, \qquad -iH_{\mathsf{Abs}}x + C^*y = i\beta x, \qquad -Cx + A_{\mathcal{Y}}y = i\beta y. \tag{227}$$

*On the exhaustive cell $g_* > 0$, this reduces to $u = (x, 0)$, $Cx = 0$, and $H_{\mathsf{Abs}} x = -\beta x$. Consequently that cell has no imaginary-axis eigenvalue exactly when the corresponding $H_{\mathsf{Abs}}$-eigenspaces have zero intersection with* $\ker C$*. On the complementary cell $g_* = 0$, equation (227) retains the geometric kernel rather than discarding it. In both cells the commutator Gramian of Definition II.7.1, applied to*

$$S = \begin{pmatrix} 0 & 0 \\ 0 & G \end{pmatrix}, \qquad A = \begin{pmatrix} -iH_{\mathsf{Abs}} & C^* \\ -C & A_{\mathcal{Y}} \end{pmatrix},$$

*has its Part I extended reduced gap. A positive value supplies the existing twisted-carrier estimate; the zero value returns its target-visible bracket kernel as the typed dark/rank-defect branch. These alternatives exhaust the record and introduce no coercivity or bracket assumption.*

*Proof.* Taking real parts of $\langle u, \mathcal{L}_W u \rangle = i\beta \|u\|^2$ and using (217) gives $\langle y, Gy \rangle = 0$. Since $G \geq 0$, this is equivalent to $G^{1/2} y = 0$, hence to $y \in \ker G$, and the two block equations give (227). The derived inequality $G \geq g_* I$ makes $y = 0$ on the cell $g_* > 0$. The remaining claims are the two values of the Part I bracket-resistance and twisted-carrier alternative for the displayed $S + A$ decomposition. □

### II.11.3 Authority-constrained structural action

The bridge in (212) is an operator. Its action record uses the actual edge or current representation induced by that same operator. This prevents a favorable target current from replacing an imposed PDE current.

**Definition II.11.11** (Wick evaluation of the inherited bridge action)**.** Fix a pre-contact state of $\mathscr{C}_I$. Let $(\mathfrak{J}, \langle \cdot, \cdot \rangle_{\mathbb{M}^{-1}})$, the target-normalized potential $D$, the actual authorized bridge current $J_C$, and its authority envelope $\mathfrak{A}$ be the corresponding Part I fields. Put

$$J_D := -\mathbb{M} \nabla D,$$

and define

$$\mathcal{A}_{\mathsf{Caus}}(J_C) := \|J_C\|^2_{\mathbb{M}^{-1}}, \qquad \mathcal{E}_{\mathrm{aim}}(D) := \|J_D\|^2_{\mathbb{M}^{-1}}, \tag{228}$$

$$\mathcal{P}_{\mathsf{Caus}}(J_C, D) := \langle J_C, J_D \rangle_{\mathbb{M}^{-1}}, \quad \mathscr{W}_{\mathfrak{A}}(C; D) := \frac{1}{2} \mathcal{A}_{\mathsf{Caus}}(J_C) + \frac{1}{2} \mathcal{E}_{\mathrm{aim}}(D) - \mathcal{P}_{\mathsf{Caus}}(J_C, D). \tag{229}$$

All fields and transports in this display remain those of $\mathscr{C}_I$. The Wick map changes none of them; it only permits the displayed action to be evaluated jointly with the transformed spectral response.

**Theorem II.11.12** (Inner least-action bridge coordinate)**.** *Every inherited bridge action evaluated after the Wick transform satisfies*

$$\mathscr{W}_{\mathfrak{A}}(C; D) = \frac{1}{2} \|J_C - J_D\|^2_{\mathbb{M}^{-1}} \geq 0. \tag{230}$$

*Let $\mathfrak{J}_{\mathfrak{A}}$ be the possibly empty set of represented currents induced by legal bridges, and use* $\mathrm{dist}(J_D, \varnothing) := +\infty$*. Then*

$$\inf_{C \in \mathfrak{A}} \mathscr{W}_{\mathfrak{A}}(C; D) = \frac{1}{2} \operatorname{dist}_{\mathbb{M}^{-1}} \big(J_D, \overline{\mathfrak{J}_{\mathfrak{A}}}\big)^2. \tag{231}$$

(i) *On the inherited closed-convex authority cell, the infimum is attained at the metric projection* $J_C = \Pi_{\mathfrak{J}_{\mathfrak{A}}} J_D$. *On every other nonempty cell, the Part I search transcript returns an* $\varepsilon$*-minimizer and charges its cover and optimization error. The empty cell returns the authority-obstruction value* $+\infty$.

(ii) *Under free authority, the infimum is zero precisely when* $J_D \in \overline{\mathfrak{J}_{\mathfrak{A}}}$*, and zero waste is attained precisely when* $J_D \in \mathfrak{J}_{\mathfrak{A}}$. *Thus density and exact representability are not conflated.*

(iii) *Under hybrid or RL authority, the same distance is evaluated on the declared control family. Its convex, nonconvex, nonattained, and empty cells are handled by the preceding exhaustive rule.*

(iv) *Under imposed PDE authority,* $\mathfrak{J}_{\mathfrak{A}}$ *is a singleton and the action evaluates its fixed current.*

*For a declared positive target power* $p$*, every legal current with* $\mathcal{P}_{\mathsf{Caus}}(J_C, D) \ge p$ *obeys*

$$\mathcal{A}_{\mathsf{Caus}}(J_C) \ge \frac{p^2}{\mathcal{E}_{\mathrm{aim}}(D)} \tag{232}$$

*on the cell* $\mathcal{E}_{\mathrm{aim}}(D) > 0$. *Equality holds exactly for* $J_C = (p/\mathcal{E}_{\mathrm{aim}}(D))J_D$ *when this current is authorized. On the complementary cell* $\mathcal{E}_{\mathrm{aim}}(D) = 0$*, no current can satisfy* $\mathcal{P}_{\mathsf{Caus}}(J_C, D) \ge p$*; the power request therefore returns the authority/target obstruction rather than a missing denominator. This minimization evaluates the action coordinate* $\mathbf{c}_{\mathrm{act}}(\mathscr{C}_W)$. *Zero action is one favorable coordinate; structural optimality is the subsequent saturated-profile comparison of Definition II.11.29.*

*Proof.* Expand the square in (230); taking the infimum gives (231) because distance is unchanged by closure. The Hilbert projection theorem gives the closed-convex cell, while the definition of an infimum supplies $\varepsilon$-minimizers on every other nonempty cell. The authority statements are its free, control-family, and singleton specializations. Cauchy–Schwarz gives $p^2 \le \mathcal{A}_{\mathsf{Caus}}(J_C)\mathcal{E}_{\mathrm{aim}}(D)$, with the displayed equality condition; when the second factor is zero, the left-hand side cannot be positive. □

**Corollary II.11.13** (Authority monotonicity)**.** *The Part I authority order compares envelopes only at fixed quotient, geometry, target, and cost convention. Every ordered pair* $\mathfrak{A}_1 \preceq \mathfrak{A}_2$*, equivalently* $\mathfrak{J}_{\mathfrak{A}_1} \subseteq \mathfrak{J}_{\mathfrak{A}_2}$*, satisfies*

$$\inf_{C \in \mathfrak{A}_2} \mathscr{W}_{\mathfrak{A}_2}(C; D) \le \inf_{C \in \mathfrak{A}_1} \mathscr{W}_{\mathfrak{A}_1}(C; D).$$

*The corresponding saturated attainable profile is ordered in the opposite direction: imposed* $\le$ *hybrid* $\le$ *free whenever the represented families are nested with no larger implementation cost.*

### II.11.4 Positive response kernel and quantitative transfer

The quotient response $F_W(\lambda)^{-1}$ need not be normal. Statistical learning therefore uses a positive transform of the full response rather than a diagonalization of $F_W$.

**Definition II.11.14** (Wick response kernel)**.** Fix $\lambda > 0$. The complement and full resolvents, and the invertibility of $F_W(\lambda)$, are consequences of Theorem II.11.7 on every ordinary realization, both in the positive-self-energy and general complement cells. There define

$$T_{W,\lambda} := \left(I + F_W(\lambda)^* F_W(\lambda)\right)^{-1}, \qquad A_{W,\lambda} := T_{W,\lambda}^{-1} = I + F_W(\lambda)^* F_W(\lambda). \tag{233}$$

The operator $T_{W,\lambda}$ is a positive contraction. On a nonordinary realization, $T_{W,\lambda}$ is the unavailable response-kernel value and all response-dependent upper bounds equal $+\infty$. For every ordinary kernel, Part I supplies each finite evaluation carrier $\mathscr{E}$—an empirical sample, mixing-block sample, or sequential path/tree—has its represented evaluation map $E_{\mathscr{E}}$, with the normalization of its Part I theorem, and finite positive response matrix

$$T^{\mathscr{E}}_{W,\lambda} := E_{\mathscr{E}} T_{W,\lambda} E^*_{\mathscr{E}}, \qquad \operatorname{Tr}_{\mathscr{E}} T_{W,\lambda} := \operatorname{Tr} T^{\mathscr{E}}_{W,\lambda} < \infty. \tag{234}$$

This is the trace used by the Part I empirical Rademacher bounds. It is finite because $\mathscr{E}$ is finite, independently of every compactness property of the ambient response. Its empirical ridge effective dimension is exactly the Part I Gram-matrix quantity

$$\widehat{N}_{W,\mathscr{E}}(\lambda,\rho) := \operatorname{Tr}\Big[T^{\mathscr{E}}_{W,\lambda}(T^{\mathscr{E}}_{W,\lambda}+\rho I)^{-1}\Big] \le \dim \mathscr{E}. \tag{235}$$

For ridge parameter $\rho > 0$, define the population effective dimension as the extended positive trace

$$\begin{aligned} N_W(\lambda,\rho) &:= \operatorname{Tr}_{\mathrm{ext}}\Big[T_{W,\lambda}(T_{W,\lambda}+\rho I)^{-1}\Big] \\ &:= \sup_{P:\,P=P^*=P^2,\ \mathrm{rank}\,P<\infty} \operatorname{Tr}\Big[P T_{W,\lambda}(T_{W,\lambda}+\rho I)^{-1}P\Big] \in [0,\infty], \end{aligned} \tag{236}$$

$$m_W(\lambda,\rho;f) := \frac{\|T_{W,\lambda}(T_{W,\lambda}+\rho I)^{-1}f\|^2}{\|f\|^2}. \tag{237}$$

The capture ratio is defined to be zero at $f = 0$, as in the Part I target-capture convention. Every represented nested dense Galerkin family $P_n \uparrow I$ satisfies

$$N_W(\lambda,\rho) = \sup_n \operatorname{Tr}\Big[P_n T_{W,\lambda}(T_{W,\lambda}+\rho I)^{-1}P_n\Big]. \tag{238}$$

Thus finiteness of the population coordinate is a derived value. When the supremum is infinite, the effective-dimension candidate equals $+\infty$ and the Part I recordwise certificate minimum uses another valid same-loss bound or returns its existing capacity/budget obstruction. This operation forms the risk coordinate of the record; the saturated profile retains the structural comparison.

**Proposition II.11.15** (Part I trace accounting without an ambient trace gate)**.** *Put $R_{W,\rho} = T_{W,\lambda}(T_{W,\lambda}+\rho I)^{-1}$. For every represented nested finite-rank family $P_n \uparrow I$,*

$$\operatorname{Tr}_{\mathrm{ext}} R_{W,\rho} = \lim_{n\to\infty} \operatorname{Tr}(P_n R_{W,\rho} P_n) \in [0,\infty]. \tag{239}$$

*The value is finite exactly when $R_{W,\rho}$ is trace class; trace-class behavior is therefore a conclusion about this coordinate, not an admissibility hypothesis. Every empirical Rademacher, block, and sequential trace remains finite on its own evaluation carrier even when the value in* (239) *is infinite.*

*Proof.* Choose an orthonormal basis adapted to the nested ranges of $P_n$. Since $R_{W,\rho} \ge 0$, the finite traces are the increasing partial sums of $\sum_j \langle e_j, R_{W,\rho} e_j\rangle$. Their limit is the extended positive trace, independently of the adapted basis. A positive bounded operator is trace class precisely when this sum is finite. The final statement follows because every evaluation compression in (234) is a finite positive matrix; (235) additionally bounds its ridge mode count by the carrier dimension. □

**Proposition II.11.16** (Exhaustive modal and nonnormal response formulas)**.** *On the positive-self-energy cell $A_{\mathcal{Y}} = 0$, put $B = \lambda I + \Sigma_C(\lambda) = B^*$. Evaluate the derived spectral- commutation status: its ordinary commuting value means that the spectral measures of $H_{\mathsf{Abs}}$ and $B$ commute,* *and its complementary value means that they do not. On the commuting cell, their joint spectral representation has scalar response weight*

$$\eta^W(\omega, \sigma) = \frac{1}{1 + (\lambda + \sigma)^2 + \omega^2}; \tag{240}$$

*in a joint pure-point cell this reads* $H_{\mathsf{Abs}}\phi_j = \omega_j\phi_j$, $\Sigma_C(\lambda)\phi_j = \sigma_j\phi_j$, *and* $\eta_j^W = \eta^W(\omega_j, \sigma_j)$. *On the noncommuting cell,*

$$A_{W,\lambda} = I + B^2 + H_{\mathsf{Abs}}^2 + i[B, H_{\mathsf{Abs}}], \tag{241}$$

*as a quadratic-form identity on the inherited common form domain, and the singular values of* $F_W(\lambda)$, *rather than separate eigenvalues, determine* $T_{W,\lambda}$.

*On the general cell* $A_{\mathcal{Y}} \neq 0$, *set* $B = \lambda I + \Sigma_C(\lambda)$, *which need not be self-adjoint. The always-valid precision identity is*

$$A_{W,\lambda} = I + B^*B + H_{\mathsf{Abs}}^2 + i(B^*H_{\mathsf{Abs}} - H_{\mathsf{Abs}}B). \tag{242}$$

*Thus every record has an exact singular-value formula; the modal formula is an additional ordinary value derived only on the recorded commuting positive-self-energy cell, and its absence removes no record.*

*Proof.* The joint spectral theorem applied on the commuting cell gives (240). On the positive-self-energy cell, $(B + iH_{\mathsf{Abs}})^*(B + iH_{\mathsf{Abs}}) = (B - iH_{\mathsf{Abs}})(B + iH_{\mathsf{Abs}})$, which gives (241). Without self-adjointness of $B$, the same product expands instead to $B^*B + H_{\mathsf{Abs}}^2 + i(B^*H_{\mathsf{Abs}} - H_{\mathsf{Abs}}B)$, proving (242). □

**Theorem II.11.17** (Master Wick learning transfer)**.** *For each complete Wick record, evaluate every Part I learning-certificate candidate already carried by that record. In each ordinary candidate, substitute*

$$T_k = T_{W,\lambda}, \qquad A = A_{W,\lambda}, \qquad E_q = E_{W,\Delta}, \tag{243}$$

*and retain its sample law, loss, source cell, confidence allocation, and authority record. Every conclusion of that Part I candidate then holds with* $N_{\text{eff}}(\rho) = N_W(\lambda, \rho)$ *and capture* $m_k(\rho) = m_W(\lambda, \rho; f_*)$. *A candidate whose Part I record fields are nonordinary has value* $+\infty$. *The recordwise minimum over valid same-loss candidates forms the Part I statistical-radius coordinate, and the typed failure mechanism is unchanged. Across complete records, the inherited saturated profile remains the comparison rule.*

*In particular, for the energy ball* $\mathcal{F}_W(R) = \{f : \langle f, A_{W,\lambda}f\rangle \le R^2\}$, *let* $S$ *be the declared* $m$*-point evaluation carrier and use its Part I compression* $T_{W,\lambda}^S$ *from* (234). *Then*

$$\widehat{\mathfrak{R}}_S(\mathcal{F}_W(R)) \le \frac{R}{m}\sqrt{\operatorname{Tr}_S T_{W,\lambda}}, \tag{244}$$

*and, on the Part I* $[0, 1]$*-valued* $L_\ell$*-Lipschitz loss cell, the corresponding candidate satisfies with probability at least* $1 - \delta$,

$$L(f) \le L_S(f) + \frac{2L_\ell R}{m}\sqrt{\operatorname{Tr}_S T_{W,\lambda}} + 3\sqrt{\frac{\log(2/\delta)}{2m}} + \varepsilon_{\text{task}}. \tag{245}$$

*On the named Part I rate cell* $f_* = T_{W,\lambda}^r g$, $0 < r \le 1$, *and* $\eta_j^W \asymp j^{-2\beta}$, $\beta > 1/2$, *its certified kernel-ridge oracle inequality gives*

$$\|\widehat{f}_{\rho_m} - f_*\|_{L^2}^2 \lesssim m^{-4\beta r/(4\beta r+1)}, \qquad \rho_m \asymp m^{-1/(2r+1/(2\beta))}. \tag{246}$$

*The constants and probability event are exactly those of the fixed-kernel oracle certificate;* (246) *is asserted only for the named eigenvalue and source class.*

*Proof.* The operator $T_{W,\lambda}$ is positive and self-adjoint, so the algebraic gate of the Part I kernel and energy-ball candidates is derived rather than assumed. Its empirical compression $T^S_{W,\lambda}$ is a finite positive matrix, so the Part I ellipsoid support calculation gives (244), and symmetrization and contraction give (245). The source residual is $O(\rho^{2r})$, while $N_W(\lambda, \rho) \asymp \rho^{-1/(2\beta)}$; balancing variance and bias gives (246). This named eigenvalue class makes the extended effective dimension finite as a conclusion. The remaining transfers are substitutions in positive-operator statements and preserve their recorded cells and units. □

**Proposition II.11.18** (Learned-operator stability)**.** *On every finite Part I evaluation carrier, let two represented positive-normal-form records have the derived error coordinates*

$$\varepsilon_H := \|H - \widehat{H}\|, \qquad \varepsilon_C := \|C - \widehat{C}\|, \qquad \varepsilon_G := \|G - \widehat{G}\|.$$

*Put* $r = \max\{\|(\lambda + G)^{-1}\|, \|(\lambda + \widehat{G})^{-1}\|\}$ *and* $c = \max\{\|C\|, \|\widehat{C}\|\}$*. Then*

$$\|F_W - \widehat{F}_W\| \le \varepsilon_H + 2cr\varepsilon_C + c^2 r^2 \varepsilon_G =: \varepsilon_F, \tag{247}$$

$$\|T_{W,\lambda} - \widehat{T}_{W,\lambda}\| \le (\|F_W\| + \|\widehat{F}_W\|)\varepsilon_F. \tag{248}$$

*Here* $r \le \lambda^{-1}$ *is derived from positivity, so every finite-carrier quantity in the display is finite. On a continuum record, the same formulas are evaluated with its Part I form-to-resolvent error coordinates in* $[0, \infty]$*; a nonordinary comparison gives the valid value* $+\infty$ *rather than an admissibility restriction. The general nonnormal branch has the same formula with* $\varepsilon_G$ *replaced by the derived same-form-norm distance between* $(G - A_{\mathcal{Y}}) - (\widehat{G} - \widehat{A}_{\mathcal{Y}})$ *and with both complement resolvent norms retained.*

*Proof.* Add and subtract the two mixed self-energy terms and use $R - \widehat{R} = R(\widehat{G} - G)\widehat{R}$ for $R = (\lambda + G)^{-1}$. This proves (247). Apply the resolvent identity to $(I + F^*F)^{-1}$ and use $\|F^*F - \widehat{F}^*\widehat{F}\| \le (\|F\| + \|\widehat{F}\|)\|F - \widehat{F}\|$. □

### II.11.5 Statistically calibrated training objectives

The response kernel determines the regularizer, but it does not by itself specify how the block generator is identified or how a selectable bridge is trained. The three inherited authority/data cells therefore evaluate three objectives: a dynamics loss on target-free data, a response-kernel loss on supervised data, and, for controlled systems, a Bellman/policy loss on the represented deployment law. Their samples and confidence allocations are fields of the Part I record.

**Definition II.11.19** (Structure-preserving dynamics loss)**.** On each represented finite-dimensional chart supplied by the Part I learner, quotienting by its recorded carrier gauge gives identifiable coordinates. Let $Z_j = (x_j, y_j, \dot{x}_j, \dot{y}_j)$, $1 \le j \le n_{\mathrm{dyn}}$, be dynamics-only observations, and let $W_{x,j}$ and $W_{y,j}$ be represented positive precision matrices. For $\vartheta = (H, G, A_{\mathcal{Y}}, C)$ satisfying

$$H = H^* \succeq 0, \qquad G = G^* \succeq 0, \qquad A^*_{\mathcal{Y}} = -A_{\mathcal{Y}},$$

define

$$\mathcal{L}_{\mathrm{dyn}}(\vartheta) := \frac{1}{2n_{\mathrm{dyn}}} \sum_{j=1}^{n_{\mathrm{dyn}}} \left( \|\dot{x}_j + iHx_j - C^* y_j\|^2_{W_{x,j}} + \|\dot{y}_j + Cx_j + Gy_j - A_{\mathcal{Y}} y_j\|^2_{W_{y,j}} \right). \tag{249}$$

This is the generalized least-squares loss for the two block equations. On the inherited conditional-Gaussian channel cell with inverse covariance $W_{x,j} \oplus W_{y,j}$, it is also the negative log-likelihood up to a parameter-independent constant.

When derivatives are unavailable, raw transitions use the likelihood loss

$$\mathcal{L}_{\mathrm{tr}}(\vartheta) := \frac{1}{2n_{\mathrm{dyn}}} \sum_j \|u_{j+1} - e^{\Delta_j \mathcal{L}_W(\vartheta)} u_j\|^2_{\Omega_j^{-1}} + \frac{1}{2n_{\mathrm{dyn}}} \sum_j \log\det \Omega_j, \tag{250}$$

with the conditional mean and covariance replaced by the represented transition law when it is non-Gaussian. A discretization defect, derivative estimation error, or misspecified noise covariance is retained as a separate model coordinate. Neither loss uses target labels or rewards. The fitted parameters and $\varepsilon_{\mathrm{opt}}$ are inner training coordinates in the sense of Definition II.11.29. Comparison between complete records is deferred to the saturated profile.

**Proposition II.11.20** (Identification and response error)**.** *In the linear derivative chart of Definition II.11.19, write*

$$r_j(\vartheta_* + h) = \xi_j + \mathcal{D}_j h, \qquad Q_n = \frac{1}{n_{\mathrm{dyn}}} \sum_j \mathcal{D}_j^* W_j \mathcal{D}_j, \qquad s_n = \frac{1}{n_{\mathrm{dyn}}} \sum_j \mathcal{D}_j^* W_j \xi_j,$$

*where $W_j = W_{x,j} \oplus W_{y,j}$, and where $\vartheta_*$ is any represented comparator; its residual $\xi_j$ is retained, so realizability is not presumed. Define from the observed design*

$$\kappa_{\mathrm{id}} := \lambda_{\min}(Q_n), \qquad P_{\mathrm{id}} := \mathbf{1}_{(0,\infty)}(Q_n),$$

*and let $\kappa_{\mathrm{id}}^+$ be the least positive eigenvalue of $Q_n$, with $\kappa_{\mathrm{id}}^+ = +\infty$ when $Q_n = 0$. Every represented $\varepsilon_{\mathrm{opt}}$-minimizer compared with $\vartheta_*$ obeys*

$$\|P_{\mathrm{id}}(\widehat{\vartheta} - \vartheta_*)\| \le \frac{2\|s_n\|}{\kappa_{\mathrm{id}}^+} + \sqrt{\frac{2\varepsilon_{\mathrm{opt}}}{\kappa_{\mathrm{id}}^+}}. \tag{251}$$

*On the derived cell $\kappa_{\mathrm{id}} > 0$, $P_{\mathrm{id}} = I$ and this is the full-parameter bound. On the cell $\kappa_{\mathrm{id}} = 0$,* $\ker Q_n$ *is exactly the unidentifiable design/gauge direction and remains an obstruction coordinate; no excitation hypothesis is imposed.*

*One Part I score-concentration candidate is the conditional sub-Gaussian cell. On that recorded cell, in a $p_{\mathrm{id}}$-dimensional chart, the design certificate gives, with probability at least $1 - \delta_{\mathrm{id}}$,*

$$\|s_n\| \le c_{\mathrm{id}} \sigma_{\mathrm{id}} \sqrt{\frac{p_{\mathrm{id}} + \log(1/\delta_{\mathrm{id}})}{n_{\mathrm{dyn}}}}. \tag{252}$$

*The constant $c_{\mathrm{id}}$, the score norm, and the sampling/dependence certificate are fields of the record. Off that cell, the Part I minimum uses its mixing, martingale, or finite-description same-loss candidate; this is a recordwise radius coordinate, while the saturated profile compares records. An unavailable score radius equals $+\infty$. Componentwise application of Proposition II.11.18 converts* (251) *into certified bounds for $F_W$, $T_{W,\lambda}$, effective dimension, capture, and every operator-Lipschitz training coordinate.*

*Proof.* The loss difference is a quadratic with Hessian $Q_n$ and linear score $s_n$. Moreover $s_n \perp \ker Q_n$, because every vector in $\ker Q_n$ is killed by each weighted design map. Writing $h = \widehat{\vartheta} - \vartheta_*$, comparison with the represented comparator and the optimization tolerance gives $\langle h, Q_n h \rangle / 2 \le$

$\|s_n\|\|P_{\mathrm{id}}h\|+\varepsilon_{\mathrm{opt}}$. The spectral inequality $Q_n \succeq \kappa_{\mathrm{id}}^{+}P_{\mathrm{id}}$ and the quadratic formula imply (251). The kernel component does not enter the loss and is therefore exactly unidentifiable. The concentration display is the Part I sub-Gaussian score candidate on its named cell. The last assertion follows from (247)– (248) and the response-coordinate Lipschitz bounds in Corollary II.11.26. □

**Definition II.11.21** (Unrestricted-Caus spectral training loss)**.** For a legal bridge $C$, let

$$\mathcal{H}_{W,C} := \operatorname{ran} T_{W,\lambda}(C)^{1/2}, \qquad \|f\|_{W,C}^2 := \|T_{W,\lambda}(C)^{\dagger/2} f\|^2.$$

On a finite represented score space this norm is $\langle f, A_{W,\lambda}(C) f\rangle$. Given a target-training split $S_{\mathrm{tr}} = \{(X_i, Y_i)\}_{i=1}^{m_{\mathrm{tr}}}$, define the spectral kernel-ridge estimator

$$\widehat{f}_{C,\rho} := \operatorname*{arg\,min}_{f \in \mathcal{H}_{W,C}} \left\{ \frac{1}{2m_{\mathrm{tr}}} \sum_{i=1}^{m_{\mathrm{tr}}} |f(X_i) - Y_i|^2 + \frac{\rho}{2}\|f\|_{W,C}^2 \right\}. \tag{253}$$

Thus the correct spectral penalty is the closed quadratic form of the *inverse* response, equivalently $A_{W,\lambda}$ on its form domain, rather than $T_{W,\lambda}$ itself. On the ordinary Part I evaluation cell, the represented evaluation map is bounded on $\mathcal{H}_{W,C}$; the positive ridge term therefore makes (253) coercive in its response norm and gives its unique minimum. On a finite score carrier this is the positive-definite normal equation in (338). An unavailable evaluation map or nonordinary form retains its inherited value and makes the corresponding upper-risk candidate $+\infty$.

Let $\mathfrak{A}_{\mathrm{dyn}}$ be the dynamics confidence set obtained from an independent target-free split. For a designable control component it is the full declared bridge family; for a native or imposed component it is the identified confidence set. On an independent validation split define

$$\boxed{\widehat{\mathcal{J}}_{\mathrm{free}}(C,\rho) = \widehat{R}_{\mathrm{val}}(\widehat{f}_{C,\rho}) + \lambda_{\mathscr{W}} \mathscr{W}_{\mathfrak{A}}(C; D) + \mathrm{pen}_{\mathrm{sel}}(C,\rho) + \iota_{\mathfrak{A}_{\mathrm{dyn}}}(C),} \tag{254}$$

where $\iota$ is zero on the confidence set and $+\infty$ outside it. The selection penalty is a valid bridge/kernel selection deviation obtained from a split sample, finite union, PAC–Bayes mixture, or uniform cover. Operator estimation and optimization are recorded separately, so no radius is charged twice. Under unrestricted authority the zero-waste fiber

$$\mathfrak{A}_D^0 := \{C \in \mathfrak{A}_{\mathrm{dyn}} : J_C = J_D\} \tag{255}$$

is evaluated rather than presumed nonempty. It is nonempty exactly when the represented legal family realizes $J_D$. On that cell, exact least-action training minimizes (254) over $\mathfrak{A}_D^0$. On the empty cell, the distance (231) is the irreducible representation/authority gap. The penalized form trades statistical risk against physical action when such a trade is part of the declared task. The kernel-ridge fit and bridge search are inner candidate-generating optimizations. Their risk, action, cover charge, and optimization error enter the complete certificate; the saturated profile and its strict inverse select among the resulting structural records according to Definition II.11.29.

**Theorem II.11.22** (Oracle bound for unrestricted-Caus learning)**.** *Consider the Part I split-validation candidate of a complete record. Its fields are a charged finite $\varrho$-net $\mathfrak{A}_\varrho$ of the declared legal-bridge and ridge family $\mathfrak{A}_{\mathrm{dyn}} \times \mathcal{P}$, of cardinality $N_\varrho$, the recorded loss range $L_{\mathrm{loss}}$, the independent validation carrier of size $m_{\mathrm{val}}$, and the simultaneous fixed-kernel oracle values*

$$\mathcal{B}_C(\rho) := C_{\mathrm{var}} \frac{\sigma^2 N_W^C(\lambda,\rho)}{m_{\mathrm{tr}}} + C_{\mathrm{bias}} B_C(\rho; f_*) + \varepsilon_{\mathrm{task}} + L_{\mathrm{op}} \varepsilon_T(C), \tag{256}$$

*where $N_W^C$ has the extended meaning of* (236)*. Thus $\mathcal{B}_C(\rho)$ is a well-defined value in $[0,\infty]$; a divergent mode count removes this candidate from the recordwise finite oracle minimum rather than excluding the model from the framework. The bias term is*

$$B_C(\rho; f_*) = \|[I - T_C(T_C + \rho I)^{-1}]f_*\|_{L^2}^2.$$

*Let*

$$r_{\mathrm{val}} := L_{\mathrm{loss}}\sqrt{\frac{\log(2N_\varrho/\delta_{\mathrm{val}})}{2m_{\mathrm{val}}}}. \tag{257}$$

*Take* $\mathrm{pen}_{\mathrm{sel}} = r_{\mathrm{val}}$ *on this net; a candidate-dependent valid radius gives the analogous oracle bound with that radius inside the infimum. Let $L_{\mathrm{sel}} \in [0,\infty]$ be the Part I net-transport modulus of candidate risk plus action, and let $\varepsilon_{\mathrm{out}} \in [0,\infty]$ be the recorded optimization gap of the returned pair $(\widehat{C}, \widehat{\rho})$. On the candidate's simultaneous event,*

$$\begin{aligned} &R(\widehat{f}_{\widehat{C},\widehat{\rho}}) - R(f_*) + \lambda_{\mathscr{W}}\mathscr{W}_{\mathfrak{A}}(\widehat{C}; D) \\ &\quad\le \inf_{C\in\mathfrak{A}_{\mathrm{dyn}},\,\rho\in\mathcal{P}} \{\mathcal{B}_C(\rho) + \lambda_{\mathscr{W}}\mathscr{W}_{\mathfrak{A}}(C; D)\} + 2r_{\mathrm{val}} + L_{\mathrm{sel}}\varrho + \varepsilon_{\mathrm{out}}. \end{aligned} \tag{258}$$

*An absent split, loss-range, simultaneous-oracle, or finite net-transport field sets this candidate to $+\infty$; the inherited Rademacher and PAC–Bayes candidates remain in the Part I same-loss certificate minimum. The choice $\mathcal{P} = (0,\infty)$ is permitted only on a Part I record carrying a charged cover and transport modulus for that continuum; otherwise $\mathcal{P}$ is exactly its represented finite ridge grid. That minimum supplies the risk coordinate subsequently restricted in the saturated profile. On the cell $\mathfrak{A}_D^0 \neq \varnothing$, restriction to that fiber removes the action term. Within its named eigenvalue/source rate cell $\eta_j^W(C) \asymp j^{-2\beta_C}$ and $f_* = T_C^{r_C} g_C$, and whose charged ridge family contains the balancing scale or a net point with the displayed transport error, optimization of $\rho$ gives the certified inner rate $m_{\mathrm{tr}}^{-4\beta_C r_C/(4\beta_C r_C+1)}$ at that fixed record, plus the displayed bridge-selection and operator errors. Outside that rate cell,* (258) *remains the quantitative conclusion.*

*Proof.* Hoeffding's inequality and a union bound give a validation deviation at most $r_{\mathrm{val}}$ for every net point. Insert the validation risk on both sides of the approximate-minimizer inequality. The two deviations, net-transport term, and optimization error give (258); then substitute the simultaneous candidate oracle certificates. The zero-waste statement is Theorem II.11.12. The rate follows by balancing $\rho^{2r_C}$ with $m_{\mathrm{tr}}^{-1}\rho^{-1/(2\beta_C)}$. □

**Definition II.11.23** (Action-safe Wick actor–critic loss)**.** Let $P_{\mathcal{A}}$ project onto the represented legal-action span and let $P_{\mathrm{safe}}$ project onto its failure-killed safe subspace. Define

$$T_C^{\mathcal{A},\mathrm{safe}} := P_{\mathrm{safe}}P_{\mathcal{A}}T_{W,\lambda}(C)P_{\mathcal{A}}P_{\mathrm{safe}}|_{\mathrm{ran}\,P_{\mathrm{safe}}}, \qquad N_C^{\mathcal{A},\mathrm{safe}}(\rho) := \mathrm{Tr}_{\mathrm{ext}}\left[T_C^{\mathcal{A},\mathrm{safe}}(T_C^{\mathcal{A},\mathrm{safe}} + \rho I)^{-1}\right]. \tag{259}$$

The trace has the extended Part I meaning of (236); every empirical critic calculation uses its finite evaluation compression from (234). The Part I control record supplies a real-valued observable-score map. Thus every critic $q$ below ranges over its inherited real score slice inside the complexified response space. Hermitian response energies are restricted to that slice, while maxima, exponentials, rewards, and policy probabilities are real-valued. This is the existing control-score type, not an additional regularity condition on the Wick response. For an executed bridge action $a \mapsto C(h,a)$, put

$$w_D(h,a) := \frac{1}{2}\|J_{C(h,a)} - J_D(h)\|_{\mathbb{M}^{-1}}^2.$$

On the action-charged authority cell, its stage reward is $r_\alpha(h,a) = r(h,a) - \alpha w_D(h,a)$. On the hard-action-constraint cell, the legal action set carries that constraint and $\alpha = 0$; the same action is not charged a second time.

At fitted Bellman iteration $k$, freeze a copy $q_k^-$ of the preceding critic. A delayed target network retains the lag $\varepsilon_{\mathrm{lag},k} := \|q_k^- - \widehat{q}_k\|_\infty$. Set

$$Y_i^{(k)} = r_\alpha(h_i, a_i) + \chi_i^{\mathrm{cont}} \gamma \max_{a' \in \mathcal{A}_{\mathrm{safe}}(h_i')} q_k^-(h_i', a'). \tag{260}$$

Here $\chi_i^{\mathrm{cont}} \in \{0,1\}$ is the terminal-continuation field of the complete Part I dynamical record. Before taking the maximum, that record either certifies a nonempty safe successor row or returns its existing authority/terminal status; no maximum over an empty set is inserted into the loss. For represented behavior-to-deployment weights $0 \le \omega_i \le \bar{\omega}$, the critic loss is

$$\mathcal{L}_{\mathrm{crit}}^{(k)}(q; C) := \frac{1}{2m_k} \sum_{i=1}^{m_k} \omega_i (q(h_i, a_i) - Y_i^{(k)})^2 + \frac{\rho}{2} \|(T_C^{\mathcal{A},\mathrm{safe}})^{\dagger/2} q\|^2. \tag{261}$$

Thus critic regularization again uses the inverse positive response. The frozen target makes (261) a supervised regression loss and avoids the double-sampling defect of a same-sample squared Bellman residual.

For the neutral action row $\nu(a \mid h)$ carried by the Part I control record, define on its represented support

$$\begin{aligned} \pi_q^\tau(a \mid h) &= \frac{\nu(a \mid h) e^{q(h,a)/\tau}}{\sum_{a' \in \mathcal{A}_{\mathrm{safe}}(h)} \nu(a' \mid h) e^{q(h,a')/\tau}}, \quad a \in \mathcal{A}_{\mathrm{safe}}(h), \\ \mathcal{L}_{\mathrm{actor}}(\pi; q) &= \mathbb{E}_{d_{\mathrm{dep}}} D_{\mathrm{KL}}(\pi(\cdot \mid h) \| \pi_q^\tau(\cdot \mid h)). \end{aligned} \tag{262}$$

The policy is zero outside $\mathcal{A}_{\mathrm{safe}}(h)$. The displayed denominator is positive on the ordinary Part I support cell. If the neutral row has zero total safe mass, the Gibbs row and actor certificate take their typed support-obstruction values. A zero mass on any required safe action is retained by the support coordinate $\nu_{\min}$ in Theorem II.11.24. The canonical alternating objective is

$$\boxed{\mathcal{L}_{W,\mathrm{RL}}^{(k)}(q, \pi, C) = \mathcal{L}_{\mathrm{crit}}^{(k)}(q; C) + \lambda_\pi \mathcal{L}_{\mathrm{actor}}(\pi; q) + \lambda_{\mathrm{dyn}} \mathcal{L}_{\mathrm{dyn}}(\vartheta) + \lambda_{\mathrm{num}} \mathcal{L}_{\mathrm{num}}(q, \pi, C).} \tag{263}$$

The critic is fitted first with $q_k^-$ fixed; the actor is then fitted to the new critic. Importance, occupancy, filter, numerical, bridge-selection, and target-network costs belong to the complete RL record. The critic, actor, and bridge updates are inner optimization transcripts. Their completed record enters the saturated profile, which supplies the structural order across policies, responses, and authority cells.

**Theorem II.11.24** (RL critic and policy calibration)**.** *Work on the inherited discounted or contractive Part I control cell, with recorded modulus $0 \le \gamma < 1$. A finite-horizon or noncontractive control record retains its own Part I dynamic-programming candidate and does not use the denominators below. For each fitted critic step, let $e_k^2 \in [0, \infty]$ be the recordwise minimum of the applicable simultaneous Part I critic-oracle candidates. It is the critic-radius coordinate of that fixed RL record; structural comparison uses the saturated profile. Its Wick effective-dimension candidate is*

$$\begin{aligned} \mathcal{E}_{k,\mathrm{KRR}} := &\frac{C_{\mathrm{var}} \sigma_{\mathrm{TD},k}^2 N_{C_k}^{\mathcal{A},\mathrm{safe}}(\rho_k)}{m_k} + C_{\mathrm{bias}} B_{C_k}^{\mathcal{A},\mathrm{safe}}(\rho_k) \\ &+ \varepsilon_{\mathrm{shift},k} + \varepsilon_{\mathrm{model},k} + \varepsilon_{\mathrm{sel},k} + \varepsilon_{\mathrm{opt},k}, \qquad e_k^2 \le \mathcal{E}_{k,\mathrm{KRR}}. \end{aligned} \tag{264}$$

*where the safe source residual includes $\|(I - P_{\mathrm{safe}})Q_*\|^2$. An unavailable candidate has value $+\infty$, so this definition excludes no record.*

*Let $\mathcal{K}_{\infty,k}$ be the set of same-law conversion constants certified by the Part I record and define the totalized conversion error*

$$\varepsilon_{\infty,k} := \inf_{\kappa \in \mathcal{K}_{\infty,k}} \kappa e_k \in [0, \infty], \qquad \inf \varnothing := +\infty. \tag{265}$$

*Every ordinary member of $\mathcal{K}_{\infty,k}$ certifies $\|\widehat{q}_{k+1} - \mathcal{T}_{B,\alpha} q_k^-\|_\infty \le \kappa e_k$. In the extended nonnegative reals, the restricted regularized Bellman target therefore satisfies*

$$\Gamma_{Q,K} := \|\widehat{q}_K - Q^*_{B,\alpha}\|_\infty \le \gamma^K \Gamma_{Q,0} + \sum_{k=0}^{K-1} \gamma^{K-1-k} (\varepsilon_{\infty,k} + \gamma \varepsilon_{\mathrm{lag},k}). \tag{266}$$

*Define two further derived record coordinates*

$$\nu_{\min} := \inf_{h, a \in \mathcal{A}_{\mathrm{safe}}(h)} \nu(a \mid h), \qquad \kappa_\pi := \sup_h D_{\mathrm{KL}}(\widehat{\pi}(\cdot \mid h) \| \pi^\tau_{\widehat{q}_K}(\cdot \mid h)) \in [0, \infty],$$

*and use* $\log(1/0) := +\infty$. *The actor's extended one-step action certificate is*

$$d_{\mathrm{act}} := 2\Gamma_{Q,K} + \tau \log \frac{1}{\nu_{\min}} + \mathrm{Osc}(\widehat{q}_K) \sqrt{\frac{\kappa_\pi}{2}}, \tag{267}$$

*and*

$$\|V^*_{B,\alpha} - V^{\widehat{\pi}}_\alpha\|_\infty \le \frac{d_{\mathrm{act}}}{1 - \gamma}. \tag{268}$$

*These displays are finite exactly on the Part I same-law/support/actor cells that make all their coordinates finite. The value for the unrestricted control problem additionally includes the represented budget, safe-action, and observation/partial-information gaps. When the same-law conversion set is empty, $\varepsilon_{\infty,k} = +\infty$: the $L^2(d_{\mathrm{dep}})$ critic certificate remains valid, whereas the sup-norm Bellman and policy certificates take the honest value $+\infty$.*

*Proof.* The fixed-target critic is kernel ridge regression with kernel $T_C^{\mathcal{A},\mathrm{safe}}$, so its effective-dimension oracle certificate is (264). Apply the converter, insert the target-network lag, and iterate the $\gamma$-contraction to obtain (266). The Gibbs variational identity bounds its soft-greedy defect by $\tau \log(1/\nu_{\min})$; Pinsker's inequality gives the final actor term in (267), and replacement of the true critic by $\widehat{q}_K$ costs $2\Gamma_{Q,K}$. The discounted performance-difference identity sums the uniform one-step defect and gives (268). □

**Corollary II.11.25** (Action-safe RL rate cell)**.** *On the named Part I rate cell in which the nonzero eigenvalues of $T_C^{\mathcal{A},\mathrm{safe}}$ obey $\eta_j \asymp j^{-2\beta}$, $\beta > 1/2$, the Bellman-optimal critic of that fixed RL record has source order $0 < r \le 1$, the safe approximation residual is zero, the charged ridge family contains the balancing scale or a net point with no larger transport error, and the shift, model, selection, and optimization coordinates have no larger order, the general oracle value specializes to*

$$e_k^2 = O\Big(m_k^{-4\beta r/(4\beta r+1)}\Big), \qquad e_k = O\Big(m_k^{-2\beta r/(4\beta r+1)}\Big). \tag{269}$$

*On the subcell carrying uniformly bounded conversion constants $\kappa_{\infty,k} \in \mathcal{K}_{\infty,k}$, the fitted-iteration residual has converged, $\tau \to 0$, and the derived $\kappa_\pi \to 0$, the policy gap has order*

$$O\bigg(\frac{m^{-2\beta r/(4\beta r+1)}}{(1-\gamma)^2}\bigg), \tag{270}$$

*in addition to the exact budget, safe-action, and observation approximation gaps. Every complementary cell retains the extended oracle* (266)–(268)*; it is not excluded. A direct sup-norm critic oracle may replace the square-root conversion and yields its own certified rate. Bellman optimality here is an inner control coordinate of the fixed RL record; the risk-qualified saturated profile supplies the structural optimum across RL records.*

**Corollary II.11.26** (Specialized quantitative ledger)**.** *The Part I confidence allocator constructs one simultaneous event for every ordinary candidate below. On that event the following bounds specialize to a Wick certificate as indicated; every nonordinary candidate is assigned $+\infty$ and remains in the record.*

(i) *On each nonempty represented Part I stage, let $\{\mathcal{H}^W_\alpha\}_{\alpha\in\mathfrak{A}^{\mathrm{src}}_W}$ be its exact source cells. For the recorded $[-1,1]$-loss candidate, source selection obeys*

$$\widehat{\mathfrak{R}}_S\left(\bigcup_\alpha \mathcal{H}^W_\alpha\right) \le \max_\alpha \widehat{\mathfrak{R}}_S(\mathcal{H}^W_\alpha) + \sqrt{\frac{2\log|\mathfrak{A}^{\mathrm{src}}_W|}{m}}. \tag{271}$$

*For a cell-mixture prior and posterior,*

$$\mathrm{KL}(Q\|P) = \mathrm{KL}(q\|w) + \sum_\alpha q_\alpha \, \mathrm{KL}(Q_\alpha\|P_\alpha). \tag{272}$$

*Thus the selection of a Wick source, bridge family, or authority cell is charged once by a union, mixture, split-sample, or uniform-cover certificate.*

(ii) *For every fixed bridge and ridge scale, the complete kernel-ridge oracle candidate is*

$$\|\widehat{f}_{C,\rho} - f_*\|^2_{L^2} \le C_{\mathrm{var}} \frac{\sigma^2 N^C_W(\lambda,\rho)}{m} + C_{\mathrm{bias}} B_C(\rho; f_*). \tag{273}$$

*The source rate* (246)*, a minimax lower rate, or a condition-number lower capacity bound transfers only on its declared eigenvalue/source class. In particular, on an $m$-point score space,*

$$\mathrm{TaskCap}_{A_W}(y) \ge \frac{\|y\|_2}{\sqrt{m\,\kappa(A_W)}}. \tag{274}$$

*Dark, target-null, and authority-excluded components remain exact approximation floors rather than statistical terms.*

(iii) *On the inherited moving-bridge/feature cell, a terminal kernel $T_{\widehat{\theta}}$ selected from its charged $\varrho$-net has feature-learning certificate*

$$\begin{aligned}\mathrm{ExcessRisk} &\le C_{\mathrm{var}} \frac{\sigma^2 N^{\widehat{\theta}}_W(\lambda,\rho)}{m} + C_{\mathrm{bias}} B_{\widehat{\theta}}(\rho; f_*) \\ &\quad + r_{\mathrm{sel}} + L_{\mathrm{sel}}\varrho + \varepsilon_{\mathrm{est}} + \varepsilon_{\mathrm{opt}},\end{aligned} \tag{275}$$

*where, for an independent bounded-loss audit of size $m_a$, one may take*

$$r_{\mathrm{sel}} = L_{\mathrm{loss}} \sqrt{\frac{\log(2\mathcal{N}(\Theta,\varrho)/\delta)}{2m_a}}.$$

*The fixed-response/lazy regime is the specialization with no kernel movement; a neural or nonlinear readout adds its own represented margin or covering certificate and is combined only in common risk units.*

(iv) *A learned family* $\{T_{W,\lambda}(\theta)\}$ *uses* (248) *in the learned-operator covering bound. Since* $A_{W,\lambda} \geq I$, *its selection radius is*

$$\inf_{\varrho>0}\left[\frac{R}{m}\sqrt{\sup_{\theta}\operatorname{Tr}_S T_\theta}+R\sqrt{\frac{L_T\varrho}{m}}+\frac{R}{m}\sqrt{2\log\mathcal{N}(\Theta,\varrho)}\right]. \tag{276}$$

*Here* $\varrho$ *is the charged cover resolution, not the ridge scale* $\rho$.

(v) *On the represented positive defect/bracket range* $R_{E,W}$, *the proper Gaussian structural prior has precision* $\mathbf{M}_{E,W} = P_{E,W}A_{W,\lambda}P_{E,W}|_{R_{E,W}}$. *Its PAC–Bayes radius is*

$$\sqrt{\frac{\mathrm{KL}^{\mathrm{str}}_{E,W}(Q)+\log(2\sqrt{m}/\delta)}{2(m-1)}}. \tag{277}$$

*This candidate is ordinary for* $m \geq 2$*; at* $m < 2$ *it equals* $+\infty$. *Null and dark directions retained by the model receive a separate proper label-independent prior and KL charge.*

(vi) *The Part I resistance-query candidate uses* $\langle E_{W,\Delta}, Q^{+}_{K,S}E_{W,\Delta}\rangle$. *Rayleigh monotonicity gives monotonicity under grounding. The existing submodularity coordinate either certifies the* $1 - 1/e$ *greedy batch factor or assigns that factor-candidate* $+\infty$*; the resistance query itself remains available.*

(vii) *Put* $\varepsilon_T := \|\widehat{T} - T\|$, *let* $g_* \geq 0$ *be the derived target-band separation, and define*

$$d_{\mathrm{DK}} := \begin{cases} 2\varepsilon_T/g_*, & g_* > 2\varepsilon_T, \\ 1, & g_* \leq 2\varepsilon_T. \end{cases}$$

*Davis–Kahan gives the first value, while the universal distance bound for orthogonal projectors gives the second [3]. Thus the projector discrepancy is always bounded by* $d_{\mathrm{DK}}$, *without a spectral-gap assumption. Smoothed bands use the recordwise minimum of this value and their represented operator-Lipschitz value. The result propagates to capture, effective dimension, action-safe capacity, and every declared Lipschitz certificate coordinate.*

(viii) *On the homogeneous symmetric label-noise cell* $\eta < 1/2$, *every clean-risk radius for a flip-symmetric bounded loss is divided by* $\chi_\eta := 1 - 2\eta$. *For squared loss the debiased Wick estimator satisfies*

$$\|\widehat{f}_{C,\rho} - f_*\|^2_{L^2} \leq C_{\mathrm{var}}\frac{v^2_\eta N^C_W(\lambda,\rho)}{\chi^2_\eta m} + C_{\mathrm{bias}}B_C(\rho; f_*). \tag{278}$$

*Wick source ancestry and operator-selection costs are unchanged. General reward, transition, action, and observation corruption enters through its typed same-norm channel converter, not through the binary factor.*

(ix) *On the Part I stationary* $\beta$*-mixing cell of length* $N$, *block length* $a$, *and* $\mu_a = \lfloor N/(2a)\rfloor$, *put* $\delta_a = \delta - 2(\mu_a - 1)\beta(a)$. *For every block length with* $\mu_a \geq 1$ *and* $\delta_a > 0$, *the represented block response kernel* $T_{W,a}$ *gives*

$$|R(f) - \widehat{R}_N(f)| \leq \frac{2R_A}{\mu_a}\sqrt{\operatorname{Tr}_{\mathrm{blk(a)}} T_{W,a}} + 3\sqrt{\frac{\log(4/\delta_a)}{2\mu_a}} + \frac{2a}{N}. \tag{279}$$

*Here* $\operatorname{Tr}_{\mathrm{blk(a)}}$ *is the finite compression to the* $\mu_a$*-block evaluation carrier. Block lengths with* $\mu_a = 0$ *or* $\delta_a \leq 0$ *have value* $+\infty$, *so the recordwise minimum over all represented* *block lengths is a valid bound coordinate. For a positive represented horizon $T$, the Part I predictable-path cell uses*

$$\mathfrak{R}_T^{\mathrm{seq}} \le R_A \mathfrak{q}_T(A_W) \le \frac{R_A}{T}\sqrt{\operatorname{Tr}_{\mathrm{seq}} T_{W,\lambda}} \tag{280}$$

*where $\operatorname{Tr}_{\mathrm{seq}}$ is the Part I supremum of the finite $T$-step path/tree compression traces. For a fixed response this gives the displayed bound; a selectable response uses the learned-operator sequential cover, and the Bayesian branch uses the martingale PAC–Bayes term*

$$\frac{\mathrm{KL}^{\mathrm{str}}_{E,W}(Q) + \log(1/\delta)}{\xi T} + \frac{\xi}{8}. \tag{281}$$

*for its represented $\xi > 0$; other coefficient values are nonordinary. The online regret bound is $2L_\ell T \sup_{\mathbf{z}} \mathfrak{R}_T^{\mathrm{seq}}$. Each complementary dependence cell uses its own Part I candidate or the value $+\infty$; a dark space alone supplies no dependence coordinate.*

(x) *Define the acquisition coordinate $\Gamma_{\mathrm{acq}} := \sup_{h,a} |\widehat{g} - g| \in [0,\infty]$. The recorded $\varepsilon_{\mathrm{opt}}$-maximizer loses at most*

$$\sup_a g(h,a) - g(h,\widehat{a}(h)) \le 2\Gamma_{\mathrm{acq}} + \varepsilon_{\mathrm{opt}}. \tag{282}$$

*This maximization is the operational action choice within a fixed acquisition record. Its error and action cost enter that record, after which the saturated profile compares complete acquisition structures. Over a horizon $H$, the action-value loss is the survival-weighted sum of these stepwise terms. Resistance-greedy selection obtains the $1 - 1/e$ batch factor on the positive submodularity-certificate cell in the earlier querying item; on its complement that factor-candidate is $+\infty$.*

(xi) *For RL authority, compress $T_{W,\lambda}$ first to the represented action span and then to the safe subspace. Effective-dimension monotonicity is unconditional. On the named Part I safe source-order cell, the exact source residual has the displayed upper bound:*

$$N_W^{\mathcal{A},\mathrm{safe}}(\rho) \le N_W^{\mathcal{A}}(\rho) \le N_W(\lambda,\rho), \tag{283}$$

$$B_W^{\mathcal{A},\mathrm{safe}}(\rho; f_*) \le C_r \|g_{\mathrm{safe}}\|^2 \rho^{2r} + \|(I - P_{\mathrm{safe}})f_*\|^2. \tag{284}$$

*The critic and actor objectives are (261) and (262). Define the extended coordinates $\varepsilon_M$ as the same-norm model defect and $\varepsilon_B$ as the learned Bellman residual. On the Part I discounted Bellman/contraction cell, the Bellman candidate is*

$$\|V^* - v\|_\infty \le \frac{\varepsilon_B + \varepsilon_M}{1-\gamma}, \tag{285}$$

$$\|V^* - V^{\widehat{\pi}_v}\|_\infty \le \frac{2\gamma(\varepsilon_B + \varepsilon_M)}{(1-\gamma)^2} + \frac{2\varepsilon_M}{1-\gamma}. \tag{286}$$

*Here $0 < \gamma < 1$ is the declared discount or contraction modulus. The distance from $J_D$ to the legal policy-current family is the irreducible authority/action gap; it cannot be removed by additional value samples. The complementary control-status cell assigns these Bellman candidates $+\infty$ and retains its typed control defect.*

(xii) *For a complete target-qualified Wick record, source-resolved statistical, information, structural, and dynamical candidates combine only after conversion to the same deployed target advantage $g_W$:*

$$g_W \le \min\left\{1,\ U_W^{\mathrm{stat}\to\mathrm{tgt}},\ \beta_W^+ + \sqrt{\frac{\log 2}{2} J_W^{\mathrm{src}}},\ eH_B \sum_a \Lambda_W^{(a)},\ U_W^{\mathrm{dyn}\to\mathrm{tgt}}\right\}. \tag{287}$$

*Here $\Lambda_W^{(a)}$ is the contextual charge assigned to source a; it is distinct from the generator $\mathcal{L}_W$. For a finite sample-independent output codebook of logarithmic size $L_{\text{out}}$, the fallback deviation candidate is*

$$2\sqrt{\frac{2L_{\text{out}}}{m_{\text{eff}}}} + 3\sqrt{\frac{\log(2/\delta_{\text{eff}})}{2m_{\text{eff}}}} + \varepsilon_{\text{dep}}. \tag{288}$$

(xiii) *Galerkin or finite-element records use* (247) *and* (248) *after their form-to-resolvent error has been certified. Every finite Galerkin carrier uses its finite Part I trace. Along a represented nested dense family, the population effective dimension is the monotone extended value in* (238)*. Operator approximation errors on the $d_n$-dimensional carrier are charged by* (290)*. A represented tail estimate may supply a finite upper envelope; without one, the upper value is $+\infty$. No compactness or trace-class premise is inserted.*

(xiv) *On every finite empirical band, define $\varepsilon_H := \|\widehat{H} - H\|$, $\varepsilon_C := \|\widehat{C} - C\|$, its derived separation $g_\Delta \geq 0$, and*

$$d_\Delta := \begin{cases} 2\varepsilon_H / g_\Delta, & g_\Delta > 2\varepsilon_H, \\ 1, & g_\Delta \leq 2\varepsilon_H. \end{cases}$$

*With $c = \max\{\|C\|, \|\widehat{C}\|\}$,*

$$\big|\|CP_\Delta\| - \|\widehat{C}\widehat{P}_\Delta\|\big| \leq \varepsilon_C + c d_\Delta. \tag{289}$$

*Thus zero is excluded from this interval precisely when the data certify a nonzero quasi-lumpability defect. On every common certified reduced subspace, define $\varepsilon_Q := \|\widehat{Q}_K - Q_K\|$; Weyl's inequality bounds the reduced-gap discrepancy by $\varepsilon_Q$. In quotient dimension $d_Q$, putting $\varepsilon_T = \|\widehat{T} - T\|$ gives*

$$|\widehat{N}_W(\lambda, \rho) - N_W(\lambda, \rho)| \leq \frac{d_Q}{\rho} \varepsilon_T, \tag{290}$$

$$|\widehat{m}_W(\lambda, \rho; f) - m_W(\lambda, \rho; f)| \leq \frac{2}{\rho} \varepsilon_T \tag{291}$$

*for fixed normalized $f$. Data-dependent readouts add their separately certified sample-splitting or uniform-selection radius.*

*Each item retains the approximation floor, target-readout miss, source-cell selection cost, model-channel perturbations, confidence allocation, and deployment conversion of its Part I antecedent. An absent antecedent is the nonordinary value $+\infty$, never an omitted record or hidden assumption.*

### II.11.6 Part I model choice and the saturated certified frontier

No new simplicity functional is introduced. The model family, selection penalties, and comparison order are inherited from Part I. Source choice is charged by its exact-cell Rademacher union term or by the cell-mixture PAC–Bayes divergence. A label-dependent operator choice is charged by the learned-operator cover, and coefficient selection is charged by the declared finite coefficient family. At fixed geometry, readout size is chosen by the Part I target-subspace coverage criterion. Across complete records, simplicity is the Pareto order of the existing saturated resource and action vector $\mathbf{c}$, with optimality interpreted throughout by Definition II.11.29.

| Part I quantitative family | Wick specialization | Record field / nonordinary output |
|---|---|---|
| Identification and operator stability | (251), (247), (248) | quotient gauge; derived design range/kernel; extended same-norm error |
| Fixed supervised learning | (244), (273), (246) | positive response is derived; named source/eigenvalue cell or general oracle |
| Bridge or feature selection | (258), (275), (276) | inherited split/cover candidate; absent candidate equals $+\infty$ |
| Bayesian and noisy learning | (277), (278), (272) | inherited prior/channel cell; absent candidate equals $+\infty$ |
| Dependent and online learning | (279), (280), (281) | inherited dependence cell; other cells use their own candidate |
| Active learning | resistance querying and (282) | extended resistance; submodularity coordinate controls only batch factor |
| Controlled/RL learning | (264)– (268), (285)–(286) | safe/action and converter coordinates, each totalized in $[0, \infty]$ |
| Saturated-profile working frontier | (293), (310), (313) | risk-restricted saturated profile and strict Pareto inverse; inner minima remain record coordinates; empty feasible set is typed |
| Deployment and continuum | (287), empirical intervals, resolvent/trace transfer | extended converter, allocated simultaneous event, finite-carrier or extended-trace accounting |

Table 6: Exhaustive location of the quantitative learning families used by the Wick specialization. Every row is evaluated on one complete selected generator. A dependent upper-error or risk field returns $+\infty$, a dependent lower-usefulness field returns zero, and every other unavailable coordinate retains its typed nonordinary value; none is replaced by an omitted theorem or numerical surrogate.

**Proposition II.11.27** (Inner minimum-response-energy coordinate)**.** *Let $R_C : \mathcal{Z}_C \to \mathcal{Y}_C$ be a bounded operator between Hilbert spaces and let $T_C = R_C R_C^*$. Define the totalized response energy*

$$\mathcal{E}_{T_C}(f) := \begin{cases} \|T_C^{\dagger/2} f\|^2, & f \in \operatorname{ran} T_C^{1/2}, \\ +\infty, & f \notin \operatorname{ran} T_C^{1/2}. \end{cases} \tag{292}$$

*Then* $\operatorname{ran} T_C^{1/2} = \operatorname{ran} R_C$ *and, for every* $f \in \mathcal{Y}_C$*,*

$$\mathcal{E}_{T_C}(f) = \inf_{z \in \mathcal{Z}_C :\, R_C z = f} \|z\|^2_{\mathcal{Z}_C}, \qquad \inf \varnothing := +\infty. \tag{293}$$

*For* $f \in \operatorname{ran} R_C$*, the infimum is attained uniquely in* $(\ker R_C)^\perp$*, at*

$$z_f = R_C^\dagger f. \tag{294}$$

*In particular, take $R_C = T_{W,\lambda}(C)^{1/2}$ on its represented support. Then the inverse-response penalty in* (253) *is exactly the least squared response effort among all latent lifts producing $f$. The same statement holds with $T_C^{\mathcal{A},\mathrm{safe}}$ for the RL critic. Nonclosed continuum ranges are included: unattainable targets receive $+\infty$, while every attainable target has the minimum lift above. This minimum is the response-energy coordinate of a fixed complete record. It enters the resource/action ledger used by the saturated profile and does not compare different response kernels or source cells.*

*Proof.* The polar decomposition $R_C = T_C^{1/2} V_C$ gives the operator-range identity $\operatorname{ran} R_C = \operatorname{ran} T_C^{1/2}$. For an attainable $f$, the affine fiber $R_C^{-1}\{f\}$ is a nonempty closed set of the form $z_f + \ker R_C$. Its unique element perpendicular to $\ker R_C$ is the Moore–Penrose solution $z_f = R_C^\dagger f$. Functional calculus and the polar decomposition give $\|z_f\| = \|T_C^{\dagger/2} f\|$. Pythagoras then yields $\|z_f + k\|^2 = \|z_f\|^2 + \|k\|^2$ for every $k \in \ker R_C$. For an unattainable $f$, both sides of (293) equal $+\infty$ by definition. ☐

**Definition II.11.28** (Inherited Wick family-selection certificate)**.** Fix a resource–action cell $(B, a)$. Let $\mathfrak{A}_W$ be the exact Part I source cells represented in that cell. For $\alpha \in \mathfrak{A}_W$, let $\Theta_\alpha$ be its declared finite-coverable Wick generator/bridge family, let $\Lambda$ be the declared finite set of training-coefficient vectors, let $\mathfrak{K}$ be the finite capacity grid used in Part I, and let $\mathcal{H}^W_{\alpha,\theta,\lambda,K}$ be the resulting readout or policy-loss class at capacity $K$. Let $\mathfrak{I}_W \subseteq \mathfrak{A}_W \times \Lambda \times \mathfrak{K}$ be the represented family indices. Put

$$\mathcal{H}_W(B, a) := \bigcup_{(\alpha,\lambda,K) \in \mathfrak{I}_W} \ \bigcup_{\theta \in \Theta_\alpha} \mathcal{H}^W_{\alpha,\theta,\lambda,K}. \tag{295}$$

When $\mathfrak{I}_W = \varnothing$, the union is empty, every upper certificate below is defined to be $+\infty$, and the learner returns the typed source/authority/budget obstruction. The remaining formulas are evaluated on each nonempty finite Part I stage; this is a case distinction, not a restriction on the underlying family. Every union index, net resolution, terminal kernel, coefficient vector, training transcript, and deployment map is part of the complete record. Equations (297) and (299) are respectively the simultaneous specializations of Eqs. (271) and (276) and Eqs. (272) and (277).

For the Rademacher branch, define the within-family learned-response radius

$$\mathfrak{r}^W_{\alpha,\lambda,K} := \inf_{\varrho > 0} \left[ \frac{R}{m} \sqrt{\sup_{\theta \in \Theta_\alpha} \operatorname{Tr}_S T_\theta} + R \sqrt{\frac{L_{T,\alpha}\varrho}{m}} + \frac{R}{m} \sqrt{2 \log \mathcal{N}(\Theta_\alpha, \varrho)} \right] \tag{296}$$

The supremum and covering number are extended values; no attainment or compactness is used. The complete exact-family union radius is

$$\mathfrak{r}^W_{\mathrm{fam}} := \max_{(\alpha,\lambda,K)\in\mathfrak{I}_W} \mathfrak{r}^W_{\alpha,\lambda,K} + \sqrt{\frac{2\log|\mathfrak{I}_W|}{m}}. \tag{297}$$

Thus, for a $[0,1]$-valued $L_\ell$-Lipschitz loss, the inherited Rademacher upper certificate is

$$U^W_{\mathrm{Rad}}(h) := \widehat{R}_S(h) + 2L_\ell \mathfrak{r}^W_{\mathrm{fam}} + 3\sqrt{\frac{\log(2/\delta_{\mathrm{Rad}})}{2m}} + \varepsilon_{\mathrm{task}} + \varepsilon_{\mathrm{op}} + \varepsilon_{\mathrm{dep}}. \tag{298}$$

Here $L_{T,\alpha}$ is supplied by Proposition II.11.18; a fixed operator uses the fixed-response radius in (245).

For the Bayesian branch, let $\mathfrak{I}_{W,\mathrm{PB}}$ be the finite hierarchical index of exact source cell, represented operator-net point, and capacity. Use exactly the Part I mixture prior $P = \sum_{\iota\in\mathfrak{I}_{W,\mathrm{PB}}} w_\iota P_\iota$, fixed independently of the labeled sample, and posterior $Q_\lambda = \sum_\iota q_{\lambda,\iota} Q_{\lambda,\iota}$. Put

$$\mathrm{KL}_W(Q_\lambda \| P) := \mathrm{KL}(q_\lambda \| w) + \sum_{\iota\in\mathfrak{I}_{W,\mathrm{PB}}} q_{\lambda,\iota} \mathrm{KL}(Q_{\lambda,\iota} \| P_\iota), \tag{299}$$

$$U^W_{\mathrm{PB}}(Q_\lambda) := \widehat{R}_S(Q_\lambda) + \sqrt{\frac{\mathrm{KL}_W(Q_\lambda\|P) + \log(2\sqrt{m}|\Lambda|/\delta_{\mathrm{PB}})}{2(m-1)}} + \varepsilon_{\mathrm{task}} + \varepsilon_{\mathrm{op}} + \varepsilon_{\mathrm{dep}}. \tag{300}$$

The PAC–Bayes expression is an ordinary candidate for $m \geq 2$. At $m < 2$, or when its prior/posterior fields are unavailable, it has value $+\infty$ and does not affect the recordwise same-loss minimum. A label-dependent $\theta$-selection is either an index in the fixed hierarchical prior or is paid by the learned-operator cover; it is not treated as a label-independent geometry. A split-validation certificate such as (258) is an alternative candidate for the same selection step; its radius is not added again to a union or mixture candidate. Let $U_W(M)$ be the minimum of (298), (300), and the other applicable candidates in Corollary II.11.26, after all have been converted to the same population loss on one confidence event. An unavailable candidate has value $+\infty$. Let $\Gamma_W(M) \in [0,\infty]$ denote the smallest applicable two-sided Part I radius on that same event, including the identical family index, cover, channel, operator, and deployment charges. Thus $U_W(M) \leq \mathcal{R}(M) + 2\Gamma_W(M)$ in the extended reals; absence of a two-sided comparator certificate is exactly $\Gamma_W(M) = +\infty$. Both minima are recordwise statistical certificate coordinates. They implement the same-loss Rademacher/PAC–Bayes choice of Part I; the saturated profile below, rather than either scalar radius, orders complete records.

**Definition II.11.29** (Saturated-profile semantics of Wick optimality)**.** The terms *optimal*, *minimal*, and *simplest* retain the Part I hierarchy. An optimization performed with a complete record fixed—including least-action projection, minimum response lift, ridge fitting, critic fitting, or minimum full-coverage readout—is an *inner optimization.* Its value, optimizer transcript, and optimization error are coordinates of that record. Only the saturated profile orders two different structural records.

Structural optimization is defined only by the inherited saturated profile, instantiated for Wick records in (314), and its Pareto-ordered strict inverse (315). Let $\ell$ denote a fixed Part I population-law, estimand, norm, and loss-scale identifier after every declared converter has been applied. Let $\mathscr{R}^{(n)}_{W,\ell}(B,a)$ denote that typed partition of the charged finite stage produced by the Part I learner, and suppress $\ell$ from the notation below by writing $\mathscr{R}^{(n)}_W(B,a)$. Records with different identifiers define separate profiles and are never compared through a statistical minimum. Let $\underline{V}_W(M)$ be the simultaneous Part I lower confidence bound for $V_M^W$. At requested population-risk accuracy $\epsilon$, the computable saturated profile and strict inverse within the fixed typed partition are

$$\widehat{G}^{\mathrm{sat},\leq\epsilon}_{Q,W,\mathfrak{A},n}(B,a) := \sup_{\substack{M\in\mathscr{R}_W^{(n)}(B,a)\\ U_W(M)\leq\epsilon}} \underline{V}_W(M), \tag{301}$$

$$\widehat{\mathfrak{P}}^{>,\leq\epsilon}_{Q,W,\mathfrak{A},n}(\theta) := \{(B,a) : \widehat{G}^{\mathrm{sat},\leq\epsilon}_{Q,W,\mathfrak{A},n}(B,a) > \theta\}. \tag{302}$$

Thus structural optimality at fixed resources is the saturated supremum of complete visibility. A record is called an optimizer only on a cell where it attains that supremum; otherwise the Part I search transcript retains its $\varepsilon$-maximizers and nonattainment status. A simplest working record is a Pareto-minimal point of the appropriate strict inverse. The population risk-qualified profile is (316); Proposition II.11.34 proves that it is a nested restriction of the unrestricted Part I profile. Every earlier or later inner minimizer is used only to evaluate the visibility, risk, action, or resource coordinates that enter these profiles.

**Proposition II.11.30** (Part I fixed-response capacity coordinate)**.** *Fix one Wick response and let $P_W$ be the orthogonal projector onto its represented response support. For the declared target-score subspace $\mathcal{Y}_{\mathrm{tgt}}$, put*

$$\mathcal{V}_W^* := \overline{\mathrm{ran}(P_W P_{\mathcal{Y}_{\mathrm{tgt}}})}, \qquad d_W^* := \dim \mathcal{V}_W^* \in \mathbb{N}\cup\{\infty\}. \tag{303}$$

*For a reachable readout subspace $\mathcal{R}_K$ with $\dim \mathcal{R}_K \leq K$, define*

$$\mathcal{C}_W(y) := \frac{\|P_{\mathcal{V}_W^*} y\|^2}{\|y\|^2}, \qquad \mathcal{E}_W(\mathcal{R}_K; y) := \frac{\|P_{\mathcal{R}_K} P_{\mathcal{V}_W^*} y\|^2}{\|y\|^2}.$$

*As in Part I, both ratios are defined to be zero at $y = 0$. Then*

$$\mathcal{C}_W(y) - \mathcal{E}_W(\mathcal{R}_K; y) = \frac{\|(I - P_{\mathcal{R}_K}) P_{\mathcal{V}_W^*} y\|^2}{\|y\|^2}. \tag{304}$$

*Full structural coverage is equivalent to*

$$P_{\mathcal{R}_K} P_{\mathcal{V}_W^*} = P_{\mathcal{V}_W^*}. \tag{305}$$

*It forces $K \geq d_W^*$. On the finite-dimensional cell, any represented family containing $\mathcal{V}_W^*$ attains it at $K = d_W^*$; on the cell $d_W^* = \infty$, no finite-capacity readout has full coverage. Hence $d_W^*$ is the minimum readout capacity that covers the complete target-active response subspace, exactly as in the Part I model-family criterion. It is an inner minimum at fixed response. Across responses and source cells, capacity is a component of* $\mathbf{c}$*, and selection is governed by Definition II.11.29.*

*Proof.* The orthogonal decomposition $v = P_{\mathcal{R}_K} v + (I - P_{\mathcal{R}_K})v$, with $v = P_{\mathcal{V}_W^*} y$, proves (304). Identity (305) holds precisely when $\mathcal{V}_W^* \subseteq \mathcal{R}_K$, which implies $d_W^* \leq \dim \mathcal{R}_K \leq K$. On the finite-dimensional cell, taking $\mathcal{R}_{d_W^*} = \mathcal{V}_W^*$ proves attainability on exactly the represented reachable-family cell containing that subspace. Infinite $d_W^*$ rules out finite $K$ directly. □

**Theorem II.11.31** (Part I oracle selection and the simplest certified Wick record)**.** *Use the Part I confidence allocation $\delta_{\mathrm{Rad}} + \delta_{\mathrm{PB}} + \delta_{\mathrm{other}} \leq \delta$ in Definition II.11.28. At represented stage $n$, let $\mathscr{R}_W^{(n)}(B,a)$ be the charged finite family of complete Wick records produced by Part I, and let $\mathcal{R}(M)$*

*denote their common population loss. For a requested risk accuracy* $\epsilon$ *and the inherited visibility threshold* $\theta$*, define*

$$\mathscr{F}_W(\epsilon,\theta;B,a) := \{M \in \mathscr{R}_W^{(n)}(B,a) : U_W(M) \le \epsilon,\ \underline{V}_W(M) > \theta\}. \tag{306}$$

*The selected Pareto set is*

$$\widehat{\mathscr{M}}_W(\epsilon,\theta;B,a) := \mathrm{Min}_{\preceq_{\mathbf{c}}} \mathscr{F}_W(\epsilon,\theta;B,a). \tag{307}$$

*Here* $\mathrm{Min}_{\preceq_{\mathbf{c}}}$ *denotes all Pareto-minimal records in the complete resource–action order already used by the saturated profile. These recordwise frontiers are attached to feasible grid points of the certified risk-qualified strict inverse* (302)*; its full represented grid boundary is identified in* (v)*. Then, with probability at least* $1-\delta$*:*

(i) *every* $\widehat{M} \in \widehat{\mathscr{M}}_W(\epsilon,\theta;B,a)$ *satisfies*

$$\mathcal{R}(\widehat{M}) \le \epsilon, \qquad V_{\widehat{M}}^W > \theta; \tag{308}$$

(ii) *for every returned* $\widehat{M}$*, no certified* $(\epsilon,\theta)$*-working record in* $\mathscr{R}_W^{(n)}(B,a)$ *has* $\mathbf{c}(M) \prec \mathbf{c}(\widehat{M})$*;*

(iii) *every comparator* $M_0 \in \mathscr{R}_W^{(n)}(B,a)$ *whose extended two-sided radius and lower visibility satisfy*

$$\mathcal{R}(M_0) + 2\Gamma_W(M_0) \le \epsilon, \qquad \underline{V}_W(M_0) > \theta, \tag{309}$$

*belongs to* $\mathscr{F}_W(\epsilon,\theta;B,a)$*, and some* $\widehat{M} \in \widehat{\mathscr{M}}_W(\epsilon,\theta;B,a)$ *obeys*

$$\mathbf{c}(\widehat{M}) \preceq \mathbf{c}(M_0); \tag{310}$$

(iv) *when* $\mathscr{R}_W^{(n)}(B,a) \neq \varnothing$*, an* $\varepsilon_{\mathrm{opt}}$*-minimizer of* $U_W$ *satisfies the Part I oracle inequality*

$$\mathcal{R}(\widehat{M}_{\mathrm{or}}) \le \inf_{M \in \mathscr{R}_W^{(n)}(B,a)} \{\mathcal{R}(M) + 2\Gamma_W(M)\} + \varepsilon_{\mathrm{opt}}. \tag{311}$$

*This is the Part I risk-oracle comparison at a fixed represented stage. It supplies the risk restriction in* (301)*; structural selection is its saturated visibility supremum and Pareto strict inverse.*

(v) *across the Part I resource–action grid, the learner reports the returned record frontiers at precisely the Pareto-minimal grid points of the certified strict region*

$$\widehat{\mathfrak{P}}^{>,\le\epsilon}_{Q,\overline{W},\mathfrak{A},n}(\theta).$$

*If the feasible set is empty, the output is the existing insufficient- confidence, target-miss, authority, or budget status. If several records remain statistically unresolved and Pareto-minimal, the learner returns their certified frontier; no uniqueness criterion is added. The directed sequence of stages carries the Part I net remainder. Its infimum and upward feasibility region are therefore defined without compactness; Pareto-minimal continuum records are reported exactly on stages or limit cells where they exist, and nonattainment remains in the search transcript.*

*Proof.* The learned-response Rademacher bound and exact-source union bound give (298) uniformly over (295). The PAC–Bayes chain rule gives (299); applying the structural PAC–Bayes bound to every $\lambda \in \Lambda$ at confidence $\delta_{\mathrm{PB}}/|\Lambda|$ gives (300). The confidence allocation and union bound put all applicable candidates on one event. Their recordwise minimum is valid because they bound the same population loss after the declared converters.

On that event, $\mathcal{R}(M) \le U_W(M)$ and $\underline{V}_W(M) \le V_M^W$, proving (i). Statement (ii) is the definition of a Pareto-minimal element of the finite feasible set. A two-sided radius gives $U_W(M_0) \le \mathcal{R}(M_0) + 2\Gamma_W(M_0)$, so (309) places $M_0$ in the feasible set. Every member of a finite partially ordered set dominates a minimal member, proving (iii). Finally,

$$\mathcal{R}(\widehat{M}_{\mathrm{or}}) \le U_W(\widehat{M}_{\mathrm{or}}) \le \inf_M U_W(M) + \varepsilon_{\mathrm{opt}} \le \inf_M \{\mathcal{R}(M) + 2\Gamma_W(M)\} + \varepsilon_{\mathrm{opt}},$$

which proves (iv). By (301)– (302), a grid point belongs to the certified strict inverse exactly when it contains a record in $\mathscr{F}_W(\epsilon, \theta; B, a)$. The grid-level reporting rule selects exactly the Pareto-minimal such points and returns the recordwise Pareto frontier there, proving (v). Net stability errors are already components of $\Gamma_W$ and $\mathbf{c}$. □

**Corollary II.11.32** (Saturated-profile minimum working models for free Caus and RL)**.** *For unrestricted Caus, take the candidates produced by* (253) *over the exact source/operator/coefficient family in* (295)*. Select them by the Rademacher certificate* (298)*, the PAC–Bayes certificate* (300)*, or their valid same-loss minimum to form $U_W$, and then apply the risk-qualified saturated profile at the inherited threshold $\theta$. Then* (308)– (310) *hold, and the response-energy coordinate is the exact minimum lift from Proposition II.11.27. On the zero-waste fiber $\mathfrak{A}_D^0$, its action coordinate is zero. At fixed response, the smallest full-coverage readout has $K = d_W^*$ by Proposition II.11.30; across responses, the inherited cover or mixture charge enters $\mathbf{c}$, and the risk-qualified saturated profile decides the tradeoff.*

*For RL, let $g_{\mathrm{auth}}, g_{\mathrm{obs}} \in [0, \infty]$ be the derived budget/safe-set and observation gaps and define the extended policy certificate*

$$U_{\mathrm{RL}}(M) := \frac{2\Gamma_{Q,K} + \tau \log(1/\nu_{\min}) + \mathrm{Osc}(\widehat{q}_K)\sqrt{\kappa_\pi/2}}{1-\gamma} + g_{\mathrm{auth}} + g_{\mathrm{obs}}. \tag{312}$$

*Fix the Part I environment, value norm, reward convention, and deployment-law identifier. Over the resulting typed partition of represented RL candidates, with the nonordinary converter/support values totalized as in Theorem II.11.24, let*

$$\widehat{\mathscr{M}}_{\epsilon,\theta}^{\mathrm{RL}} := \mathrm{Min}_{\preceq \mathbf{c}}\{R : U_{\mathrm{RL}}(R) \le \epsilon,\ \underline{V}_W(R) > \theta\},$$

*where every critic, bridge, coefficient, and policy-family selection term in $\Gamma_{Q,K}$ is the corresponding inherited Rademacher or PAC–Bayes charge. Every $\widehat{R} \in \widehat{\mathscr{M}}_{\epsilon,\theta}^{\mathrm{RL}}$ satisfies*

$$\|V_\alpha^* - V_\alpha^{\widehat{\pi}}\|_\infty \le \epsilon, \qquad V_{\widehat{R}}^W > \theta, \qquad \nexists R:\ U_{\mathrm{RL}}(R) \le \epsilon,\quad \underline{V}_W(R) > \theta,\quad \mathbf{c}(R) \prec \mathbf{c}(\widehat{R}). \tag{313}$$

*Thus the output is precisely the Part I Pareto-minimal certified safe-policy frontier in the risk-qualified strict inverse at the requested value accuracy and visibility threshold.*

*Proof.* The unrestricted statement is Theorem II.11.31 with the exact energy identity from Proposition II.11.27; zero action on $\mathfrak{A}_D^0$ follows from Theorem II.11.12. For RL, Theorem II.11.24 and the two declared approximation gaps give $\|V_\alpha^* - V_\alpha^{\pi_M}\|_\infty \le U_{\mathrm{RL}}(M)$. Feasibility therefore proves the first inequality in (313); the simultaneous lower visibility bound proves its second conclusion, and Pareto minimality in the saturated strict inverse proves the last statement. □

### II.11.7 Certified learner and primal–dual profile

**Definition II.11.33** (Complete Wick certificate)**.** A complete represented Wick certificate is an extension

$$\mathscr{C}_W = (\mathscr{C}_I, \mathfrak{W}_{\mathsf{Abs}}, E_{W,\Delta}, \Sigma_C, F_W, T_{W,\lambda}, \mathscr{S}, \mathscr{T}, \mathscr{L}, \mathbf{c}),$$

where $\mathscr{C}_I$ is the complete selected Part I compensated quotient–geometry record. In particular, it already contains $U, P_\Delta, H_{\mathsf{Abs}}, G, A_{\mathcal{Y}}, C, \mathfrak{A}, D, J_C$, their source and authority provenance, and their structural resource coordinates. The map $\mathfrak{W}_{\mathsf{Abs}}$ is (211); it adds no source or resource field. The derived ledger $\mathscr{S}$ contains the self-energy, dark-space, bracket, quasi-lumpability, and continuum-transfer certificates; $\mathscr{T}$ contains the dynamics, free-Caus, or RL loss, its data split, optimization error, selected bridge/kernel, and behavior-to-deployment comparison, together with the exact source cell, declared operator and coefficient families, Rademacher cover or PAC–Bayes mixture charge, target-subspace coverage identity, and certified risk tolerance; $\mathscr{L}$ contains the target capture, statistical confidence, source, noise, and deployment records; and $\mathbf{c}$ is the inherited complete Pareto resource vector together with the evaluated Caus action. Every derived object is computed from the Wick image of the same selected Part I record. The optimization entries in $\mathscr{T}$ are inner transcripts; only the saturated profile below compares complete records. Its Wick visibility is the existing master-certificate visibility evaluated with the full response and authority data; write it $V^W_{\mathscr{C}_W}$. Its dynamic stability coordinate is the existing general-resolvent factor

$$S_W(\lambda) = \max\{0, 1 - \Delta_W(\lambda)\},$$

where $\Delta_W$ is the represented target-functional discrepancy between the true compressed response and $F_W(\lambda)^{-1}$. It vanishes for the exact block model of Theorem II.11.7; empirical and continuum model residuals enter through the resolvent identity.

For an authority envelope $\mathfrak{A}$, resource ceiling $B$, and action ceiling $a$, let $\mathcal{C}^{\mathfrak{A}}_{Q,W}(B,a)$ be the represented family of complete certificates satisfying both ceilings and having a nonzero quotient defect. The zero-defect boundary belongs to the existing exact-quotient cell. Put

$$G^{\mathrm{sat}}_{Q,W,\mathfrak{A}}(B,a) := \sup_{\mathscr{C}_W \in \mathcal{C}^{\mathfrak{A}}_{Q,W}(B,a)} V^W_{\mathscr{C}_W}, \qquad \sup\varnothing := 0. \tag{314}$$

For a threshold $\theta$, its strict inverse is the upward-closed resource–action feasibility region

$$\mathfrak{P}^{>}_{Q,W,\mathfrak{A}}(\theta) := \{(B,a) : G^{\mathrm{sat}}_{Q,W,\mathfrak{A}}(B,a) > \theta\} = \bigcup_{\mathscr{C}_W : V^W_{\mathscr{C}_W} > \theta} \{(B,a) : \mathbf{c}_{\mathrm{res}}(\mathscr{C}_W) \preceq B, \quad \mathbf{c}_{\mathrm{act}}(\mathscr{C}_W) \le a\}. \tag{315}$$

For a requested population-risk accuracy $\epsilon \in [0,\infty]$, fix one Part I population-law, estimand, norm, and loss-scale identifier. After all declared converters have been charged, let $\mathcal{C}^{\mathfrak{A}}_{Q,W,\ell}(B,a)$ denote the corresponding typed partition of $\mathcal{C}^{\mathfrak{A}}_{Q,W}(B,a)$; suppress $\ell$ in the display below. The same Part I saturation restricted to working records is

$$G^{\mathrm{sat},\le\epsilon}_{Q,W,\mathfrak{A}}(B,a) := \sup_{\substack{\mathscr{C}_W \in \mathcal{C}^{\mathfrak{A}}_{Q,W}(B,a) \\ \mathcal{R}(\mathscr{C}_W) \le \epsilon}} V^W_{\mathscr{C}_W}, \qquad \sup\varnothing := 0, \tag{316}$$

and its strict inverse is

$$\mathfrak{P}^{>;\le\epsilon}_{Q,W,\mathfrak{A}}(\theta) := \{(B,a) : G^{\mathrm{sat},\le\epsilon}_{Q,W,\mathfrak{A}}(B,a) > \theta\}. \tag{317}$$

Its Pareto-minimal elements are reported when they exist. In particular, they exist on the finite resource–action grid used by the learner below; the region itself, rather than a possibly empty set of minimal elements, is the exact generalized inverse on an arbitrary ordered resource space.

**Proposition II.11.34** (Risk qualification is a restriction of Part I saturation)**.** *For* $0 \le \epsilon_1 \le \epsilon_2 \le +\infty$*,*

$$G^{\mathrm{sat},\le\epsilon_1}_{Q,W,\mathfrak{A}}(B,a) \le G^{\mathrm{sat},\le\epsilon_2}_{Q,W,\mathfrak{A}}(B,a) \le G^{\mathrm{sat}}_{Q,W,\mathfrak{A}}(B,a), \tag{318}$$

$$\mathfrak{P}^{>,\le\epsilon_1}_{Q,W,\mathfrak{A}}(\theta) \subseteq \mathfrak{P}^{>,\le\epsilon_2}_{Q,W,\mathfrak{A}}(\theta) \subseteq \mathfrak{P}^{>}_{Q,W,\mathfrak{A}}(\theta). \tag{319}$$

*At* $\epsilon = +\infty$*, both rightmost inclusions are equalities. The same identities hold at every represented stage after replacing* $G$ *and* $\mathfrak{P}$ *by the certified quantities in* (301)– (302) *and taking the unrestricted certified profile to be the same supremum without the condition* $U_W(M) \le \epsilon$*. Consequently risk accuracy restricts the record family inside the inherited saturated optimization; it introduces no second structural order.*

*Proof.* The record sets in each supremum are nested because $\{\mathcal{R} \le \epsilon_1\} \subseteq \{\mathcal{R} \le \epsilon_2\} \subseteq \mathcal{C}^{\mathfrak{A}}_{Q,W}(B,a)$. Suprema preserve inclusion, proving (318). Taking strict superlevel sets proves (319). Every extended risk satisfies $\mathcal{R} \le +\infty$, which gives equality at $\epsilon = +\infty$. Replacing $\mathcal{R}$ by the simultaneous upper certificate $U_W$ gives the identical finite-stage proof. □

**Theorem II.11.35** (Bayesian comparison for the Wick saturated learner)**.** *Fix a represented stage* $n$*, a common confidence event, and one population loss. For a complete Wick record* $M$*, let* $U^W_{\mathrm{PB}}(M)$ *denote the PAC–Bayes certificate in* (300)*, evaluated on its carried posterior and assigned* $+\infty$ *when that candidate is unavailable. Define the PAC–Bayes-only saturated lower profile and strict inverse by*

$$\widehat{G}^{\mathrm{PB},\le\epsilon}_{Q,W,\mathfrak{A},n}(B,a) := \sup_{\substack{M\in\mathscr{R}^{(n)}_W(B,a)\\ U^W_{\mathrm{PB}}(M)\le\epsilon}} \underline{V}_W(M), \qquad \sup\varnothing := 0, \tag{320}$$

$$\widehat{\mathfrak{P}}^{\mathrm{PB},>,\le\epsilon}_{Q,W,\mathfrak{A},n}(\theta) := \{(B,a) : \widehat{G}^{\mathrm{PB},\le\epsilon}_{Q,W,\mathfrak{A},n}(B,a) > \theta\}. \tag{321}$$

*Then*

$$U_W(M) \le U^W_{\mathrm{PB}}(M), \tag{322}$$

$$\widehat{G}^{\mathrm{PB},\le\epsilon}_{Q,W,\mathfrak{A},n}(B,a) \le \widehat{G}^{\mathrm{sat},\le\epsilon}_{Q,W,\mathfrak{A},n}(B,a), \tag{323}$$

$$\widehat{\mathfrak{P}}^{\mathrm{PB},>,\le\epsilon}_{Q,W,\mathfrak{A},n}(\theta) \subseteq \widehat{\mathfrak{P}}^{>,\le\epsilon}_{Q,W,\mathfrak{A},n}(\theta). \tag{324}$$

*The profile inequality is strict at* $(B,a)$ *whenever some* $M \in \mathscr{R}^{(n)}_W(B,a)$ *satisfies*

$$U_W(M) \le \epsilon < U^W_{\mathrm{PB}}(M), \qquad \underline{V}_W(M) > \widehat{G}^{\mathrm{PB},\le\epsilon}_{Q,W,\mathfrak{A},n}(B,a). \tag{325}$$

*Let* $\mathfrak{B}(\mathscr{C}_W)$ *be the universal Bayesianization of Part I applied to the active execution carried by a complete Wick record. Embed the Wick resource vector in the complete Bayesian resource lattice by filling the belief, likelihood, normalization, and belief-memory slots with zero. The Part I compiler and forgetting map give*

$$\mathcal{R}(\mathfrak{B}(\mathscr{C}_W)) = \mathcal{R}(\mathscr{C}_W), \qquad V(\mathfrak{B}(\mathscr{C}_W)) = V^W_{\mathscr{C}_W}, \qquad \mathbf{c}(\mathscr{C}_W) \preceq \mathbf{c}(\mathfrak{B}(\mathscr{C}_W)), \tag{326}$$

*and the Caus action coordinate is unchanged. The resource inequality is strict precisely when the canonical Bayesian annotation has a positive belief-specific resource slot. Hence the direct Wick realization reaches the same risk–visibility point no later in the saturated Pareto inverse than its universal Bayesian annotation.*

*For the full record classes, the inherited Part I comparison remains*

$$\mathrm{BayesArr}^{\mathrm{sat}}_{I,H}(B) \le \mathrm{ActArr}^{\mathrm{sat}}_{I,H}(B) \le \mathrm{BayesArr}^{\mathrm{sat}}_{I,H}(\Psi_{\mathrm{uBayes}}(B,H,n)). \tag{327}$$

*Thus Bayesian optimality is universal after the Part I transfer ceiling, whereas* (323) *is a same-family finite-sample domination and* (326) *is a same-execution resource comparison. The Wick class satisfies*

$$\mathcal{C}^{\mathfrak{A}}_{Q,W}(B,a) \subseteq \mathcal{C}^{\mathfrak{A}}_{Q,\mathrm{nr}}(B), \qquad G^{\mathrm{sat}}_{Q,W,\mathfrak{A}}(B,a) \le G^{\mathrm{sat}}_{Q,\mathrm{nr},\mathfrak{A}}(B). \tag{328}$$

*These relations give the complete comparison: pointwise certificate domination on the common Wick family, a resource comparison for the same execution, and transfer-ceiling equivalence for the universal classes.*

*Proof.* By definition, $U_W(M)$ is the recordwise minimum of every applicable same-loss certificate, including $U^W_{\mathrm{PB}}(M)$. This proves (322). Consequently

$$\{M : U^W_{\mathrm{PB}}(M) \le \epsilon\} \subseteq \{M : U_W(M) \le \epsilon\}$$

inside the same charged family $\mathscr{R}^{(n)}_W(B,a)$. Taking suprema of the same lower visibility proves (323); taking strict superlevel sets proves (324). The record in (325) lies only in the second feasible set and has lower visibility above the first supremum, proving strictness.

The universal Bayesianization theorem of Part I retains every query, transition, action, continuation, stopping, and deployment kernel of the original active record and appends a constant represented belief, unit likelihood and normalizer, and identity update. It therefore preserves the execution law, deployed population risk, and target visibility. Its belief-specific costs are nonnegative. The Part I forgetting compiler deletes exactly those slots with no increase in cost. Together with the exact Wick inheritance in Proposition II.11.2, this proves (326) and its strictness criterion. The two profile inequalities in (327) are respectively the Part I forgetting and universal Bayesianization inequalities. Finally, the Wick record inclusion in the nonreversible family gives the stated specialized-family relation. □

**Definition II.11.36** (Two-stage Wick structural learner)**.** The learner receives the candidate records produced by the Part I structural learner from a represented empirical generator or closed-form approximation, with their confidence radii, finite-coverable quotient, geometry, and authority families, resolvent and ridge grids, and total resource ceiling. It later receives a represented target/reward sample. The Wick stage neither retypes nor enlarges those candidate families. Raw trajectories enter through a represented generator/form estimator and its dependence certificate. An exact PDE operator is the zero-estimation- error specialization of the same interface.

(L1) On dynamics-only data, enumerate the inherited charged finite net. For every Part I candidate, minimize (249), or the raw-path likelihood (250), within that record's existing structural and authority constraints. Fit its represented parameters and residual, then apply $\mathfrak{W}_{\mathsf{Abs}}$ to the selected quotient block. Certify $E_{W,\Delta}$, the derived complement-resolvent bounds, self-energy, dark dimension, bracket gap, the derived design range/kernel, and the identification/perturbation bounds of Propositions II.11.18 and II.11.20.

(L2) Freeze the quotient/geometry selection or retain a simultaneous uniform cover. On split target/control data, choose the target band and readout. Under unrestricted authority, minimize (253) and (254), restricting to the zero-waste fiber when least action is a hard requirement. Under RL authority, alternate the frozen-target critic and Gibbs actor steps in (263). In a hybrid envelope, optimize only the declared control part. Every selected bridge current is certified by Theorem II.11.12; an imposed bridge remains fixed. For each fixed response, compute $\mathcal{V}_W^*$, set $K = d_W^*$ whenever the reachable family verifies (305), and otherwise retain the exact gap (304).

(L3) Compute the lower-confidence master visibility and every applicable entry of Corollary II.11.26. At fixed $(B, a)$, return an $\varepsilon$-maximizer of the saturated profile. Ties within the certified tolerance use the inherited Part I Pareto order on the complete resource–action vector; the declared candidate order only enumerates unresolved equals. When population-risk accuracy $\epsilon$ and visibility threshold $\theta$ are requested, construct $U_W$ from the inherited Rademacher union/cover or PAC–Bayes mixture/coefficient certificate in Definition II.11.28, and apply (306) to the certified risk-qualified saturated profile; on the RL branch use (312). An empty certified feasible set is reported by its typed failure status.

(L4) For a requested threshold $\theta$, enumerate the inherited budget/action grid, return the certified strict region (315), or (302) when risk-qualified, and report its Pareto-minimal grid points.

The output is a certificate and one of the following typed statuses: certified profile, target spectral miss, uncompensated model defect, target-visible dark mode, authority obstruction, insufficient statistical confidence, continuum-transfer defect, or budget obstruction.

**Theorem II.11.37** (Correctness and termination of the Wick learner)**.** *At every represented Part I stage, the candidate set is the charged finite net constructed by that learner, and its perturbation remainder is the extended value supplied by Proposition II.11.18. Consequently the Wick stage terminates by finite enumeration. The Part I confidence allocation and union/PAC–Bayes theorem construct a simultaneous event of probability at least* $1 - \delta$*; an unavailable radius is* $+\infty$*. On that event the budget-first output has certified visibility at least*

$$\left[ \sup_{\mathscr{C}_W \in \mathcal{C}^{\mathfrak{A}}_{Q,W}(B,a)} V^W_{\mathscr{C}_W} - \varepsilon - \varepsilon_{\mathrm{net}} - \varepsilon_{\mathrm{stat}} \right]_+ , \tag{329}$$

*where the last two terms are those derived extended perturbation and confidence radii and* $[r]_+ := \max\{0, r\}$*, with an infinite subtracted radius giving zero. Its threshold-first output contains every truly feasible strict budget/action point outside these radii and excludes every truly infeasible point outside them; intersecting intervals are reported as insufficient confidence.*

*On the unrestricted-Caus branch, the returned predictor satisfies the extended oracle inequality* (258)*. On the RL branch, every critic iteration satisfies* (264)*, and the output satisfies the extended inequalities* (266) *and* (268)*. A nonordinary converter or actor coordinate makes only the corresponding bound infinite and returns the typed confidence, shift, or policy-conversion status. For the minimum-working-model option, the empty Part I-certified feasible set returns its typed status; every record returned on the complementary cell satisfies* (308)*–* (310)*; the RL specialization satisfies* (313)*. These records are exactly the recordwise frontiers reported at Pareto-minimal points of the certified risk-qualified strict inverse. At fixed response, full structural coverage uses the minimum capacity* $K = d_W^*$ *whenever* (305) *is verified; this capacity remains an inner coordinate of the saturated-profile comparison.*

*Moreover, Theorem II.11.35 gives* (328)*, and*

$$G^{\mathrm{sat}}_{Q,W,\mathfrak{A}}(B,a) > \theta \quad \Longleftrightarrow \quad (B,a) \in \mathfrak{P}^{>}_{Q,W,\mathfrak{A}}(\theta). \tag{330}$$

*For every* $\epsilon$*, the risk-qualified specialization likewise satisfies*

$$G^{\mathrm{sat},\leq\epsilon}_{Q,W,\mathfrak{A}}(B,a) > \theta \quad \Longleftrightarrow \quad (B,a) \in \mathfrak{P}^{>,\leq\epsilon}_{Q,W,\mathfrak{A}}(\theta). \tag{331}$$

*Proof.* Finite enumeration at each Part I stage and the declared tie rule give termination. The allocated confidence event and the extended net Lipschitz bounds compare every population certificate with its evaluated representative; maximizing the simultaneous lower bound gives (329). The interval rule proves the inverse claim. Sample splitting and uniform selection invoke Theorem II.11.22 on the free branch; fitted Bellman contraction and Gibbs calibration invoke Theorem II.11.24 on the RL branch. The model-family selector invokes Proposition II.11.30, Theorem II.11.31, and Corollary II.11.32. Every record in $\mathcal{C}^{\mathfrak{A}}_{Q,W}$ has a nonzero quotient defect evaluated through the full nonreversible resolvent and therefore is a record in the existing Part I $Q$, nr family, proving (328). Finally, (330) is the generalized-inverse identity for a monotone saturated profile and its strict feasibility region; the same argument proves (331) after restriction to the risk-qualified record family. □

*Remark* II.11.38 (Totalized structural multipliers). The inherited optimization-status field separates the finite-dimensional candidate problems into exhaustive cells. On the convex constraint-qualified cell, the KKT multipliers of the resource, action, capacity, and confidence constraints are shadow prices for those declared structural resources. The other cells retain their nonconvexity or failed-qualification status and make no multiplier claim. Identification of an ordinary multiplier with a measured physical coupling constant is a distinct representation question, not a structural assumption or a conclusion of the present construction.

### II.11.8 Empirical implementation by differentiable linear algebra

The preceding learner is directly implementable on each finite chart in the Part I stage. This subsection specifies the tensor representation, the differentiable solves, and the separation between inner training and outer structural selection. Every branch uses the complete source-resolved learning ledger of Part I. The RL branch additionally uses its complete dynamical learning record; when a candidate is active or structural Bayesian, it also uses the complete neural implementation record. Thus no record type is silently enlarged. In particular, the optimizer output is one represented Wick record; backpropagation does not replace the saturated profile by a scalar training objective.

**Definition II.11.39** (Finite empirical Wick interface)**.** Fix one candidate specification $s$ in the charged finite stage $\mathscr{R}^{(n)}_W(B,a)$. Its Part I record supplies the following data.

(i) The quotient and complement charts $\mathbb{R}^{d_Q}$ and $\mathbb{R}^{d_Y}$, their gauges, evaluation maps, precision convention, and source and authority fields.

(ii) A bridge chart $\mathrm{Dec}_{\mathfrak{A},s} : \Phi_{C,s} \to \mathbb{R}^{d_Y \times d_Q}$. Its image is the bridge family carried by $s$. On the unrestricted branch it is the full declared bridge chart; on the RL branch it is the inherited legal-action compiler.

(iii) Positive grids for $\lambda, \rho, \tau$, the loss weights, the target evaluation maps, the legal and safe action masks, and the complete confidence allocation. The record also carries its population estimand, probability-law, norm, and loss identifiers and every declared converter between them. These are the grids and types already charged in the Part I candidate, rather than new continuous hyperparameter families.

(iv) Identifiers and the declared overlap relation for the dynamics, target-training, validation, RL, and audit carriers used by that candidate. Independence is required exactly when the selected Part I split-sample certificate uses it. A shared carrier instead uses its simultaneous union, cover, PAC–Bayes, sequential, or martingale certificate. A carrier may be absent when its associated certificate is not invoked.

The dynamics carrier contains one of the two records

$$Z_j = (x_j, y_j, \dot{x}_j, \dot{y}_j, W_{x,j}, W_{y,j}), \qquad Z_j^{\mathrm{tr}} = (u_j, u_{j+1}, \Delta_j, \Omega_j),$$

used in (249) and (250), respectively. A supervised carrier stores $(\Phi_{\mathrm{tr}}, y_{\mathrm{tr}})$ and $(\Phi_{\mathrm{val}}, y_{\mathrm{val}})$, where each $\Phi$ is the represented evaluation map from quotient scores to observations. An RL carrier stores $(h_i, a_i, r_i, h_i', \omega_i)$, its terminal indicator, and its legal and safe masks. When the record contains an encoder, raw observations first pass through that inherited map; its reconstruction and deployment errors remain separate certificate coordinates.

The terminal empirical object is

$$\widehat{\mathscr{C}}_{W,s} = (s, \widehat{\vartheta}_s, \widehat{f}_s, \widehat{q}_s, \widehat{\pi}_s, \mathsf{Opt}_s, \mathsf{Audit}_s, \mathbf{c}_s, \mathsf{Status}_s). \tag{332}$$

Here absent supervised or RL fields are null slots. The optimizer transcript $\mathsf{Opt}_s$ stores the initialization, iterates, stochastic gradients, moments, clipping, stopping rule, terminal weights, and the represented optimization gap. The audit field stores split identifiers, all evaluated residuals and intervals, the same-loss risk candidates, lower master visibility, precision and solve residuals, and the confidence allocation. The status belongs to the typed list in Definition II.11.36, augmented only by the numerical status already present in the Part I implementation record.

**Definition II.11.40** (Structure-preserving tensor compiler)**.** Let $L_H, L_G, K$ be unconstrained real trainable matrices of the appropriate sizes and let $\phi_C \in \Phi_{C,s}$. Define

$$H = L_H L_H^{\mathsf{T}}, \qquad G = L_G L_G^{\mathsf{T}}, \qquad A_{\mathcal{Y}} = \frac{1}{2}(K - K^{\mathsf{T}}), \qquad C = \mathrm{Dec}_{\mathfrak{A},s}(\phi_C). \tag{333}$$

This displayed parameterization is the dense trainable cell. If $s$ carries additional sparsity, tying, locality, or coefficient restrictions, the factors range only over its inherited chart and are passed through its exact structure compiler. A finite or nondifferentiable Part I chart is enumerated and then held fixed during the differentiable subproblem. Thus (333) does not enlarge the candidate family. The displayed square factors parameterize the full dense positive-semidefinite cones. A rectangular factor is used only when the inherited candidate already carries the corresponding rank ceiling. Nonuniqueness of the factors is an optimizer-coordinate gauge, whose storage and optimization are charged in the record; all identification and perturbation bounds concern the compiled operators $H, G, A_{\mathcal{Y}}, C$, not a choice of factor. The real matrices are converted to the declared complex precision only when the Wick generator or its response is evaluated. For $\lambda > 0$, put

$$B_C = \lambda I_{d_Y} + G - A_{\mathcal{Y}}, \qquad R_C = \mathrm{solve}(B_C, C), \tag{334}$$

$$\Sigma_C = C^* R_C, \qquad F_C = \lambda I_{d_Q} + iH + \Sigma_C, \tag{335}$$

$$A_C = I_{d_Q} + F_C^* F_C, \qquad \mathrm{Resp}_C(V) = \mathrm{solve}(A_C, V). \tag{336}$$

Thus $\mathrm{Resp}_C(V) = T_{W,\lambda}(C)V$, without materializing a matrix inverse. For an evaluation map $E$, its finite response Gram matrix is

$$K_{C,E} = E\,\mathrm{Resp}_C(E^*). \tag{337}$$

If $E$ is represented by a redundant row system, its Part I range compression is applied before the solve. A Moore–Penrose calculation uses the same declared finite-precision range decomposition and stores the range residual. No eigenvalue is clipped and no diagonal jitter is added unless that modification and its induced operator error are fields of $s$.

For a coefficient vector $f \in \mathbb{C}^{d_Q}$, the finite supervised inner problem is

$$\widehat{f}_{C,\rho} = \operatorname{solve}\left(\frac{1}{m_{\mathrm{tr}}}\Phi_{\mathrm{tr}}^{*}\Phi_{\mathrm{tr}} + \rho A_C, \frac{1}{m_{\mathrm{tr}}}\Phi_{\mathrm{tr}}^{*}y_{\mathrm{tr}}\right). \tag{338}$$

This is the normal equation of (253). Kernel form $(K_{C,E} + m_{\mathrm{tr}}\rho I)\alpha = y_{\mathrm{tr}}$ is equivalent on the represented evaluation range. The primal form makes the inverse-response penalty $f^{*}A_C f$ explicit.

**Proposition II.11.41** (Faithfulness and differentiability of the tensor compiler)**.** *For every finite empirical specification $s$, the compiler of Definition II.11.40 has the following properties.*

(i) *It satisfies $H = H^{*} \succeq 0$, $G = G^{*} \succeq 0$, $A_{\mathcal{Y}}^{*} = -A_{\mathcal{Y}}$, and $C \in \operatorname{im}\operatorname{Dec}_{\mathfrak{A},s}$ at every optimizer iterate.*

(ii) *The matrix $B_C$ is invertible and $\|B_C^{-1}\| \le \lambda^{-1}$. The matrix $A_C$ is Hermitian positive definite, $A_C \succeq I$, and $\|A_C^{-1}\| \le 1$. Hence* (334)– (338) *have unique ordinary values in exact arithmetic for every $\lambda, \rho > 0$.*

(iii) *The compiled block generator is exactly* (212) *and satisfies* (217)*. Its computed self-energy and response are exactly the finite-chart restrictions of* (221) *and* (233)*.*

(iv) *The tensor core is differentiable in $H, G, A_{\mathcal{Y}}, C$ at every ordinary finite-precision evaluation point. Its composition with an inherited parameter decoder is differentiated on the decoder's represented differentiable cell; a finite or nondifferentiable decoder is handled by the inherited enumeration/search transcript. For $X = B^{-1}D$, its differential is*

$$\mathrm{d}X = B^{-1}(\,\mathrm{d}D - (\,\mathrm{d}B)X). \tag{339}$$

*For the transition loss, the differential of the matrix exponential is the Fréchet formula*

$$D\exp(M)[\dot{M}] = \int_0^1 e^{(1-t)M}\dot{M}e^{tM}\ \mathrm{d}t. \tag{340}$$

*Reverse automatic differentiation through matrix multiplication, linear solves, and the matrix exponential therefore evaluates the derivatives of the mathematical losses on the differentiable cell. Since the trainable parameters are real and the terminal loss is real, this is the real pullback of the complex intermediate calculation.*

*A failed finite-precision solve, a nonfinite loss, or a residual exceeding the declared tolerance produces the numerical-status branch of* (332)*. Every dependent upper-error, risk, or policy-gap candidate receives $+\infty$, while every dependent lower visibility candidate receives zero. A raw derived coordinate that has no ordered extended value retains its typed nonordinary status. The record remains in the finite stage.*

*Proof.* The first assertion follows from (333), the exact structured compiler when present, and the range of the inherited authority decoder. For $v \in \mathbb{C}^{d_Y}$,

$$\operatorname{Re}\langle v, B_C v\rangle = \lambda\|v\|^2 + \langle v, Gv\rangle \ge \lambda\|v\|^2.$$

Finite-dimensional accretivity gives injectivity, hence invertibility, and the displayed inverse bound. Moreover, for $w \in \mathbb{C}^{d_Q}$, $\langle w, A_C w\rangle = \|w\|^2 + \|F_C w\|^2$, proving the claims about $A_C$. Substitution in the block generator proves the energy identity, and block elimination gives the two response identities. Differentiating $BX = D$ proves (339); the standard variation-of-constants identity proves (340). Composition and the chain rule give the automatic-differentiation assertion on the represented differentiable cell; the other chart types retain their Part I search rule. The last statement is the extended-value convention of Proposition II.11.3 applied to the numerical field of the complete implementation record. $\square$

**Dynamics fit.** The following PyTorch-style pseudocode describes the inner dynamics step.

```
for spec in part1_stage:
model = WickModule(spec)
for batch in spec.dynamics_loader:
H = Lh @ Lh.T;  G = Lg @ Lg.T
Ay = (K - K.T) / 2;  C = spec.decode_caus(phi_c)
Lw = wick_generator(cplx(H), cplx(G), cplx(Ay), cplx(C))
L = derivative_loss(batch, H, G, Ay, C)
# or:  L = transition_loss(batch, matrix_exp(dt * Lw))
optimizer.zero_grad();  L.backward();  optimizer.step()
store_optimizer_transcript(model, optimizer)
```

The two branches evaluate (249) and (250), respectively. The terminal design matrix $Q_n$, score $s_n$, range projector $P_{\mathrm{id}}$, kernel, and optimization gap are then evaluated from the stored trajectory; they are not training assumptions.

**Unrestricted-Caus fit.** For every charged $(s, \lambda, \rho)$, response and readout fitting use the same bridge parameter. Gradients of the validation term pass through (338).

```
for spec, lam, rho in charged_free_candidates:
for outer_step in spec.bridge_steps:
C = spec.decode_caus(phi_c)
B = lam * Iy + G - Ay;  R = linalg.solve(B, C)
F = lam * Iq + 1j * H + C.mH @ R
A = Iq + F.mH @ F
M = Phi_tr.mH @ Phi_tr / m_tr + rho * A
f = linalg.solve(M, Phi_tr.mH @ y_tr / m_tr)
action = squared_current_distance(C, target_current, spec.metric) / 2
penalty = spec.selection_penalty(C, rho)
loss = validation_risk(f) + action_weight * action + grad_part(penalty)
bridge_optimizer.zero_grad();  loss.backward()
bridge_optimizer.step()
detach_and_audit(spec, C, f)
```

The selection penalty and the indicator of $\mathfrak{A}_{\mathrm{dyn}}$ in (254) are certificate and parameter-domain terms. They are added to the displayed loss when they vary differentiably within the candidate. Otherwise the Part I finite net or search transcript compares the detached terminal candidates using their full outer value, including the penalty; `grad_part` is then constant during that inner step. If zero action is a hard task requirement, Part I first evaluates the represented fiber $\mathfrak{A}_D^0$. On its nonempty cell the decoder is restricted to that fiber and the action is identically zero. On its empty cell the candidate receives the distance in (231); training does not replace that value by a penalty surrogate. If the task declares a risk–action tradeoff, the displayed action weight is the declared $\lambda_{\mathscr{W}}$ of (254).

**Action-safe RL fit.** Let $\mathscr{E}_{\mathrm{RL}}$ be the complete finite critic-evaluation carrier, with the weights and normalization of its selected Part I statistical certificate. Its represented safe row map $E_{\mathscr{E}}^{\mathcal{A},\mathrm{safe}} : \mathcal{Q} \to \mathbb{C}^{r_{\mathscr{E}}}$ is the full-carrier evaluation map restricted to the legal, failure-killed action rows. Apply the inherited exact row-range compression when those rows are redundant. The finite safe response is

$$K_{C,\mathscr{E}}^{\mathrm{safe}} = E_{\mathscr{E}}^{\mathcal{A},\mathrm{safe}} \operatorname{Resp}_C\big((E_{\mathscr{E}}^{\mathcal{A},\mathrm{safe}})^*\big). \tag{341}$$

This is exactly the finite compression in (234) applied to (259). It is positive definite after the declared row-range compression: for $z \neq 0$,

$$z^* K^{\text{safe}}_{C,\mathscr{E}} z = \left\langle (E^{\mathcal{A},\text{safe}}_{\mathscr{E}})^* z, T_{W,\lambda}(C)(E^{\mathcal{A},\text{safe}}_{\mathscr{E}})^* z \right\rangle > 0.$$

Without compression it is positive semidefinite and the already declared Moore–Penrose range calculation is used. A minibatch matrix is a principal compression of this one full-carrier matrix, not a separately selected kernel. It may be used when the selected Part I trace or batch certificate explicitly calls for that compression. The inverse-response regularizer is not separable across principal submatrices, so its solve is performed against the full safe-carrier matrix. Consequently its trace, effective dimension, and critic norm are precisely the empirical Part I quantities, rather than per-history or per-minibatch surrogates. One fitted iteration is:

```
q_target.load_state_dict(q.state_dict());  q_target.eval()
require(nonempty_safe_rows, status=authority_obstruction)
with no_grad():
next_q = q_target(next_history)
next_safe_value = masked_max(next_q, safe_mask)
target = reward - alpha * bridge_action
+ (1 - terminal) * gamma * next_safe_value
with no_grad():
Ksafe_full = safe_response(model, spec.full_safe_basis).detach()
for critic_batch in fitted_iteration:
q_taken = q(critic_batch.history).gather(critic_batch.action)
q_safe_full = q(spec.full_safe_rows).gather_safe()
effort = real(q_safe_full.mH
@ linalg.solve(Ksafe_full, q_safe_full))
critic_loss = weighted_square(q_taken, target[critic_batch])
+ rho * effort / 2
critic_optimizer.zero_grad();  critic_loss.backward()
critic_optimizer.step()
with no_grad():  gibbs = safe_gibbs(q, neutral_row, tau)
actor_loss = lambda_pi * mean(kl(actor.safe_row, gibbs))
actor_optimizer.zero_grad();  actor_loss.backward()
actor_optimizer.step()
if spec.differentiable_model_cell:
model_loss = frozen_critic_component(q, C)
+ lambda_dyn * dynamics_loss(model, dynamics_batch)
+ lambda_num * numerical_loss(model, numerical_batch)
model_optimizer.zero_grad();  model_loss.backward()
model_optimizer.step()
store_lag_support_shift_and_optimizer_errors()
```

The safe mask is applied before both the target maximum and Gibbs normalization. The target tensor and the Gibbs target are detached in their respective alternating steps. The represented masked reduction is defined only after the nonempty-safe-row check and therefore introduces no artificial infinite logit. The bridge action is charged either in the reward or as a hard legal-action constraint, exactly as in Definition II.11.23; it is never counted twice. When the bridge or dynamics block is trainable, its permitted update uses (263) after the critic and actor steps, with the critic and actor arguments frozen as specified there, and stores the induced target-network lag and model-selection error. Every component and displayed weight is retained in the terminal objective. Backpropagation is used only on its represented differentiable slice; masks, validation rules, finite decoders, and nondifferentiable audit coordinates are held fixed or searched by the inherited finite transcript and are evaluated again in the detached terminal record.

**Definition II.11.42** (Detached audit and saturated empirical selector)**.** After every terminal inner fit, detach all terminal tensors and evaluate the complete record on its assigned audit carriers. The audit performs the following operations in order.

(A1) Evaluate the generator residual, solve residuals, lumpability defect, dark and bracket coordinates, self-energy, response, empirical effective dimension, target capture, action, and target-subspace coverage gap.

(A2) Evaluate every applicable Part I statistical candidate after conversion to the same population loss, and set $U_W$ to their recordwise minimum as in Definition II.11.28. The Rademacher union/cover, PAC–Bayes mixture, split-validation, mixing, sequential, and channel candidates retain their own applicability and confidence fields. All candidates and every data-dependent terminal selection are covered by one output-selection-independent simultaneous event allocated before selection. A confidence interval chosen only after observing the terminal output is not a certificate. An unavailable candidate equals $+\infty$. Records with different population laws, estimands, norms, or unconverted loss units remain in different typed partitions and their upper bounds are never minimized against one another.

(A3) Evaluate the existing Part I master visibility with the Wick stability, full response, actual authorized bridge, and same-record interval fields:

$$V_M^W = \overline{A}_q S_W(\lambda) M_{\mathfrak{G}_q} C_{\mathfrak{G}_q} \frac{R_{\mathfrak{G}_q}}{1+\rho_{\mathfrak{G}_q}}. \tag{342}$$

Apply the Part I monotone interval evaluator to obtain $\underline{V}_W$. A missing qualification or unauthorized current gives lower visibility zero, as in the master certificate.

(A4) Store the complete resource vector, Caus action, risk and visibility intervals, confidence allocation, and typed status. The computation, architecture, parameter, training, optimization, precision, storage, deployment, evaluation, and upper-audit costs are stored in the corresponding slots of the Part I complete learner–deployment ledger. On an active or structural-Bayesian candidate these slots are exactly the complete neural implementation cost; on an RL candidate they also populate the complete dynamical learning record.

For each fixed supervised loss type $\ell$, let $\mathscr{R}^{(n)}_{W,\ell}(B,a)$ be its same-law, same-estimand, same-norm partition. At fixed $(B,a,\ell)$, return the $\varepsilon$-maximizers of (301) restricted to that partition. Across its charged resource–action grid, return all Pareto-minimal points of (302). The RL value-performance certificate is already the separately calibrated $U_{\mathrm{RL}}$ of (312). Thus a requested $(\epsilon,\theta)$ has the two typed returns

$$\widehat{\mathscr{M}}^{\mathrm{free}}_{W,\ell} := \mathrm{Min}_{\preceq \mathbf{c}}\{M \in \mathscr{R}^{(n)}_{W,\ell}(B,a) : U_W(M) \le \epsilon,\ \underline{V}_W(M) > \theta\}, \tag{343a}$$

$$\widehat{\mathscr{M}}^{\mathrm{RL}}_{W} := \mathrm{Min}_{\preceq \mathbf{c}}\{M \in \mathscr{R}^{(n)}_{W,\mathrm{RL}}(B,a) : U_{\mathrm{RL}}(M) \le \epsilon,\ \underline{V}_W(M) > \theta\}. \tag{343b}$$

The two sets may be identified only when a converter already carried by Part I places both guarantees on the same population law, estimand, norm, and loss scale. In that case the converted records form one named partition; the conversion error is charged before the recordwise minimum. An empty set returns its inherited typed status.

The outer computation can be expressed without differentiating through the structural order:

```
records = []
for spec in part1_stage:
```

| Empirical output | Evaluated Wick coordinate | Quantitative conclusion |
|---|---|---|
| Dynamics fit | $Q_n, s_n, P_{\mathrm{id}}, \ker Q_n$, generator and compression residuals | (251), (247), and (248) |
| Response audit | $E_{W,\Delta}, \Sigma_C, F_C, T_C$, dark/bracket status, empirical traces, capture and coverage | (289)– (291), with the full alternatives in Corollary II.11.26 |
| Unrestricted Caus | inverse-response ridge fit, actual action, bridge cover and validation radius | (258), including operator, selection, net, and optimization errors |
| RL | safe effective dimension, critic residual, converter, target lag, policy KL, authority and observation gaps | (264), (266), (268), and (312) |
| Statistical audit | Rademacher union/cover, PAC–Bayes mixture, dependence and channel candidates in one typed same-loss partition | their recordwise minimum $U_W$, the source-resolved minimum (287), and Theorem II.11.35 |
| Structural return | $\underline{V}_W$, typed $U_W$ or $U_{\mathrm{RL}}$, $\mathbf{c}_{\mathrm{res}}, \mathbf{c}_{\mathrm{act}}$ | (301), (302), and (343a)– (343b) |

Table 7: Empirical outputs and their quantitative certificate coordinates. The table is an execution map for the exhaustive ledger in Corollary II.11.26; it introduces no additional bound.

```
terminal = fit_inner_wick_record(spec)
records.append(certify_detached(terminal, spec.audit_splits))
for loss_id, same_loss_records in typed_partitions(records):
for B, action_ceiling in charged_grid(same_loss_records):
feasible = [r for r in same_loss_records
if resource_leq(r.cost, B) and r.action <= action_ceiling
and working_ucb(r) <= epsilon
and r.visibility_lcb > theta]
report(loss_id, pareto_minimal(
feasible, key=complete_resource_action))
```

Here `working_ucb` dispatches to $U_W$ on a supervised partition and to $U_{\mathrm{RL}}$ on an RL value-performance partition. The partition key contains the population law, estimand, norm, and loss scale, so the code cannot compare certificates with different meanings. No weighted sum of risk, visibility, capacity, action, and computation occurs in this outer loop. A weighted loss may generate a record only when that tradeoff is declared inside the task. The record is still compared by the same saturated profile and Pareto order as every other Part I candidate.

**Theorem II.11.43** (Certified empirical realization of the Wick learner)**.** *Apply the preceding procedure to the charged finite Part I stage. Then:*

(i) *every ordinary terminal execution compiles to a complete source-resolved learning ledger and a complete Wick certificate. An RL execution also compiles to a complete dynamical learning record, and an active or structural-Bayesian execution compiles to a complete neural implementation record. Every nonordinary execution compiles to the corresponding typed records;*

(ii) *the dynamics and unrestricted-Caus tensor losses equal* (249) *or* (250), (253)– (254), *respectively, on their represented finite carriers. The alternating RL steps are exactly the critic, actor, dynamics, and numerical components of* (263), *with their displayed weights and frozen arguments;*

(iii) *an Adam or block-preconditioned update has the represented form*

$$\theta_{k+1} = \theta_k - \alpha_k W_k^{-1} \widehat{m}_k, \tag{344}$$

*and is a diagonal or block instance of the Part I certified WAdam record. Its stationarity and global-gap conclusions are exactly those of the Part I WAdam descent theorem when that record certifies its alignment, second-moment, smoothness, and, for the global conclusion, PL coordinates. Otherwise the actual terminal optimization gap remains an extended record coordinate. Even on the global-gap cell, that objective gap enters the saturated error only through the objective-to-action, objective-to-risk, or objective-to-visibility converter already carried by Part I; without an ordinary converter it is only an optimization coordinate;*

(iv) *on the simultaneous Part I confidence event, the returned free-Caus records satisfy* (258), *and the returned RL records satisfy* (264)– (268)*; and*

(v) *within each typed same-loss partition, the outer return is exactly the Part I Pareto-minimal certified risk-qualified Wick frontier. The supervised return satisfies* (308)– (310)*; the RL return satisfies* (313). *Relative to a PAC–Bayes-only supervised selector, its feasible profile and strict inverse obey* (323) *and* (324)*; and*

(vi) *the empirical compiler adds no admission premise. Its charts, laws, losses, converters, masks, confidence event, costs, and search order are fields of the Part I record; the Wick generator, solves, losses, derivatives, and audit coordinates are finite compositions or totalized evaluations of those fields.*

*Consequently the empirical use of backpropagation changes only the method used to produce and record inner candidates. The meaning of optimal, minimal, and simplest remains the saturated-profile meaning of Definition II.11.29.*

*Proof.* The tensor compiler proposition proves structural fidelity and identifies the finite response with the response used in the theoretical losses. Expansion of the residual and ridge quadratic forms proves the loss identities. The full-carrier safe compression in (341) is (234) applied to (259); its principal minibatch compressions therefore retain the same response and normalization. Detaching the target produces the fitted supervised critic problem in (261), and the actor step is (262).

The stored optimizer state and update have the form (344); the Part I WAdam conclusions follow precisely on the cells where its derived certificate coordinates are ordinary. Part I's calibration step, rather than the optimizer theorem, then determines whether its objective gap contributes to an action, risk, or visibility bound. The detached audit uses the same data splits, preallocated simultaneous confidence event, source index, authority record, response, and loss as the terminal model. It therefore constructs the quantitative bounds of Corollary II.11.26 and Theorems II.11.22 and II.11.24. The supervised selector (343a) is the selector in Theorem II.11.31; the free and RL conclusions follow from Corollary II.11.32, with the latter using (343b). The PAC–Bayes comparison is Theorem II.11.35. Typed partitions prevent invalid cross-loss minima. Exact inheritance and no-new-premise closure are Propositions II.11.2 and II.11.3; finite linear algebra merely evaluates their derived coordinates. Extended values and typed statuses give the complementary numerical and statistical cells. □

### II.11.9 Executable refinement contract

An implementation must refine one charged finite Part I stage. It is not a procedure for constructing an uncharged continuum family. The following contract fixes the remaining programming choices and makes the return value of every execution branch explicit.

**Definition II.11.44** (Typed program objects)**.** For each candidate $s \in \mathscr{R}_W^{(n)}(B, a)$, the program stores six typed objects. `WickSpec` is immutable; every other object becomes an immutable serialized value when it is attached to the terminal record.

(i) `WickSpec` stores the stage and candidate identifiers, dimensions, real and complex dtypes, device, authority decoder, all represented grids, loss and law identifiers, confidence allocation, carrier identifiers and overlap relation, numerical tolerances, maximum iteration counts, and tie rule.

(ii) `WickParameters` stores the real trainable tensors $L_H, L_G, K, \phi_C$. Any inherited encoder, readout, critic, or actor parameters occupy separately named fields.

(iii) `WickOperators` stores the compiled tensors $H, G, A_{\mathcal{Y}}, C, B_C, R_C, \Sigma_C, F_C, A_C$, their dtypes and shapes, and no optimizer state.

(iv) `CarrierBundle` stores the represented dynamics, training, validation, RL, and audit carriers together with their identifiers, masks, weights, terminal fields, and normalization constants.

(v) `FitTranscript` stores the initialization, random seeds, minibatch order, optimizer states, iterates, stopping reason, terminal parameters, objective values, and optimization-gap candidate.

(vi) `AuditRecord` stores the detached residuals, statistical candidates, confidence event identifier, lower visibility, resource and action vectors, and typed status. A successful audit constructs (332); a failed audit constructs the same record with its typed failure status.

The immutable specification is hashed and its identifier is copied into every other object. Thus a tensor, loss, or certificate produced under one specification cannot be attached to another candidate without a type failure.

The dense finite-chart shapes are fixed by the following table. A structured decoder may use smaller parameter tensors, but its output has the displayed shape.

| Object | Shape and dtype | Mathematical value |
|---|---|---|
| $L_H, H, F_C, A_C$ | $d_Q \times d_Q$, real for $L_H, H$, complex otherwise | $H = L_H L_H^\top$, (335)– (336) |
| $L_G, G, K, A_{\mathcal{Y}}, B_C$ | $d_Y \times d_Y$, real before compilation | $G = L_G L_G^\top$, $A_{\mathcal{Y}} = (K - K^\top)/2$ |
| $C, R_C$ | $d_Y \times d_Q$, real $C$, complex $R_C$ | authority-decoded bridge and $B_C^{-1} C$ |
| $\Sigma_C$ | $d_Q \times d_Q$, complex | $C^* R_C$ |
| $E$ | $r \times d_Q$, complex | represented evaluation map |
| $K_{C,E}$ | $r \times r$, complex Hermitian | $E A_C^{-1} E^*$ |
| $\Phi_{\mathrm{tr}}, y_{\mathrm{tr}}$ | $m_{\mathrm{tr}} \times d_Q$, $m_{\mathrm{tr}}$ | supervised design and target |
| $q_{\mathrm{safe}}$ | $r_{\mathscr{E}}$, real | inherited observable-score slice |

Table 8: Code-level shape contract. The complex dtype has the same precision as the declared real dtype. In particular, float64 is paired with complex128. Every adjoint in the response calculation is a conjugate transpose.

The code boundary from the complex response space to supervised real outputs is the observation map carried by $s$. A target may instead be represented as a complex observation, in which case the loss is the modulus-squared loss in (253). The RL observation map has real range by Definition II.11.23; the program applies a maximum or exponential only after that map. Calling `real` on an arbitrary complex score is not an admissible replacement for this typed observation map.

**Definition II.11.45** (Typed numerical interface)**.** The numerical core exposes the following interfaces; each output retains the input specification identifier.

```
compile(spec, parameters) -> WickOperators | TypedFailure
response(spec, operators, evaluation_map) -> ResponseGram | TypedFailure
fit_free(spec, operators, carriers) -> FitTranscript | TypedFailure
fit_rl(spec, operators, carriers) -> FitTranscript | TypedFailure
audit(spec, operators, transcript, carriers) -> AuditRecord
select(stage, audit_records, epsilon, theta) -> ParetoFrontier | TypedFailure
```

Here `compile` evaluates (333) and (334)– (336); `response` evaluates (337) or (341); `fit_free` evaluates (253)–(254); and `fit_rl` evaluates the frozen critic, actor, dynamics, and numerical blocks of (263). The audit and selector are Definition II.11.42. These functions do not infer a grid, authority envelope, loss converter, safe mask, confidence split, or stopping rule from the observed objective values. The compiler, response, audit, and selector are pure functions of their stated inputs. The two fitting interfaces are deterministic when the stored seed, minibatch order, optimizer backend, and deterministic-backend flag are fixed; otherwise their complete realized transcript is part of the return value.

The runtime data-access validator checks every requested carrier identifier against the overlap relation in `WickSpec`. A split-sample candidate requires the declared disjoint carriers. A shared carrier is accepted only for the corresponding simultaneous, uniform, sequential, martingale, or PAC–Bayes candidate. The confidence allocation and candidate index are sealed before the first target-bearing carrier is read.

**Definition II.11.46** (Residual acceptance and finite-precision transfer)**.** Let tildes denote the tensors returned by the numerical linear algebra. The audit stores certified upper enclosures of the absolute residuals

$$r_B := \|B_C\widetilde{R}_C - C\|, \qquad r_A(V) := \|A_C\widetilde{X} - V\|, \tag{345}$$

$$r_f := \left\| \left( \frac{\Phi_{\mathrm{tr}}^*\Phi_{\mathrm{tr}}}{m_{\mathrm{tr}}} + \rho A_C \right) \widetilde{f} - \frac{\Phi_{\mathrm{tr}}^* y_{\mathrm{tr}}}{m_{\mathrm{tr}}} \right\|, \tag{346}$$

$$r_{\mathrm{herm}} := \|A_C - A_C^*\|, \qquad r_{\mathrm{gram}} := \|K_{C,E} - K_{C,E}^*\|. \tag{347}$$

Such an enclosure may be computed by directed rounding, interval arithmetic, or a proved backend error bound. A residual evaluated only in working precision is stored as a diagnostic and cannot by itself certify acceptance. The audit also stores scale-normalized versions of these residuals for diagnostics. Acceptance uses the declared absolute or mixed absolute-relative tolerances in `WickSpec`; no hard-coded tolerance enters the mathematical record. Put $\mathrm{Herm}(M) = (M + M^*)/2$. The structural audit additionally evaluates

$$\begin{gathered} r_H = \|H - H^*\|, \qquad r_G = \|G - G^*\|, \qquad r_{\mathcal{Y}} = \|A_{\mathcal{Y}} + A_{\mathcal{Y}}^*\|, \\ \lambda_{\min}(\mathrm{Herm}\, H), \qquad \lambda_{\min}(\mathrm{Herm}\, G), \qquad \lambda_{\min}(\mathrm{Herm}\, A_C). \end{gathered} \tag{348}$$

Let

$$M_f := \frac{\Phi_{\mathrm{tr}}^*\Phi_{\mathrm{tr}}}{m_{\mathrm{tr}}} + \rho A_C.$$

Validated eigensolver or interval bounds produce the record fields

$$\underline{\alpha}_B \le \lambda_{\min}(\mathrm{Herm}\, B_C), \qquad \underline{\alpha}_A \le \lambda_{\min}(\mathrm{Herm}\, A_C), \qquad \underline{\alpha}_f \le \lambda_{\min}(\mathrm{Herm}\, M_f). \tag{349}$$

The compiler's matrix-product rounding envelopes are included before these lower bounds are formed. In exact arithmetic the structural parameterization gives $\underline{\alpha}_B = \lambda$, $\underline{\alpha}_A = 1$, and $\underline{\alpha}_f = \rho$ as valid choices. Finite-precision code may use the sharper validated values. Every nonfinite value, shape or dtype mismatch, failed safe-row check, or residual outside tolerance produces `TypedFailure`. The same rule applies when one of the three lower bounds in (349) is unavailable or nonpositive. Jitter, clipping, projection, or eigenvalue truncation is permitted only when the modified operator and its perturbation radius are stored as fields of a new candidate.

The validated coercivity bounds give the a posteriori estimates

$$\|\widetilde{R}_C - B_C^{-1}C\| \le \frac{r_B}{\underline{\alpha}_B}, \qquad \|\widetilde{X} - A_C^{-1}V\| \le \frac{r_A(V)}{\underline{\alpha}_A}, \qquad \|\widetilde{f} - \widehat{f}_{C,\rho}\| \le \frac{r_f}{\underline{\alpha}_f}. \tag{350}$$

The audit propagates these quantities through (247)– (248) and stores the result as numerical operator and optimization error. Thus numerical acceptance never converts a residual into an exact identity.

**Definition II.11.47** (Finite executable learner)**.** The reference program executes the following finite outer algorithm. Its enumeration and selection control flow is deterministic under the declared tie rule; any permitted backend nondeterminism is confined to a fitting transcript.

(E1) Load one charged stage and validate every immutable specification, carrier identifier, overlap declaration, grid, converter, and confidence allocation before fitting.

(E2) For each $s$, train or enumerate the dynamics candidate according to its decoder type and its declared finite stopping rule. Compile the terminal generator and evaluate the identification and response-error coordinates.

(E3) For each charged $(\lambda, \rho, \tau)$ and coefficient vector, call `response`. Dispatch to `fit_free`, `fit_rl`, or the imposed-PDE evaluation branch according to the authority tag.

(E4) Stop every optimization loop at its declared maximum iteration count or stopping predicate and store the actual stopping reason and terminal gap. A divergent or interrupted loop returns a terminal transcript with its optimization status.

(E5) Detach the terminal tensors and call `audit`. The audit recomputes all structural objects from the terminal parameters rather than trusting cached training values. It then evaluates every predeclared applicable Part I bound on the common confidence event.

(E6) Partition accepted and nonordinary records by population law, estimand, norm, and loss identifier. Within each partition call `select` and return the risk-qualified saturated Pareto frontier (343).

All enumerated sets and iteration limits are finite at stage $n$, so this algorithm terminates. Candidate failures remain in the finite audit ledger; only a malformed stage that cannot be parsed produces a stage-level failure. Write $\mathsf{C}_W = \mathsf{CompleteWickRecord}$. Its total return type is

$$\mathsf{WickReturn} := \big[\mathsf{FiniteLedger}(\mathsf{C}_W) \times \mathsf{ParetoFrontier}(\mathsf{C}_W)\big] \sqcup \mathsf{TypedStageFailure}. \tag{351}$$

The directed Part I stage iterator is external to this finite call and retains its existing limit and nonattainment semantics.

**Theorem II.11.48** (Program refinement and certificate soundness)**.** *Let an execution conform to Definitions II.11.44 to II.11.47. Then:*

(i) *in exact arithmetic, every successful numerical-core output equals the finite-chart mathematical object named by its interface;*

(ii) *in finite precision, every accepted output has the a posteriori errors in* (350)*, and every dependent certificate includes their propagated operator or optimization error;*

(iii) *every returned ordinary record satisfies Theorem II.11.43, while every rejected or unavailable branch returns the corresponding totalized record; and*

(iv) *the outer return is the Part I Pareto-minimal risk-qualified saturated frontier within each typed partition.*

*Consequently a conforming implementation is a refinement of the mathematical learner: it may change optimization method, batching, device, or linear-algebra backend only through fields recorded in the transcript and numerical audit. It may not change the structural selection rule or the meaning of optimality.*

*Proof.* The tensor identities and exact-arithmetic solve properties are Proposition II.11.41. The free and RL dispatches use the same tensors, losses, frozen arguments, masks, and normalizations as Definitions II.11.21 and II.11.23; hence their successful exact-arithmetic outputs are the stated finite candidate objects. For finite precision, subtract the exact equations from (345)–(346) and use the three validated inverse bounds from (349). This proves (350). The perturbation bounds then produce the stored response errors.

The sealed specification and data-access validator preserve the candidate index, authority, loss type, sample law, and simultaneous confidence event. The detached audit therefore satisfies the hypotheses of Theorem II.11.43. Its totalized failure rules are Proposition II.11.3. Finally, steps (E5) and (E6) are exactly Definition II.11.42; the saturated-profile conclusion follows from Theorem II.11.31 and Corollary II.11.32. □

The definitions above determine a direct implementation order: first implement the immutable records and compiler, then the two solve routines, the branch-specific fits, the detached audit, and finally the typed saturated selector. Each layer has a local mathematical oracle in Table 9; no end-to-end training run is required to diagnose a failed algebraic layer.

| Required automated check | Acceptance criterion |
|---|---|
| Shape, dtype, and authority property tests | Every tensor satisfies Table 8; decoded bridges lie in the represented authority image. |
| Compiler identity tests | Symmetry, positivity, skew symmetry, energy cancellation, and residuals in (348) meet the declared tolerances; the outward-validated lower bounds in (349) are positive. |
| Block-elimination tests | Direct full-block solves agree with the compressed Feshbach solve within the propagated residual bound. |
| Kernel tests | Gram matrices are Hermitian positive semidefinite; range-compressed matrices are positive definite; empirical effective dimensions lie in $[0, r]$. |
| Free-Caus loss tests | Primal and dual KRR predictions agree on the represented evaluation range; automatic gradients agree with directional derivatives of the full outer loss within the declared derivative-test tolerance. |
| RL semantic tests | Terminal rows remove continuation, unsafe rows receive zero policy mass, empty safe rows return the authority status, targets are detached during critic fitting, and the full-carrier response is reused across minibatches. |
| Selection tests | Synthetic records with known partial order return exactly all and only the Pareto-minimal feasible records, separately for every typed loss partition. |
| Failure-injection tests | Nonfinite solves, excessive residuals, missing converters, unsupported carrier overlaps, zero safe support, and unavailable certificates return the specified typed status with upper bound $+\infty$ and lower visibility zero where ordered. |
| Reproducibility tests | The specification hash, seed, carrier identifiers, minibatch order, terminal tensors, and audit output are stored; deterministic backends reproduce the declared tolerance class. |

Table 9: Minimum verification suite for a conforming implementation. These tests verify program refinement and failure polarity. Statistical coverage is provided by the selected Part I certificate on its allocated event, not by a software test.

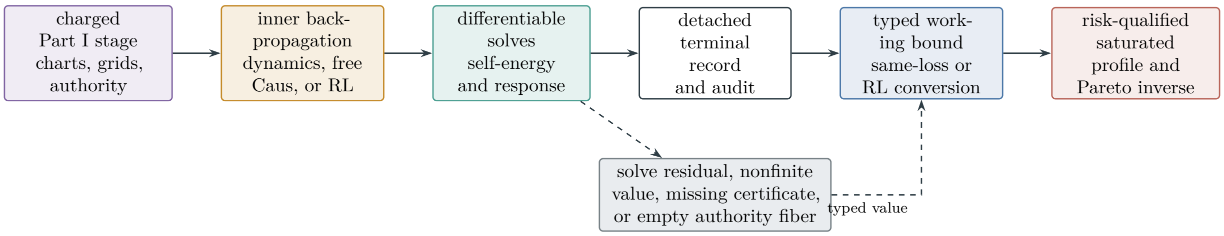


Figure 19: Empirical realization of the Wick learner. Differentiation is confined to the inner candidate fit. The detached record is audited under the inherited confidence allocation, all applicable same-loss bounds are combined by their valid minimum, the RL bound is separately calibrated into value units, and complete records are ordered only within their typed risk-qualified saturated profile and Pareto strict inverse.

### II.11.10 A solvable two-sector model

**Example II.11.49** (Scalar Wick quotient)**.** Let $\mathcal{Q} = \mathcal{Y} = \mathbb{C}$, $H_{\mathsf{Abs}} = \omega \geq 0$, $G = g > 0$, $A_{\mathcal{Y}} = 0$, and $C = c \in \mathbb{R}$. Then

$$\mathcal{L}_W(c) = \begin{pmatrix} -i\omega & c \\ -c & -g \end{pmatrix}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}(|x|^2 + |y|^2) = -2g|y|^2. \tag{352}$$

The quotient defect is $|c|$, the dark space is all of $\mathbb{C}$ when $c = 0$ and zero otherwise, and

$$\Sigma_c(z) = \frac{c^2}{z+g}, \qquad F_c(z) = z + i\omega + \frac{c^2}{z+g}, \qquad \eta_c^W(\lambda) = \frac{1}{1 + \left(\lambda + \dfrac{c^2}{\lambda + g}\right)^2 + \omega^2}. \tag{353}$$

The poles are the roots of

$$(z + i\omega)(z + g) + c^2 = 0. \tag{354}$$

Take the represented current coordinate to be $J_c = c$, the mobility to be $a > 0$, and $J_D = -a\nabla D$. Then

$$\mathscr{W}(c; D) = \frac{(c - J_D)^2}{2a}.$$

Free authority selects $c = J_D$; a restricted interval $[c_-, c_+]$ selects the Euclidean projection of $J_D$ onto that interval; and an imposed PDE coefficient $c = c_0$ retains the fixed waste $(c_0 - J_D)^2/(2a)$. The same $c$ occurs in the defect, self-energy, response weight, and action record.

*Remark* II.11.50 (Relation to familiar physics)*.* The exact correspondences proved here are spectral Wick rotation, Feshbach–Schur elimination, Mori–Zwanzig memory, conservative coupling, and real geometric dissipation. They explain why the resulting structural presentation has the same operator shapes as effective Hamiltonians, system–bath models, and Euclidean-to-spectral descriptions. Quantum measurement postulates, complete positivity of density-matrix evolution, gauge-field reconstruction, and completeness of a finite set of physical constants require additional axioms and are outside these theorems.

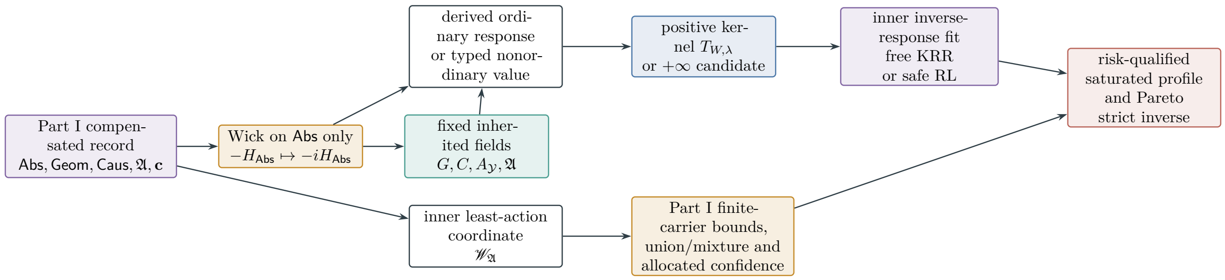


Figure 20: Wick learning as a derived operation on a Part I record. The Wick map changes only the spectral Abs evolution. The real Geom, actual authorized Caus bridge, authority envelope, provenance, and charges are fixed inherited fields. On every ordinary Part I realization, Feshbach elimination produces the self-energy and positive response kernel for every $\lambda > 0$, as a consequence of maximal dissipativity; a nonordinary realization returns its typed value and $+\infty$ response candidate. The unchanged bridge supplies the action record. Inverse-response KRR or action-safe RL trains the applicable authority branch. The Part I Rademacher union/cover or PAC–Bayes mixture/coefficient rule then uses finite evaluation-carrier traces to form the risk coordinate. The risk-qualified saturated usefulness profile then returns the Pareto-minimal points of its strict resource–action inverse at the inherited visibility threshold.

# Part III

# Structural Reachability and Regularity

## III.1 Singular targets and cascade compilers

The algebra and normalized shell identity first evaluate whether target-aligned structure exists. Its absence excludes contact directly. On the surviving aligned branch, contact itself generates the canonical infinite disjoint shell family, and (367) generates its sharp lower-price series in the single physical measure. Divergence excludes the source cell; finite total price generates zero-price tails and activates the compact successor relation. Shell accountability, cascade extraction, physical pricing, and ordered proof/model evaluation are consecutive constructive operations.

**Definition III.1.1** (Complete continuation target)**.** Let $\mathsf{Cont}^{\mathrm{tot}}$ be the total continuation relation of Definition I.1.1. For a finite maximal orbit, its terminal germ is the tuple of all limits of the generated evaluation coordinates along times increasing to $T_*$; compact metrizability supplies such germ subsequences. Let $\mathscr{B}_{\mathrm{cont}} \subset X_{\mathrm{ev}}$ be the set of terminal germs for which $\mathsf{Cont}^{\mathrm{tot}}$ has no ordinary regular extension. Define

$$\Sigma_{\mathrm{sing}} := \overline{\mathscr{B}_{\mathrm{cont}}}^{X_{\mathrm{ev}}}. \tag{355}$$

This is the complete singular target. Contact means convergence of a preterminal subsequence to $\Sigma_{\mathrm{sing}}$. Every finite maximal noncontinuation is therefore contact with this target. An empty local-solution fiber remains the separate form defect $\perp_{\mathrm{loc}}$; it is not target avoidance.

**Proposition III.1.2** (Terminal coverage)**.** *Every ordinary maximal orbit has exactly one of the following outcomes: $T_* = \infty$, regular continuation through its terminal germ, or contact with $\Sigma_{\mathrm{sing}}$. In particular there is no continuation remainder outside the singular target.*

*Proof.* Compact metrizability gives a terminal germ subsequence at finite $T_*$. The total continuation relation has either an ordinary extension value or its failure value. In the latter case the germ belongs to $\mathscr{B}_{\mathrm{cont}} \subseteq \Sigma_{\mathrm{sing}}$. An ordinary extension contradicts maximality unless it is recorded as continuation. These values are disjoint by totalization. $\square$

**Definition III.1.3** (Singularity-formation support)**.** For an ordinary maximal orbit $u$, let $\mathscr{D}_\Sigma(u)$ be a family of pre-contact shell records

$$R_{r,m} = (I_r, \widehat{\phi}_{r,m,I_r}, (Q^c_{r,m,I_r})_{c\in\mathfrak{C}})$$

attached to that same orbit. The family is *shell-complete* when contact implies that it contains such a record for every sufficiently small $r$. Every record generated by contact then satisfies

$$\mathrm{Prog}_\Sigma(R_{r,m}) = \sum_{c\in\mathfrak{C}} Q^c_{r,m,I_r}(1) = 1. \tag{356}$$

The singularity-formation support is

$$\mathfrak{S}_\Sigma(u) := \bigcup_{R\in\mathscr{D}_\Sigma(u)} \mathrm{Align}_\Sigma(R) \subseteq \mathfrak{C}. \tag{357}$$

It answers the target-relative question: which inherited source currents actually have positive pairing with the covectors descending toward the singular set? It does not ask which terms of the PDE are large in a target-independent norm.

**Proposition III.1.4** (A singularity requires aligned structure)**.** *If $u$ is generated by the compiled continuum algebra and $\mathscr{D}_\Sigma(u)$ is shell-complete, then*

$$u \text{ contacts } \Sigma \quad\Longrightarrow\quad \mathfrak{S}_\Sigma(u) \neq \varnothing. \tag{358}$$

*Consequently $\mathfrak{S}_\Sigma(u) = \varnothing$ implies target avoidance. This implication is independent of the size or compactness of the full orbit.*

*Proof.* Contact supplies normalized records satisfying (356). By Theorems I.5.2 and I.5.14, each has a nonempty aligned support, contained in the union (357). The contrapositive gives target avoidance. $\square$

**Theorem III.1.5** (Canonical disjoint shell cascade)**.** *Let a compiled ordinary maximal branch start outside $\Sigma$ and contact $\Sigma$ through a cofinal pre-contact sequence at its first contact or maximal time, and let $(U_r, \Phi_r)$ be a target-shell presentation. There are scales $r_j \downarrow 0$ and pairwise disjoint, chronologically ordered, strictly pre-contact intervals $I_j = [a_j, b_j]$ such that*

$$u(a_j) \notin U_{2r_j}(\Sigma), \qquad u(b_j) \in U_{r_j}(\Sigma), \qquad \mathrm{Prog}_\Sigma(I_j, \Phi_{r_j}) = 1. \tag{359}$$

*Every interval therefore has a nonempty aligned support and a source occurrence with positive target progress. On every ordinary or boundary current graph, one of the five source-resolved contributions has positive part at least $1/5$. No compactness, price gap, or contact margin is used in this extraction.*

*Proof.* Set $b_0 = 0$ and $r_0 = 1$. Since every strictly pre-contact state is outside $\Sigma = \bigcap_{r>0} U_{2r}(\Sigma)$, after $b_{j-1}$ choose a regular time $a_j > b_{j-1}$ and then a scale $r_j < \min\{r_{j-1}/4, 1/j\}$ for which $u(a_j) \notin U_{2r_j}(\Sigma)$. Contact supplies a later strictly pre-contact time at which the orbit belongs to $U_{r_j}(\Sigma)$; choose one as $b_j$. This inductively gives $b_{j-1} < a_j < b_j$, hence pairwise disjoint intervals. No continuity premise is needed: the two membership choices give $\Phi_{r_j}(u(a_j)) = 0$ and $\Phi_{r_j}(u(b_j)) = 1$ directly. Apply Theorem I.5.2 on each interval to get the final assertions. $\square$

**Definition III.1.6** (Generated renormalization-scale row)**.** Let $z_*$ be a contact point or germ. The normalization grammar is the least family containing the identity and the space, time, amplitude, gauge, and quotient transformations for which the fixed PDE's totalized covariance graphs have ordinary values. Each failed covariance value is a $\mathsf{Lift}$- or $\mathsf{Abs}$-tagged graph defect. Applying every ordinary grammar value to the canonical shell cascade produces scales $r_j \downarrow 0$, physical pre-contact events $Q_j = Q(z_*, r_j)$, and records

$$V_j = \mathcal{S}_{z_*,r_j} u \in X_{\mathrm{eval}},$$

where every $\mathcal{S}_{z,r}$ has its generated $\mathsf{Abs}/\mathsf{Lift}$ ancestry. Pass to the least infinite constant source-support and realization-mode subsequence in the fixed enumeration; it exists because both alphabets are finite. Let $\mathfrak{b}$ denote the resulting same-origin active branch signature. All closures and prices in this row are restricted to $X_{\mathfrak{b}}$. The row returns:

(R1) the geometric-separation value obtained, after the canonical dyadic subsequence, from $r_{j+1} \le \vartheta r_j$ for some $0 < \vartheta < 1$ and the events $Q_j$ are pairwise disjoint;

(R2) the totalized target-persistence inclusion $\overline{\{V_j\}} \subseteq B_{\Sigma,\mathfrak{b}}$, where $\mathfrak{b}$ is the active branch extracted from this same-origin cascade; the complementary value is a target-persistence graph defect;

(R3) the total-graph value of the physical pullback relation

$$\mu_u(Q_j) \ge c_{\mathrm{sc}}^* r_j^{\kappa} \overline{d}_{\mathrm{sc}}(V_j), \tag{360}$$

Here $c_{\mathrm{sc}}^*$ is the sharp transfer infimum, $\kappa$ is the exponent returned by the covariance graph, and $\overline{d}_{\mathrm{sc}}$ is the canonical lower-semicontinuous relaxation. If no scalar exponent represents the pullback, the row returns the full scale-dependent weight sequence or its graph defect. Thus scale covariance is evaluated rather than postulated.

**Theorem III.1.7** (Scale-cascade price compiler)**.** *Evaluate the totalized scale interface on its generated active branch $\mathfrak{b}$. Its normalized price value is*

$$\delta_{\mathrm{sc}} := \inf_{V \in B_{\Sigma,\mathfrak{b}}} \overline{d}_{\mathrm{sc}}(V) \in [0, \infty]. \tag{361}$$

*For the (j)-th generated scale stratum, define its sharp physical lower price exactly as in* (367) *and denote it by $\underline{p}_{\mathrm{sc},j}$. The exact scale-price value is*

$$\mathscr{S}_\Sigma := \sum_{j=1}^{\infty} \underline{p}_{\mathrm{sc},j} \in [0, \infty]. \tag{362}$$

*On the ordinary scalar-pullback branch, whenever the displayed factors are finite,* (360) *gives*

$$\underline{p}_{\mathrm{sc},j} \ge c_{\mathrm{sc}}^* r_j^{\kappa} \delta_{\mathrm{sc}}. \tag{363}$$

*The branch $\mathscr{S}_\Sigma = +\infty$ gives*

$$B_{\mathrm{phys}} \ge \mu_u\left(\bigsqcup_{j=1}^{\infty} Q_j\right) = \sum_{j=1}^{\infty} \mu_u(Q_j) \ge \sum_{j=1}^{\infty} \underline{p}_{\mathrm{sc},j} = \mathscr{S}_\Sigma = \infty, \tag{364}$$

*contradicting a finite physical budget. The stronger condition $a_* := \inf_j r_j^{\kappa} > 0$ gives the uniform event price $\mu_u(Q_j) \ge c_{\mathrm{sc}}^* a_* \delta_{\mathrm{sc}} > 0$ on the positive transfer and price branches. On the positive transfer and normalized-price branches, after restricting to $r_j \le 1$, divergence is automatic when*

*$\kappa \le 0$, and the scale-critical case $\kappa = 0$ transfers the normalized gap without loss. When $\kappa > 0$, the exact criterion is divergence of the generated lower-price series; a normalized gap alone supplies no such conclusion. The complementary value $\mathscr{S}_\Sigma < \infty$ produces vanishing-price windows in its inherited cell and is routed by Theorem II.2.46. In particular, a positive dissipative pullback exponent cannot be converted into a scale-independent price: the scale motion is retained in the signed carrier cohomology and then tested by the successor-saturation algebra.*

*Proof.* The definition of each sharp lower price gives $\mu_u(Q_j) \ge \underline{p}_{\mathrm{sc},j}$. Sum over the pairwise disjoint events to obtain (364); countable additivity and the finite physical budget give the contradiction. On the scalar pullback branch, insert (361) into (360) to obtain (363). If $0 < r_j \le 1$, then $r_j^\kappa \ge 1$ for $\kappa \le 0$. For $\kappa > 0$, the individual weights tend to zero, so their summability must be decided from the actual scale sequence. □

**Proposition III.1.8** (Canonical blow-up invariant image)**.** *Every contact cascade and every ordinary value of the generated normalization grammar has the canonical invariant image*

$$\left(\vartheta, \overline{\{V_j\}}, \mathfrak{H}^{\Sigma}_{\{V_j\}}, \kappa, \delta_{\mathrm{sc}}\right). \tag{365}$$

*Here $\vartheta \le 1/4$ is produced by the canonical shell subsequence, the closure is taken in the generated evaluation compactification, $\mathfrak{H}^{\Sigma}_{\{V_j\}}$ records whether the cascade remains target-visible, $\kappa$ or its full weight-sequence replacement is the physical pullback value, and $\delta_{\mathrm{sc}}$ is the canonical price infimum. A failed normalization or pullback occupies the corresponding* $\mathsf{Lift}/\mathsf{Abs}$ *graph-defect cell. Thus every blow-up outcome has an image; none of the five coordinates is additional problem data.*

**Definition III.1.9** (Normalized cascade refinement)**.** For a contact orbit, combine the canonical intervals from Theorem III.1.5 with every ordinary value of the generated normalization grammar, then pass to the least infinite constant source-support and realization-mode subsequence. Let $\mathfrak{b}$ be its same-origin active branch signature. The resulting normalized cascade refinement consists of

- (C1) pairwise disjoint measurable pre-contact events $Q_1, Q_2, \ldots$ in the domain of the same physical expenditure measure $\mu_u$;
- (C2) generated normalizations $\mathcal{N}_{Q_j}$, carrying their $\mathsf{Abs}/\mathsf{Lift}$ ancestry, such that $w_j := \mathcal{N}_{Q_j} u \in B_{\Sigma,\mathfrak{b}}$, for the persistent active branch $\mathfrak{b}$ selected from this same-origin cascade;
- (C3) for every $j$, a normalized shell-crossing record $R_j$ carried by $w_j$ for which

$$\mathrm{Prog}_\Sigma(R_j) = 1, \qquad \max_{c \in \mathfrak{C}} \left(\mathrm{Prog}^{c}_\Sigma(R_j)\right)_+ \ge \frac{1}{5}. \tag{366}$$

By Theorem I.4.6 and the applicable prospective shell-accountability theorem, every occurrence in clause (C3) has a nonempty aligned source support $A_j := \mathrm{Align}_\Sigma(R_j) \subseteq \mathfrak{C}$. Every nonordinary normalization value is already a typed graph-defect record. The refinement asserts no positive price; it generates the family on which the price row is evaluated.

**Proposition III.1.10** (Persistent channel extraction)**.** *For every cascade generated by Definition III.1.9, some nonempty support $A \subseteq \mathfrak{C}$ occurs for infinitely many indices. In particular, at least one of the five elementary source labels belongs to the ancestry of infinitely many target-approaching windows.*

*Proof.* There are only $2^5 - 1$ nonempty supports. The infinite pigeonhole principle gives an infinite constant-support subsequence. Any member of that support has the second asserted property. □

**Definition III.1.11** (Target-aligned source ledger)**.** Fix an ordinary maximal branch $u$ and a normalized cascade family $(Q_j, R_j)_{j\geq 1}$. For every nonempty $A \subseteq \mathfrak{C}$, let $\mathcal{F}_A(u)$ be the family of source-current occurrences extracted from those $R_j$ for which $\mathrm{Align}_\Sigma(R_j) = A$. The source ledger is generated as follows. For $j \in I_A := \{j : \mathrm{Align}_\Sigma(R_j) = A\}$, let the compiler first evaluates the total graph of the target-aligned charging relation (90). On its ordinary admitted value, let $\mathcal{Q}^{\mathrm{pay}}_{A,j}$ be the family of all generated ordinary events under the same physical ceiling and in the same propagated source/mode, shell/scale, and carrier-factorization stratum as $Q_j$, normalized to the same unit target progress. Equality of stratum means equality of every transcript entry preceding and including the charging row, so the equivalence relation is canonical. Define the sharp admissible physical lower price

$$\underline{p}_{A,j} := \inf\{\mu_v(Q) : (v, Q) \in \mathcal{Q}^{\mathrm{pay}}_{A,j}\} \in [0, \infty], \tag{367}$$

with $+\infty$ for an empty stratum. The stratum contains $(u, Q_j)$, so $\mu_u(Q_j) \geq \underline{p}_{A,j}$. Each source row has exactly one of the following exhaustive entries:

(L1) a *null entry*,

$$\mathfrak{H}^{\Sigma}_{\mathcal{F}_A(u)} = 0; \tag{368}$$

(L2) a *divergent-price entry*, when the target quotient is nonzero and

$$|I_A| = \infty \Longrightarrow \sum_{j\in I_A} \underline{p}_{A,j} = \infty; \tag{369}$$

(L3) a *finite-price successor entry*, when the target quotient is nonzero and either the charging relation has a nonordinary or failed graph value, or its ordinary admitted value satisfies

$$\sum_{j\in I_A} \underline{p}_{A,j} < \infty. \tag{370}$$

The target quotient, the totalized charging relation, and the dichotomy for a nonnegative series make the list exhaustive. A charging defect retains the same source, shell covector, and event origin and enters the successor algebra; it is never assigned price $+\infty$. Quotient, symmetry, conservation, carrier, resistance, compactness–rigidity, equality, scale, and entropy calculations are evaluators of these exact entries. A uniform positive lower price is a special divergent-price value. Since $|\mathfrak{C}| = 5$, the ledger has at most $2^5 - 1 = 31$ nonempty rows even when the cascade contains infinitely many scales.

**Theorem III.1.12** (Physical expenditure exclusion)**.** *Let $\mu$ be a nonnegative countably additive measure with $\mu(\Omega) \leq B_{\mathrm{phys}} < \infty$. Let $Q_j \subseteq \Omega$ be pairwise disjoint measurable sets and let $\delta_j \geq 0$. If*

$$\mu(Q_j) \geq \delta_j \quad \textit{for every } j, \qquad \sum_{j=1}^{\infty} \delta_j = \infty, \tag{371}$$

*then such a family cannot exist. The uniform-price criterion $\delta_j \geq \delta > 0$ is a special case.*

*Proof.* For every $N$, finite additivity and monotonicity give

$$B_{\mathrm{phys}} \geq \mu\left(\bigsqcup_{j=1}^{N} Q_j\right) = \sum_{j=1}^{N} \mu(Q_j) \geq \sum_{j=1}^{N} \delta_j.$$

The partial sums on the right are unbounded by (371), contradicting the fixed upper bound $B_{\mathrm{phys}}$. $\square$

**Corollary III.1.13** (Shell progress implies expenditure exclusion)**.** *On one ordinary same-origin active contact branch $\mathfrak{b}$, let $R_j$ be the normalized shell records on pairwise disjoint pre-contact events $Q_j$. On the ordinary admitted value of the target-local charging row,*

$$\mathrm{Prog}_\Sigma(R_j) = 1 \quad \Longrightarrow \quad \mu_u(Q_j) \geq \delta_j \geq 0. \tag{372}$$

*If $\sum_j \delta_j = \infty$, then $u$ cannot contact $\Sigma$.*

*Proof.* Contact supplies the unit identities by Theorem I.5.2. The price implication is a structural fact on $X_\mathfrak{b}$ by Theorems II.2.4 and II.2.18. It gives (371), and Theorem III.1.12 gives the contradiction. □

**Theorem III.1.14** (Finite structural-ledger exclusion)**.** *Let $u$ have the canonical contact cascade of Theorem III.1.5. On the fast-track value where every persistent target-aligned support is either zero in the target quotient or has a divergent physical-price series, $u$ does not contact $\Sigma$.*

*Proof.* On the contact branch, Proposition III.1.10 gives a nonempty $A \subseteq \mathfrak{C}$ for which $I_A := \{j : \mathrm{Align}_\Sigma(R_j) = A\}$ is infinite. If $A$ has a null entry, then every target projection of the corresponding source-current occurrences vanishes. This contradicts $A = \mathrm{Align}_\Sigma(R_j) \neq \varnothing$. If $A$ has a divergent-price entry, the pairwise disjoint events $(Q_j)_{j \in I_A}$ carry the divergent lower-price schedule $(\underline{p}_{A,j})_{j \in I_A}$, contradicting Theorem III.1.12. The ledger alternatives are exhaustive, so contact is impossible. □

**Corollary III.1.15** (Estimate-to-invariant reduction)**.** *In the setting of Theorem III.1.14, the canonical transcript evaluates each persistent source support as null, divergent-price, or finite-price successor. By Theorem II.1.19, no global estimate of the full orbit remains in the verdict after these row values are known. In particular, arbitrarily large target-null motion is absent from the reduced calculation. The last value is not a verdict: it activates Theorem II.2.46.*

**Corollary III.1.16** (Entropy exclusion is physical expenditure exclusion)**.** *On the entropy-cocycle, finite-error, carrier-domination, and positive-production rows of one same-origin active contact branch $\mathfrak{b}$, $\mathfrak{D}_j(I_j) \geq \delta > 0$ on every selected window and contact is impossible.*

*Proof.* By Corollary II.10.6, the glued production is a finite physical measure on this branch. Its use as the price of the selected shell events is admitted by Theorem II.2.4. Apply Theorem III.1.12 to its restrictions to the disjoint same-origin events $Q_j$. □

## III.2 The continuous structural regularity theorem

The five-channel conclusion is derived before any regularity estimate. The only equation-dependent step in target exclusion is the physical evaluation of the target-aligned class exposed by contact.

**Definition III.2.1** (Totalization of a continuum interface)**.** Let $F : D(F) \subset X_F \to Y_F$ be any partial interface used by the continuum argument: a constitutive realization, a gradient potential map, a target-shell derivative, a physical-price comparison, or a continuation map. Adjoin a failure value $\perp_F$ and define

$$F^{\mathrm{tot}}(x) = \begin{cases} F(x), & x \in D(F), \\ \perp_F, & x \notin D(F). \end{cases}$$

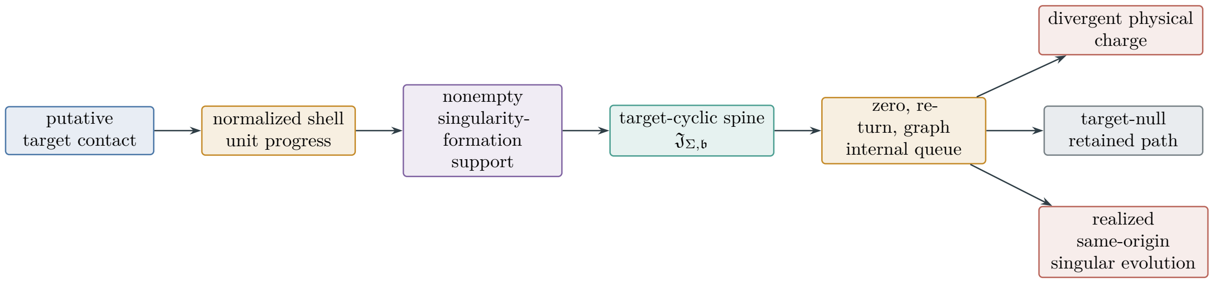


Figure 21: Compressed singularity-formation accounting. Contact exposes one target-aligned source cell. Projection-first saturation and signed gluing produce the target-cyclic feeding–storage spine. Zero-kernel restrictions, signed-dyadic return maps, and graph-boundary successors are internal queue transitions. The queue has three outputs: divergent physical charge, target-nullity, or a realized same-origin singular evolution.

Embed the graph of this total map in the structural test compactification and set

$$\partial F := \overline{\operatorname{graph} F^{\mathrm{tot}}} \setminus \operatorname{graph} F. \tag{373}$$

Thus $\partial F$ contains both limiting graph defects and the explicit failure records $(x, \perp_F)$. The *totalized interface record* of an input is its graph value when the input lies in $D(F)$, and otherwise its class in the closed defect cell generated by $\partial F$. The defect cell retains the ancestry of the approximating graph records; failure at a chart, weak-limit, or terminal interface appends $\mathsf{Lift}$ or $\mathsf{Bdry}$ according to Definition I.2.1. Its target-null part is quotiented by $\mathscr{N}_\Sigma$; its nonzero target quotient is a target-visible defect class in the same five-channel algebra.

**Definition III.2.2** (Total continuum interface algebra)**.** The target-relative continuum interface algebra is the finite coproduct

$$\mathfrak{H}^{\Sigma}_{\mathrm{cont}} := \mathfrak{H}^{\Sigma}_{\mathrm{form}} \sqcup \mathfrak{H}^{\Sigma}_{\mathrm{grad}} \sqcup \mathfrak{H}^{\Sigma}_{\mathrm{chain}} \sqcup \mathfrak{H}^{\Sigma}_{\mathrm{price}} \sqcup \mathfrak{H}^{\Sigma}_{\mathrm{cov}}. \tag{374}$$

Here the five cells are respectively the target-active quotients of:

(D1) the graph boundary of the local-solution map and of the native constitutive or sector-form realization;

(D2) $\overline{\operatorname{Ran} G} \setminus \operatorname{Ran} G$, together with the target-active part of $\ker G^*$, after the exact target-null quotient;

(D3) the source-resolved derived Stieltjes shell currents of (34), including their ordinary, Lebesgue, jump, Cantor, oscillation, concentration, exterior, and variation-boundary graph coordinates;

(D4) the total carrier graph, including failure to produce the finite physical measure dictated by the equation, and the zero-price limit of every finite-expenditure shell cascade;

(D5) the generated chart, gauge, scale, and quotient-covariance graphs, including every commutator defect created when an operation fails to descend.

Every cell is source-tagged before its target-null quotient is taken. A missing property is therefore an input to the activity lift and successor operator, not a terminal verdict. Regular continuation is not a sixth interface cell: by Definition III.1.1 and Proposition III.1.2, a finite maximal germ without an ordinary continuation is itself a point of the complete target and is processed by the same shell and successor machinery.

**Theorem III.2.3** (Defect totalization and five-channel closure)**.** *For every fixed continuum evolution, target, and physical carrier, each of the five interfaces in Definition III.2.2 has an* *ordinary total-graph value or a graph-boundary value. Every target-visible value has ancestry in* $\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}$*; every target-invisible value lies in* $J_{\mathrm{res}}$*. Every target-visible value, ordinary or defective, is converted into an activity measure and a witness-indexed successor. Hence lack of closed range, a classical chain rule, positive price, or covariance is not a stopping point: it is a concrete successor mechanism in the same closed algebra. Finite noncontinuation is target contact, not an unprocessed remainder.*

*Proof.* For each partial map, the total value is either an ordinary graph value or the failure value $\perp_F$. Closing the total graph adds its limiting defects, so (373) exhausts both kinds of failure. Sourcewise closure and Theorem I.3.4 retain the ancestry of every approximating record. The gradient cell is exhausted by Theorem I.3.8. For the shell interface, the finite approximation grammar gives the exact polynomial Stieltjes identity (34). Sourcewise graph closure passes that equality to all Radon and variation-boundary values and preserves the ancestry of every summand; there is no untyped chain-rule remainder. A finite-price cascade enters Definition II.2.30; a carrier-graph failure is lifted by the same total-graph coordinate. Covariance and descent failures retain the tags of the operation and commutator that generated them. Finally quotient each cell by its intersection with $\mathscr{N}_\Sigma$. The kernel is residual. Every nonzero quotient value enters $\mathcal{R}_\eta$, and Definition II.2.39 iterates it until it is eliminated or lies on an infinite zero-price path. Terminal coverage handles finite noncontinuation. This proves all assertions. $\square$

**Theorem III.2.4** (No favorable analytic premise in constructive closure)**.** *Fix only the canonical weak-PDE signature, its complete continuation target, and its equation-generated physical carrier. The source compiler, sourcewise closure, target quotient, total interface graphs, activity lift, successor relation, and finite-depth viability sets are defined without requiring any of the following properties:*

$$\begin{gathered}\textit{a gradient/sector realization, local continuation, compactness,}\\ \textit{continuation of the full branch, or closed gradient range,}\\ \textit{a classical chain rule, smooth shells, coercivity, a positive price gap, rigidity, polarity,}\\ \textit{finite resistance, summable tails, orbit realization, or continuation.}\end{gathered} \tag{375}$$

*Each listed property is tested by a total graph generated from the equation. Both its ordinary and boundary values are supplied to the same activity lift, successor relation, and finite-depth viability recursion. A branch therefore cannot terminate at the sentence that a property fails: it is removed by the target-null quotient, charged by the physical carrier, or retained as an explicit zero-price recurrent mechanism. Consequently none of the listed properties is a premise of the totalization, contact-accountability, or constructive regularity theorems.*

*Proof.* The compiler is structural induction on the fixed equation syntax. Sourcewise closure is defined separately in each tagged cell. Quotienting by the common target kernel is exact by definition. Totalization replaces every partial interface by the disjoint union of its graph values, explicit failure values, and graph-boundary values. The realization modes partition the resulting ordinary and defect records. The positive target-current part of each record defines its activity measure, whose normalized lift defines $\mathcal{R}_\eta$. Iterating $Z_{n+1} = \mathcal{V}_\eta(Z_n)$ removes every record without arbitrarily long zero-price descendants. Compactness turns survival at every finite depth into an infinite path and a stationary path law. Thus both sides of every analytic relation undergo an explicit structural operation, and no favorable side of a dichotomy is selected. $\square$

**Definition III.2.5** (Physical reachability profile)**.** For a finite number $B_{\text{phys}}$ arising from the carrier identity, let $\mathscr{O}^{\text{ord}}_{\le B_{\text{phys}}}$ be the ordinary maximal solution branches whose derived expenditure satisfies $\mu_u(\mathfrak{Q}_u) \le B_{\text{phys}}$. When this set is nonempty, define

$$\mathrm{Reach}^{\text{phys}}_{\Sigma}(B_{\text{phys}}) := \sup_{u \in \mathscr{O}^{\text{ord}}_{\le B_{\text{phys}}}} \mathbf{1}_{\{u \text{ contacts } \Sigma\}}. \tag{376}$$

There is no empty-supremum convention. If the ordinary solution stratum is empty, the total local-solution or carrier graph returns its explicit defect record. Thus $\mathrm{Reach}^{\text{phys}}_{\Sigma}(B_{\text{phys}}) = 0$ means that ordinary maximal branches exist and none contacts the target. The ceiling is not a second datum: it is only a stratum of the one physical expenditure generated by the equation.

**Definition III.2.6** (Constructive zero-price mechanism cells)**.** Fix a derived finite physical stratum $B_{\text{phys}}$. For nonempty $A \subseteq \mathfrak{C}$ and $m \in \mathfrak{M}$, let $Z^{\Sigma}_{\infty,A,m}(B_{\text{phys}})$ be the part of $Z^{\Sigma}_{\infty}$ whose inherited support is exactly $A$ and whose primary mode is exactly $m$. Full ancestry and all nonprimary flags remain coordinates of the record. Define

$$\mathfrak{F}^{\Sigma}_{A,m}(B_{\text{phys}}) := \overline{\operatorname{span}} \left\{ \Pi_{\Sigma}(\pi_0)_{\#}\boldsymbol{\nu} : \boldsymbol{\nu} \text{ stationary on } \mathsf{Path}(Z^{\Sigma}_{\infty,A,m}(B_{\text{phys}})) \right\}, \tag{377}$$

$$\mathfrak{F}^{\text{phys}}_{\Sigma}(B_{\text{phys}}) := \bigoplus_{\substack{\varnothing \ne A \subseteq \mathfrak{C} \\ m \in \mathfrak{M}}} \mathfrak{F}^{\Sigma}_{A,m}(B_{\text{phys}}). \tag{378}$$

These cells are not defined from contact orbits. They are the finite source/mode decomposition of the prospectively generated zero-price fixed point. The direct sum is provenance-tagged: analytically equal representatives with different ancestry remain distinct. Every generator carries its stationary path law, full weak equation or recurrent graph boundary, target pairing, and diagonal cascade-origin witnesses.

**Theorem III.2.7** (Constructive source decomposition of zero-price dynamics)**.** *For every derived finite physical stratum $B_{\text{phys}}$,*

$$\boxed{\mathfrak{H}^{\Sigma}_{\text{succ}}(B_{\text{phys}}) = \mathfrak{F}^{\text{phys}}_{\Sigma}(B_{\text{phys}}).} \tag{379}$$

*The right side has at most $31 \times 5$ provenance-tagged cells. Every finite-expenditure contact orbit produces a nonzero generator in one of them. Consequently, whenever the ordinary stratum is nonempty,*

$$\mathfrak{F}^{\text{phys}}_{\Sigma}(B_{\text{phys}}) = 0 \quad \Longrightarrow \quad \mathrm{Reach}^{\text{phys}}_{\Sigma}(B_{\text{phys}}) = 0. \tag{380}$$

*A nonzero cell is not a label awaiting a further estimate: its generator is a stationary zero-price successor law carrying the weak equation, recurrent graph defects, target pairing, source ancestry, and cascade-origin witness sequence; an ordinary contact origin makes these actual physical events.*

*Proof.* Along a successor path the inherited support can only increase. Because the source alphabet is finite, it is eventually constant; stationarity makes it constant almost surely. The primary mode is likewise monotone in a finite ordered set and is stationary only when constant almost surely. Decomposing every stationary path law along these invariant finite parts proves (379).

If an ordinary branch contacts $\Sigma$, the total shell identity gives the disjoint unit-progress cascade without a chain-rule premise. Theorem II.2.46 converts that same cascade into a stationary zero-price law with nonzero target projection and diagonal origin-event witnesses. The finite tag argument places a nonzero component in one $(A, m)$-cell. The contrapositive proves (380); the stated data are supplied by Theorem II.2.43 and Lemma II.2.33. □

**Theorem III.2.8** (Finite-depth structural evaluator)**.** *For every source/mode cell, start with its generated target-active zero-price records and form*

$$Z_{0,A,m} \supseteq Z_{1,A,m} \supseteq \cdots, \qquad Z_{n+1,A,m} = \mathcal{V}_\eta(Z_{n,A,m}). \tag{381}$$

*Exactly one of the following constructive outcomes occurs:*
(E1) *at some finite cylinder-moment level the dual proof cone contains* $-1$*; the finite ordered identity produces the pruning transcript of Theorem II.2.41 and closes the cell;*
(E2) *every finite moment level is feasible; diagonal reconstruction produces an infinite successor path law and a nonzero generator of* $\mathfrak{F}^{\Sigma}_{A,m}(B_{\mathrm{phys}})$*.*
*Applying the generated zero-cost equations to outcome* (E2) *removes each target-null component and recursively expands each ordinary or graph-boundary component. Thus every nonzero row value undergoes a further structural operation.*

*Proof.* Apply Theorem II.2.38 to the path algebra restricted to cell $(A, m)$. Its dual branch is (E1) and its primal projective-limit branch is (E2). The ordered ideal contains the target floor, zero price, same-origin successors, and source/mode coordinates, so reconstruction retains all of them. The final reduction is Theorem II.2.43 applied to every support component. □

**Theorem III.2.9** (End-to-end constructive closure)**.** *For a derived finite physical stratum* $B_{\mathrm{phys}}$*, apply source compilation, the target quotient, total interface graphs, shell activity lifting, witness-indexed successors, finite-depth viability pruning, and the generated zero-cost equations, in that order. The output is*

$$\boxed{\mathfrak{H}^{\Sigma}_{\mathrm{out}}(B_{\mathrm{phys}}) = \mathfrak{H}^{\Sigma}_{\mathrm{succ}}(B_{\mathrm{phys}}) = \bigoplus_{A,m} \mathfrak{F}^{\Sigma}_{A,m}(B_{\mathrm{phys}}).} \tag{382}$$

*On every ordinary local finite-carrier branch,*

$$\bigoplus_{A,m} \mathfrak{F}^{\Sigma}_{A,m}(B_{\mathrm{phys}}) = 0 \quad \Longrightarrow \quad \mathrm{Reach}^{\mathrm{phys}}_{\Sigma}(B_{\mathrm{phys}}) = 0. \tag{383}$$

*The compiler never consults an actual-contact image. A contact orbit itself constructs a nonzero cell. Conversely, a cell that survives the evaluator comes with an infinite zero-price path, stationary law, full generated saturation equations, source support, target pairing, and diagonal origin events. A finite exclusion carries its rational ordered derivation. A surviving cell carries the positive-functional model of all enumerated relations; coherent approximation-origin paths are retained separately from relaxed models.*

*Proof.* Totalization inserts every target-visible local-solution or carrier boundary witness into the same successor graph. On an ordinary finite-carrier branch, source and mode tags partition the zero-price path laws and Theorem III.2.7 identifies their direct sum with $\mathfrak{H}^{\Sigma}_{\mathrm{succ}}$, proving (382). If contact occurs, shell accountability, disjoint cascade extraction, and Theorem II.2.46 construct a nonzero stationary path law in one cell. Its contrapositive proves (383). The finite-depth evaluator and saturation-equation compiler supply the listed data for every surviving generator, without an actual-contact filter. □

**Theorem III.2.10** (Unconditional continuum contact mechanism)**.** *Fix the canonical physical continuum presentation and close its orbit algebra sourcewise. Every target-active current, total-graph defect, and target-visible limit has ancestry in* $\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}$*, while*

$$G^{J_{\mathrm{res}},\mathrm{src},\mathrm{sat}}_{B_{\mathrm{phys}},\Sigma} = 0. \tag{384}$$

*Every contact orbit has a canonical sequence of pairwise disjoint pre-contact shell intervals $I_j$ with unit target progress. After passage to a subsequence, one nonempty support $A \subseteq \mathfrak{C}$ occurs on every $I_j$ and contributes positive target progress there. This statement does not require a constitutive realization, classical chain rule, or price gap: every failed relation contributes through its derived typed graph current with the ancestry of the failed operation.*

*Proof.* The continuum normal-form compiler and source-preserving completion give the five-channel statement for ordinary values. Derived graph currents give the same statement for every boundary value. The target quotient gives (384): by construction residual means target-null. The canonical disjoint shell-cascade theorem supplies the intervals. On each interval the derived shell identity has increment one. Hence one source-resolved current has positive target pairing. The infinite pigeonhole principle selects a persistent support. $\square$

**Theorem III.2.11** (Constructive physical-price evaluation)**.** *In the setting of Theorem III.2.10, restrict to the derived finite physical stratum $\mu_u(\mathfrak{Q}_u) \leq B_{\mathrm{phys}} < \infty$ of ordinary maximal branches beginning outside $\Sigma$. For each persistent support use the canonical sharp prices of* (367)*. Then:*

(P1) *if*

$$\sum_{j=1}^{\infty} \underline{p}_{A,j} = \infty. \tag{385}$$

*then that support cannot sustain contact;*

(P2) *if the series is finite, the corresponding shell tails enter the zero-price ordered path algebra. A checked identity $-1 \in \mathcal{K}_{\mathfrak{b}}$ excludes the active cell; otherwise the order-unit alternative supplies a stationary zero-price law in $\mathfrak{F}^{\Sigma}_{A,m}(B_{\mathrm{phys}})$.*

*If every persistent support is removed either by the divergent-price argument or by a finite ordered identity, then*

$$\boxed{\mathrm{Reach}^{\mathrm{phys}}_{\Sigma}(B_{\mathrm{phys}}) = 0.} \tag{386}$$

*Thus a finite lower-price series is never the end of the target-reachability track.*

*Proof.* If an ordinary maximal branch contacted $\Sigma$, Theorem III.2.10 would supply pairwise disjoint intervals and a persistent support $A$. On their selected subsequence, the definition of the sharp price and countable additivity give

$$B_{\mathrm{phys}} \geq \sum_j \mu_u(I_j) \geq \sum_j \underline{p}_{A,j} = \infty,$$

a contradiction in case (P1). In the finite-series case, use the actual prices as the price coordinate in the activity lift. Cofinal tails have price tending to zero, so Theorems II.2.36 and II.2.46 give precisely the two outcomes in (P2). If every persistent support is eliminated, contact has no possible support, contradicting shell accountability. This proves (386). $\square$

*Remark* III.2.12 (Orbitwise source ledgers)*.* The divergent-price branch is evaluated only for the source support that persists along a putative singular cascade. Its canonical lower-price schedule may depend on the orbit stratum and may decay with scale. A uniform positive price is stronger and yields uniform window counts. Theorem III.2.11 uses only divergence of this generated series; five-channel exhaustion and cascade existence were proved earlier.

**Theorem III.2.13** (Total constructive continuum regularity evaluator)**.** *Let $u_0$ be a datum in the public state class of the fixed equation. Totalize the local-solution and carrier graphs and run the constructive closure of Theorem III.2.9 on every ordinary maximal branch. Put $\eta_0 = 1/5$. Exactly one of the following algebraic outcomes holds:*

(i) *every contact-generated same-origin branch reaches a local target-null, continuation, class-incompatibility, or divergent physical-charge cell; then no ordinary maximal branch from $u_0$ with finite physical expenditure contacts $\Sigma_{\text{sing}}$. Terminal coverage consequently supplies continuation for every such ordinary branch;*

(ii) *diagonal target-aligned saturation reconstructs an explicit singular mechanism, including its closed reduced five-channel equation, physical ledger, origin tag, and actual cofinal shell contact.*

*Finite noncontinuation is already contact with the complete continuation target. It never produces a separate continuation class. An empty ordinary local-solution fiber is retained as the form value $\perp_{\text{loc}}$ and is not converted into either regularity or singularity. Positive stationary functionals, equality kernels, and typed defects are intermediate successor states and do not constitute a third outcome.*

*Proof.* On an ordinary branch, finite maximal time without continuation is contact by Proposition III.1.2. Contact then gives disjoint unit-progress shells and a persistent five-channel support by Theorem III.2.10. A divergent sharp price contradicts the finite carrier. A finite-price cascade enters the zero-price successor algebra. The normalized shell identity gives a source contribution at least 1/5, so no additional target-floor parameter is needed. Every finite local closing cell contradicts the retained target floor or the finite physical ledger, so if all generated contact branches close, contact is impossible. Terminal coverage then turns that target-local conclusion into continuation of each ordinary branch. Otherwise the ordered proof/model alternative produces the initial target-retaining compiler state. Apply Theorem III.4.34. Its charge and null outputs close the contact branch; its realized output is precisely (ii). Nonvacuity and realization follow from Proposition I.1.11 and Theorem I.1.4. □

**Theorem III.2.14** (Target-aligned physical fast track)**.** *Fix a generated finite physical stratum and let $\Gamma_\Sigma$, $\mathscr{J}_\Sigma$, and the sourcewise shell currents be as above. For each reachable active branch $\mathfrak{b}$, let $\mathcal{K}_{\mathfrak{b}}$ be the ordered proof cone generated by Definitions II.2.22 and II.2.35. Then*

$$\boxed{\operatorname{Reach}^{\text{phys}}_{\Sigma}(B_{\text{phys}}) = 0 \quad \textit{or} \quad \mathfrak{S}_\Sigma(B_{\text{phys}}) \neq \varnothing,} \tag{387}$$

*where every member of $\mathfrak{S}_\Sigma(B_{\text{phys}})$ is an explicit singular mechanism generated by zero-price resaturation. At an intermediate finite level, failure of an ordered identity constructs a normalized stationary positive-functional successor satisfying*

$$\Pi_\Sigma\left(\ker Q_{\text{phys}} \cap \ker I_{\mathfrak{b}} \cap \bigcap_{w \in \mathscr{W}_{\mathfrak{b}}} \{P_w = 0\}\right) \neq 0 \tag{388}$$

*for some reachable $\mathfrak{b}$. The successor is reprocessed by Definition III.4.23; support, compactness, passivity, fidelity, and realization values are internal queue coordinates. By Theorem III.4.34, the successor is never a third mathematical output.*

*Proof.* Suppose contact occurs. The derived shell-current theorem gives infinitely many disjoint unit-progress events. Finite source exhaustion gives a persistent nonempty support $A$, and event normalization gives the universal target floor 1/5. If the physical prices have divergent sum, countable additivity contradicts the finite physical budget. Otherwise a subsequence has price tending to zero and the same-origin successor construction gives a stationary target-aligned zero-price law in an active branch $\mathfrak{b}$. This law defines the positive functional in Theorem II.2.36; weak duality forbids $-1 \in \mathcal{K}_{\mathfrak{b}}$. Run the target-aligned recurrent closure compiler on this state. Its divergent-charge and target-null outputs contradict contact; its remaining output is the explicit mechanism in $\mathfrak{S}_{\Sigma}(B_{\mathrm{phys}})$. This proves (387).

Conversely, failure of the ordered derivation gives the positive functional by Theorem II.2.36. Its ideal relations are the weak PDE, source-adapted zero-production, successor, and stationarity equations, while its target floor makes the target projection nonzero. This is (388). Approximation realization is exactly Theorem I.1.4; it distinguishes a coherent generalized solution path from its next typed successor. Applying the three-output current compiler to every typed or ordinary outcome gives the two displayed mathematical alternatives after the finite physical ledger removes the divergent-charge cell. ☐

**Corollary III.2.15** (Generated PDE reachability output)**.** *Let $\mathfrak{U}_{\mathrm{pub}}$ be the datum class in the public PDE presentation. Derive the physical expenditure from the carrier identity and run the source, signed-carrier, witness-successor, viability, and saturation compilers on every resulting finite stratum. For each stratum the generated output is exactly*

$$\boxed{\mathrm{Reach}^{\mathrm{phys}}_{\Sigma_{\mathrm{sing}}} = 0 \quad \textit{or} \quad \textit{an explicit same-origin singular mechanism is reconstructed.}} \tag{389}$$

*The second output includes the closed reduced equation, ordinary realization of every total graph, the physical and origin ledgers, and actual contact with a cofinal shell family. Positive functionals and compatible equality successors are intermediate resaturation states and do not appear in* (389)*. Neither output is a premise supplied with the PDE presentation.*

*Proof.* Apply Theorem III.2.14 at the derived finite ceiling of each branch. The finite ordered hierarchy initializes the local queue, and Theorem III.4.34 evaluates all its descendants. Charge or target-nullity gives the first output; the compatible realized queue limit gives the explicit singular mechanism in the second. ☐

**Corollary III.2.16** (Quantitative finite-cascade bound)**.** *Fix an active branch $\mathfrak{b}$. If the physical expenditure satisfies $\mu_u(\mathfrak{Q}_u) \le B_{\mathrm{phys}}$ and each event in a disjoint same-origin family has normalized window in $B_{\Sigma,\mathfrak{b}}$ and admissible target-local price at least $\delta_{\mathrm{phys}}(\mathfrak{b}) > 0$, then that family has cardinality at most*

$$\left\lfloor \frac{B_{\mathrm{phys}}}{\delta_{\mathrm{phys}}(\mathfrak{b})} \right\rfloor. \tag{390}$$

*Proof.* The proof of Theorem III.1.12 gives $N\delta_{\mathrm{phys}}(\mathfrak{b}) \le B_{\mathrm{phys}}$ for every finite subfamily. Eligibility of each price is exactly Theorem II.2.4; events outside $B_{\Sigma,\mathfrak{b}}$ do not enter this count. ☐

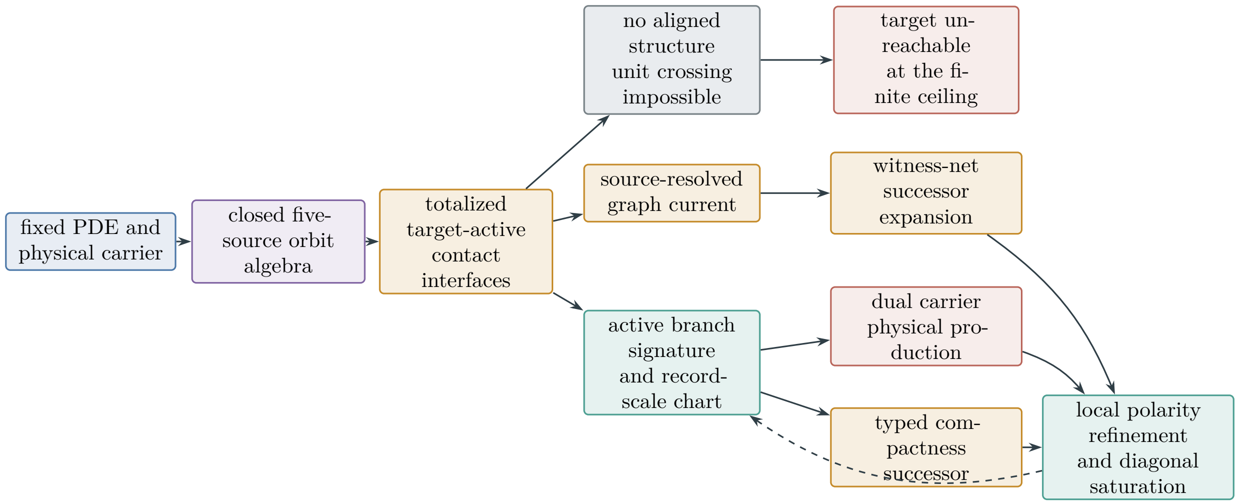


Figure 22: Constructive singularity evaluator. Only the target-active quotient is analyzed. If it contains no singularity-aligned structure, normalized shell contact is impossible. Otherwise contact fixes an active branch signature and, when needed, a record-scale chart. The source-adapted dual carrier either charges target progress to physical production or creates a typed compactness successor. The next unresolved coordinate is split only on the retained target cell; every nonnull polarity cell re-enters the same compiler with larger resolved-prefix rank. A cofinal refinement is saturated diagonally, and a realized same-origin contact path is the sole singular terminal.

## III.3 Structural evaluator coverage and failure semantics

The following table is the coverage proof for the fast track. Every classical analytic question is assigned a generated invariant and every opposite value activates a structural operation. No row accepts a desired analytic conclusion as input or returns a bare label.

| Analytic question | Generated structural invariant | Successor action |
|---|---|---|
| Does a native, chain, trace, or covariance relation hold? | target-active boundary of its total graph | activity lift of the typed graph witness |
| Does a reduction descend through later operations? | commutator of the operation with the quotient map | typed descent defect |
| Is the gradient range closed where the target sees it? | $\mathfrak{H}^{\Sigma}_{\mathrm{grad}}$ | inherited closure or harmonic class |
| Is a tail structurally closed? | $\mathfrak{H}^{\Sigma}_{\mathrm{tail}}$ | target-visible tail cokernel |
| Is target feeding dominated? | $\alpha_*$ | supercritical or positive null-cost feeding profile |
| Is normalized target progress physically priced? | $\delta_*$ | zero-price sequence in the same source cell |
| Can the source be routed? | $\mathcal{R}(\Sigma)$ | target-visible harmonic non-routable class |
| Does a Wick quotient admit compensated target response? | $F_W(z)$, $T_{W,\lambda}$, dark space, and bracket gap | certified response/action profile or a typed dark, authority, or transfer obstruction |
| Is the target polar? | $\mathrm{Cap}(\Sigma)$ | positive-capacity target-compatible class |
| Does a scale cascade exceed finite expenditure? | $\mathscr{S}^{\Sigma}$ | finite-price successor chain and recurrent saturation row |
| Do internal signed currents create net target feed? | carrier cohomology $[F_{\mathrm{sc}}-D-M]$ | nonzero scale-feed cohomology successor |
| Does a failed interface persist dynamically? | recurrent witness boundary $\partial_\infty F$ | target-null loss or recurrent failure mechanism |
| Does a sharp branch close? | $\mathfrak{H}^{\Sigma}_{\mathrm{sat}}$ | nonzero saturation/equality remainder |
| Are stochastic fluctuations invisible or routed? | $\chi^{\Sigma}$ and fluctuation resistance | target-visible covariance/bracket class |
| Does a weak-solution graph value persist? | recurrent witness boundary $\partial_\infty F$ | finite-depth removal or recurrent weak-equation path |
| Can zero-price target progress persist? | ordered cones $\mathcal{K}_{\mathfrak{b}}$ and finite moment levels | finite rational derivation or a reconstructed stationary law |

*Remark* III.3.1 (Exact input closure). The fast-track data are constructed from the public equation, generated target, source grammar, and native physical carrier. A specialization cannot append compactness, coercivity, rigidity, nonpolarity, continuation, or a desired invariant value. Each relation and its complement are graph values. The complementary value is converted into a typed successor on the surviving target-active quotient, and its descendants are processed by finite-depth viability within the same generated recursion.

*Remark* III.3.2 (Separation of invariant roles). The logical interfaces enforce ten distinctions.

(N1) An imposed $\mathsf{Caus}$ term is transported by a chart and is not optimized toward the target. A free or hybrid authority envelope may optimize only its declared represented control slot.
(N2) A killed spectral resolvent is mapped to arrival only through the totalized terminal-flux realization row; a nonzero realization defect is retained.
(N3) Finiteness of a basic energy does not imply a positive cost at every singular scale; the validity or failure of (191) is the physical-price row.
(N4) Local entropy monotonicity does not yield an orbitwise production bound without the reset relation; its failure is the corresponding cocycle graph defect.
(N5) Preservation of a nonlinear badness predicate under weak limits is the boundary-defect value in the canonical evaluation closure.
(N6) Failure of an interface creates its totalized graph-defect record. Target visibility makes that defect part of the singularity-formation algebra; target invisibility places it in $J_{\mathrm{res}}$.
(N7) Large or noncompact motion in $\mathscr{N}_\Sigma$ requires no downstream estimate because it cannot increase a declared prospective target functional.
(N8) A quotient is propagated through the zero descent-defect branch; failure to factor is retained as a target-active descent defect.
(N9) Summability of a constraint tail closes the analytic tail but supplies a physical price only through a carrier comparison such as (202).
(N10) A positive normalized scale gap excludes an infinite cascade when the pulled-back lower prices have the divergent total mass required by (362); a nonvanishing infimum is a stronger sufficient condition.

## III.4 A reusable PDE instantiation protocol

For a particular deterministic PDE, the specialization datum is only its public textbook presentation from Proposition I.1.7 and Corollary I.1.10. Recording that datum requires no new mathematical result. The compiler then performs the following generated operations.

(I1) Read the displayed equation, datum class, domain, boundary conditions, weak test core, declared regularity class, native transformations, and formal carrier multiplier. Enumerate the exhaustive rational approximation grammar, its tested residuals, and its signed carrier records. Generate the total solution graph, continuation graph, and complete singular target.
(I2) Form the graph-complete ordered cylinder algebra. Every nonlinear operation carries its input, output, oscillation, concentration, tail, and first-failure coordinates. Ordinary states embed as evaluation states, and zero-defect approximation paths satisfy the public weak equation.
(I3) Expand the formal carrier multiplier against the weak equation, glue common faces with their signs, and take the Jordan decomposition. This generates the one measure $\mu_u$ and its derived ceiling. Every nonordinary value is inserted into the activity lift; no replacement measure is requested.
(I4) Run the normal-form source compiler. It labels native mobility or reversible form $\mathsf{Geom}$, fixed directed transport $\mathsf{Caus}$, quotient operations $\mathsf{Abs}$, generated charts and augmented variables $\mathsf{Lift}$, and terminal or exterior fluxes $\mathsf{Bdry}$.
(I5) Form the sourcewise mathematical closure of every dynamically generated weak limit, concentration, oscillation, tail, defect, and blow-up germ.
(I6) Generate finite-cylinder target shells and their source-resolved Stieltjes currents. Contact supplies unit shell progress, the canonical disjoint cascade, a persistent five-source support, its detecting covector, and the same-origin active signature. The signature also contains

the canonical record-scale chart when the selected scales collapse. Directions annihilated by the shell covector are removed immediately in $\mathscr{N}_\Sigma$. Apply projection-first saturation; every nonzero descent or normalization defect retains the ancestry of its failed operation.

(I7) On the resulting target spine, generate the forward ancestry module and the backward target-cyclic covector module. Compute their finite incidence cores and pass to the minimal target-active realization $\mathsf{M}_{\Sigma,\mathfrak{b}}$. Delete every equation, constraint, chart, defect, tail, and carrier coordinate outside a complete path from a source occurrence to the shell covector. On the retained realization, restrict the public equation, constraints, gradient structure, native carrier, charts, and total graphs. Glue compatible currents with their signs and form the single aggregate row $\mathfrak{J}_{\Sigma,\mathfrak{b}}$. This step simultaneously generates the retained target-cyclic covectors, nonnegative productions, constraint tails, commutator squares, compactness coordinates, feeding cohomology, and exact storage transfer. Only carrier-dominated production can spend the public physical measure.

(I8) Run the target-aligned recurrent closure compiler of Definition III.4.23. Its internal queue computes the recurrent circulation price and processes zero kernels, signed-dyadic return laws, and first graph-boundary values in one fixed order. Positive price returns a generated carrier potential; zero price returns the next circulation support. By Theorem III.4.34, the output is a divergent physical charge, target-nullity on the retained path, or an actual public-equation-faithful same-origin singular evolution. Hence no separate compactness, equality, tail, pressure, scale, or rigidity stage is part of the specialization protocol. At each finite depth, the incidence fast track precedes the ordered moment calculation, so no source-transverse coordinate enters that calculation.

This protocol is equation-agnostic. Different continuum equations generate different row values in steps (I2), (I5), and (I7); none supplies desired values as additional input. The closed source algebra, local target restriction, charging rule, and failure semantics do not change. Universal analytic identities used in any step are activated only through Theorem II.2.18.

### III.4.1 Target-cyclic degeneracy compilation

The equality case of a physical carrier law must itself be evaluated by the closed algebra. It is not an auxiliary rigidity hypothesis. The following construction extracts all consequences that the public equation forces from a target covector and from vanishing physical production.

**Definition III.4.1** (Target-cyclic covector module)**.** Fix an active branch $\mathfrak{b}$, a selected target shell $\Phi$, and the five-channel current

$$J = J^{\mathsf{Geom}} + J^{\mathsf{Caus}} + J^{\mathsf{Abs}} + J^{\mathsf{Lift}} + J^{\mathsf{Bdry}}.$$

Let $\mathscr{C}^0_{\mathfrak{b}}$ be the rational localization module generated by $D\Phi$ on the retained cylinder algebra. Inductively, $\mathscr{C}^{n+1}_{\mathfrak{b}}$ is obtained from $\mathscr{C}^n_{\mathfrak{b}}$ by adjoining every ordinary value of

(C1) the adjoint transports $(DJ^c)^*\lambda$, the covector Lie derivatives $\mathcal{L}_{J^c}\lambda$, and all iterated commutators of these operations, for $c \in \{\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}\}$;

(C2) multiplication by rational cylinder cutoffs, restriction to record scales, pullback by generated charts, and descent through the generated constraint, gauge, and boundary quotients;

(C3) polarization of every multilinear operation in the public weak equation and of every defect form generated by its total graph; and

(C4) the source-adapted carrier and square-completion operations of Definition II.2.22.

Here $DJ^c$ denotes the total graph of rational difference quotients on the cylinder core. Its ordinary value is the derivative when that derivative exists; no differentiability of the weak current is presumed. Every operation is totalized. Its first nonordinary value is routed with its source ancestry and the covector that exposed it; it is not adjoined as an ordinary word. Define

$$\mathscr{C}_{\mathfrak{b}}^{\infty} := \overline{\bigcup_{n\geq 0} \mathscr{C}_{\mathfrak{b}}^{n}}^{\mathrm{cyl}}. \tag{391}$$

This is the least covector module forced by the public dynamics and the target shell. In particular, no multiplier, compactness statement, or rigidity assertion is supplied as specialization data.

**Definition III.4.2** (Generated production family and joint kernel)**.** For $\lambda \in \mathscr{C}_{\mathfrak{b}}^{\infty}$, expand the exact weak identity obtained by pairing $\lambda$ with the public equation. Glue signed internal currents before taking positive parts. The *target-cyclic production family* $\mathscr{P}_{\mathfrak{b}}$ consists of every nonnegative term produced algebraically by that expansion: mobility squares, Dirichlet and commutator forms, positive graph defects, boundary production, and their polarized positive localizations. A member of $\mathscr{P}_{\mathfrak{b}}$ is charged to the physical ledger only when the same expansion proves its domination by the public carrier measure. Otherwise it is retained as an equality-kernel coordinate. Put

$$X_{\mathfrak{b}}^{n} := X_{\mathfrak{b}} \cap \bigcap_{\substack{P\in\mathscr{P}_{\mathfrak{b}} \\ \mathrm{depth}(P)\leq n}} \{P=0\}, \qquad X_{\mathfrak{b}}^{\infty} := \bigcap_{n\geq 0} X_{\mathfrak{b}}^{n}. \tag{392}$$

The zero set is interpreted in the total ordered graph, so failure of lower semicontinuity, closedness, trace descent, or constraint descent is itself a nonzero typed coordinate with inherited source. Thus a missing analytic property enlarges the generated structure; it is never presumed away.

**Definition III.4.3** (Forward structural ancestry module)**.** Fix an active signature $\mathfrak{b}$. Let $\mathscr{R}_{\mathfrak{b}}^{0}$ consist of the five projected source-current occurrences in its unit shell identity, with their origin and carrier rows. Inductively, $\mathscr{R}_{\mathfrak{b}}^{n+1}$ is obtained from $\mathscr{R}_{\mathfrak{b}}^{n}$ by adjoining the ordinary outputs and first nonordinary total-graph values of every public operation that accepts a word already present: weak evolution, localization, multilinear polarization, constraint and gauge descent, chart transport, boundary trace, signed interface gluing, carrier formation, return, and same-origin succession. Put

$$\mathscr{R}_{\mathfrak{b}}^{\infty} := \overline{\bigcup_{n\geq 0} \mathscr{R}_{\mathfrak{b}}^{n}}^{\mathrm{cyl}}. \tag{393}$$

Each word retains its complete source ancestry. This is the forward closure of what the public equation can generate from the source occurrence; it is formed without a norm bound or compactness premise.

**Definition III.4.4** (Target-incidence core and minimal realization)**.** For finite depths $m, n$, form the bipartite total-incidence graph $\mathsf{G}_{\Sigma,\mathfrak{b}}^{m,n}$. Its left vertices are the words of $\mathscr{R}_{\mathfrak{b}}^{m}$, its right vertices are the covectors of $\mathscr{C}_{\mathfrak{b}}^{n}$, and it contains:

(I1) an ordinary edge $(R, \lambda)$ for every nonzero signed-dyadic cell of the pairing $\langle \lambda, R\rangle$;

(I2) a typed edge for the first nonordinary value of that pairing or of an operation needed to form it; and

(I3) the directed constructor edges recording forward ancestry and backward adjoint transport; and

(I4) one typed frontier vertex for each well-typed public constructor schema whose next instance has not yet been expanded, with edges from its admitted input types to its possible output and adjoint types.

Delete every vertex that lies on no directed path from one of the five source roots to the root target covector $D\Phi$, allowing paths through frontier vertices, and repeat deletion until stable. A frontier vertex is replaced by its ordinary, zero, signed-dyadic, and graph-boundary values when its instance enters the enumeration. Denote the surviving finite graph by $\mathsf{M}_{\Sigma,\mathfrak{b}}^{m,n}$. Its diagonal closure

$$\mathsf{M}_{\Sigma,\mathfrak{b}} := \overline{\bigcup_{k\geq 0} \mathsf{M}_{\Sigma,\mathfrak{b}}^{k,k}}^{\mathrm{gc}} \tag{394}$$

is the *minimal target-active realization*; frontier vertices are absent from this diagonal closure because every fixed constructor instance is eventually expanded. A pairing value zero in the ordinary graph is deleted; failure of the zero pairing to descend through a constructor is its typed incidence edge and survives precisely when it lies on a source–target path.

**Theorem III.4.5** (Target-incidence compression)**.** *For every active signature $\mathfrak{b}$:*

(M1) *every contribution to a shell crossing, every target-visible limit defect, and every target-visible first failure belongs to $\mathsf{M}_{\Sigma,\mathfrak{b}}$;*

(M2) *deletion of the complementary forward words and backward covectors does not change the shell-current identity, the admitted physical price, the aggregate feeding–storage row, the recurrent circulation price, or the set of realized singular mechanisms;*

(M3) *the reduction is minimal: every retained nonfrontier vertex occurs on a source–target constructor path or on a typed prefix that can extend through the current frontier, while removing it deletes at least one algebraically possible target-current word at that depth; every spurious frontier path is removed at the first depth that resolves its failed extension;*

(M4) *if $\mathsf{M}_{\Sigma,\mathfrak{b}} = \varnothing$, then the signature is target-null. If it is nonempty, each of its terminal source–target paths has one of the five source ancestries, and the sum of their signed pairings is the unit shell progress.*

*Thus structural analysis of singular formation may be performed entirely on $\mathsf{M}_{\Sigma,\mathfrak{b}}$.*

*Proof.* Start with the exact unit shell-current identity. Every term has a source root in $\mathscr{R}_{\mathfrak{b}}^{0}$ and the target root $D\Phi$. Apply one public constructor. On its ordinary graph, the chain rule or weak duality moves the constructor forward on the current or backward by its adjoint on the covector, preserving the pairing. On a nonordinary graph value, source-preserving totalization inserts the failed operation as the typed incidence edge. Induction over constructor depth proves (M1).

A deleted vertex has no complete source–target path. Its ordinary pairing with every target-cyclic descendant is zero, and every failed descent that could change this conclusion would itself be a typed path and hence would survive. Removing all such vertices therefore changes neither the shell identity nor any object generated from its target-visible terms. Physical price is admitted only through those terms, signed feeding and storage are their glued images, circulation uses their return edges, and diagonal realization uses their same-origin paths. This proves (M2).

Finite graph pruning retains exactly the vertices on source–target paths or on their well-typed unresolved frontier prefixes. Exhaustive enumeration resolves every fixed frontier instance, proving (M3). Compact diagonal closure preserves the realized paths and all typed boundary values. Empty core annihilates every source pairing, so prospective accountability forbids shell crossing. On a nonempty core, group the retained root paths by their inherited source labels and recover the five-source shell identity, proving (M4). $\square$

**Theorem III.4.6** (Commutation of the continuum compiler with minimal target realization)**.** *Let $\mathsf{P}_{\mathsf{M}}$ denote restriction to $\mathsf{M}_{\Sigma,\mathfrak{b}}$, and let $|\mathsf{M}_{\Sigma,\mathfrak{b}}|$ denote the closed cylinder state support on which all retained incidence paths are realized. Target quotient, sourcewise completion, generated production,* *signed carrier gluing, joint-kernel restriction, first-return formation, circulation duality, circulation elimination, and finite target branch-and-bound commute with* $\mathsf{P}_{\mathsf{M}}$*. In particular,*

$$\delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}} = \delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}}|_{\mathsf{M}_{\Sigma,\mathfrak{b}}}, \tag{395}$$

$$X_\infty = X_\infty \cap |\mathsf{M}_{\Sigma,\mathfrak{b}}|, \tag{396}$$

$$\mathscr{U}_n = \mathsf{P}_{\mathsf{M}_n}\mathscr{U}_n, \tag{397}$$

*where* $\mathsf{M}_n$ *is the finite incidence core at the coordinates used at depth* $n$*. Consequently the finite evaluator never generates a moment, multiplier, defect, tail, or graph coordinate outside a complete source–target path.*

*Proof.* Each listed operation is generated functorially from pairings and public constructor edges. By Theorem III.4.5, its target-visible inputs and every failure of its descent lie in the incidence core. Restricting before or after the operation therefore gives the same ordinary values and the same typed boundary values. The physical objective and target-return normalization are unchanged, proving (395). Return, origin, and realization edges are among the constructor paths, giving (396). At finite depth, graph pruning is a finite preliminary step and the remaining moment rows are identical, which proves (397). □

**Corollary III.4.7** (Finite incidence fast track)**.** *At cylinder depth* $n$*, the minimal target-active realization is computed before any ordered moment problem by two finite graph searches: forward from the five source roots and backward from* $D\Phi$*. Their intersection is the conservative incidence core* $\mathsf{M}^{n,n}_{\Sigma,\mathfrak{b}}$*, including its finitely many typed constructor-frontier vertices. If the finite incidence graph has* $V_n$ *vertices and* $E_n$ *constructor or pairing edges, this pruning uses* $O(V_n + E_n)$ *graph operations. The depth-* $n$ *moment program then contains only coordinates and localizing rows indexed by the intersection. Empty intersection closes the finite target cell by the zero-pairing identities attached to the deleted edges; nonempty intersection supplies the exact list of currently generated target-aligned rows to price or refine, together with the frontier schemas that must be expanded before any further deletion.*

*Proof.* A vertex lies on a source–target path exactly when it is both forward reachable from a source root and backward reachable from the target root. Breadth-first or depth-first search computes these two sets in linear graph time. The commutation theorem permits deletion before production and moment generation. Frontier vertices prevent deletion of a prefix that a later well-typed constructor can connect. The stored zero or nonordinary pairing value on each resolved boundary edge supplies the stated local identity or typed successor. □

**Definition III.4.8** (Covariant production valuation)**.** Let $r$ be a generated record scale. For every carrier term $K$, production $P$, and target current $Q$, the compiler obtains their scale weights by substituting the public scaling chart into the displayed weak identity:

$$K = r^{\kappa_K}\widetilde{K}, \qquad P = r^{\kappa_P}\widetilde{P}, \qquad Q = r^{\kappa_Q}\widetilde{Q}.$$

These are valuations of equation terms, not analytic estimates. Terms of equal weight are first combined with their signs. If the resulting carrier law has critical weight, it is used directly. If a positive power of $r$ would erase normalized production as $r \downarrow 0$, the scale velocity $a = \dot{r}/r$ is retained and Proposition II.2.26 is applied. Exact scale coboundaries are absorbed into the covariant carrier; nonexact scale cohomology remains an explicit $\mathsf{Caus}$- or $\mathsf{Lift}$-current. A failure of the chart, valuation, conjugation, or coboundary identity is the corresponding total-graph successor. In particular, the compiler never infers $\widetilde{P} = 0$ from $r^{\kappa_P}\widetilde{P} \to 0$.

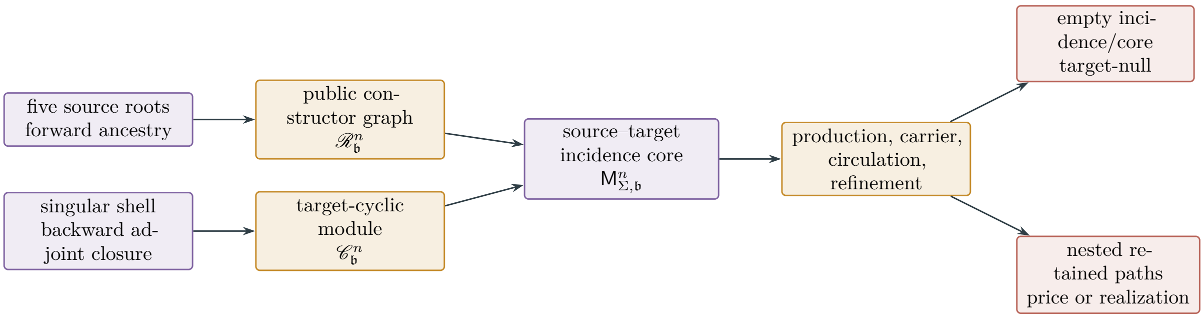


Figure 23: Bidirectional target-incidence fast track. Forward closure records what the equation can generate from the five source occurrences. Backward closure records what the singular shell can detect through adjoint and total-graph descendants. Only their incidence core is passed to physical pricing, recurrent circulation, and finite refinement.

**Definition III.4.9** (Defect-polarization compiler)**.** Suppose a public $d$-linear operation $B$ has total weak graph coordinate

$$B(u_\nu, \dots, u_\nu) \rightharpoonup B(u, \dots, u) + \mathcal{R}_B.$$

The compiler reconstructs the symmetrized multilinear defect from its rational diagonal values by the polarization identity. When an ordered nonsymmetric component is used by the weak equation, the rational grammar also adjoins all mixed-input graph coordinates; these determine that component separately. It then applies the following exclusive rules.

(D1) A positive combination is admitted only when positivity follows from the public positive cone, an exact square, or lower semicontinuity of an already positive carrier form. It is added to $\mathscr{P}_{\mathfrak{b}}$.

(D2) A signed defect is retained as a signed current and glued across all compatible faces before its Jordan decomposition.

(D3) Failure of positivity, polarization, gluing, localization, or descent is retained as the first nonordinary graph value with the ancestry of $B$.

Thus oscillation and concentration are generated structural coordinates of the weak graph. They cannot be discarded as errors and cannot be promoted to physical expenditure without a carrier identity.

**Proposition III.4.10** (Functorial constraint and boundary descent)**.** *The target-cyclic construction commutes with every ordinary constraint, gauge, chart, trace, and boundary quotient generated by the public presentation. More precisely, if $q$ is such an ordinary quotient, then*

$$q^*\mathscr{C}^n_{q\mathfrak{b}} \subseteq \mathscr{C}^n_{\mathfrak{b}}, \qquad q(X^n_{\mathfrak{b}}) \subseteq X^n_{q\mathfrak{b}}$$

*on its ordinary graph domain. Equality holds only on coordinates for which the generated lifting graph is ordinary and onto. A missing reverse inclusion is therefore recorded by its cokernel coordinate. Outside the ordinary domain, the least failed commutation, tail, trace, chart, or gauge coordinate is routed as a target-null value, a physically charged value, or the next target-cyclic successor. No surjectivity, closed range, compact trace, or tail decay is assumed.*

*Proof.* The assertion holds at depth zero because the target shell is pulled back through the same quotient. If it holds at depth $n$, naturality of pullback gives the claim for adjoint transport and Lie derivatives; the multilinear polarization identity gives it for weak defects; signed gluing gives it for interfaces; and the public covariance identity gives it for record-scale charts. These are exactly the generators in Definition III.4.1. Induction proves the ordinary inclusions. An ordinary onto lifting graph supplies the reverse inclusions. Totalization supplies a value when naturality or lifting fails, and the source-preserving graph closure retains the failed operation and the target covector that detected it. The three stated routes are then the exhaustive source-accountability alternatives. □

**Theorem III.4.11** (Target-cyclic joint-kernel descent)**.** *Let $\mathfrak{b}$ be a same-origin target-active branch obtained from a legal orbit with finite physical ledger. Run Definitions III.4.1, III.4.2, III.4.8 and III.4.9 in the fixed enumeration of rational cylinders and public graph operations. Exactly one of the following occurs, with priority given to the first finite event in that enumeration.*

(K1) *A generated covariant carrier law charges a disjoint cofinal family of shell crossings by a nonnegative physical measure whose required total mass diverges. This contradicts the finite physical ledger.*

(K2) *On the retained same-origin path support in $X_{\mathfrak{b}}^{\infty}$, every source current selected by the generated target-cyclic covectors is target-null:*

$$\langle \lambda, J^c \rangle = 0 \quad (\lambda \in \mathscr{C}_{\mathfrak{b}}^{\infty} \text{ along that support}, c \in \{\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}\}).$$

*Hence that retained path cannot cross the selected shell. No assertion is made about the rest of $X_{\mathfrak{b}}^{\infty}$.*

(K3) *A production, defect, commutator, scale-cohomology, constraint, boundary, chart, or total-graph coordinate is nonzero. Its least such value, together with its zero, signed dyadic, or graph-boundary polarity, detecting covector, and inherited origin, is the next proper target-active successor; its resolved-prefix rank strictly increases.*

(K4) *All finite coordinates are resolved, compatible, and ordinary as operations of their successively induced graphs; their inverse limit induces a closed five-channel equation on $X_{\mathfrak{b}}^{\infty}$ containing a same-origin path that crosses every cofinal target shell. This is a dynamically realized singular subsystem, not a formal equality model.*

*No compactness, coercivity, Liouville property, closed-range statement, or equality classification occurs among the hypotheses.*

*Proof.* Contact gives unit progress across each normalized shell. By prospective accountability, before each crossing at least one of the five source pairings is nonzero. Starting from that shell covector, differentiate the pairing through the public equation. The chain rule gives the adjoint transports and Lie derivatives in (C1); localization, quotient descent, weak passage, and carrier completion give (C2)–(C4). Hence every term exposed by any finite iteration is either an exact signed current, a generated nonnegative production, or a total-graph value. The signed terms are glued before their positive parts are formed, and Definition III.4.8 prevents collapsing scales from erasing a production. This proves that the enumeration omits no descendant of the initial target pairing.

If a carrier domination charges the cofinal disjoint crossings with divergent total mass, (K1) holds. Restrict next to the actual same-origin path support. If every generated target pairing vanishes there, the shell derivative vanishes along that path, so the fundamental theorem of calculus forbids its shell crossing and (K2) holds. No global annihilation of the joint kernel is used. Otherwise choose the least nonzero or nonordinary coordinate and apply its generated polarity split. Source-preserving totalization, Proposition III.4.10 and Theorem II.2.15 give (K3) without an analytic exclusion assumption.

It remains to consider the case in which every finite coordinate is resolved and compatible in its induced graph. For each $N$, let $C_N$ be the closed finite-cylinder set of paths satisfying the first $N$ public equation, joint-kernel, covariance, defect, origin, and shell-crossing coordinates. The original contact branch and its first $N$ successor restrictions make $C_N$ nonempty; the total ordered graph makes each finite observer factor compact; and forgetting the last coordinates maps $C_{N+1}$ into $C_N$. The inverse limit is therefore nonempty. Any path in it has resolved values for every public operation and ordinary operations in the induced graph, lies in $X_{\mathfrak{b}}^{\infty}$, retains the same origin, and crosses the cofinal shell family. The conclusion is exactly the nonnull case of Theorem II.2.16; a missing induced operation would instead be a finite event of type (K3). Thus (K4) holds. The fixed priority rule makes the four outputs disjoint. □

**Corollary III.4.12** (Generated mobility-annihilator test)**.** *If the restriction of the target-cyclic module to the retained same-origin path support annihilates all five induced currents there, then that path cannot contact the target. Otherwise the least nonzero restricted pairing, together with its generated polarity cell, is the initial covector of the next successor. Thus the covector needed to continue the equality analysis is generated by the equation and the target; it is never an external multiplier or a user-supplied rigidity certificate.*

### III.4.2 Target-relative resaturation of zero-price dynamics

A zero-price equality law is not a terminal object. It is a new public dynamical stratum on which prospective accountability is run again. This is the continuum counterpart of forming the saturated closure before taking the residual in Part I.

**Definition III.4.13** (Zero-price resaturation successor)**.** Let $\mathfrak{b}_n = (A_n, m_n, \Gamma_n, \eta_n, \mathbf{R})$ be a reachable active branch with ordered ideal $I_n$, dual-carrier module $\mathscr{W}_n$, and zero-price recurrent law $L_n$. Its *resaturation successor* is constructed by the following generated operations.

(R1) Starting from the selected shell covector, generate $\mathscr{C}_{\mathfrak{b}_n}^{\infty}$ and $\mathscr{P}_{\mathfrak{b}_n}$ by Definitions III.4.1 and III.4.2. Adjoin every production annihilated by $L_n$, and form the closed ordered kernel

$$X_{n+1} := X_n \cap \bigcap_{P \in \mathscr{P}_{\mathfrak{b}_n}} \{P = 0\}. \tag{398}$$

(R2) Restrict the public weak equation, its constraints, carrier, charts, and total graphs to $X_{n+1}$. Recompute the five source projections there; no old source tag or origin coordinate is discarded.

(R3) Apply the prospective shell identity on the restricted pre-contact path. If contact remains, select a persistent source support and target covector with the normalized floor 1/5.

(R4) Apply the covariant production valuation and defect-polarization compiler. Adjoin every resulting adjoint, quotient, localization, signed-gluing, covariance, commutator, defect, and square-completion relation to $I_n$, and adjoin its detecting covector to $\mathscr{W}_n$. Exact scale coboundaries are absorbed and nonexact scale cohomology is retained as a source current.

(R5) Totalize every operation used above. Its first nonordinary value is a typed successor. If every value is ordinary, the restriction gives the next zero-price branch $\mathfrak{b}_{n+1}$.

(R6) Apply the generated polarity split to the next unresolved coordinate on the retained same-origin target stratum. Prune a cell only by its finite local ordered identity; every positive-moment cell appends its polarity relation to the ideal and renormalizes the selected target current before any physical charge is made.

The successor is *proper* when either its ordered ideal or its observable module strictly increases after reduction by the preceding relations. Every proper successor has strictly larger resolved-prefix rank.

**Proposition III.4.14** (Monotone resaturation)**.** *Along a zero-price resaturation chain,*

$$I_0 \subseteq I_1 \subseteq \cdots, \qquad \mathscr{W}_0 \subseteq \mathscr{W}_1 \subseteq \cdots, \qquad X_0 \supseteq X_1 \supseteq \cdots, \qquad \rho(\mathfrak{b}_0) < \rho(\mathfrak{b}_1) < \cdots \tag{399}$$

*Every retained branch has the same origin tag and target floor* $1/5$*. At a limit stage use*

$$I_\infty := \overline{\bigcup_n I_n}^{\mathrm{ord}}, \qquad \mathscr{W}_\infty := \overline{\bigcup_n \mathscr{W}_n}^{\mathrm{cyl}}, \qquad X_\infty := \bigcap_n X_n. \tag{400}$$

*These closures create no source: every limit relation retains the ancestry of the finite words that generate it.*

*Proof.* Kernel restriction only adds zero relations, while dual-carrier compilation only adds words and their consequences. The polarity step appends the next resolved coordinate, proving the three inclusions and the rank inequality. The shell selector is applied to the same pre-contact path, so the origin tag is unchanged and normalization restores the floor $1/5$. The two closures in (400) are the inverse-limit closures of the countable rational cylinder grammar. Source-preserving completion therefore gives the final assertion. □

### III.4.3 Recurrent passivity and support transfer

The resolved-prefix rank detects a first failed operation, but rank growth alone does not evaluate a current which repeatedly reappears after renormalization. The carrier must be followed through the same-origin successor chain. The following construction performs that step before a limit path is declared to be a singular mechanism.

**Definition III.4.15** (Target-local marked carrier complex)**.** Fix a same-origin active branch $\mathfrak{b}$ and its compact retained path space. A marked carrier cell is a tuple

$$C = (w, w^+, \omega, K^-, K^+, P, H, E),$$

where $w\mathcal{R}_\eta w^+$, $\omega$ is the retained successor witness, $K^\pm$ are the endpoint values of one generated carrier, $P \geq 0$ is the sum of the generated nonnegative productions in that cell, $H$ is the complete signed interior, chart, scale, constraint, and interface current after compatible faces have been glued, and $E$ is the true exterior or physical-input current. Every coordinate is restricted to $X_\mathfrak{b}$ and carries the detecting shell covector. Its total graph contains the exact relation

$$P = K^- - K^+ + H + E. \tag{401}$$

Let $\mathscr{B}_\mathfrak{b}$ be the closed linear space generated by shift coboundaries $F \circ S - F$, paired internal-face currents, exact chart and gauge currents, and currents annihilated by the selected target quotient. The *target-local feeding class* is

$$[H]_\mathfrak{b} \in \mathscr{H}_\mathfrak{b} := \overline{\operatorname{span}\{H\}}/\mathscr{B}_\mathfrak{b}. \tag{402}$$

Exterior input is not put in this quotient. Its ordinary carrier-dominated part is charged by (90); every other value is its $\mathsf{Bdry}$- or carrier-domination successor.

**Proposition III.4.16** (Stationary carrier reduction)**.** *Let $\varpi$ be a shift-invariant probability law on a concatenated marked carrier path in one active branch. On the ordinary finite-integral value,*

$$\int P \, \mathrm{d}\varpi = \int [H]_{\mathfrak{b}} \, \mathrm{d}\varpi + \int E \, \mathrm{d}\varpi. \tag{403}$$

*Here the first integral on the right means evaluation of the cohomology class by the stationary law. If an endpoint, concatenation, integrability, quotient, or exterior coordinate is nonordinary, its least total-graph value is the corresponding proper successor.*

*Proof.* Sum (401) over a finite consecutive block. Concatenation cancels every common endpoint carrier. Divide by the block length and pass to the empirical path law. Bounded endpoint storage vanishes, as does every shift coboundary. Paired faces and exact chart or gauge currents already vanish in the quotient. This gives (403). Each operation used in this calculation is totalized, so its first nonordinary value gives the stated successor instead of an identity. □

**Theorem III.4.17** (Target-aligned passivity–support alternative)**.** *Let an infinite zero-price successor path retain the active signature $\mathfrak{b}$, and form its stationary marked carrier law. Exactly one of the following events occurs in the fixed total-graph order.*

(V1) *A true exterior or input current is admitted by the target-local carrier graph. Its restrictions to the diagonal original events are charged to the public physical measure; divergent required mass contradicts the finite ledger, and finite mass enters the zero-input tail successor.*

(V2) *A marked-carrier, gluing, quotient, support-transfer, or realization coordinate is nonordinary and gives its typed proper successor.*

(V3) *The ordinary feeding class has nonzero positive stationary value. The normalized positive measure*

$$\mathrm{d}\lambda_H := \frac{([H]_{\mathfrak{b}})_+ \, \mathrm{d}\varpi}{\int ([H]_{\mathfrak{b}})_+ \, \mathrm{d}\varpi} \tag{404}$$

*has a target-visible support projection. Its first atomic, separated, nested, diffuse, tail, rough, gauge, constraint, or recurrent projection is the next same-origin target-active return record, retaining the source word of the current that finances $P$. It is a proper successor when it resolves a new coordinate; otherwise its positive-measure signed-dyadic cell defines the induced recurrent-current return map.*

(V4) *The ordinary exterior value and feeding cohomology vanish. Then*

$$\int P \, \mathrm{d}\varpi = 0, \qquad P = 0 \quad \varpi\text{-almost surely,} \tag{405}$$

*and the path law is restricted to the target-cyclic joint kernel generated by these zero productions.*

*Thus lack of passivity is itself a target-aligned structural current. It cannot be retained as an unevaluated equality law.*

*Proof.* Apply Proposition III.4.16. The exterior coordinate is processed first by the local charging rule, giving (V1) or its successor. Any nonordinary operation gives (V2). On the remaining ordinary zero-exterior cell, nonnegativity of $P$ gives

$$\int [H]_{\mathfrak{b}} \, \mathrm{d}\varpi = \int P \, \mathrm{d}\varpi \geq 0.$$

If the value is positive, (404) is a probability measure on the already compact activity observer space. Apply Theorem II.2.31. If every support projection were target-null, the target-local functional defining $[H]_{\mathfrak{b}}$ would integrate to zero. Hence one projection is target-visible and supplies (V3) with its same-origin witness. On an equivalent support cell, Poincare recurrence for the least positive signed-dyadic set supplies the induced return map, so equivalence does not discard the current. If the value is zero, (403) gives $\int P \, \mathrm{d}\varpi = 0$. Since $P \geq 0$, it vanishes almost surely, proving (V4). $\square$

**Theorem III.4.18** (Target-visible fidelity defects generate carrier successors)**.** *Let $F$ be a public equation, constitutive, nonlinear, constraint, trace, chart, or realization operation, and let $\mathcal{D}_F$ be a nonzero value of its total-graph defect on an active branch $\mathfrak{b}$. For a target-cyclic covector $\lambda$, exactly one of the following occurs.*

(D1) $\langle \lambda, \mathcal{D}_F \rangle = 0$*; this occurrence of the defect lies in the target kernel.*

(D2) *The normalized variation of $\langle \lambda, \mathcal{D}_F \rangle$, localized by the rational cylinder core, produces an atomic, separated, nested, diffuse, tail, or graph-boundary support record with the same origin and the ancestry of $F$. A support record inequivalent to the parent is a proper target-active successor.*

(D3) *Every positive support record is equivalent to the parent in the resolved coordinates. Its event-count law is recurrent. The defect pairing is then retained, with its sign, as either a nonzero component of $\Pi_{\Sigma,\mathfrak{b}}[H]_{\mathfrak{b}}$ or an exact carrier-storage transfer in $\mathscr{B}_{\mathfrak{b}}$. Both values are evaluated by Theorem III.4.22; an exact transfer is not identified with a target-null current.*

*Consequently no target-visible fidelity defect can disappear at a recurrent limit: it is a proper support successor, a recurrent feeding current, or a recurrent storage-transfer current.*

*Proof.* Apply the scalar distribution $T = \langle \lambda, \mathcal{D}_F \rangle$ to the enumerated rational local test core. Either every value is zero, which is (D1), or there is a least test with nonzero value. The total variation of its rational localizations is a nonzero positive finite measure on each finite cylinder; if its variation is infinite, use the normalized-variation boundary coordinate. Normalize and apply compact activity-profile exhaustion. Source-preserving completion retains the ancestry of $F$, and the same-origin selector retains the original event indices. This gives (D2).

If one support record is inequivalent to the parent, its witness is a proper successor. Otherwise take empirical laws of the equivalent support records. Successor recurrence gives a shift-invariant law. Pairing the public weak equation with the transported localizations of $\lambda$ places $T$ in the complete signed current $H$; paired internal occurrences cancel before the quotient. If its class is nonzero, its target projection is retained as a feeding coordinate. If its class is zero, the definition of the closed coboundary space supplies bounded cylinder storage functions whose coboundaries converge to the pairing. Their finite-block sums are endpoint transfers. The support law retains those endpoints and therefore routes the value as a carrier-storage coordinate. Only annihilation by every target-cyclic covector gives (D1). This proves (D3) without the false implication that nonzero variation has nonzero stationary mean. $\square$

**Definition III.4.19** (Target-active recurrent component)**.** Let $\Omega_{\mathfrak{b}}$ be the compact path space of same-origin successor chains and let $S$ be the left shift. A *target-active recurrent component* is the support of an ergodic $S$-invariant probability law $\varpi$ obtained from a contact-generated chain and satisfying the retained shell floor. Every generated scalar coordinate has a fixed zero, signed-dyadic, or graph-boundary value on the corresponding measurable cell. A target-visible value is retained even when its support successor is equivalent in the previously resolved coordinates. In that case it is a recurrent current coordinate and remains an active operation of the compiler.

Such a component exists for every infinite chain: take a weak limit of its empirical shift laws and then an ergodic component with nonzero target mass. This construction asserts recurrence of structural records only; an actual PDE path is asserted only after the same-origin realization graph has an ordinary value.

**Theorem III.4.20** (Target-relative recurrent stratification)**.** *Every infinite same-origin zero-price resaturation route has exactly one of the following constructive outputs.*

(T1) *a finite target-null, continuation, class-incompatibility, or divergent physical-charge relation;*

(T2) *a proper target-active successor produced by a compactness mode, a fidelity defect, or a previously unresolved feeding or storage current;*

(T3) *a shift-invariant equality law satisfying* (405)*, followed by the target-cyclic mobility-annihilator test;*

(T4) *a target-active recurrent component. On it, every nonzero aggregate feeding or storage current is selected by a signed-dyadic return set and becomes the next recurrent-current successor. If all such currents vanish, the component lies in the generated zero-production joint kernel. A cofinal chain of recurrent-current successors has a diagonal inverse limit, which is either target-null or a public-equation-faithful explicit singular mechanism.*

*A nonzero public-equation, constitutive, constraint, boundary, fidelity, support-transfer, feeding, or storage coordinate is therefore an operation on* (T4)*, never a name for an unresolved terminal. In* (T3)*, a target-null joint kernel closes; a nonzero target-cyclic pairing is the next successor. Consequently an unending classification chain is replaced by a recurrent current system with a definite structural evolution.*

*Proof.* Apply compact activity-profile exhaustion to the diagonal same-origin path. An atomic, separated, nested, diffuse, tail, rough, gauge, constraint, or equality value is a coordinate of its single activity law. Process its first target-visible value. The carrier values are then exhausted by Theorem III.4.17. Cases (V1)– (V2) give (T1)–(T2); case (V3) gives a proper support successor when it resolves a new coordinate and otherwise gives the recurrent-current return system in (T4); case (V4) gives (T3).

Repeat on every proper successor. Strict resolved-prefix growth and compact diagonal saturation give a same-origin inverse-limit path space. By Theorem II.2.34 it carries an invariant law; select an ergodic component of nonzero target mass. Apply Theorem III.4.18 to every retained public-equation defect. A target-visible defect returns to (T2) or becomes a recurrent feeding or storage coordinate; every remaining defect is target-null.

For a nonzero recurrent coordinate choose the least signed-dyadic set $A = \{sF \in [2^{-k}, 2^{-k+1})\}$ of positive $\varpi$-measure. Poincare recurrence makes the first-return map $S_A$ finite almost everywhere, and the normalized restriction $\varpi(A)^{-1}\varpi|_A$ is $S_A$-invariant. This is the recurrent-current successor in (T4); it retains the same-origin witnesses and source ancestry of $F$. If it resolves a new coordinate it is (T2). If equivalent successors recur indefinitely, their compact diagonal limit is processed by Theorem II.2.16: it is target-null or it realizes the public equation and cofinal shell contact. If no nonzero coordinate remains, stationary carrier reduction gives (T3). This proves exhaustiveness without asserting that an increasing refinement chain has a sink. □

**Corollary III.4.21** (Local criterion excluding a recurrent singular component)**.** *Suppose the generated passivity row of every reachable active signature has target-null zero-production kernel, and every recurrent feeding, fidelity, storage, or support-transfer current is either charged by the public physical carrier or has target-null return law. Then no target-active recurrent component contacts the continuation target. Hence every finite-physical- expenditure ordinary branch is regular and continuable. These statements are outputs of the local ordered cones, return maps, and total graphs of the active signatures; they are not specialization data.*

*Proof.* A contact branch produces an infinite zero-price route after the divergent- price cases have been removed. Apply Theorem III.4.20. The first three outputs close or continue. In the recurrent output, every nonzero current is charged or has target-null return law by hypothesis. The remaining component lies in the zero-production kernel, which is target-null by the generated local ordered identity, contradicting its shell floor. □

**Theorem III.4.22** (Target-local passivity fast track)**.** *For a reachable active signature $\mathfrak{b}$, after exterior input has been processed, the full recurrent carrier analysis reduces to one aggregate target-current row,*

$$\mathfrak{J}_{\Sigma,\mathfrak{b}} := (\Pi_{\Sigma,\mathfrak{b}}[H]_{\mathfrak{b}}, \Pi_{\Sigma,\mathfrak{b}} H_{\mathfrak{b}}^{\mathrm{stor}}), \tag{406}$$

*where $H_{\mathfrak{b}}^{\mathrm{stor}}$ is the endpoint-transfer coordinate of the closed coboundary graph. Its total ordered graph has the following exhaustive evaluation.*

(F1) *A finite identity in $\mathcal{K}_{\mathfrak{b}}$ proves $\Pi_{\Sigma,\mathfrak{b}}[H]_{\mathfrak{b}} = 0$. Stationary carrier reduction forces the generated production to vanish. A nonzero storage coordinate is then selected by its signed-dyadic return set; if storage also vanishes, the target-cyclic joint kernel is evaluated on that same cell.*

(F2) *The class has a nonzero signed-dyadic value. Its positive support law* (404)*, source word, detecting covector, and same-origin witness define a positive-density return-map record. The compiler applies the five-source normal form and prospective accountability directly to this record; it does not separately estimate the individual pressure, transport, chart, cutoff, or scale terms from which the class was assembled.*

(F3) *The projection, gluing, support, or realization graph is nonordinary. Its first graph-boundary value, with the same data, is the next active record.*

*Thus a specialization evaluates one aggregate target-aligned row. Exact storage and nonexact feeding are distinguished, but neither is discarded. Atomic, separated, nested, diffuse, and tail modes occur only as support coordinates of the selected return law and introduce no additional proof stage.*

*Proof.* Signed carrier gluing constructs $[H]_{\mathfrak{b}}$ and its storage graph before a positive part is taken. Apply the target-local factorization and the generated polarity split to (406). The zero cohomology cell is (F1) by Proposition III.4.16 and Theorem III.4.17. On that cell the exact component is a block endpoint transfer and is split by the same signed-dyadic return construction. Positive and negative dyadic feeding cells retain the signed current; reverse the orientation on a negative cell and apply the positive support construction, giving (F2). Poincare recurrence supplies the induced return law. Totalization gives (F3). Compact activity exhaustion is applied to the single support measure in (F1), (F2), or (F3), which proves the final assertion. □

**Definition III.4.23** (Target-aligned recurrent closure compiler)**.** For a reachable active signature $\mathfrak{b}$, a compiler state is

$$\mathfrak{q} = (X, I, \mathscr{C}, \mathscr{P}, \varpi, \mathfrak{J}_{\Sigma,\mathfrak{b}}, \delta_{\Sigma,\mathfrak{b}}^{\mathrm{cyc}}, \mathbf{R}), \tag{407}$$

where $X$ is the current same-origin target cell, $I$ its ordered ideal, $\mathscr{C}$ the target-cyclic covector module, $\mathscr{P}$ the generated production family, $\varpi$ its recurrent law when present, $\mathfrak{J}_{\Sigma,\mathfrak{b}}$ the aggregate feeding–storage row, and $\delta_{\Sigma,\mathfrak{b}}^{\mathrm{cyc}}$ its sharp recurrent circulation price when the return graph has been formed, and $\mathbf{R}$ the original-event witnesses. Before enqueueing a state, restrict all of these coordinates to the minimal target-active realization $\mathsf{M}_{\Sigma,\mathfrak{b}}$. Initialize the queue with the algebraically reachable shell state. Process its least generated coordinate by the following fixed transition order:

(Q1) Dovetail the finite identities in the local cone and the dual circulation-price hierarchy. A verified $-1$-identity removes the cell. A carrier-domination identity records the restriction of the public physical measure; a cofinal family with divergent total charge removes the cell. A positive recurrent circulation price produces the dual carrier potential and removes the recurrent cell by infinite return charge.

(Q2) On the zero cell of $\Pi_{\Sigma,\mathfrak{b}}[H]_{\mathfrak{b}}$, adjoin every forced nonnegative production to the zero-production joint kernel. Apply prospective accountability on the restricted same-origin path. Vanishing of all target-cyclic pairings removes the cell in $\mathscr{N}_{\Sigma}$; otherwise the least nonzero pairing is appended to the queue.

(Q3) On the zero-price circulation supplied by the primal hierarchy, select a signed-dyadic feeding or storage cell of positive recurrent mass, form its first-return map and normalized induced law, and append the sign, dyadic level, source ancestry, detecting covector, and return-time graph. This induced state is the next queue element even when its previously resolved coordinates agree with those of its parent.

(Q4) On a graph-boundary value, append the first failed operation, its total graph, source ancestry, target covector, and same-origin support law. Apply the same transition order to that typed state.

Only descendants generated from the selected shell covector are enqueued. Every other coordinate is removed by the target quotient. Compatible signed currents are glued before (Q2)–(Q3), so the queue contains aggregate target currents rather than separate pressure, transport, chart, cutoff, scale, and interface estimates.

**Definition III.4.24** (Target-cyclic structural spine)**.** The *target-cyclic structural spine* of an active signature $\mathfrak{b}$ is the least closed subgraph

$$\mathscr{T}_{\Sigma,\mathfrak{b}} \subseteq \mathrm{Sat}_{\mathsf{G}_{\Sigma}}(q_{\Sigma}\mathscr{J}) \tag{408}$$

that contains the five projected source currents paired with the selected shell covector and is closed under target-cyclic adjoint transport, projected total graphs, signed carrier gluing, zero-production restriction, signed-dyadic first return, same-origin succession, and diagonal realization. The physical valuation on the spine is the restriction of the public carrier measure. No coordinate with zero target projection belongs to the spine.

Equivalently, $\mathscr{T}_{\Sigma,\mathfrak{b}}$ is generated by starting from the unit shell-current identity and repeatedly adjoining only the next coordinate selected by (Q2)–(Q4). It is therefore enumerated from the public presentation, target shells, and physical carrier; it is not an additional specialization object.

**Proposition III.4.25** (Incidence presentation of the target spine)**.** *Let* $\mathrm{pr}_{\mathscr{R}}$ *forget the covector and incidence-edge coordinates while retaining current ancestry, return, origin, and total-graph values. Then*

$$\mathrm{pr}_{\mathscr{R}}\, \mathsf{M}_{\Sigma,\mathfrak{b}} = \mathscr{T}_{\Sigma,\mathfrak{b}}. \tag{409}$$

*Thus the minimal realization is the bidirectional, finitely prunable presentation of the target-cyclic structural spine.*

*Proof.* Every retained incidence path starts at a projected source occurrence and is closed under the target-cyclic adjoint, total-graph, gluing, kernel, return, origin, and realization operations that define the spine. Its current projection is therefore contained in the spine. Conversely, structural induction over the least spine closure attaches to every spine word the target-cyclic covector and constructor path that generated it. This gives a vertex of the incidence realization with that current projection. The two inclusions prove (409). □

**Proposition III.4.26** (Spine restriction of the recurrent compiler)**.** *Every nonclosing state of the target-aligned recurrent closure compiler lies in $\mathscr{T}_{\Sigma,\mathfrak{b}}$. Conversely, every finite spine word is a finite compiler state with the same ordered ideal, source ancestry, physical valuation, and origin witness. Hence restricting the compiler to the spine changes none of its outputs.*

*Proof.* The initial shell-current state is a spine generator. Transition (Q2) applies a target-cyclic adjoint or joint-kernel restriction; (Q3) applies a first-return constructor; and (Q4) applies a projected total graph. These are precisely the closure operations in Definition III.4.24. Induction places every queued state in the spine. Conversely, structural induction over the least closed subgraph expresses every finite spine word as a finite composition of those transitions. Projection-first saturation supplies every descent defect, and source-preserving completion retains its ancestry. Carrier and origin coordinates are unchanged by Theorem I.4.7. □

**Definition III.4.27** (Target-current circulation problem)**.** Let $E_{\Sigma,\mathfrak{b}}$ be the compact total-graph edge space of $\mathscr{T}_{\Sigma,\mathfrak{b}}$, with source and range maps $\pi_-, \pi_+$. The selected signed-dyadic orientation defines a bounded nonnegative target-return coordinate $a_\Sigma$; normalize each active return cell so that $a_\Sigma \geq \eta > 0$. Let $p_{\mathrm{phys}} \geq 0$ be the lower-semicontinuous physical valuation obtained from the local carrier-domination graph. Its continuous cylinder lower envelopes $p_N \uparrow p_{\mathrm{phys}}$ are generated by the totalized carrier coordinate; an infinite endpoint is routed directly to physical charge. A *target-current circulation* is a finite positive Borel measure $\gamma$ on $E_{\Sigma,\mathfrak{b}}$ satisfying

$$(\pi_-)_\# \gamma = (\pi_+)_\# \gamma, \qquad \int a_\Sigma \, \mathrm{d}\gamma = 1, \tag{410}$$

and all ordered ideal, source, origin, and total-graph relations of the active signature. Its sharp recurrent physical price is

$$\delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}} := \inf_{\gamma} \int p_{\mathrm{phys}} \, \mathrm{d}\gamma \in [0, +\infty], \tag{411}$$

with value $+\infty$ when the circulation set is empty.

**Theorem III.4.28** (Target-current circulation duality)**.** *Write $\mathcal{K}^0_{\mathfrak{b}} := I_{\mathfrak{b}} + M_{\mathfrak{b}}$ for the local ordered cone before shift coboundaries are adjoined. All functions below are functions on the target quotient; hence no current-null space is added to this scalar cone. On every reachable active signature,*

$$\delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}} = \sup \left\{ \delta \in \mathbb{R} : \begin{array}{l} \text{there are } N \text{ and a generated continuous cylinder potential } V \text{ with} \\ p_N + V \circ \pi_- - V \circ \pi_+ - \delta a_\Sigma \in \overline{\mathcal{K}^0_{\mathfrak{b}}} \end{array} \right\}. \tag{412}$$

*Exactly one of the following values is produced.*

(D1) $\delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}} > 0$. *A rational $0 < \delta < \delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}}$ and a finite cylinder approximation of $V$ give a finite local ordered identity. On the first $N$ returns it gives*

$$\sum_{j<N} p_{\mathrm{phys}}(e_j) \geq \delta \sum_{j<N} a_\Sigma(e_j) - 2\|V\|_\infty.$$

*Thus a recurrent sequence with unbounded accumulated normalized target return has infinite public physical charge.*

(D2) $\delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}} = 0$. *Compatible minimizing cylinder moments reconstruct a zero-price target-current circulation. Its ergodic components and positive support initialize the next zero-production, return, or graph-boundary transition of Definition III.4.23.*

(D3) $\delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}} = +\infty$. *A finite ordered identity excludes the normalized circulation constraints, so the recurrent cell is empty.*

*The potential $V$ is generated by the cylinder hierarchy. It is not a multiplier supplied with the PDE presentation.*

*Proof.* Let $\mathcal{M}_{\mathfrak{b}}$ be the convex set of positive edge measures satisfying the first relation in (410) and the active ordered relations. On the normalized slice, $a_\Sigma \geq \eta$ bounds total mass; the slice is therefore weak-* compact. First minimize each continuous lower envelope $p_N$. Linear-programming separation on the compact slice gives its continuous dual. Compactness of minimizing measures and lower semicontinuity show that these minima increase to the minimum of $p_{\mathrm{phys}}$: otherwise a convergent subnet of minimizers would violate the Portmanteau inequality. The annihilator of equal source and range marginals is the closed space of coboundaries $V \circ \pi_- - V \circ \pi_+$. The active ordered relations contribute $\overline{\mathcal{K}^0_{\mathfrak{b}}}$. The edge space was formed after the target quotient, so target-null currents have already disappeared and cannot be added to the scalar dual cone. This proves (412) by strong separation.

The rational cylinder functions are uniformly dense in the generated evaluation algebra. If the optimum is positive, choose rational $\delta$ strictly below it and approximate the dual potential. Archimedean order-unit completion absorbs the approximation error and the finite cylinder-moment hierarchy returns the corresponding finite identity. Its sum along a finite path telescopes the coboundary, and its integral against a circulation cancels it exactly. Hence $\int p_{\mathrm{phys}} \geq \delta \int a_\Sigma = \delta$, proving (D1).

If the optimum is zero, choose minimizing feasible moments at each cylinder depth. Compact projective consistency reconstructs a circulation with zero physical valuation. Ergodic decomposition preserves flow conservation, zero price, and positive target mass on at least one component. Its support law is precisely the next recurrent compiler state, proving (D2). If the feasible slice is empty, Archimedean separation supplies the finite $-1$-identity in (D3). $\square$

**Definition III.4.29** (Circulation-elimination operator)**.** For a target-spine cell $X$, define $\mathcal{E}_\Sigma(X)$ by computing $\delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}}$ in the nested cylinder hierarchy. Remove $X$ in cases (D1) and (D3). In case (D2), restrict to the zero-price circulation support, adjoin all vanishing productions, and inspect the least unresolved target-cyclic covector or total graph in the fixed public enumeration. Return its finite zero, signed-dyadic, and graph-boundary polarity partition after deleting the target-null members. Each returned cell is embedded in the universal spine by retaining its parent coordinates; hence it is a closed refinement of $X$. Iterate

$$X_0 := \mathscr{T}_{\Sigma,\mathfrak{b}}, \qquad X_{n+1} := \mathcal{E}_\Sigma(X_n), \qquad X_\infty := \bigcap_{n\geq 0} X_n, \tag{413}$$

with diagonal saturation at limit stages. This operator is generated solely from the target spine, public physical valuation, and ordered equation graph.

**Definition III.4.30** (Target-aligned structural branch-and-bound)**.** Fix the enumeration of rational target-cylinder coordinates, public graph relations, and source descendants. At depth $n$, partition the first $n$ coordinates into rational closed boxes of diameter at most $2^{-n}$ in the canonical evaluation metric. Only coordinates in the generated target-cyclic spine are partitioned. For every surviving parent box $C$, perform the following finite operations.

(B1) Adjoin the first $n$ equation, origin, carrier, polarity, and total-graph rows, and use the continuous price envelope $p_n$.

(B2) Solve the resulting rational truncated circulation problem and its ordered dual. Record the certified dual lower price $\underline{\delta}_n(C)$ and the compact set $\mathsf{P}_n(C)$ of feasible truncated moments.

(B3) Delete $C$ when a finite $-1$-identity makes $\mathsf{P}_n(C) = \varnothing$, or when a positive dual identity gives $\underline{\delta}_n(C) > 0$. Attach that finite identity to the deleted box.

(B4) Otherwise subdivide the least unresolved target-visible cylinder coordinate. A zero value adjoins the corresponding kernel relation; a nonzero value is divided into signed dyadic cells; and a nonordinary value is divided by its typed total-graph boundary. Origin and source ancestry are copied to every child.

Let $\mathscr{U}_n$ be the union of the surviving boxes, intersected with their surviving parents, and let

$$\mathscr{U}_\infty := \bigcap_{n \geq 0} \mathscr{U}_n. \tag{414}$$

The depth $n$ is an algebraic refinement index, not a second physical budget. Every operation uses rational cylinder arithmetic and the public ordered graph; no multiplier, compactness certificate, or analytic estimate is supplied by the user.

**Theorem III.4.31** (Stagewise computation of the target mechanism core)**.** *The branch-and-bound construction has the following properties.*

(C1) *Every depth is a finite exact calculation. Its output consists of a finite target-cylinder cover $\mathscr{U}_n$, rational ordered identities for every deleted cell, and rational lower bounds for every retained cell. The covers are compact and decreasing.*

(C2) *The construction loses no target-active mechanism and retains no spurious infinite branch:*

$$\mathscr{U}_\infty = X_\infty. \tag{415}$$

*Thus $\mathscr{U}_n = \varnothing$ at some finite depth if and only if the target mechanism core is empty.*

(C3) *Let $F$ be any generated bounded continuous target-aligned structural observable. From the finite boxes one obtains certified intervals*

$$I_n(F) := \left[ inf_{\mathscr{U}_n} F_n - 2^{-n}, \sup_{\mathscr{U}_n} F_n + 2^{-n} \right],$$

*where $F_n$ is its generated rational cylinder approximation. After replacing each interval by its intersection with the preceding ones,*

$$I_n(F) \downarrow \left[ \min_{X_\infty} F, \max_{X_\infty} F \right]. \tag{416}$$

*Consequently every continuous invariant of the surviving target structure is computed by certified two-sided enclosures.*

(C4) *If the core is nonempty, the complete stream $(\mathscr{U}_n)_n$, together with the compatible moment tables and origin rows, is a canonical name of the compact set of realized target mechanisms. Each infinite nested box word reconstructs its same-origin path through the diagonal realization map. Hence the nonempty output is structural data that can be further queried and refined, not the statement that all finite tests happened to remain feasible.*

*The algorithm is target local: refining any source-transverse or target-null coordinate leaves every $\mathscr{U}_n$ and every interval $I_n(F)$ unchanged.*

*Proof.* At fixed depth there are finitely many boxes, rows, monomials, and moment coordinates. Rational semialgebraic feasibility and the corresponding ordered dual are decidable by exact elimination; a displayed rational dual identity is checked by direct arithmetic. This proves (C1).

Every genuine circulation satisfies every finite relaxation, so it is never deleted. Conversely, choose a point or compatible moment table from each box along an infinite nested word. Compactness of the bounded total-graph coordinates gives a diagonal limit. Because the enumeration is exhaustive, each equation, carrier, polarity, origin, and graph relation is imposed from some finite depth onward. The least-unresolved-coordinate rule therefore gives precisely the circulation-elimination limit, proving (415). If the intersection is empty, compactness of the decreasing covers gives a finite empty depth; the reverse implication is immediate.

Uniform cylinder approximation gives $\|F - F_n\|_{\mathscr{U}_n} \le 2^{-n}$. Extrema of a continuous function over decreasing nonempty compact sets converge to the extrema over their intersection. This proves (416). Finally, the compatible moments reconstruct a measure on the total-graph inverse limit, and the origin and cofinal-shell rows survive. Resolving every generated graph coordinate invokes diagonal target-aligned saturation and yields the paths in (C4). All partition coordinates lie in the target quotient by construction, proving target locality. □

**Theorem III.4.32** (Completeness of circulation elimination)**.** *The sets in* (413) *are closed, target-active, and decreasing. Moreover,*

$$X_\infty = \varnothing \quad \Longleftrightarrow \quad \mathsf{Sing}^{\mathrm{real}}(\mathscr{T}_{\Sigma,\mathfrak{b}}) = \varnothing. \tag{417}$$

*If the intersection is empty, compactness gives a finite depth at which it is empty; concatenating the circulation-dual identities gives a finite target-local regularity derivation. If it is nonempty, every finite production and graph descendant has been evaluated, and diagonal realization produces a same-origin explicit singular mechanism.*

*Proof.* Closedness follows from weak-* compactness of the circulation slices and total-graph closure. The operator removes exactly the cells having no normalized recurrent circulation or having positive recurrent physical price. On a zero-price cell it resolves the least target-cyclic descendant. Consequently membership in every $X_n$ is equivalent to compatibility with the first $n$ circulation, production, polarity, graph, origin, and cofinal-shell rows. By Proposition III.4.26 and Theorem III.4.34, such a fixed point is the target projection of an explicit singular mechanism, and every explicit singular mechanism supplies a zero-price circulation at every finite depth: its finite total physical charge divided by the unbounded accumulated return tends to zero, and a weak-* limit of its empirical edge measures is invariant. This proves (417).

For decreasing closed subsets of the compact spine, empty intersection implies that one finite intersection is empty. The fixed enumeration records the finitely many dual identities used before that depth. In the nonempty case, compact diagonal saturation preserves the origin and all ordinary public graphs and therefore supplies the stated realized path. □

**Definition III.4.33** (Explicit singular mechanism)**.** An *explicit singular mechanism* is a same-origin path $u_*$ in a closed reduced public state space $X_*$ such that:

(E1) every weak-equation, constraint, carrier, chart, covariance, boundary, support-transfer, and realization coordinate has an ordinary value on its induced graph, and every public-equation fidelity defect vanishes; in particular the projection of $u_*$ solves the original public weak equation rather than an equation with an independent unresolved defect forcing;

(E2) the induced five-channel equation on $X_*$ is closed under the resaturation successor and the recurrent passivity row has been evaluated; every nonzero feeding or storage current carries its signed-dyadic return law, detecting target covector, source ancestry, and same-origin successor, whereas every vanishing production is recorded in the joint kernel;

(E3) the path has the inherited physical ledger and origin coordinate; and

(E4) the shell functions satisfy $\sup_t \Phi_r(u_*(t)) = 1$ for every member of a cofinal target-shell family.

Thus an ordered positive functional, invariant law, formal profile, or equality kernel is not an explicit singular mechanism. A graph-boundary coordinate becomes part of one only after its descendants close to an ordinary induced equation and the same-origin path realizes contact.

**Theorem III.4.34** (Three-output target-aligned current compiler)**.** *Run Definition III.4.23 on a contact-selected active signature. Every maximal compiler execution has exactly one of the following outputs:*

(C1) *a cofinal collection of disjoint return events is charged to the public physical measure with divergent total mass;*

(C2) *the retained same-origin path lies in $\mathscr{N}_\Sigma$, so it cannot cross the selected target shells;*

(C3) *the compatible inverse limit of the queue is an explicit singular mechanism in the sense of Definition III.4.33.*

*Compactness modes, fidelity defects, equality kernels, feeding classes, storage transfers, and nonordinary graph values are internal queue coordinates. None is an additional output. Consequently the full continuum regularity track on a finite physical stratum is the target-local decision*

$$\boxed{\textit{contact} \implies \mathsf{Charge}_\infty \,\dot\cup \mathscr{N}_\Sigma \,\dot\cup \mathsf{Sing}^{\mathrm{real}}_\Sigma.} \tag{418}$$

*The first two cells contradict contact under a finite physical ledger. Thus the compiler proves regularity precisely when every reachable queue branch ends in* (C1) *or* (C2)*. Each such ending contains its finite local carrier or ordered identity; failure of all finite endings constructs the realized path in* (C3) *rather than an unevaluated case.*

*Proof.* At a finite queue state, target-current circulation duality, target-local stratification, and the ordered proof–model alternative give (Q1) or one of (Q2)–(Q4). Positive cycle price or infeasibility supplies the finite dual identity in (Q1); zero cycle price supplies the primal circulation processed by the remaining transitions. The local charging rule proves that only (Q1) spends physical measure. The joint-kernel descent proves that (Q2) either gives (C2) or supplies another target-aligned coordinate. Recurrent stratification and Poincare recurrence prove that (Q3) retains every positive-density current, including an exact storage transfer. Fidelity-defect support transfer proves that (Q4) retains every target-visible failed operation. These are the complete polarity and total-graph values of the least coordinate, so no finite execution has another outgoing edge.

For an infinite execution, resolved-prefix monotonicity and compactness of the bounded total-graph factors give a compatible inverse-limit path space. The same-origin witnesses survive by successor no-double-counting. Apply recurrent stratification to its empirical shift laws. A nonzero feeding, storage, fidelity, or graph coordinate invokes (Q3) or (Q4) and therefore is already represented on the cofinal queue. When these transitions continue, diagonal target-aligned saturation resolves all their generated descendants. The target-null diagonal is (C2). On the complementary diagonal, public-equation fidelity, ordinary induced operations, the physical ledger, origin witnesses, and cofinal shell contact are exactly conditions (E1)–(E4); this is (C3). A divergent accumulated carrier row is (C1). The fixed transition priority and disjoint target quotient make the outputs mutually exclusive. □

**Corollary III.4.35** (Target-spine criterion for regularity)**.** *On an ordinary branch with finite public physical ledger,*

$$\boxed{\mathrm{Reach}^{\mathrm{phys}}_\Sigma > 0 \quad\iff\quad \mathsf{Sing}^{\mathrm{real}}(\mathscr{T}_{\Sigma,\mathfrak{b}}) \neq \varnothing \textit{ for some reachable } \mathfrak{b}.} \tag{419}$$

*Consequently*

$$\mathrm{Reach}^{\mathrm{phys}}_\Sigma = 0 \quad\iff\quad \mathsf{Sing}^{\mathrm{real}}(\mathscr{T}_{\Sigma,\mathfrak{b}}) = \varnothing \textit{ for every reachable } \mathfrak{b}. \tag{420}$$

*Both sides are properties of the projected generated algebra. Evaluating the full orbit space, a source-transverse norm, or any target-null motion cannot change either value. Here "reachable"* *means that the corresponding same-origin ordered cell survives the generated finite relations; target contact is a conclusion only for a realized member of that cell. Equivalently, the right side vanishes precisely when circulation elimination has empty core $X_\infty$ on every reachable signature. This supplies the canonical computation of the spine condition.*

*Proof.* If contact occurs, initialize the compiler with its shell current. The finite physical ledger excludes output (C1) of Theorem III.4.34, and shell crossing excludes (C2). Output (C3) therefore gives a realized singular path. By Proposition III.4.26 it lies in the target spine. Conversely, a member of $\mathsf{Sing}^{\mathrm{real}}(\mathscr{T}_{\Sigma,\mathfrak{b}})$ contains an actual same-origin path crossing every cofinal target shell, hence gives contact by Definition III.4.33. This proves (419); negation gives (420). □

**Theorem III.4.36** (No terminal zero-price model)**.** *Apply resaturation to any target-active zero-price successor. Exactly one of the following generated events occurs.*

(Z1) *a finite dual-carrier row charges infinitely many disjoint target crossings to positive physical production or exterior input, contradicting the finite physical ledger;*

(Z2) *a finite relation makes the branch target-null, regular, continuable, or incompatible with the public solution class;*

(Z3) *a first nonordinary total-graph value, a nonzero target-local feeding class, its positive support record, or a proper kernel restriction creates the next target-active successor, to which resaturation is reapplied;*

(Z4) *recurrent stratification reduces the compatible limit of the resaturation chain to a target-active recurrent current system. Diagonal realization of all its ordinary public graphs reconstructs an explicit singular mechanism in the sense of Definition III.4.33.*

*The alternatives are read in the fixed enumeration of graph coordinates and dual words: the first finite closing or successor event is selected, and* (Z4) *occurs only when no finite event occurs. In particular, a positive stationary functional or target-visible equality kernel can occur only in* (Z3) *or as an intermediate object in the reconstruction of* (Z4)*; neither is terminal by itself.*

*Proof.* At each stage run the target-cyclic joint-kernel descent of Theorem III.4.11. Its case (K1) is (Z1), and (K2) is (Z2). A least nonzero or nonordinary coordinate in (K3) is exactly the proper successor (Z3) specified in Definition III.4.13. Because its origin and detecting covector are retained, the theorem applies again without adding specialization data. Monotonicity at finite and limit stages is Proposition III.4.14.

If no finite closing or charged event occurs, apply Theorems III.4.20 and III.4.22 to the diagonal limit supplied by strict prefix growth. Every nonzero feeding, fidelity, storage, support-transfer, or nonordinary coordinate returns to (Z3) or to its induced recurrent-current successor. If these successors continue cofinally, diagonal target-aligned saturation gives either target-nullity or a component with ordinary public operations, zero fidelity defect, the inherited origin, and a path crossing a cofinal target-shell family. Stationary carrier reduction supplies its evaluated production and feeding coordinates. These are precisely conditions (E1)–(E4), so (Z4) holds. The priority convention in the joint-kernel theorem gives exclusivity. □

**Corollary III.4.37** (Post-resaturation residual and terminal exhaustiveness)**.** *After zero-price resaturation, the residual is target-null. The only mathematical terminals are a proved regularity contradiction* (Z1)–(Z2) *or a reconstructed explicit singular mechanism* (Z4)*. Classification names, compactness defects, formal invariant laws, and equality kernels are successor data, not conclusions.*

### III.4.4 Diagonal first-failure closure

The preceding protocol becomes a regularity method only after every routed row is followed to a mathematical conclusion. A row name records the first failed structural test; it is not a terminal output.

**Definition III.4.38** (Equation-generated closure graph)**.** Fix a public PDE presentation and one persistent target-aligned source support $A$. Its *closure graph* $\mathfrak{G}_A$ has the following vertices: the ordinary retained cells and every first-failure cell generated in steps (I1)–(I8). There is a directed edge $v \to w$ precisely when the weak equation, a signed carrier identity, a source-preserving limit operation, or a target-shell successor sends the witness defining $v$ to the witness defining $w$. The terminal vertices of the regularity branch are exactly

(T1) a target-null cell;
(T2) a regular or continuable cell;
(T3) a class-incompatible cell;
(T4) a cell whose disjoint descendants have divergent total physical price;

Every zero-price equality vertex instead enters the resaturation successor of Definition III.4.13. The only complementary conclusive output is

$$\text{an explicit singular mechanism satisfying (E1)–(E4).} \tag{S}$$

The graph and every edge relation are generated from the public equation. They are not additional specialization data.

At depth $k$, the canonical closure procedure evaluates the first $k$ generated graph coordinates and the first $k$ dual-carrier words. A finite first failure creates its displayed successor. A compatible cofinal sequence retains all earlier coordinates and its origin tag. Thus no named row is terminal merely by being classified, and no closure rank is supplied with the PDE presentation.

**Theorem III.4.39** (Diagonal structural closure)**.** *For every target-aligned contact branch, the canonical finite-depth closure procedure has exactly one of the following outcomes:*

(G1) *a finite stage reaches a terminal of type* (T1)–(T4)*;*
(G2) *compatible stages exist at every depth, and diagonal compactness constructs a same-origin infinite path satisfying every generated graph and dual-carrier relation;*
(G3) *resaturation reconstructs the explicit singular mechanism* (S)*.*

*On contact-generated input, the path in* (G2) *retains the original contact witnesses. Every nonzero approximation, descent, chart, or realization coordinate on that path is the next nonterminal successor; if all such coordinates vanish, the path is passed through zero-price resaturation. Hence* (T1)–(T4) *contradict contact, while* (G3) *is a proved singularity conclusion rather than a formal model. Absence of singular contact on the generated finite-physical class is equivalent to exclusion of the explicit singular output on every algebraically reachable active branch. Target-local kernel identities are generated sufficient tests, but a kernel name or positive functional never ends the procedure. The criterion requires no user-provided closure rank.*

*Proof.* Contact produces a persistent nonempty support $A$ by Theorem III.2.10. Run the finite cylinder and dual-word enumeration on its active branch. A finite terminal gives (G1) or (G3). Otherwise every finite depth has a compatible retained value. The observer spaces are compact and the bonding maps preserve all earlier coordinates, so the diagonal inverse-limit argument constructs (G2). The origin coordinate preserves the original contact events. Totalization selects the first nonzero defect as the next successor; if no defect remains, the path is an ordinary or zero-defect solution path and enters Theorem III.4.36. The contradictions associated with (T1)–(T4) are respectively the target floor, terminal coverage, the public branch class, and finite physical expenditure. Finally, Theorem II.2.29 identifies the only recurrent nonterminal support with a target-local equality kernel. Its resaturation is exhaustive by Theorem III.4.36. It either closes the branch or reconstructs (S). Conversely, failure of regularity supplies that realized singular mechanism through the contact-generated diagonal construction and realization graph. □

**Definition III.4.40** (PDE backend)**.** A *PDE backend* is the homomorphic compilation of a public presentation into the source, shell, carrier, total-graph, successor, and ordered-relation signatures used to construct $\mathfrak{G}_{\mathfrak{b}}$. It may compute equation-specific coefficients, scaling weights, constraint tails, and closure edges. It may not add a source family, a target-relative premise, a physical measure not generated by the equation, or a terminal declaration without a proved relation of the ordered algebra. Here $\mathfrak{G}_{\mathfrak{b}}$ is the induced subgraph of $\mathfrak{G}_A$ obtained by fixing the complete active signature $\mathfrak{b} = (A, m, \Gamma, \eta, \mathbf{R})$ and its earlier zero coordinates.

**Proposition III.4.41** (Backend conservativity)**.** *Every PDE backend preserves the five-channel normal form, prospective contact accountability, source-preserving limit closure, the target-null meaning of $J_{\mathrm{res}}$, and the single physical-budget ledger. Therefore a backend can validate or refute regularity only by evaluating the vertices and closure edges of $\mathfrak{G}_{\mathfrak{b}}$; it cannot change the continuum theory that generated them.*

*Proof.* The backend is a homomorphism into the signature already constructed in Section III.4. Preservation of source normal forms follows from Theorem I.1.6; preservation of contact accountability and the residual quotient follows from Theorem III.2.10; preservation of the physical ledger follows from signed carrier gluing and Theorem II.3.5. The prohibited additions are not symbols of the target signature, so they cannot occur in its image. □

## III.5 Specialization examples

The examples in this section test whether distinct public equations compile into the same closure calculus. They are consequences and evaluations of the general theory. No example supplies a premise used by Theorem III.4.39 or changes its source alphabet, physical ledger, successor semantics, or terminal alternatives. In particular, the Navier–Stokes calculation below has the role of a specialization example, just as LPN does in Part I; it is not part of the definition or proof of the continuum calculus.

**Proposition III.5.1** (Specialization noninterference)**.** *For every public PDE presentation $\mathfrak{P}$, specialization is the homomorphic substitution*

$$\mathsf{Compile}_{\mathrm{cont}}(\mathfrak{P}) \longrightarrow (I_{\mathfrak{b}}, \mathscr{W}_{\mathfrak{b}}, \mathcal{E}_{\mathfrak{b}}, \mathcal{K}_{\mathfrak{b}}, \mathfrak{G}_{\mathfrak{b}}) \tag{421}$$

*for its reachable active branches. The definitions and proofs of the record-scale selector, dual-carrier grammar, compactness routing, modulation conjugation, selector-accountability closure, and ranked closure theorem are independent of $\mathfrak{P}$. A backend computes only the images of its equation, source operators, native carrier, constraints, symmetries, and target-shell covectors. It cannot add a branch premise or alter a closing rule. Hence every backend example has the same logical status as an application of the saturated calculus of Part I.*

*Proof.* All objects on the right of (421) are functorial constructions from the public weak equation: ideal generation, adjoint closure, chart pullback, signed gluing, target quotient, ordered completion, and total-graph completion. The general theorems quantify over these generated objects and contain no equation-specific coefficient. Substitution therefore preserves their identities and alternatives. Backend conservativity Proposition III.4.41 excludes every additional source, physical ledger, or target-relative premise. □

### III.5.1 Specialization example: three-dimensional incompressible Navier–Stokes

Consider on $\mathbb{R}^3$ or the periodic box

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu\Delta u + f, \qquad \nabla \cdot u = 0. \tag{422}$$

For the regularity validation take $f = 0$, a smooth divergence-free datum of finite kinetic energy, and the usual maximal strong branch inside the suitable weak-solution graph. The complete continuation target consists of terminal germs at which that strong branch has no continuation. This target is generated by the continuation graph; no blow-up norm or profile is inserted as target data.

The public backend contains the distributional equation, the divergence-free rational test core, Galerkin and mollification words, the pressure constraint, Galilean and parabolic covariance graphs, and the tested kinetic-energy carrier. On every ordinary unforced branch the carrier relation is

$$\frac{1}{2}\|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 \,\mathrm{d}s + \mu_{\mathrm{suit}}(\Omega \times [0,t]) = \frac{1}{2}\|u_0\|_2^2, \tag{423}$$

where $\Omega$ denotes the spatial domain and $\mu_{\mathrm{suit}} \geq 0$ is the totalized suitable-energy defect. This generates the single physical measure

$$\mathrm{d}\mu_{\mathrm{NS}} := \nu|\nabla u|^2 \,\mathrm{d}x\,\mathrm{d}t + \mathrm{d}\mu_{\mathrm{suit}}.$$

This is the sole budget used below. Its restriction to a parabolic event is raw physical expenditure. It becomes the price of singular-shell progress only after the local energy identity is restricted to a generated target-active branch, decomposed into the five source currents, and admitted by (90).

For a rational $E > 0$, denote by $\mathscr{X}_{\mathrm{NS}}(E)$ the closed physical stratum generated by

$$\frac{1}{2}\|u_0\|_2^2 \leq E, \qquad \mu_{\mathrm{NS}}(\Omega \times [0,T]) \leq E \quad (T < T_*). \tag{424}$$

The viscosity is reduced to $\nu = 1$ by the ordinary deterministic scaling of the unforced equation; retaining $\nu > 0$ as a positive public parameter gives the same ordered graph. The number $E$ indexes level sets of the single physical ledger and is not a second budget. Every smooth finite-energy datum lies in $\mathscr{X}_{\mathrm{NS}}(E)$ for some rational $E$. The finite target programs below operate uniformly on this stratum: origin-cylinder values are variables constrained by the public datum class and (424), rather than unlisted real coefficients of the program.

**Proposition III.5.2** (Navier–Stokes public input ledger)**.** *The specialization uses only the following seed relations:*

(B1) *the distributional momentum equation and incompressibility;*

(B2) *the whole-space kinetic-energy inequality and the local suitable-energy inequality, totalized by their nonnegative defects;*

(B3) *the pressure equation $-\Delta p = \partial_i\partial_j(u_i u_j)$, modulo functions of time;*

(B4) *Galilean covariance, parabolic change of variables, rational cutoffs, Galerkin approximation, and mollification; and*

(B5) *integration by parts on the rational test core and weak lower semicontinuity of the local $L^2$ norm.*

*Target shells, source labels, record scales, the adjoint module, defect coordinates, production forms, carrier cohomology, joint kernels, and branch closures are generated after these seeds are inserted. None belongs to the input ledger.*

*Proof.* Items (B1)–(B3) are the standard weak and suitable-energy formulation of (422). Item (B4) consists of algebraic transformations and the declared approximation grammar. Item (B5) is part of the canonical weak test calculus. Every later Navier–Stokes object is defined below by applying the general continuum constructors to these relations. □

**Proposition III.5.3** (Navier–Stokes physical entropy cocycle)**.** *On an ordinary unforced suitable branch define, for $0 \leq s \leq t < T_*$,*

$$\mathcal{M}_{\mathrm{NS}}[s,t] := \frac{1}{2}\|u(s)\|_2^2 - \frac{1}{2}\|u(t)\|_2^2 = \mu_{\mathrm{NS}}(\Omega \times [s,t]). \tag{425}$$

*using the right-continuous kinetic-energy representative; equivalently, first define the identity at common Lebesgue times and take its Stieltjes extension. The totalized defect makes this an equality on every graph value. It obeys the cocycle law*

$$\mathcal{M}_{\mathrm{NS}}[s,t] + \mathcal{M}_{\mathrm{NS}}[t,r] = \mathcal{M}_{\mathrm{NS}}[s,r] \qquad (s \leq t \leq r), \tag{426}$$

*and its localized, moving-chart descendant is the signed covariant carrier* (448)*. After restriction to the minimal state $X^{\min}_{\mathrm{NS},\mathfrak{b}}$ of Definition III.5.8, only incidence-retained production can pair with a singular shell. Such production is charged by $\mathcal{M}_{\mathrm{NS}}$ exactly when the local carrier-domination row is ordinary; every other value is a typed successor.*

*Proof.* Subtract (423) at times $s$ and $t$. Countable additivity of viscous expenditure and the suitable defect gives (425) and (426). Push the local suitable-energy equality through the moving parabolic chart and test by the localized kinetic carrier to obtain (448). Target-incidence compression and the charging rule give the last assertion. □

**Definition III.5.4** (Navier–Stokes active and successor signatures)**.** A Navier–Stokes active signature is the full framework record

$$\mathfrak{b} = (A, m, \Gamma, 1/5, \mathbf{R}),$$

together with its ordered list of earlier zero polarity and total-graph coordinates. It is *algebraically reachable* when each of its finite prefixes survives the public ordered relations and the same-origin successor graph. This condition is enumerated from the presentation and makes no reference to an actual singular orbit. If contact occurs, prospective accountability places its induced signature in this enumeration. For the physical stratum $\mathscr{X}_{\mathrm{NS}}(E)$, write $\mathfrak{B}^{\mathrm{reach}}_{\mathrm{NS}}(E)$ for the resulting generated set of signatures. For $\xi \in \{\mathrm{eq}, \mathrm{nl}, \mathrm{pr}, \mathrm{ext}, \mathrm{sc}, \mathrm{cyc}, \mathrm{ord}\}$, let $\mathfrak{b}^{\xi}$ be the canonical successor signature obtained by adjoining ordinary values of every earlier Navier–Stokes interface and the zero, signed-dyadic, or graph-boundary value of the first interface of type $\xi$. Write $Z_{\mathrm{NS},\mathfrak{b}^{\xi}}$ for its target-local stratum and $\mathcal{K}_{\mathrm{NS},\mathfrak{b}^{\xi}}$ for the image of the general cone $\mathcal{K}_{\mathfrak{b}^{\xi}}$ under Navier–Stokes specialization. These signatures retain the shell family, realization mode, origin, scale record, and detecting covector; the label $\xi$ alone never defines a cell.

**Proposition III.5.5** (Navier–Stokes source and scaling compilation)**.** *For* (422)*, viscosity has* Geom *ancestry, the fixed nonlinear transport and pressure-constraint transport have* Caus *ancestry, Galilean and pressure-constant quotients append* Abs*, parabolic charts and scale motion append* Lift*, and exterior or terminal flux appends* Bdry*. Under*

$$V(y,s) = r\,u(x_0+ry, t_0+r^2 s), \qquad P(y,s) = r^2 p(x_0+ry, t_0+r^2 s),$$

*the viscous physical price satisfies*

$$\nu \int_{Q_r(x_0,t_0)} |\nabla u|^2 \,\mathrm{d}x\,\mathrm{d}t = \nu r \int_{Q_1} |\nabla V|^2 \,\mathrm{d}y\,\mathrm{d}s. \tag{427}$$

*Hence a normalized dissipation gap alone cannot give a scale-independent admissible price. Every finite-expenditure contact cascade is routed to the target-aligned scale-carrier successor cell.*

*Proof.* The source assignments follow directly from the constructor grammar. For (427), use $\nabla_x u = r^{-2}\nabla_y V$ and $\mathrm{d}x\,\mathrm{d}t = r^5\,\mathrm{d}y\,\mathrm{d}s$. The resulting factor is $r$. A geometric scale sequence can have finite sum, so the physical-price theorem routes the finite branch to zero-price saturation rather than asserting a uniform lower bound. □

**Definition III.5.6** (Navier–Stokes target-cyclic module)**.** Write the moving-frame vector field on divergence-free fields as

$$\mathcal{N}_{a,b}(V) := \nu\Delta V - \mathbf{P}_\sigma \nabla\cdot(V\otimes V) + a\Lambda V + b\cdot\nabla V,$$

where $\mathbf{P}_\sigma$ is the Leray projection, interpreted through its local, harmonic, and exterior total graphs. Fix an algebraically reachable Navier–Stokes active branch $\mathfrak{b} = (A, m, \Gamma, 1/5, \mathbf{R})$ and a generated shell $\Phi \in \Gamma$. The raw forward and backward words are generated on $X_{\mathfrak{b}}$; after Definition III.5.8, every downstream use is restricted to $X^{\min}_{\mathrm{NS},\mathfrak{b}}$. Let $\mathscr{C}^0_{\mathrm{NS},\mathfrak{b}}(\Phi)$ be the rational localization module generated by $D\Phi$. The next level adjoins localizations, derivatives, chart pullbacks, constraint descents, pressure-tail coordinates, defect polarizations, the source-adapted carrier and square-completion words whose native carrier covector is $DK_\chi(V) = \chi V$, and the ordinary values of

$$\mathcal{A}^*_{V,a,b}\lambda := \nu\Delta\lambda + \mathbf{P}_\sigma\Big((V\cdot\nabla)\lambda - (\nabla V)^{\mathsf{T}}\lambda\Big) - a(2\lambda + y\cdot\nabla\lambda) - b\cdot\nabla\lambda. \tag{428}$$

Iterating this rule and all commutators gives

$$\mathscr{C}^\infty_{\mathrm{NS},\mathfrak{b}}(\Phi) := \overline{\bigcup_{n\ge 0} \mathscr{C}^n_{\mathrm{NS},\mathfrak{b}}(\Phi)}^{\mathrm{cyl}}. \tag{429}$$

Every derivative in this definition is the total graph of rational difference quotients. A failed Leray projection, pressure split, localization, chart, derivative, or commutator is retained at its first nonordinary value.

**Proposition III.5.7** (Adjoint image of the Navier–Stokes current)**.** *On the ordinary divergence-free test core,*

$$D\mathcal{N}_{a,b}(V)^*\lambda = \mathcal{A}^*_{V,a,b}\lambda.$$

*Consequently* $\mathscr{C}^\infty_{\mathrm{NS},\mathfrak{b}}(\Phi)$ *is exactly the target-local image of the general target-cyclic module under the Navier–Stokes specialization homomorphism. It is generated by* (422) *and* $\Phi$*; no additional multiplier is part of the backend.*

*Proof.* Linearization of the nonlinear current in a divergence-free direction $W$ gives $-\mathbf{P}_\sigma((W\cdot\nabla)V+(V\cdot\nabla)W)$. Integration by parts gives $(V\cdot\nabla)\lambda-(\nabla V)^{\mathsf{T}}\lambda$. The Laplacian is self-adjoint, $(b\cdot\nabla)^*=-b\cdot\nabla$, and in three dimensions $(1+y\cdot\nabla)^*=-(2+y\cdot\nabla)$. This proves (428). The remaining generators are the images of localization, quotient, covariance, polarization, and commutator operations in Definition III.4.1. □

**Definition III.5.8** (Navier–Stokes minimal target-active realization)**.** Let $\mathscr{R}^n_{\mathrm{NS},\mathfrak{b}}$ be the forward module generated from the viscous, nonlinear/pressure, quotient, parabolic-chart, and exterior-current roots by the operations in the public ledger. Pair it with $\mathscr{C}^n_{\mathrm{NS},\mathfrak{b}}(\Phi)$ from (429). The finite Navier–Stokes incidence core $\mathsf{M}^n_{\mathrm{NS},\Sigma,\mathfrak{b}}$ consists exactly of vertices lying both:

(R1) downstream of one of those five roots through the momentum, pressure, localization, chart, defect, carrier, return, or realization graphs; and

(R2) upstream of $D\Phi$ through a nonzero signed pairing, a typed pairing boundary, or an adjoint/commutator edge.

Its graph-complete diagonal closure is denoted $\mathsf{M}_{\mathrm{NS},\Sigma,\mathfrak{b}}$. In particular, a pressure tail, Reynolds component, scale current, annular flux, or commutator word is retained exactly when it lies on such a source–shell path. Write

$$X^{\min}_{\mathrm{NS},\mathfrak{b}}:=X_{\mathfrak{b}}\cap|\mathsf{M}_{\mathrm{NS},\Sigma,\mathfrak{b}}|,\qquad \mathscr{C}^{n,\min}_{\mathrm{NS},\mathfrak{b}}:=\mathscr{C}^n_{\mathrm{NS},\mathfrak{b}}\big|_{X^{\min}_{\mathrm{NS},\mathfrak{b}}}. \tag{430}$$

Likewise set $\mathcal{K}^{\min}_{\mathrm{NS},\mathfrak{b}}:=\mathcal{K}_{\mathrm{NS},\mathfrak{b}}\big|_{X^{\min}_{\mathrm{NS},\mathfrak{b}}}$.

**Proposition III.5.9** (Navier–Stokes incidence reduction)**.** *The Navier–Stokes specialization maps the abstract minimal realization onto the one in Definition III.5.8:*

$$\mathsf{S}_{\mathrm{NS}}(\mathsf{M}_{\Sigma,\mathfrak{b}})=\mathsf{M}_{\mathrm{NS},\Sigma,\mathfrak{b}}. \tag{431}$$

*Restriction to this set preserves the signed carrier identity, scale-balance row, admitted local-energy price, cycle price, circulation-elimination core, and every finite target cover. At finite depth the retained coordinate list is computed by forward and backward graph search before any Navier–Stokes moment or localizing matrix is formed.*

*Proof.* The forward public constructors specialize to the weak momentum, pressure, Leray, localization, Galilean, parabolic, carrier, defect, return, and realization graphs listed above. The backward constructors specialize to (428) and its localized, constrained, and polarized descendants by Proposition III.5.7. Specialization preserves their pairing and every first nonordinary value. It therefore preserves forward reachability, backward reachability, and their intersection, proving (431). The remaining assertions are the specialized commutation identities of Theorem III.4.6. □

**Definition III.5.10** (Target-active Navier–Stokes Hamiltonian response)**.** On an ordinary point of $X^{\min}_{\mathrm{NS},\mathfrak{b}}$, let

$$H_{\mathrm{NS}}W:=-\nu\mathbf{P}_\sigma\Delta W, \tag{432}$$

$$A_{V,a,b}W:=-\mathbf{P}_\sigma\big((V\cdot\nabla)W+(W\cdot\nabla)V\big)+a\Lambda W+b\cdot\nabla W, \tag{433}$$

$$L_{V,a,b}:=-H_{\mathrm{NS}}+A_{V,a,b}. \tag{434}$$

The divergence-free and pressure quotients are included in the generated $\mathsf{Abs}$ graph. Let $P_{\mathscr{I}}$ be the cylinder projection onto the retained forward source words and let $P_{\mathscr{O}}$ be the cylinder projection onto the retained target-cyclic covectors. At an ordinary resolvent point define

$$\Psi_{\mathrm{NS},\Sigma}(\lambda;V,a,b):=\left\|P_{\mathscr{O}}(\lambda-L_{V,a,b})^{-1}P_{\mathscr{I}}\right\|. \tag{435}$$

The resolvent, its norm epigraph, and its input/output projections are total-graph coordinates. Their first nonordinary values are target-cyclic successors with the ancestry of the operation that failed.

**Proposition III.5.11** (Compressed Navier–Stokes resolvent factorization)**.** *The response* (435) *is the Navier–Stokes specialization of* (173) *after minimal incidence reduction. For every retained source variation $K^c$, its ordinary derivative is*

$$P_{\mathscr{O}}(\lambda - L)^{-1}K^c(\lambda - L)^{-1}P_{\mathscr{I}}. \tag{436}$$

*Consequently every target-visible pseudospectral amplification factors through a retained* $\mathsf{Geom}$*,* $\mathsf{Caus}$*,* $\mathsf{Abs}$*,* $\mathsf{Lift}$*, or* $\mathsf{Bdry}$ *word. If the resolvent relation, pressure descent, domain identification, or input/output projection is nonordinary, its typed value is the next incidence edge. At finite depth, only matrix elements indexed by $\mathsf{M}^n_{\mathrm{NS},\Sigma,\mathfrak{b}}$ enter the ordered target program.*

*Proof.* The Stokes form is the $\mathsf{Geom}$ part of the specialized generator; linearized convection is fixed $\mathsf{Caus}$, the Leray and pressure graphs are $\mathsf{Abs}$, chart velocity is $\mathsf{Lift}$, and exterior domain values are $\mathsf{Bdry}$. Compression by $P_{\mathscr{I}}$ and $P_{\mathscr{O}}$ is exactly the source–target incidence restriction. Apply Proposition II.6.2 on the ordinary resolvent graph to obtain (436). Source-preserving totalization supplies every listed boundary value. The finite statement is Corollary III.4.7 under Navier–Stokes specialization. □

**Definition III.5.12** (Navier–Stokes generated productions)**.** For $\lambda \in \mathscr{C}^{\infty,\min}_{\mathrm{NS},\mathfrak{b}}(\Phi)$, pair the moving equation with every rational localization of $\lambda$, glue common faces, and complete every algebraic square. The resulting production cone $\mathscr{P}_{\mathrm{NS},\mathfrak{b}}(\Phi)$ is the least closed positive cone containing

(P1) for rational $0 \le \chi \le 1$,

$$D_\chi(V) = \nu \int \chi^2 |\nabla V|^2, \qquad M^{(r)}_\chi = r^{-1} \int \chi^2 \Big(\frac{x - x_0}{r}, \frac{t - t_0}{r^2}\Big)\, \mathrm{d}\mu_{\mathrm{suit}}(x,t);$$

(P2) the positive Reynolds-defect polarizations generated by weak limits of $V \otimes V$;

(P3) the Dirichlet forms $\Gamma_{\nu,\chi}(\lambda) = \nu \int |\nabla(\chi\lambda)|^2$ and every commutator square generated from $\nu\Delta$, $\mathcal{A}^*_{V,a,b}$, rational localization, and the Leray constraint; and

(P4) positive boundary or scale terms only when the signed carrier identity proves that positivity after all compatible faces have been glued.

Each member retains the source word of the expansion that generated it. Only members dominated by $\mu_{\mathrm{NS}}|_Q$ through the target-local signed carrier identity are admissible physical charges for the selected shell event. All other members are joint-kernel or carrier-domination successor coordinates. Define

$$X^{\infty}_{\mathrm{NS},\mathfrak{b},\Phi} := X^{\min}_{\mathrm{NS},\mathfrak{b}} \cap \bigcap_{P \in \mathscr{P}_{\mathrm{NS},\mathfrak{b}}(\Phi)} \{P = 0\}. \tag{437}$$

**Proposition III.5.13** (Navier–Stokes defect polarization)**.** *Let $V_k \rightharpoonup V$ in $L^2_{\mathrm{loc}}$ along a total-graph sequence in the minimal target-active stratum $X^{\min}_{\mathrm{NS},\mathfrak{b}}$. Exactly one of the following occurs: the tensor coordinate has a first nonordinary value, producing the nonlinear successor $Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{nl}}}$; or it has an ordinary weak measure value and*

$$\mathcal{R}_{ij} := \mathrm{w}\text{–}\lim\nolimits_k (V_k)_i (V_k)_j - V_i V_j$$

*is positive semidefinite in the sense that*

$$\int \chi\, \zeta_i \mathcal{R}_{ij} \zeta_j \ge 0 \qquad (\chi \ge 0, \zeta \in \mathbb{Q}^3) \tag{438}$$

*for every rational compactly supported cutoff. Its pressure descendant is the total solution of*

$$-\Delta q_{\mathcal{R}} = \partial_i \partial_j \mathcal{R}_{ij}, \tag{439}$$

*split into local singular-integral, harmonic, and exterior coordinates. Thus a nonlinear weak defect yields a positive production coordinate and its signed* Caus/Abs/Bdry *descendants. Failure of the tensor limit or of any pressure coordinate is the corresponding first-failure successor. The production coordinate is not thereby declared to be physical expenditure: on the active branch it is charged to $\mu_{\mathrm{NS}}$ only if the target-local signed carrier factorization proves the domination row* (90)*. A nonordinary domination value is retained in the carrier successor coordinate of the same signature.*

*Proof.* For fixed $\chi$ and $\zeta$, weak lower semicontinuity of the $L^2(\chi\, \mathrm{d}y\, \mathrm{d}s)$ norm gives

$$\int \chi |\zeta \cdot V|^2 \leq \liminf_k \int \chi |\zeta \cdot V_k|^2,$$

which is (438). Rational polarization determines the symmetric tensor coordinate. Taking the divergence of the weak momentum equation gives (439); the standard local Poisson decomposition gives the three displayed descendants. The total graph supplies their boundary values without a convergence premise. □

**Theorem III.5.14** (Navier–Stokes prospective contact compiler)**.** *If an ordinary finite-energy Navier–Stokes branch contacts the complete continuation target, then it generates pairwise disjoint pre-contact events $Q_j$, normalized shell records $R_j$, and sourcewise currents such that*

$$1 = \mathrm{Prog}_{\Sigma_{\mathrm{sing}}}(R_j) = \sum_{c \in \mathfrak{C}} Q_{c,j}, \qquad \max_{c \in \mathfrak{C}} Q^+_{c,j} \geq \frac{1}{5}. \tag{440}$$

*After passage to a subsequence one nonempty support $A \subseteq \mathfrak{C}$, one realization mode, and one same-origin cascade record occur for every $j$, producing the active signature $\mathfrak{b} = (A, m, \Gamma, 1/5, \mathbf{R})$. The residual current has zero pairing with every shell differential and cannot occur in $A$. Moreover*

$$\sum_j \mu_{\mathrm{NS}}(Q_j) \leq \mu_{\mathrm{NS}}(\mathbb{R}^3 \times (0, T_*)) \leq \frac{1}{2} \|u_0\|_2^2. \tag{441}$$

*The quantities in* (441) *are raw event expenditures. They are admissible prices for the sourcewise shell currents only on the ordinary carrier-domination value of the target-local fact normal form; every other value is retained as a successor of $\mathfrak{b}$.*

*Proof.* Apply Theorem I.5.2 to the generated finite-cylinder shells and then Theorem III.1.5. Source preservation under continuum completion is Theorem I.3.4, and the target-null residual statement is (384). Since there are five sources, (440) gives the floor 1/5, and finite source and realization-mode exhaustion gives the constant-signature subsequence. Countable additivity of the measure in (423) gives (441). □

**Definition III.5.15** (Scale-aware Navier–Stokes expenditure record)**.** For a parabolic event $Q_r(x_0, t_0)$, write

$$D_R(V) := \nu \int_{Q_R} |\nabla V|^2 \, \mathrm{d}y \, \mathrm{d}s, \tag{442}$$

$$M_{r,R} := r^{-1}\mu_{\text{suit}}(Q_{rR}(x_0,t_0)), \tag{443}$$

$$p_{r,R} := \mu_{\text{NS}}(Q_{rR}(x_0,t_0)), \tag{444}$$

and retain $r, D_R, M_{r,R}$ as separate graph coordinates. The raw event-expenditure coordinate satisfies

$$p_{r,R} = r\big(D_R(V) + M_{r,R}\big). \tag{445}$$

The product relation, rather than either factor separately, belongs to the physical carrier ideal. Its use as target price additionally requires the local charging relation for the selected shell current.

**Lemma III.5.16** (Exact fixed-scale closure)**.** *Let a zero-event-expenditure Navier–Stokes successor law be supported on a stratum $r \geq r_* > 0$. Then it is supported on $D_R = M_{r,R} = 0$ for every rational $R$. Its ordinary weak-equation component is spatially constant. It is zero on $\mathbb{R}^3$ under finite energy and is constant modulo the Galilean quotient on the periodic box. Hence its ordinary target projection vanishes.*

*Proof.* Equation (445), nonnegativity, and $r \geq r_*$ give

$$D_R + M_{r,R} \leq r_*^{-1} p_{r,R} = 0.$$

An exhaustion by rational cylinders gives $\nabla V = 0$ distributionally. The weak fundamental theorem on connected domains makes $V$ spatially constant, as in Corollary II.2.49. A spatial constant in $L^2(\mathbb{R}^3)$ is zero; on the periodic box it is removed by the generated Galilean quotient. Such states lie in $\mathscr{N}_{\Sigma_{\text{sing}}}$. ☐

**Lemma III.5.17** (Scale-collapse degeneracy is algebraic)**.** *On the closed stratum $r = 0$, the physical-price relation* (445) *imposes no upper bound on $D_R + M_{r,R}$. Consequently zero physical price alone cannot activate Corollary II.2.49 on a parabolically normalized singular cascade.*

*Proof.* In the ordered coordinate algebra the relation becomes $p_{0,R} = 0 \cdot (D_R + M_{0,R}) = 0$. For a sequential realization, choose any $d_n \geq 0$, put $r_n = (n(1+d_n))^{-1}$, and set $D_R(V_n) + M_{r_n,R} = d_n$. Then $r_n \to 0$ and $p_{r_n,R} \leq n^{-1} \to 0$, while $d_n$ is arbitrary. Thus no inequality $D_R + M_R \leq C p_{r,R}$ with scale-independent $C$ is a consequence of the carrier relation. The failure is the vanishing coefficient of a displayed equation. It activates the covariant microscope below; it is not an endpoint of the structural calculation. ☐

**Definition III.5.18** (Covariant parabolic microscope)**.** Let $r(\tau) > 0$ and $x(\tau) \in \mathbb{R}^3$ be absolutely continuous and choose physical time by

$$\dot{t}(\tau) = r(\tau)^2.$$

For $y = (x - x(\tau))/r(\tau)$, set

$$V(y,\tau) = r(\tau)u(x(\tau) + r(\tau)y, t(\tau)), \qquad P(y,\tau) = r(\tau)^2 p(x(\tau) + r(\tau)y, t(\tau)), \tag{446}$$

and write

$$a = \frac{\dot{r}}{r}, \qquad b = \frac{\dot{x}}{r}, \qquad \Lambda V = V + y \cdot \nabla V.$$

The coordinates $(a,b)$ are chart velocities, not additional controls and not additional physical expenditure. Their currents retain Lift ancestry and the ancestry of the equation transported by the chart.

**Proposition III.5.19** (Exact moving-microscope equation)**.** *On every ordinary part of the solution graph, the fields in* (446) *satisfy*

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu\Delta V + a\Lambda V + b \cdot \nabla V, \qquad \nabla \cdot V = 0. \tag{447}$$

*On a boundary value of any displayed operation, the same formula holds in the total graph with the corresponding typed boundary current. Thus passage to a collapsing scale does not discard the normalized equation: it adds the explicit scale and translation currents in* (447)*.*

*Proof.* Differentiate (446), use $\dot{t} = r^2$, and substitute (422). The derivatives of the amplitude, dilation, and center are respectively $aV$, $ay \cdot \nabla V$, and $b \cdot \nabla V$. The spatial divergence is preserved by the chart. Totalization commutes with this algebraic calculation by Theorem I.3.4; a failed product, derivative, pressure, or chart relation is therefore retained with its typed ancestry. □

**Theorem III.5.20** (Signed covariant kinetic carrier)**.** *Let $\phi \in C_c^\infty(\mathbb{R}^3)$ be nonnegative and independent of $\tau$, and put*

$$e = \frac{|V|^2}{2}, \qquad K_\phi(V) = \int e\phi, \qquad D_\phi = \nu \int |\nabla V|^2 \phi.$$

*The pushed-forward suitable-energy defect defines a nonnegative Stieltjes measure $\mathrm{d}M_\phi$, and the exact local carrier identity is*

$$\mathrm{d}K_\phi + D_\phi \, \mathrm{d}\tau + \mathrm{d}M_\phi = \left(F_{\mathrm{int},\phi} + F_{\mathrm{sc},\phi} + F_{\mathrm{tr},\phi}\right) \mathrm{d}\tau, \tag{448}$$

*where*

$$F_{\mathrm{int},\phi} = \int (e + P)V \cdot \nabla\phi + \nu \int e\,\Delta\phi, \tag{449}$$

$$F_{\mathrm{sc},\phi} = -a \int e(\phi + y \cdot \nabla\phi), \tag{450}$$

$$F_{\mathrm{tr},\phi} = -\int e\, b \cdot \nabla\phi. \tag{451}$$

*The three terms are signed currents. In particular, neither $F^+_{\mathrm{sc},\phi}$ nor $|F_{\mathrm{sc},\phi}|$ is declared to be physical expenditure. Scale motion is first glued with the other oriented currents; only the surviving nonexact scale current enters the target-active carrier cohomology.*

*Proof.* Test (447) against $V\phi$. Incompressibility gives the convection and pressure contribution in (449); integration by parts gives the viscous terms. In dimension three,

$$\int V \cdot \Lambda V\,\phi = -\int e(\phi + y \cdot \nabla\phi), \qquad \int V \cdot (b \cdot \nabla V)\phi = -\int e\, b \cdot \nabla\phi.$$

The local suitable inequality becomes equality after adjoining its nonnegative defect measure. This proves (448). The last assertion is exactly the signed gluing order of Theorem II.2.2. □

**Proposition III.5.21** (No free shrinkage)**.** *Fix an algebraically reachable active signature $\mathfrak{b} = (A, m, \Gamma, 1/5, \mathbf{R})$, and let $\Phi \in \Gamma$ be a generated differentiable target-shell observable in the moving chart. Along an ordinary chart interval in $X^{\min}_{\mathrm{NS},\mathfrak{b}}$,*

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\Phi(V) = D\Phi(V)[\nu\Delta V - (V \cdot \nabla)V - \nabla P + a\Lambda V + b \cdot \nabla V]\,, \tag{452}$$

*and the total-graph version holds without differentiability by the derived Stieltjes current. Hence a unit shell crossing in a shrinking chart still has total source pairing one. After quotienting target-null translations and gluing internal faces, if the* Geom*,* Caus*,* Abs*, and* Bdry *pairings vanish, then the nonexact* Lift *scale current has positive target pairing. Shrinkage can move target progress into the scale channel; it cannot erase that progress or place it in* $J_{\mathrm{res}}$*.*

*Proof.* Equation (452) is the chain rule applied to (447). On a nonordinary value it is the definition of the derived shell current, so all first failures remain in the five-source sum. Integrating over a normalized crossing interval gives increment one. Signed gluing cancels exact internal and chart coboundaries. The target-local fact factorization keeps every term in the full signature $\mathfrak{b}$. The residual is target-null, so the asserted scale pairing is forced by the remaining equality; no statement about a current outside $X^{\min}_{\mathrm{NS},\mathfrak{b}}$ is used. □

**Theorem III.5.22** (Scale-cell gluing and microscope passivity)**.** *Let $C_0, \ldots, C_N$ be consecutive target-retaining parabolic cells from one physical Navier–Stokes orbit, with compatible centers, scales, cutoffs, and overlap orientations, all in one active signature* $\mathfrak{b}$*. Restrict each local-energy relation to* $X^{\min}_{\mathrm{NS},\mathfrak{b}}$*, apply the five-source fact normal form, and sum* (448) *before taking positive parts gives*

$$K_N + \sum_{j<N} (D_j + M_j) = K_0 + I_{\mathrm{ext},N} + H_{\mathrm{sc},N}. \tag{453}$$

*Here $D_j = \int_{C_j} D_\phi \,\mathrm{d}\tau$, $M_j = M_\phi(C_j)$, and $K_0, K_N$ are the oriented root and leaf carrier values. Here paired convection, pressure, cutoff, translation, and exact scale currents telescope; $I_{\mathrm{ext},N}$ is the genuine exterior* Bdry *input, and $H_{\mathrm{sc},N}$ is the nonexact scale-carrier class. After totalization, every infinite recurrent target-retaining chain has exactly one of the following generated continuations:*

(i) *a compactness, realization, pressure, overlap, exterior-tail, chart, or carrier relation has a first nonordinary value, which produces its typed target-active successor;*

(ii) *all preceding coordinates are ordinary, the exterior current either cancels in the coherent complex or satisfies the target-local charging rule for the public orbitwise carrier, and the aggregate carrier has a strict block deficit $D + M - F_{\mathrm{sc}} \geq \delta > 0$, which is incompatible with bounded $K_\phi$ and* (453)*;*

(iii) *all preceding coordinates are ordinary and a stationary target-aligned moving-microscope law satisfies*

$$\int (F_{\mathrm{sc}} - D - M) \,\mathrm{d}\varpi = 0 \tag{454}$$

*and all same-origin successor relations.*

*Thus the ordinary scale-collapse branch is reduced to the equality law* (454)*; it is not a branch on which normalized dissipation becomes unobservable.*

*Proof.* Apply Theorem II.2.2 to each finite compatible subcomplex. It gives (453) and identifies its two noninternal terms. Totalize compactness, realization, pressure, overlap, exhaustion, chart, and carrier gluing in that order. Their least nonordinary value gives (i). On the ordinary graph, common faces cancel. A remaining exterior current either satisfies the target-local carrier domination or is its typed Bdry coordinate, already covered by (i). Apply Theorem II.2.9 to the resulting recurrent law. In the strict alternative every retained block lowers the bounded carrier by at least $\delta$; summation contradicts $K_\phi \geq 0$ and the orbitwise carrier ceiling, giving (ii). The complementary ordinary law is (454), giving (iii). The calculation uses the scale-critical normalized carrier itself and never divides physical price by $r$. □

**Theorem III.5.23** (Target-local record-scale reduction with totalized tails)**.** *Let $\mathfrak{b}$ be an algebraically reachable ordinary recurrent Navier–Stokes branch whose scales collapse. Apply the canonical record-scale selector, so $a \leq 0$. Use rational cutoffs $\chi_R$ in the moving chart, retain every annular convection, pressure, diffusion, translation, and dilation term, and totalize the exhaustion $R \to \infty$. For a periodic physical domain the collapsing chart is lifted to its expanding rescaled cover before this exhaustion; the fixed torus mean is not used with the nonperiodic vector field $y \cdot \nabla$. Denote the resulting ordinary translation-invariant mean, when it exists, by $\mathfrak{m}$, and set*

$$O(V) := \mathfrak{m}(|V - \mathfrak{m}V|^2), \qquad P(V) := \nu\mathfrak{m}(|\nabla V|^2).$$

*Exactly one of the following occurs:*

(M1) *a mean, exhaustion, annular flux, pressure tail, divergence, or chart relation takes a nonordinary value and produces its typed target-aligned successor;*

(M2) *the invariant law $\varpi$ satisfies*

$$\int \left(P + \frac{a}{2}O + B_\infty\right) \mathrm{d}\varpi = 0, \tag{455}$$

*where $B_\infty$ is the ordinary total-graph value of the glued annular and exhaustion current. On the tail-free cell $B_\infty = 0$. Its $a = 0$ subcell has $P = 0$ and is Galilean constant on each connected retained component. Its $a < 0$ subcell satisfies the scale-balance relation*

$$\int P \,\mathrm{d}\varpi = \int \frac{-a}{2} O \,\mathrm{d}\varpi. \tag{456}$$

*This relation defines a scale-relative equality cell; it does not force $P$ or $O$ to vanish. The equality cell, its nonzero boundary current, and every failed exhaustion value are passed to target-cyclic circulation and least-coordinate refinement.*

*This conclusion concerns only the generated target-active branch; it makes no claim about recurrent Navier–Stokes states outside its target cell. The Navier–Stokes dilation row has the coefficient and sign in* (455)*. Hence the abstract sign identity* (112) *applies only to branches that generate that different row, while the Navier–Stokes $a < 0$ cell remains in the scale-balance stratum.*

*Proof.* Test (447) against $\chi_R(V - \mathfrak{m}_R V)$. Incompressibility converts convection and pressure into annular currents; diffusion, translation, and the derivative of the local mean produce further explicit boundary coordinates. The same calculation applies on the expanding lift of a shrinking periodic chart. If a coordinate fails to converge in the exhaustion graph, its least value is (M1). On an ordinary exhaustion, signed gluing gives the single limit current $B_\infty$.

For $W = V - \mathfrak{m}V$, the ordinary boundary-free dilation identity in three dimensions is

$$\mathfrak{m}(W \cdot (W + y \cdot \nabla W)) = -\frac{1}{2}\mathfrak{m}(|W|^2) = -\frac{1}{2}O.$$

Stationarity annihilates the time coboundary. Pairing the remaining moving equation with $W$ therefore gives (455). When $B_\infty = 0$, the cell $a = 0$ has $P = 0$; the connected gradient kernel and Galilean quotient make it target-null. On $a < 0$, the same identity is (456), so the compiler retains it as a scale-relative equality row rather than applying the gradient kernel. □

**Corollary III.5.24** (Navier–Stokes ordinary-cell resaturation)**.** *For the Navier–Stokes public presentation, the target-cyclic compiler uses viscosity as the native mobility, incompressibility and pressure as the constraint functor, Galilean and parabolic transformations as quotient and record-scale charts, and the local suitable-energy form as carrier. These substitutions generate the record-scale mean row of Theorem III.5.23; it is not additional specialization data. Consequently an ordinary zero-price stationary law is not terminal. Exactly one of the following occurs:*

(N1) *on the tail-free $a = 0$ cell,* (455) *gives $P = 0$; the connected gradient kernel and Galilean quotient make the cell target-null;*

(N2) *on $a < 0$, the scale-balance row* (456)*, or a nonzero ordinary value of $B_\infty$, is adjoined to the circulation ideal. Its least nonzero target-cyclic pairing is divided into signed dyadic cells and becomes the next scale, nonlinear, pressure, exterior, constraint, or boundary successor;*

(N3) *a mean, localization, integration-by-parts, pressure, chart, or realization operation is nonordinary and its first total-graph value is the next target-active successor.*

*In* (N2)–(N3) *the selected current is passed to Definition III.4.13. Hence the ordinary stationary law cannot remain as an unexamined Navier–Stokes equality model.*

*Proof.* Apply Theorem III.4.11. The first localization of the viscous production by the covariant spatial-mean covector is precisely the calculation in Theorem III.5.23. On the ordinary tail-free $a = 0$ cell, its joint kernel is Galilean constant and gives (N1). On $a < 0$, the exact dilation calculation gives (456), not vanishing oscillation. The generated mobility-annihilator and circulation tests select its least target-active pairing, giving (N2). A failed mean, pressure, constraint, chart, or realization operation is case (K3) and gives (N3). Thus the list is a specialization of the general descent theorem rather than an independent Navier–Stokes closure argument. □

**Theorem III.5.25** (Navier–Stokes recurrent-component reduction)**.** *On every algebraically reachable Navier–Stokes recurrent component, the target-aligned recurrent closure compiler on a full active signature $\mathfrak{b}$ has exactly one of the following outputs: a target-null fixed-scale state; a target-null tail-free $a = 0$ state; a target-active scale-balance circulation cell; a nonlinear, pressure, exterior, scale, constraint, chart, fidelity, storage, or realization successor; or a cofinal recurrent-current system whose diagonal realization is an explicit Navier–Stokes singular mechanism. In particular scale collapse never removes target-aligned structure from the compiler.*

*Proof.* Initialize Definition III.4.23 with the Navier–Stokes shell covector and signed local-energy carrier. Its aggregate row is generated on $\mathscr{T}_{\Sigma_{\mathrm{sing}},\mathfrak{b}}$ by Theorem I.4.7; it is the target projection of convection, pressure, exterior, chart, and scale currents after signed gluing. Let a target-active recurrent component be algebraically reachable. If its record scale is bounded away from zero, Lemma III.5.16 makes its zero-expenditure ordinary component spatially constant and target-null. Hence its scale must collapse. Apply the canonical nonincreasing record chart and Theorem III.5.23. Alternative (M1) is the corresponding typed successor. Alternative (M2) gives

$$\int \left( \nu \mathfrak{m} |\nabla V|^2 + \frac{a}{2}\, \mathfrak{m} |V - \mathfrak{m} V|^2 + B_\infty \right) \, \mathrm{d}\varpi = 0, \qquad a \le 0.$$

On the tail-free $a = 0$ cell, the viscous production vanishes and the component is target-null. On $a < 0$, viscous production is balanced by the scale current and defines the equality cell (456). A nonzero $B_\infty$ is its boundary successor. Target-current circulation and prospective accountability select the least target-aligned value on either cell; this is alternative (N2) or (N3) of Corollary III.5.24. Apply the signed-dyadic return construction of the general recurrent compiler. A new coordinate is a proper successor. Equivalent recurrent-current successors have a compact diagonal limit; by Theorem III.4.34 it is target-null or realizes the original public equation and cofinal shell contact. The latter is the explicit singular mechanism in the statement. These alternatives exhaust the recurrent component. □

**Corollary III.5.26** (Navier–Stokes target-spine circulation evaluator)**.** *For every algebraically reachable Navier–Stokes signature $\mathfrak{b}$, let $a_{\mathrm{NS},\Sigma}$ be the normalized shell return current on $\mathscr{T}_{\mathrm{NS},\Sigma_{\mathrm{sing}},\mathfrak{b}}$,* *represented by the incidence core $\mathsf{M}_{\mathrm{NS},\Sigma,\mathfrak{b}}$, and let $p_{\mathrm{NS,phys}}$ be the restriction of the suitable local-energy carrier admitted by the charging graph, with generated continuous lower envelopes $p_{\mathrm{NS},N} \uparrow p_{\mathrm{NS,phys}}$. Denote by $\mathcal{K}^0_{\mathrm{NS},\mathfrak{b}}$ the specialized local equation and positivity cone before edge coboundaries are adjoined. The nested Navier–Stokes cylinder hierarchy computes*

$$\delta^{\mathrm{cyc}}_{\mathrm{NS},\mathfrak{b}} = \inf_{\gamma} \int p_{\mathrm{NS,phys}} \, \mathrm{d}\gamma = \sup_{N,V,\delta} \left\{ \delta : \begin{array}{r} p_{\mathrm{NS},N} + V \circ \pi_- - V \circ \pi_+ \\ - \delta a_{\mathrm{NS},\Sigma} \in \mathcal{K}^0_{\mathrm{NS},\mathfrak{b}} \end{array} \right\}. \tag{457}$$

*Here the primal circulation includes the moving equation, incompressibility, pressure graph, signed suitable-energy carrier, scale covariance, target- cyclic productions, same-origin relations, and the six typed graph-boundary cells after restriction to $X^{\min}_{\mathrm{NS},\mathfrak{b}}$. The following computation is exhaustive:*

(N1) *positive $\delta^{\mathrm{cyc}}_{\mathrm{NS},\mathfrak{b}}$ produces a finite local carrier-potential identity and excludes recurrent singular progress;*

(N2) *infeasibility produces a finite local $-1$-identity;*

(N3) *zero price reconstructs the invariant circulation, restricts to its positive target support, and automatically initializes the next nonlinear, pressure, exterior, scale, cyclic, storage, or realization refinement.*

*Iterating these refinements computes the decreasing Navier–Stokes circulation core. At every depth the target-aligned branch-and-bound returns a finite rational cylinder cover, certified price bounds, and checked identities for all deleted cells; its interval outputs converge to the exact range of every generated Navier–Stokes target observable. By Theorem III.4.32, that core is empty if and only if the Navier–Stokes target spine contains no realized singular path. Thus the specialization contains an explicit evaluator for the remaining spine condition; it does not leave a separate absorption or Liouville theorem to be supplied.*

*Proof.* The specialization commutation theorem identifies the abstract edge space, target return, physical valuation, ordered cone, and descent defects with their Navier–Stokes counterparts. Apply Theorem III.4.28 to obtain (457). The six first-failure cells of Definition III.5.31 are precisely the projected total- graph refinements in the zero-price case. Apply Definitions III.4.29 and III.4.30 and Theorems III.4.31 and III.4.32 to their successive supports. □

**Definition III.5.27** (Finite Navier–Stokes target program)**.** Fix a rational physical stratum $\mathscr{X}_{\mathrm{NS}}(E)$ and an algebraically reachable signature $\mathfrak{b}$. At depth $n$, first compute $\mathsf{M}^n_{\mathrm{NS},\Sigma,\mathfrak{b}}$ by the two graph searches of Definition III.5.8. Then let $\mathcal{A}_{\mathrm{NS},n}(\mathfrak{b})$ be the rational cylinder algebra generated by the retained members of the first $n$ coordinates in the following fixed order:

(T1) the shell current, five source pairings, source support, realization mode, same-origin rows, and the physical inequalities (424);

(T2) the weak momentum equation, incompressibility, and rational localizations through cylinder index $n$;

(T3) the local, harmonic, exterior, and gauge-fixed pressure graphs;

(T4) the scale, center, moving-microscope, signed carrier, overlap, and physical-price graphs;

(T5) the adjoint words in $\mathscr{C}^{n,\min}_{\mathrm{NS},\mathfrak{b}}$, the polarized nonlinear defects, the compressed Hamiltonian/resolvent matrix elements (435)– (436), and the generated production words in $\mathscr{P}_{\mathrm{NS},\mathfrak{b}}$ through index $n$; and

(T6) the first $n$ signed-dyadic return, total-graph boundary, circulation, and prior-zero relations.

Partition these coordinates into rational boxes of canonical diameter at most $2^{-n}$. For a box $C$, the finite target program has variables the moments of a positive edge functional $L_C$ on monomials of degree at most $2n$. It imposes

$$0 \le L_C(1) \le 5, \qquad\qquad L_C(a_{\mathrm{NS},\Sigma}) = 1, \tag{458}$$

$$L_C(f\circ\pi_-) = L_C(f\circ\pi_+) \qquad\qquad (f\in\mathcal{A}_{\mathrm{NS},n-1}), \tag{459}$$
$$L_C(I_{\mathrm{NS},n}) = 0, \qquad\qquad L_C(M_{\mathrm{NS},n}) \geq 0, \tag{460}$$

together with the localizing matrices for $C$. Here $I_{\mathrm{NS},n}$ contains precisely the incidence-retained truncated rows in (T1)–(T6), while $M_{\mathrm{NS},n}$ contains their generated squares, nonnegative viscous and suitable-energy productions, positive Reynolds polarizations, and admitted exterior productions. The objective is

$$\underline{\delta}_{\mathrm{NS},n}(C) := \inf L_C(p_{\mathrm{NS},n}). \tag{461}$$

Its dual variables are coefficients of the same finite equation and positivity rows together with a cylinder potential $V_n\circ\pi_- - V_n\circ\pi_+$. All variables and constraints are generated from (B1)–(B5). The integer $n$ is a refinement index and has no place in the physical energy identity (423).

**Theorem III.5.28** (Complete finite-stage Navier–Stokes evaluation)**.** *Fix rational $E>0$. Apply Definition III.5.27 simultaneously to every algebraically reachable signature in $\mathscr{X}_{\mathrm{NS}}(E)$ whose defining prefix occurs among the first n generated coordinates, and retain only the target-visible boxes. There are finitely many such prefixes at each depth. At depth n, delete a box carrying a finite infeasibility identity or a strictly positive cycle-price identity, and refine every other box at its least unresolved coordinate in the order* (T1)–(T6)*. Let $\mathscr{U}_{\mathrm{NS},n}$ be the resulting compact finite union; its dependence on E is suppressed. Then:*

(E1) *the calculation at depth n is finite and exact; it returns $\mathscr{U}_{\mathrm{NS},n}$, the feasible moment sets, rational lower cycle prices, and a checked ordered identity for every deleted box;*

(E2) *the covers decrease and satisfy*

$$\bigcap_{n\geq 0}\mathscr{U}_{\mathrm{NS},n} = \mathscr{U}_{\mathrm{NS},\infty}, \qquad \mathscr{U}_{\mathrm{NS},\infty} = \bigsqcup_{\mathfrak{b}\in\mathfrak{B}^{\mathrm{reach}}_{\mathrm{NS}}(E)} \mathsf{S}_{\mathrm{NS}}(X_{\infty,\mathfrak{b}}\cap|\mathsf{M}_{\Sigma,\mathfrak{b}}|)\,; \tag{462}$$

*the diagonal realization image of $\mathscr{U}_{\mathrm{NS},\infty}$ is exactly the set of same-origin Navier–Stokes singular mechanisms in the reachable target spines;*

(E3) *if $\mathscr{U}_{\mathrm{NS},n}=\varnothing$ at a finite depth, the attached identities concatenate to a finite derivation of $\mathrm{Reach}^{\mathrm{phys}}_{\Sigma_{\mathrm{sing}}}(E)=0$ on the declared physical stratum; if all depths are nonempty, every infinite compatible box word supplies the complete moments, graphs, origin rows, and shell rows of a realized singular mechanism;*

(E4) *for every generated bounded continuous Navier–Stokes target observable F, the finite programs compute nested rational intervals*

$$I_{\mathrm{NS},n}(F)\downarrow\left[\min_{\mathscr{U}_{\mathrm{NS},\infty}} F,\ \max_{\mathscr{U}_{\mathrm{NS},\infty}} F\right]. \tag{463}$$

*In particular, pressure-tail mass, scale cohomology, Reynolds-defect mass, viscous production, shell alignment, and every generated commutator production are evaluated by two-sided structural enclosures on the surviving target core.*

*No analytic property is tested outside the target spine. A failed compactness, pressure, tail, chart, product, or realization relation changes the next box coordinate and is processed by the same finite program. Running the construction over the countable cofinal family $E\in\mathbb{Q}_{>0}$ covers the entire smooth finite-energy datum class.*

*Proof.* The rows in (T1)–(T6) are the Navier–Stokes images of the abstract target-spine enumeration. Their finite moment and localizing conditions, including the rational energy ceiling $E$, are rational semialgebraic conditions. Origin evaluations and weak integrals are moment variables constrained by graph rows rather than scalar coefficients. Exact elimination decides finite feasibility and checks the dual identities. The circulation normalization and coboundary rows are (410); the objective and its dual are (457). This proves (E1).

The specialization homomorphism preserves the target quotient, physical valuation, ordered cone, source ancestry, edge marginals, and the least unresolved total graph. Hence it maps the abstract branch-and-bound cover at every depth onto $\mathscr{U}_{\mathrm{NS},n}$. Apply Theorem III.4.31 and commute intersections of the decreasing compact covers with this graph-complete specialization to obtain (462). Diagonal target-aligned saturation then proves the realization assertion.

Compactness makes an empty limiting core empty at a finite depth; summing the finitely many attached identities proves (E3). In the nonempty case, the least-coordinate refinement resolves every public operation and retains every same-origin shell row, so the compatible inverse limit is the stated mechanism. Finally, uniform rational cylinder approximation and monotonicity of extrema on decreasing compact sets give (463). Every observable listed in (E4) is a generated cylinder coordinate or a monotone limit of such coordinates. □

**Corollary III.5.29** (Navier–Stokes output of the continuum compiler)**.** *For every rational $E > 0$, the public unforced Navier–Stokes presentation on $\mathscr{X}_{\mathrm{NS}}(E)$ satisfies the exact proof-or-mechanism alternative*

$$\mathrm{Reach}^{\mathrm{phys}}_{\Sigma_{\mathrm{sing}}}(E) = 0 \quad \textit{or} \quad \mathfrak{S}_{\mathrm{NS},E} \textit{ is a same-origin recurrent singular mechanism.} \tag{464}$$

*The singular-mechanism alternative contains an actual public weak solution path and cofinal continuation-shell contact. Taking the countable union over rational $E$ covers every ordinary maximal strong branch with smooth finite-energy datum.*

*Proof.* Contact in $\mathscr{X}_{\mathrm{NS}}(E)$ would generate a full active signature, a disjoint shell cascade, and, after the finite physical-price alternatives, an infinite same-origin zero-price resaturation route. Run Corollary III.5.26 and Theorem III.5.28. Positive cycle price and infeasibility close the active cell by finite identities. Zero cycle price constructs the next target-aligned circulation support and repeats the evaluation at its least unresolved target coordinate. A finite empty cover excludes contact, while compatible nonempty covers diagonally reconstruct the same-origin singular mechanism with all public graph rows. These are precisely the alternatives in (464). □

**Definition III.5.30** (Target-retaining Navier–Stokes carrier complex)**.** Generate the continuation target from failure of the declared strong class. Restrict the canonical parabolic chart tree to descendants with positive pairing against the singular-shell family $\Gamma_{\Sigma_{\mathrm{sing}}}$. Equivalently, take its current projection from $\mathsf{M}_{\mathrm{NS},\Sigma,\mathfrak{b}}$; typed frontier and graph-boundary edges are retained until their pairings are resolved. On each cell use the local kinetic-energy identity and the pressure split

$$-\Delta p = \partial_i \partial_j (u_i u_j)$$

with local singular-integral, harmonic-gauge, and exterior-tail graph coordinates. Sum cell identities with their orientations before taking positive parts. The remaining scale current defines

$$[F^{\mathrm{NS}}_{\mathrm{sc}}] \in \frac{\overline{\Pi_\Sigma F^{\mathrm{NS}}_{\mathrm{sc}}}}{\overline{\Pi_\Sigma \{\Psi - \Psi \circ S\}}}.$$

**Definition III.5.31** (Moving-microscope successor cells)**.** Fix an algebraically reachable active branch $\mathfrak{b}$. Let $Z^{\mathrm{coll}}_{\mathrm{NS},\mathfrak{b}}$ be the part of $Z^{\Sigma_{\mathrm{sing}}}_{\infty,A} \cap X^{\min}_{\mathrm{NS},\mathfrak{b}}$ satisfying

$$r = 0, \qquad p = 0, \qquad \mathrm{Prog}_{\Sigma_{\mathrm{sing}}} \geq \frac{1}{5}, \qquad I_{\mathrm{movNS}} = 0, \tag{465}$$

together with the same-origin parabolic successor relations and the signed carrier relation (453). Here $r = 0$ denotes the boundary of the physical chart-scale coordinate; the normalized field and its moving equation remain present through $I_{\mathrm{movNS}} = 0$, the ideal generated by (447). Order the total-graph coordinates as follows:

(N1) the normalized weak equation and incompressibility;
(N2) realization of $V \otimes V$ in the nonlinear current;
(N3) the local pressure constraint modulo time-dependent constants;
(N4) the harmonic/exterior pressure and carrier-tail coordinates;
(N5) parabolic chart covariance, the moving-microscope equation, and the scale-carrier cohomology relation.
(N6) the adjoint operator (428), its rational difference-quotient graph, the defect polarizations (438)–(439), and the generated production and joint-kernel relations (437).

The first nonordinary coordinate defines, respectively, the equation, nonlinear-concentration, pressure, exterior-boundary, or scale-covariance cell, followed by the target-cyclic cell. If all six coordinates are ordinary, the state lies in the ordinary collapsed cell. Denote the seven resulting cells by $Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{eq}}}$, $Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{nl}}}$, $Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{pr}}}$, $Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{ext}}}$, $Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{sc}}}$, $Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{cyc}}}$, and $Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{ord}}}$.

**Proposition III.5.32** (Exhaustiveness of the microscope cells)**.** *Every target-visible zero-price Navier–Stokes successor belongs to exactly one of the seven cells in Definition III.5.31, after its fixed-scale component has been removed by Lemma III.5.16. Each cell retains the complete parent signature $\mathfrak{b}$ and ancestry in its support $A \subseteq \mathfrak{C}$: nonlinear and pressure transport have* Caus *ancestry, concentration and parabolic covariance append* Lift*, pressure gauge appends* Abs*, a nonzero exterior trace appends* Bdry*, and the cyclic cell retains the union of the source words occurring in its adjoint, commutator, defect, and production coordinate. None is a residual source. In the ordinary cell, every infinite target-retaining path satisfies the scale-saturation law* (454)*; a strict carrier deficit has already been excluded by Theorem III.5.22.*

*Proof.* The scale coordinate has the exhaustive closed split $r > 0$ or $r = 0$. The first case is closed by Lemma III.5.16. On $r = 0$, totalize the six displayed interfaces and select the first nonordinary value; if none occurs, all values are ordinary. This gives a disjoint exhaustive list. The ancestry assertions follow from Proposition III.5.5 and Theorem I.3.4. The target-null quotient is taken only after these tags are fixed, so a target-visible member cannot be assigned to the residual. If all interfaces are ordinary, Theorem III.5.22 applies and leaves only its saturation alternative. □

**Theorem III.5.33** (Navier–Stokes target-cyclic stress test)**.** *Let $\mathfrak{b} = (A, m, \Gamma, 1/5, \mathbf{R})$ be an algebraically reachable target-active signature and let $\Phi \in \Gamma$. Apply the general joint-kernel descent on $X^{\min}_{\mathrm{NS},\mathfrak{b}}$ to $\mathscr{C}^{\infty,\min}_{\mathrm{NS},\mathfrak{b}}(\Phi)$, $\mathscr{P}_{\mathrm{NS},\mathfrak{b}}(\Phi)$, and $X^{\infty}_{\mathrm{NS},\mathfrak{b},\Phi}$. The specialization has the following exhaustive outputs.*

(S1) *After target-local fact factorization and admission by* (90)*, a signed local-energy, commutator, or defect identity charges a disjoint cofinal shell family by divergent $\mu_{\mathrm{NS}}$-mass, contradicting* (423)*.*
(S2) *Every restriction of $\mathscr{C}^{\infty,\min}_{\mathrm{NS},\mathfrak{b}}(\Phi)$ annihilates the viscous, nonlinear, pressure, quotient, scale, and boundary currents on $X^{\infty}_{\mathrm{NS},\mathfrak{b},\Phi}$; the shell is constant and contact is impossible.*

(S3) *The least nonordinary or nonzero descendant lies in exactly one of*

$$Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{eq}}},\quad Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{nl}}},\quad Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{pr}}},\quad Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{ext}}},\quad Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{sc}}},\quad Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{cyc}}}.$$

*Its successor datum is respectively the equation residual, Reynolds defect, pressure cokernel, exterior tail, nonexact scale current, or failed adjoint, commutator, polarization, or production coordinate, always with the detecting shell covector and original orbit tag.*

(S4) *Every finite coordinate is ordinary. The inverse limit is a closed moving-frame Navier–Stokes subsystem on $X^{\infty}_{\mathrm{NS},\mathfrak{b},\Phi}$ with zero generated production and an actual same-origin path crossing every continuation shell.*

*All compactness, pressure, tail, scale, and multiplier coordinates used here are generated inside the minimal incidence realization. The final alternative is a realized singular Navier–Stokes subsystem.*

*Proof.* By Proposition III.5.7, the module (429) is the exact Navier–Stokes image of the general cyclic module. By Proposition III.5.9, its restriction to $X^{\min}_{\mathrm{NS},\mathfrak{b}}$ is the exact image of the abstract minimal realization. The positive cone in Definition III.5.12 is the corresponding image of the general production family. The scale valuation is (427) together with the covariant scale current (450); the nonlinear polarization and pressure descent are Proposition III.5.13; and functorial constraint and boundary descent gives the local, harmonic, exterior, and Leray coordinates.

First apply Theorem II.2.18 and Corollary II.2.20 on $X^{\min}_{\mathrm{NS},\mathfrak{b}}$; this forbids charging a source occurrence from a different shell, origin, mode, or scale record. Then apply Theorem III.4.11. Its admitted physical-charge and target-null cases give (S1) and (S2). Ordering the six Navier–Stokes total interfaces as in Definition III.5.31 identifies its proper-successor case with (S3). If every finite interface is ordinary, the inverse-limit case of the general theorem preserves the moving equation, incompressibility, pressure equation, signed suitable-energy carrier, chart covariance, same-origin coordinate, and cofinal shell crossings. These are precisely (S4). The total graph accounts for each analytic failure named in the statement, so none is a premise. □

**Theorem III.5.34** (Navier–Stokes target-aligned reduction)**.** *Every finite-energy singular contact for* (422) *produces a full active signature $\mathfrak{b} = (A, m, \Gamma, 1/5, \mathbf{R})$. On its finite-expenditure successor branch, evaluation of $\mathcal{K}^{\min}_{\mathrm{NS},\mathfrak{b}}$ gives either a finite $-1$-identity, which contradicts that contact branch, or a positive functional on the same-origin ordered path algebra satisfying:*

(i) *support in $X^{\min}_{\mathrm{NS},\mathfrak{b}}$, with every retained coordinate lying on a forward five-source and backward singular-shell incidence path;*

(ii) *the normalized Navier–Stokes weak equation and incompressibility;*

(iii) *the exact moving-microscope equation and the signed local carrier identity;*

(iv) *the pressure local/harmonic/tail graph relations;*

(v) *the target-retaining scale-carrier cohomology relation and, on the ordinary recurrent cell, the saturation law $\int(F_{\mathrm{sc}} - D - M)\,\mathrm{d}\varpi = 0$, together with the totalized-tail identity* (455) *and the $a < 0$ balance* (456)*;*

(vi) *the target-cyclic adjoint relations* (428)*, Reynolds-defect polarization, and every generated production and joint-kernel coordinate; and*

(vii) *same-origin parabolic refinement and positive singular-shell pairing.*

*The fixed-scale part of this functional is target-null. Its nonzero target projection is supported on the generated disjoint union*

$$\mathscr{X}^{*}_{\mathrm{NS}} := \bigsqcup_{\mathfrak{b}\in\mathfrak{B}^{\mathrm{reach}}_{\mathrm{NS}}}\ \bigcup_{\xi\in\{\mathrm{eq},\mathrm{nl},\mathrm{pr},\mathrm{ext},\mathrm{sc},\mathrm{cyc},\mathrm{ord}\}} Z_{\mathrm{NS},\mathfrak{b}^{\xi}}, \tag{466}$$

*where* $\mathfrak{B}_{\mathrm{NS}}^{\mathrm{reach}} := \bigcup_{E\in\mathbb{Q}_{>0}} \mathfrak{B}_{\mathrm{NS}}^{\mathrm{reach}}(E)$. *Distinct origin, mode, shell, scale, or earlier polarity records remain distinct components. Consequently, finite ordered derivations*

$$-1 \in \mathcal{K}^{\min}_{\mathrm{NS},\mathfrak{b}^{\xi}}$$

*exclude that active successor cell. If every algebraically reachable signature is excluded, shell accountability excludes singular contact on the declared finite-energy datum class. On a complementary cone value, the hierarchy reconstructs the complete target-aligned scale-saturation model in that same cell and immediately applies the zero-price resaturation successor. The model is not terminal: it either reaches a finite contradiction, moves to a proper typed successor, or reconstructs an explicit Navier–Stokes singular mechanism.*

*Proof.* Contact gives the disjoint derived shell-current cascade and persistent signature $\mathfrak{b}$. Equation (423) makes the raw event expenditures summable. After target-local five-source factorization, the divergent admissible-price branch is impossible, while the finite branch has a zero-expenditure subsequence; a failed charging relation remains a typed successor. Passing to the covariant microscope preserves the shell current by Proposition III.5.21. Signed carrier gluing cancels paired interior convection, pressure, cutoff, translation, and exact chart currents; pressure gauge and exterior tails remain in their typed coordinates. By Theorem III.5.22, an infinite ordinary chain cannot have strict carrier deficit and therefore satisfies the saturation law. Successor recurrence and the branch-local alternative Theorem II.2.36 either produce the excluding identity or the functional with (i)–(vii). The fixed-scale component is target-null by Lemma III.5.16; Proposition III.5.32 gives (466). Weak duality proves exclusion separately on each $\mathfrak{b}^{\xi}$, and Theorems III.4.36 and III.5.33 and Corollary III.5.24 continues every complementary model. □

**Theorem III.5.35** (Navier–Stokes specialization commutes with the structural compiler)**.** *Let* $\mathsf{S}_{\mathrm{NS}}$ *denote substitution of the public relations* (B1)–(B5) *into the continuum presentation, and fix an algebraically reachable active signature* $\mathfrak{b}$. *Let* $\mathsf{P}_{\Sigma,\mathfrak{b}}$, $\mathsf{F}_{\mathfrak{b}}$, $\mathsf{R}_{\mathfrak{b}}$, *and* $\mathsf{O}_{\mathfrak{b}}$ *denote respectively target-local fact projection-first saturation, fact factorization, first-nonordinary successor completion, and ordered-cone completion. Let* $\mathsf{C}_{\Sigma,\mathfrak{b}}$ *be circulation formation and let* $\mathsf{B}^{n}_{\Sigma,\mathfrak{b}}$ *be depth-* $n$ *target-aligned branch-and-bound. Let* $\mathsf{I}_{\Sigma,\mathfrak{b}}$ *be bidirectional incidence minimization. Then*

$$\mathsf{S}_{\mathrm{NS}}\mathsf{P}_{\Sigma,\mathfrak{b}} = \mathsf{P}_{\mathrm{NS},\Sigma,\mathfrak{b}}\mathsf{S}_{\mathrm{NS}}, \tag{467}$$
$$\mathsf{S}_{\mathrm{NS}}\mathsf{F}_{\mathfrak{b}} = \mathsf{F}_{\mathrm{NS},\mathfrak{b}}\mathsf{S}_{\mathrm{NS}}, \tag{468}$$
$$\mathsf{S}_{\mathrm{NS}}\mathsf{R}_{\mathfrak{b}} = \mathsf{R}_{\mathrm{NS},\mathfrak{b}}\mathsf{S}_{\mathrm{NS}}, \tag{469}$$
$$\mathsf{S}_{\mathrm{NS}}\mathsf{O}_{\mathfrak{b}} = \mathsf{O}_{\mathrm{NS},\mathfrak{b}}\mathsf{S}_{\mathrm{NS}}, \tag{470}$$
$$\mathsf{S}_{\mathrm{NS}}\mathsf{C}_{\Sigma,\mathfrak{b}} = \mathsf{C}_{\mathrm{NS},\Sigma,\mathfrak{b}}\mathsf{S}_{\mathrm{NS}}, \tag{471}$$
$$\mathsf{S}_{\mathrm{NS}}\mathsf{I}_{\Sigma,\mathfrak{b}} = \mathsf{I}_{\mathrm{NS},\Sigma,\mathfrak{b}}\mathsf{S}_{\mathrm{NS}}, \tag{472}$$
$$\mathsf{S}_{\mathrm{NS}}\mathsf{B}^{n}_{\Sigma,\mathfrak{b}} = \mathsf{B}^{n}_{\mathrm{NS},\Sigma,\mathfrak{b}}\mathsf{S}_{\mathrm{NS}}. \tag{473}$$

*In particular,*

$$\mathsf{S}_{\mathrm{NS}}(\mathcal{K}_{\mathfrak{b}^{\xi}}) = \mathcal{K}_{\mathrm{NS},\mathfrak{b}^{\xi}}, \qquad \mathsf{S}_{\mathrm{NS}}(\mathscr{T}_{\Sigma,\mathfrak{b}}) = \mathscr{T}_{\mathrm{NS},\Sigma,\mathfrak{b}}, \qquad \mathsf{S}_{\mathrm{NS}}(Z_{\mathfrak{b}^{\xi}}) = Z_{\mathrm{NS},\mathfrak{b}^{\xi}}$$

*for every generated successor type* $\xi \in \{\mathrm{eq},\mathrm{nl},\mathrm{pr},\mathrm{ext},\mathrm{sc},\mathrm{cyc},\mathrm{ord}\}$. *Thus the example uses the same selector, five-source normal form, target quotient, physical charging row, same-origin refinement, circulation duality, finite target-cylinder cover, and proof-or-model recursion as the abstract calculus. In particular,*

$$\mathsf{S}_{\mathrm{NS}}(\mathscr{U}_n) = \mathscr{U}_{\mathrm{NS},n}, \qquad \mathsf{S}_{\mathrm{NS}}(I_n(F)) = I_{\mathrm{NS},n}(\mathsf{S}_{\mathrm{NS}}F). \tag{474}$$

*The specialization acts only by homomorphic substitution. After applying* (472)*, these identities restrict to*

$$\mathsf{S}_{\mathrm{NS}}\left(\mathcal{K}_{\mathfrak{b}^{\xi}}\big|_{|\mathsf{M}_{\Sigma,\mathfrak{b}}|}\right)=\mathcal{K}^{\min}_{\mathrm{NS},\mathfrak{b}^{\xi}},\qquad Z_{\mathrm{NS},\mathfrak{b}^{\xi}}\subseteq X^{\min}_{\mathrm{NS},\mathfrak{b}}.$$

*Proof.* All seven operations are least closures or finite graph restrictions under fixed constructor lists. For (467), substitution preserves every prospective shell pairing and sends each abstract descent defect (29) to its equation, pressure, exterior, chart, scale, or cyclic Navier–Stokes defect. Projection-first saturation therefore gives the commutation identity and the displayed spine identity. For (468), substitution sends the abstract weak current to the viscous $\mathsf{Geom}$ and transport/pressure $\mathsf{Caus}$ currents, sends quotient, chart, and terminal constructors to the displayed $\mathsf{Abs}$, $\mathsf{Lift}$, and $\mathsf{Bdry}$ currents, and preserves restriction to $X^{\min}_{\mathrm{NS},\mathfrak{b}}$, signed gluing, and the detecting shell covector after incidence reduction. It also sends the abstract carrier-domination graph to the local suitable energy carrier row. Hence it preserves both sides of the target-aligned charging rule, rather than identifying raw event expenditure with price.

For (469), total-graph completion commutes with a homomorphism: an ordinary value remains ordinary, while the first nonordinary equation, product, pressure, exterior, scale, or cyclic value is sent to the correspondingly typed cell with the same origin, mode, shell, scale, and earlier polarity coordinates. The remaining ordinary value is the ord cell. This is precisely Proposition III.5.32.

Finally, ideal generation, quadratic-module generation, ordered closure, and restriction to a fixed signature commute with substitution by their universal properties. The generators on the Navier–Stokes side are exactly the weak equation, moving-chart and carrier identities, pressure graphs, adjoint module, polarized defects, and same-origin relations listed above. This proves (470).

Equation (472) is Proposition III.5.9. Circulation formation uses only the specialized edge graph, equal-marginal rows, target-return normalization, and physical valuation. Substitution preserves each, proving (471). At finite depth, substitution also preserves the coordinate enumeration, rational boxes, moment and localizing matrices, dual identities, and least-unresolved coordinate. Induction on $n$ proves (473) and (474). No extra generator can occur by Proposition III.5.1, and no generator can be omitted because each member of the public ledger has been totalized. $\square$

**Theorem III.5.36** (Self-contained Navier–Stokes specialization audit)**.** *Using only the general continuum calculus and the Navier–Stokes public presentation, fix a rational physical stratum* $\mathscr{X}_{\mathrm{NS}}(E)$*. The following statements are derived:*

(A1) *singular contact forces a pre-contact source with alignment at least* $1/5$*;*

(A2) *the raw physical expenditures of its disjoint events are summable; they form the additive entropy cocycle* (425)*–*(426)*; admissible target price is determined separately by the local charging row;*

(A3) *every fixed-scale zero-price descendant is target-null;*

(A4) *every shrinking descendant remains visible in the covariant microscope, obeys* (447) *and* (453)*, and belongs to the explicitly generated seven-successor refinement of its full active signature inside* $\mathscr{Z}^{*}_{\mathrm{NS}}$ *in* (466)*;*

(A5) *the shell covector generates the explicit adjoint module* (429)*, Reynolds and pressure defects generate* (438)*–*(439)*, and forward source ancestry intersects this backward module in* $\mathsf{M}_{\mathrm{NS},\Sigma,\mathfrak{b}}$*. The common zero-production dynamics is* (437) *restricted to that minimal realization. The compressed linearized response* (435) *has the source factorization* (436)*;*

(A6) *target-local joint-kernel descent gives an admissible divergent physical charge, target-nullity, one of the six typed proper successors, or a closed same-origin singular subsystem. A stationary functional is resaturated and is never returned as a terminal answer.*

(A7) *specialization commutes with projection-first saturation, target-local factorization, total-graph successor completion, ordered-cone completion, bidirectional incidence minimization, circulation formation, and finite target-aligned branch-and-bound completion; hence every branch decision is made on the Navier–Stokes target spine inside $\mathcal{K}^{\min}_{\mathrm{NS},\mathfrak{b}^{\xi}}$, with the full algebraically generated signature retained.*

(A8) *the aggregate feeding class is evaluated before its support modes; a nonzero class creates the next target-aligned successor, while its zero cell gives* (455). *Exact storage currents are retained separately, and every cofinal recurrent-current chain is processed by the single recurrent closure queue and realized as target-null or as an explicit singular mechanism. Its sharp cycle price and carrier potential are computed by* (457)*; zero price supplies the next circulation support.*

(A9) *the finite Navier–Stokes target programs compute decreasing rational covers $\mathscr{U}_{\mathrm{NS},n}(E)$ of the complete mechanism core on $\mathscr{X}_{\mathrm{NS}}(E)$. Every deleted box carries a checked local identity, every retained box is refined at its least target-visible coordinate, and* (463) *gives two-sided values for all generated structural observables.*

*The derivation establishes the exhaustive, mutually exclusive conclusion*

$$\boxed{\mathrm{Reach}^{\mathrm{phys}}_{\Sigma_{\mathrm{sing}}}(E)=0 \quad \textit{or} \quad \mathfrak{S}_{\mathrm{NS},E} \textit{ is an explicit singular mechanism,}} \tag{475}$$

*where $\mathfrak{S}_{\mathrm{NS},E}$ includes the actual same-origin solution path, all ordinary weak and suitable-energy relations, and contact with every cofinal continuation shell. The initial seed of its resaturation is not the degenerate product stratum $p=r(D+M)=0$; it is the smaller target projection of the covariant saturation kernel*

$$\Pi_{\Sigma_{\mathrm{sing}}}\begin{pmatrix}\{r=0,p=0\}\cap\ker I_{\mathrm{movNS}}\cap\{I_{\mathfrak{b}}=0\}\cap\ker I^{\mathrm{signed}}_{\mathrm{car}}\\ \cap\ker I^{\text{same-origin}}_{\mathrm{succ}}\cap\left\{\int(F_{\mathrm{sc}}-D-M)\,\mathrm{d}\varpi=0\right\}\\ \cap\left\{\int(P+\frac{a}{2}O+B_{\infty})\,\mathrm{d}\varpi=0\right\}\cap X^{\infty}_{\mathrm{NS},\mathfrak{b},\Phi}\cap\mathscr{Z}^{\mathrm{tot}}_{\mathrm{car}}\end{pmatrix}. \tag{476}$$

*Every recurrent nonlinear, pressure, exterior, covariance, or target-cyclic boundary is kept as a typed first-failure successor and resaturated. On the ordinary row, Corollary III.5.24 removes the tail-free $a=0$ cell or sends the $a<0$ scale-balance row to circulation and least-coordinate refinement. Thus scale collapse cannot be used as a zero-cost disappearance of normalized structure, and neither a positive functional nor the displayed kernel is a terminal output. By Theorem III.5.25, every ordinary fixed-scale or tail-free $a=0$ zero mode is target-null, while every scale-relative or remaining target-visible value is retained as a typed recurrent current. Thus* (475) *is the exact output of this unit test: the compiler neither loses shrinking structure nor declares a formal invariant law to be a singular solution. Since the rational strata are cofinal, global regularity on the smooth finite-energy datum class is equivalent to the first alternative of* (475) *for every $E\in\mathbb{Q}_{>0}$.*

*Proof.* Items (A1) and (A2) are Theorem III.5.14 and Proposition III.5.3; item (A3) is Lemma III.5.16; item (A4) follows from Propositions III.5.19 and III.5.32 and Theorem III.5.20; item (A5) is Propositions III.5.7, III.5.9, III.5.11 and III.5.13; item (A6) is Theorem III.5.33; and (A7) is Theorem III.5.35; item (A8) is Theorems III.4.34 and III.5.25 and Corollary III.5.26; and item (A9) is Theorem III.5.28. If the finite target program empties every algebraically reachable box, its attached identities, including the relevant $-1\in\mathcal{K}^{\min}_{\mathrm{NS},\mathfrak{b}^{\xi}}$ identities, exclude the target-aligned functional constructed by contact and the first alternative of (475) follows. For an infinite ordinary shrinking chain, Theorem III.5.22 eliminates strict deficit and forces (454); nonordinary chains enter their first typed boundary cell. Taking the union of those totalized cells gives (476). Apply Theorems III.4.36 and III.5.28 to every complementary functional. If the finite covers become empty, their checked identities exclude contact. Otherwise their compatible recurrent-current return maps and realization rows supply $\mathfrak{S}_{\mathrm{NS},E}$, including actual shell contact. The record-scale reduction removes only its tail-free $a = 0$ gradient-kernel cells; its $a < 0$ equality cells remain in the target-aligned circulation calculation. This proves the stated proof-or-mechanism alternative. No step divides by $r$. $\square$

| Specialization row | Internally derived operation | Exact status |
|---|---|---|
| Contact and source ancestry | shell-current identity and five-source exhaustion | closed by (440) |
| Active branch | support, realization mode, shell family, alignment, origin, scale, and earlier polarity records | retained in the full signature $\mathfrak{b} = (A, m, \Gamma, 1/5, \mathbf{R})$ |
| Physical budget | raw restriction of the public orbitwise kinetic-energy carrier and its entropy cocycle (425) | summable by (441); admitted as target price only by the local charging graph |
| Noncollapsed zero-price row | gradient kernel and Galilean/finite-energy quotient | target-null by Lemma III.5.16 |
| Collapsed physical price | covariant microscope and signed carrier gluing | retains normalized dissipation and exposes the scale current by Theorem III.5.22 |
| Nonlinear weak limit | total graph of $V \otimes V$ | ordinary product or the typed cell $Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{nl}}}$ |
| Pressure constraint | local, harmonic, and exterior total graphs | ordinary pressure or the typed cells $Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{pr}}}$, $Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{ext}}}$ |
| Scale covariance | moving parabolic chart and carrier cohomology graphs | exact coboundary cancellation or $Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{sc}}}$ |
| Target-cyclic degeneracy | adjoint linearization, Reynolds polarization, commutator productions, and joint-kernel descent | target-nullity, physical charge, $Z_{\mathrm{NS},\mathfrak{b}^{\mathrm{cyc}}}$, or a realized singular subsystem |
| Minimal target realization | forward five-source ancestry intersected with backward shell-covector reachability | finite graph pruning removes every coordinate outside a complete source–shell path before production and moment generation |
| Hamiltonian response | Stokes geometry plus compressed directed linearization between retained source and shell spaces | every ordinary resolvent derivative factors through a retained source word; every nonordinary response value is a typed cyclic successor |
| Collapsed equality dynamics | covariant passivity plus ordered weak/carrier/successor algebra | strict deficit excluded; otherwise finite $-1$-identity or recursive zero-price resaturation |
| Ordered cell decision | cone generated after restriction to one full successor signature | $-1 \in \mathcal{K}_{\mathrm{NS},\mathfrak{b}^{\xi}}$ or a same-cell positive functional, followed immediately by resaturation |
| Recurrent passivity | target projection of aggregate signed feeding and exact storage, followed by circulation primal–dual and the record-scale mean identity | positive cycle price gives a carrier-potential exclusion; zero price gives the next return support; empty core is regularity and nonempty core reconstructs the singular mechanism |
| Finite target computation | rational target-cylinder boxes, truncated equation and positivity moments, circulation marginals, and least-coordinate refinement | decreasing covers $\mathscr{U}_{\mathrm{NS},n}$, checked deletion identities, two-sided structural invariant ranges, and diagonal realization of every compatible infinite word |

The refinement is constructive and internal. The weak equation first generates the moving-frame identity (447); multiplication by $V\phi$ generates (448); oriented overlap summation generates (453); and recurrent averaging generates (454). These operations replace the invalid division by $r$ with a

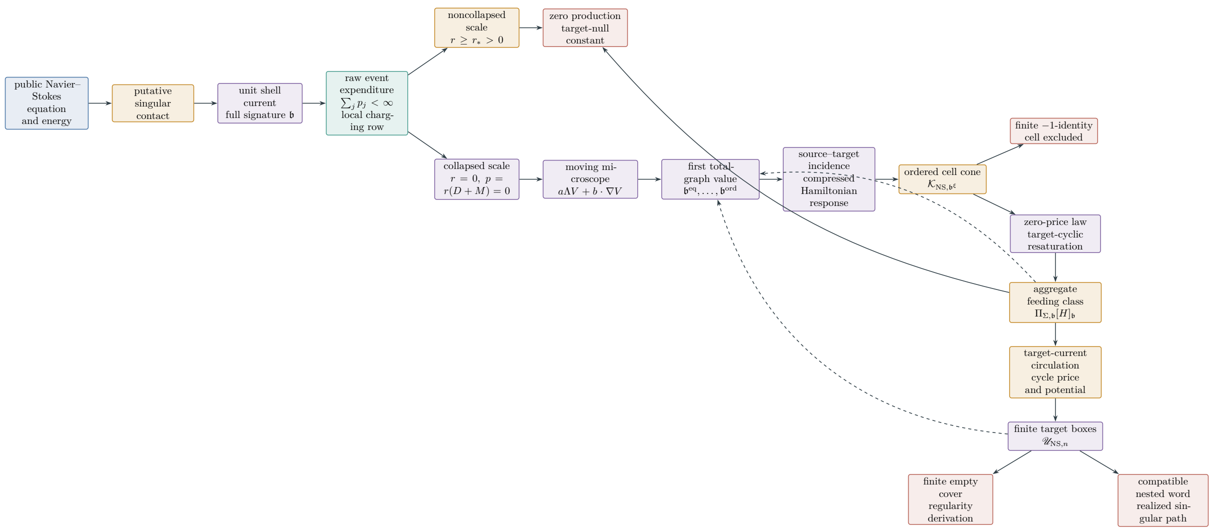


Figure 24: Covariant Navier–Stokes specialization audit. Raw physical expenditure obeys $p = r(D+M)$; it is admitted as target price only by the local target-aligned carrier graph. The collapsing branch is followed by the parabolic microscope. The induced scale and translation currents enter the exact signed carrier identity. Compatible cells are glued before positive parts are taken. The shell covector then generates the adjoint-linearized cyclic module, Reynolds and pressure defects, commutator productions, and their joint kernel. Strict carrier deficit excludes an infinite chain, while equality restricts the zero-production kernel and reruns prospective selection. Forward source ancestry and backward target-cyclic observability are intersected before the ordered program; only complete source–shell paths survive. On this intersection, the Stokes Hamiltonian and directed linearization generate the compressed resolvent (435); its variations factor through the same five sources. The aggregate feeding class is evaluated before its support is disintegrated: a nonzero class returns its support as the next successor, whereas the zero class restricts the record-scale scale-balance kernel and reruns the target-cyclic calculation. The circulation dual computes the sharp recurrent price. The finite target program then returns decreasing rational covers: an empty cover carries a finite regularity derivation, whereas a compatible nested word reconstructs a same-origin singular path. Every total-graph failure retains the complete active signature in its source-typed successor cell.

scale-critical conservation law. The ordered hierarchy is then applied to the seven successor cells generated by the six totalized interfaces, using only consequences of the displayed weak equation. A target-relative estimate, compactness assertion, or rigidity conclusion is not backend input: it must be generated by multiplier, localization, signed gluing, quotient, or completion-of-squares operations.

The cellwise refinement order is fixed. On the equation cell, localize the least failed weak coordinate and reapply the approximation first-failure map. On the nonlinear cell, polarize the failed product coordinate and pass each nonnegative concentration descendant through the target quotient. On the pressure cell, apply the Poisson constraint, quotient time-dependent constants, and separate the local and harmonic coordinates before taking a positive part. On the exterior cell, glue all common faces first; the remaining positive exterior variation is either charged to $\mu_{\mathrm{NS}}$ through the target-local charging rule or propagated as a $\mathsf{Bdry}$ successor. On the scale cell, quotient exact scale coboundaries and retain the nonexact harmonic scale coordinate. On the target-cyclic cell, apply (428), polarize the first failed nonlinear coordinate, descend its pressure defect, and append every generated Dirichlet or commutator production. On the ordinary collapsed cell, continue the same cyclic enumeration from the restricted shell covector and add every derived nonnegative square to the quadratic module. Each operation either lowers the first-failure index, removes the target projection, produces divergent physical expenditure, or strengthens the defining relations of the zero-price resaturation state. The strengthened state is immediately restricted and reselected; it is never returned as a positive-model conclusion. These are generated successor edges, not additional specialization data.

The finite computation follows the same order. At depth $n$, the program forms the target quotient first, inserts the first $n$ Navier–Stokes graph and production rows, imposes equal circulation marginals, and minimizes the generated physical-price envelope on each rational target box. A positive dual value telescopes its carrier potential over recurrent returns. A primal zero value restricts the next program to the positive support of the zero-price circulation. The least unresolved equation, nonlinear, pressure, exterior, scale, cyclic, or ordinary coordinate determines the next finite partition. Consequently every refinement has a checked local cause, and the computed covers converge to the target mechanism core in the canonical evaluation metric.

### III.5.2 Regulator-to-continuum spectral problems

The quantum mass-gap problem is a spectral-limit problem rather than a classical continuation target. Its structural presentation therefore starts with finite regulators and constructs the continuum Hilbert space.

**Definition III.5.37** (Public regulator tower)**.** A public regulator tower consists of compact finite configuration spaces $X_\lambda$, probability measures $\mu_\lambda$, reflection maps $\theta_\lambda$, gauge-invariant rational cylinder algebras $\mathcal{O}_\lambda$, positive-time subalgebras $\mathcal{O}_{\lambda,+}$, and refinement, block, and volume maps. All are generated from a finite local action and compact gauge measure. The ordered relations include normalization, gauge invariance, projective consistency, locality, and reflection positivity

$$\int \overline{\theta_\lambda F}\, F \;\mathrm{d}\mu_\lambda \geq 0, \qquad F \in \mathcal{O}_{\lambda,+}. \tag{477}$$

Ultraviolet, infrared, volume, topological, and failed-consistency values are typed graph coordinates.

**Theorem III.5.38** (Continuum reconstruction is an output)**.** *Let $L$ be a projectively consistent positive functional on the graph-complete regulator algebra satisfying locality, nontriviality, and* *reflection positivity. Quotienting the positive-time cylinder algebra by the null space of*

$$\langle F, G\rangle_{\mathrm{OS}} := L(\overline{\theta F} G)$$

*and completing gives a Hilbert space $\mathcal{H}_L$ and vacuum vector $\Omega_L = [1]$. If the generated time-translation graph is a strongly continuous contraction semigroup, it has a nonnegative self-adjoint generator $H_L$. Failure of projective consistency, reflection positivity, nontriviality, or strong continuity remains its typed regulator-limit graph value; none of these properties is assumed in a spectral conclusion.*

*Proof.* Reflection positivity makes the displayed form positive semidefinite. Quotienting its null space and completing gives $\mathcal{H}_L$; the constant cylinder defines $\Omega_L$. The semigroup property descends from the time-translation graph, and contractivity follows from its ordered relation. A strongly continuous self-adjoint contraction semigroup has the form $e^{-tH_L}$ for a unique nonnegative self-adjoint generator [11, 14]. Every failed antecedent is already a graph coordinate of the regulator tower. □

**Definition III.5.39** (Prospective gap target)**.** At finite regulator, let $\Gamma_{\mathrm{gap}}$ be the normalized cone generated by vacuum-orthogonal low-energy filters and positive-time correlation shells. The target-active quotient removes all observable and regulator directions annihilated by $\Gamma_{\mathrm{gap}}$. Gap failure is the existence of

$$\|\psi_n\| = 1, \qquad \psi_n \perp \Omega_L, \qquad \langle \psi_n, H_L \psi_n\rangle \longrightarrow 0.$$

The source compiler assigns form geometry to $\mathsf{Geom}$, directed transfer to $\mathsf{Caus}$, gauge quotient to $\mathsf{Abs}$, refinement and block motion to $\mathsf{Lift}$, and ultraviolet, infrared, and volume faces to $\mathsf{Bdry}$.

**Theorem III.5.40** (Uniform spectral-gap proof/model alternative)**.** *For the target-aligned regulator path algebra, exactly one alternative holds:*

(i) *a finite ordered derivation gives a rational $\Delta > 0$ and proves, uniformly in the retained regulator and volume coordinates,*

$$\langle \psi, H\psi\rangle \geq \Delta\|\psi\|^2, \qquad \psi \perp \Omega;$$

(ii) *compatible feasible moment levels reconstruct a projectively consistent positive functional carrying a normalized vacuum-orthogonal zero-energy spectral model, together with its ultraviolet, infrared, gauge, topological, and boundary coordinates.*

*When the first alternative is combined with a zero-defect functional satisfying Theorem III.5.38, the reconstructed Hamiltonian has a mass gap at least $\Delta$. The theorem does not infer existence of the continuum theory from a spectral derivation alone.*

*Proof.* Adjoin the Rayleigh inequality with indeterminate rational $\Delta$, vacuum orthogonality, regulator consistency, and the prospective gap floor to the ordered regulator algebra. Apply Theorems II.2.36 and II.2.38. The dual branch is the uniform ordered inequality in (i). The primal branch is the positive functional in (ii). Under continuum reconstruction, the quadratic-form inequality implies $\sigma(H_L) \cap (0, \Delta) = \varnothing$ by the spectral theorem. □

## III.6 Conclusion

The continuum extension is a physical reachability theory generated from the same five-source algebra as Part I. The graph-complete approximation tower adds the compactness needed in function space without adding a source. Sourcewise completion retains every concentration, oscillation, defect, tail, trace, and germ in the ancestry cell that generated it. The residual is the post-completion target kernel and therefore carries no singular progress.

The decisive prospective statement is exact: contact crosses every target shell, and the derived cylinder-current identity assigns the resulting unit progress to five source-resolved Stieltjes currents. One source contributes at least $1/5$. Disjoint shell extraction then confronts that aligned source with the only budget in the theory, the countably additive physical expenditure generated by the equation's carrier law. Divergent physical price excludes the cascade. Finite price forces a zero-price same-origin successor model; it does not create an unresolved source.

Recurrent evaluation uses one further target-local quotient. Compatible carrier cells are glued with signs, and their aggregate feeding current is reduced modulo shift coboundaries, paired faces, exact chart and gauge currents, and the target kernel. Projection-first saturation permits this quotient before descendant generation; every failed descent is retained in the target-cyclic spine. Only the aggregate row $\mathfrak{J}_{\Sigma,\mathfrak{b}}$ of (406) is evaluated. A nonzero value supplies its signed-dyadic return law as the next source occurrence; a nonordinary value supplies its graph witness; and zero feeding forces the generated nonnegative production to vanish in the stationary law. Exact endpoint storage is retained separately and followed by the same return construction. This single passivity fast track replaces separate compactness, pressure, tail, scale, and equality estimates whenever those modes are support coordinates of the same signed current.

The recurrent price is computed by target-current circulation duality. The primal problem searches for a flow-conserving unit target return of zero physical price. The dual problem searches the generated cylinder algebra for a carrier potential whose coboundary absorbs target return into physical production. Positive dual value closes the recurrent cell; zero value constructs the circulation support used by the next refinement. Iterated circulation elimination has empty core exactly when the realized singular spine is empty. Empty core has a finite-depth local derivation, while a nonempty core is computed as a nested sequence of finite target-cylinder covers whose continuous structural readouts converge from both sides; its compatible words reconstruct the same-origin singular evolutions.

The final evaluation is algebraic. There are only $2^5 - 1$ nonempty source supports and the shell normalization fixes the target floor at $1/5$. Each support initializes one recurrent closure queue. Before a local cone is generated, forward source ancestry and backward target observability are intersected in the minimal target-active realization. Finite graph pruning therefore removes every coordinate that cannot lie on a complete source–target path. The remaining local cone enumerates finite rational exclusions; its complementary positive functional is immediately processed by zero-kernel restriction, signed-dyadic return, or typed graph continuation. Compactness, pressure, tail, scale, fidelity, and equality values are coordinates of this queue. The three-output theorem returns divergent physical charge, target-nullity, or a public-equation- faithful same-origin path contacting every cofinal target shell. Thus the continuous proof follows only structure aligned with singular formation and never estimates the full state family.

The resulting criterion is exact on each finite physical stratum: singular reachability is equivalent to the existence of a realized path in a reachable target spine. Global regularity is the emptiness of these realized spines. The queue evaluates that property using only local target currents, their physical valuation, and their generated return or graph transitions.

The Wick map gives a quantitative learning realization of an existing Part I compensated record. It complexifies the record and replaces only $-H_{\mathsf{Abs}}$ by $-iH_{\mathsf{Abs}}$; the real dissipative $\mathsf{Geom}$, authorized $\mathsf{Caus}$ bridge, authority, ancestry, and charges are inherited unchanged. The transformed record yields an exact self-energy, memory equation, physical budget, and least-action law. The positive full- response kernel transfers the learning bounds of Part I without imposing normality. Its Rademacher traces are the finite Part I evaluation compressions, while its population effective dimension is the totalized positive trace in $[0, \infty]$. Dynamics are fitted by a constrained block likelihood; unrestricted authority uses inverse-response kernel ridge regression on the zero-waste bridge fiber; and RL uses action-safe fitted Bellman regression with a Gibbs policy loss and an explicit deployment converter. The inverse-response norm is the exact minimum latent response energy. An exact-cell Rademacher union and learned-operator cover, or the structural PAC–Bayes mixture and coefficient-selection charge, select the family by the same rule as Part I. These quantities restrict the saturated profile to the declared risk tolerance; the result is the Pareto-minimal frontier of its strict inverse at the inherited target threshold. Minimum readout capacity, established by the target-subspace projection identity, remains an inner coordinate of each fixed response. The same learner also returns the unrestricted saturated profile at a resource/action ceiling. Free, hybrid, and imposed authority differ only through their legal current families; no target-aligned current is substituted for the selected dynamics.

# A Total branch table for the main conclusions

| Generated row | Finite exclusion identity | Complementary positive model |
|---|---|---|
| Approximation tower | ordered residual identity excludes a first-failure coordinate | positive graph-complete state with its least nonzero defect |
| Five-source completion | target pairing vanishes in the inherited cell | target-visible closure, harmonic, concentration, tail, or boundary coordinate with the same ancestry |
| Shell accountability | impossible identity for unit progress from five target-null currents | source-resolved current of mass at least 1/5 |
| Signed carrier | ordered domination by the finite physical measure | target-aligned zero-price same-origin windows |
| Gradient/Hamiltonian rows | commutator, resistance, capacity, or square identity enters $M_{A,1/5}$ | positive functional on the complementary closure or harmonic directions |
| Wick compensated quotient | invertible Feshbach response, positive response kernel, authorized least-action square, and minimum-energy observable lift | target-visible dark space, non-normal transfer defect, authority obstruction, or unresolved confidence interval |
| Scale and covariance rows | scale feed is an ordered coboundary or is absorbed by generated production | positive invariant scale-cohomology coordinate |

| | | |
|---|---|---|
| Target-aligned recurrent closure | finite target-local carrier or ordered identity closes the retained branch | zero-kernel, signed-dyadic return, or typed graph transition inside the single current queue |
| PDE regularity | every target-active resaturation route reaches a target-null, continuable, class-incompatible, or divergent-price cell | explicit closed reduced equation with a realized same-origin path contacting every cofinal target shell |

# B Notation index

| Symbol | Meaning |
|---|---|
| $\mathfrak{C}$ | the five-source alphabet $\{\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}\}$ |
| $\mathfrak{M}$ | the five analytic realization flags: exact, closure, harmonic, concentration, and boundary |
| $\widehat{X}_\Sigma, U_r(\Sigma), \Phi_r$ | target state-or-germ space, normalized shell neighborhoods, and shell observables |
| $B_{\mathrm{phys}}$ | sole equation-generated physical expenditure ceiling for an ordinary maximal branch |
| $\mathscr{R}_A^{\mathrm{phys,sat}}(B_{\mathrm{phys}})$ | sourcewise saturated physical record cell with support $A$ |
| $\mathrm{Anc}(R)$, $\mathrm{Src}(R)$ | full ancestry word and its elementary source support |
| $\mathscr{N}_\Sigma$ | common kernel of all prospective target functionals |
| $q_\Sigma$, $\mathscr{T}_{\Sigma,\mathfrak{b}}$ | quotient by the target kernel and the least projected successor-invariant structural spine generated by one active shell current |
| $\mathscr{R}^n_{\mathfrak{b}}, \mathscr{C}^n_{\mathfrak{b}}, \mathsf{M}^{m,n}_{\Sigma,\mathfrak{b}}$ | forward source-ancestry words, backward target-cyclic covectors, and their finite source–target incidence core |
| $\mathsf{M}_{\Sigma,\mathfrak{b}}$ | graph-complete minimal target-active realization |
| $J_{\mathrm{res}}$ | target-null post-saturation residual coordinate; never a source |
| $\mathrm{Align}_\Sigma(R)$ | source labels with positive target progress in a prospective record |
| $Q_{c,r}(u)$, $\mathrm{Prog}^c_\Sigma(R)$ | signed contribution of source $c$ to a normalized shell crossing or another prospective factorization |
| $\mathfrak{S}_\Sigma(u)$ | union of aligned labels in the orbit's shell-complete family; its singularity-formation support |
| $\mu_u$ | finite nonnegative localizable physical expenditure measure for one orbit |
| $\alpha_*$, $\delta_*$ | sharp domination radius and normalized physical price gap on a reduced source space |
| $\mathfrak{H}^\Sigma_{\mathcal{F}}$ | closed target projection of a candidate source family |
| $\mathfrak{I}^\Sigma_{A,m}$ | fast-track invariant tuple for source/mode cell $(A, m)$ |
| $\mathscr{R}_{\mathrm{FT}}$, $\mathcal{K}^\Sigma_i$ | ordered fast-track rows and the closed physical ledger propagated through row $i$ |
| $\mathfrak{F}^\Sigma_{A,m}(B_{\mathrm{phys}})$, $\mathfrak{F}^{\mathrm{phys}}_\Sigma(B_{\mathrm{phys}})$ | source/mode cells of stationary zero-price successor laws and their provenance-tagged total algebra |
| $\mathfrak{J}_{\Sigma,\mathfrak{b}}$, $\mathfrak{q}$ | aggregate target-projected feeding–storage row and one state of its recurrent closure queue |

| | |
|---|---|
| $\gamma, a_\Sigma, p_{\mathrm{phys}}, \delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}}$ | target-current circulation, normalized target return, admitted physical edge valuation, and its sharp recurrent cycle price |
| $\mathscr{U}_n, I_n(F)$ | decreasing finite target-cylinder cover at structural refinement depth $n$ and certified range enclosure for a generated target observable |
| $\mathcal{A}_{\mathrm{NS},n}, \mathscr{U}_{\mathrm{NS},n}$ | finite Navier–Stokes target algebra and its surviving target-spine cover |
| $X^{\min}_{\mathrm{NS},\mathfrak{b}}, \mathsf{M}_{\mathrm{NS},\Sigma,\mathfrak{b}}$ | minimal Navier–Stokes target-active state space and its bidirectional incidence presentation |
| $H_{\mathrm{NS}}, L_{V,a,b}, \Psi_{\mathrm{NS},\Sigma}$ | compressed Stokes Hamiltonian, moving-frame linearized generator, and target-active resolvent response |
| $\beta^{\mathrm{red}}_*$ | reduced passivity radius of feeding flux relative to gross expenditure |
| $\mathscr{C}_I, \mathfrak{W}_{\mathsf{Abs}}$ | inherited Part I compensated record and the map $-H_{\mathsf{Abs}} \mapsto -iH_{\mathsf{Abs}}$ fixing its other fields |
| $H_{\mathsf{Abs}}, G, C$ | inherited spectral quotient, real geometric operator, and authorized bridge fields of $\mathscr{C}_I$ |
| $\mathcal{L}_W, F_W, \Sigma_C$ | Wick block generator, quotient Feshbach operator, and geometric self-energy |
| $T_{W,\lambda}, N_W(\lambda,\rho)$ | positive Wick response kernel and its ridge effective dimension |
| $\mathrm{Tr}_{\mathscr{E}}, \mathrm{Tr}_{\mathrm{ext}}$ | Part I finite-carrier trace and its totalized positive supremum over finite-rank compressions |
| $\mathcal{D}_{\mathrm{dark}}$ | largest quotient spectral subspace invisible to the bridge |
| $\mathscr{W}_{\mathfrak{A}}$ | authority-relative completed-square structural waste action |
| $\mathcal{L}_{\mathrm{dyn}}, \widehat{\mathcal{J}}_{\mathrm{free}}$ | structure-preserving block-identification loss and unrestricted-Caus spectral oracle objective |
| $\mathcal{L}^{(k)}_{\mathrm{crit}}, \mathcal{L}_{\mathrm{actor}}, \mathcal{L}^{(k)}_{W,\mathrm{RL}}$ | action-safe fitted Bellman loss, Gibbs policy loss, and their complete alternating RL objective |
| $U^W_{\mathrm{Rad}}, U^W_{\mathrm{PB}}, U_W$ | inherited Rademacher, PAC–Bayes, and same-loss minimum family-selection certificates |
| $G^{\mathrm{sat}}_{Q,W,\mathfrak{A}}, \mathfrak{P}^{>}_{Q,W,\mathfrak{A}}$ | saturated Wick usefulness profile and its strict resource–action Pareto inverse |
| $\widehat{G}^{\mathrm{PB},\le\epsilon}_{Q,W,\mathfrak{A},n}, \widehat{\mathfrak{P}}^{\mathrm{PB},>,\le\epsilon}_{Q,W,\mathfrak{A},n}$ | PAC–Bayes-only certified Wick profile and strict inverse used in Theorem II.11.35 |
| $\mathfrak{o}_{\mathrm{tail}}$ | finite-dimensional gauge-cokernel class of an elliptic constraint tail |
| $\kappa$ | physical pullback exponent for a normalized scale event |
| $B_{\Sigma,\mathfrak{b}}$ | closed normalized target-approaching class in the generated evaluation topology of active branch $\mathfrak{b}$ |
| $\mathfrak{d}$ | lower-semicontinuous physical price functional on normalized windows |
| $\delta_{\mathrm{phys}}$ | positive lower physical price for a selected event |
| $\underline{p}_{A,j}$ | canonical scale-dependent lower-price infimum for a persistent aligned source row; its series is finite or divergent |
| $g_{\mathrm{car}}, \beta_\eta$ | corrected signed carrier cocycle and its maximal invariant mean on the retained target-active class |
| $\widehat{\mathcal{R}}_\eta$ | totalized witness-successor relation for atomic, separated, nested, diffuse, tail, rough, gauge, constraint, and equality descendants |
| $\mathfrak{H}^{\Sigma}_{\mathrm{succ}}$ | target projection of stationary zero-mean successor laws after generated saturation reduction |

| | |
|---|---|
| $Z_n^{\Sigma}, Z_\infty^{\Sigma}$ | finite-depth zero-price viability sets and their greatest compact successor fixed point |
| $E_n$ | open complements of the semantic finite-depth viability sets |
| $\mathcal{K}_{\mathfrak{b}}$ | ordered proof cone for one same-origin active branch signature; membership of $-1$ is a finite local exclusion identity |
| $\mathrm{Reach}_{\Sigma}^{\mathrm{phys}}(B_{\mathrm{phys}})$ | physical target-contact profile under the single ceiling $B_{\mathrm{phys}}$ |

## References


[1] Jean-Pierre Aubin. Un théorème de compacité. *Comptes Rendus de l'Académie des Sciences de Paris*, 256:5042–5044, 1963.

[2] Guillem Duran Ballester. Presentation-relative structural complexity: Quantitative limits for general intelligent computation, 2026. Part I of this series.

[3] Chandler Davis and W. M. Kahan. The rotation of eigenvectors by a perturbation. III. *SIAM Journal on Numerical Analysis*, 7(1):1–46, 1970. doi: 10.1137/0707001.

[4] Herman Feshbach. Unified theory of nuclear reactions. *Annals of Physics*, 5(4):357–390, 1958. doi: 10.1016/0003-4916(58)90007-1.

[5] Masatoshi Fukushima, Yoichi Oshima, and Masayoshi Takeda. *Dirichlet Forms and Symmetric Markov Processes*. De Gruyter, Berlin, second revised and extended edition, 2011.

[6] Arne Jensen, Eric Mourre, and Peter Perry. Multiple commutator estimates and resolvent smoothness in quantum scattering theory. *Annales de l'Institut Henri Poincaré, Physique Théorique*, 41(2):207–225, 1984.

[7] Tosio Kato. *Perturbation Theory for Linear Operators*. Springer, Berlin, second edition, 1995.

[8] Jacques-Louis Lions. *Quelques méthodes de résolution des problèmes aux limites non linéaires*. Dunod, Paris, 1969.

[9] Zhi-Ming Ma and Michael Röckner. *Introduction to the Theory of (Non-Symmetric) Dirichlet Forms*. Springer, Berlin, 1992.

[10] Hazime Mori. Transport, collective motion, and brownian motion. *Progress of Theoretical Physics*, 33(3):423–455, 1965. doi: 10.1143/PTP.33.423.

[11] Konrad Osterwalder and Robert Schrader. Axioms for euclidean green's functions. *Communications in Mathematical Physics*, 31:83–112, 1973.

[12] El Maati Ouhabaz. *Analysis of Heat Equations on Domains*. Princeton University Press, Princeton, 2005.

[13] Grisha Perelman. The entropy formula for the ricci flow and its geometric applications, 2002.

[14] Michael Reed and Barry Simon. *Methods of Modern Mathematical Physics, Volume I: Functional Analysis*. Academic Press, San Diego, revised and enlarged edition, 1980.

[15] Konrad Schmüdgen. *The Moment Problem*. Springer, Cham, 2017.

[16] Jacques Simon. Compact sets in the space $l^p(0, t; b)$. *Annali di Matematica Pura ed Applicata*, 146:65–96, 1987.

[17] Cédric Villani. Hypocoercivity. *Memoirs of the American Mathematical Society*, 202(950), 2009.

[18] Robert Zwanzig. Ensemble method in the theory of irreversibility. *The Journal of Chemical Physics*, 33(5):1338–1341, 1960. doi: 10.1063/1.1731409.